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(54) **TREATMENT OF MOST BOTHERSOME SYMPTOM (MBS) ASSOCIATED WITH MIGRAINE USING ANTI-CGRP ANTIBODIES**

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See application file for complete search history.

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ABSTRACT

Methods for treatment of most bothersome symptom (MBS) associated with migraine are provided. Exemplary methods provide improvement in MBS associated with migraine within 1 month of administration of anti-CGRP antibodies of the invention. Also provided are methods for improvement of patient impression of change (PGIC) associated with migraine. Exemplary methods comprise administration of an anti-CGRP antagonistic antibody to a patient in need thereof.

12 Claims, 23 Drawing Sheets

Specification includes a Sequence Listing.

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Figure 1 - Heavy Chain Protein Sequence

Sequence Name: Ab6

FR1	EVQLVESGGGLVQPGGSLRLSCAVSGIDLS	SEQ ID NO: 203
CDR1	GYMNM	SEQ ID NO: 204
FR2	WVRQAPGKGLEWVG	SEQ ID NO: 205
CDR2	VIGINGATYYASWAKG	SEQ ID NO: 206
FR3	RFTISRDNSTTVYLQMNSLRAEDTAVYFCAR	SEQ ID NO: 207
CDR3	GDI	SEQ ID NO: 208
FR4	WGQGTLLTVSS	SEQ ID NO: 209
Variable Region	EVQLVESGGGLVQPGGSLRLSCAVSGIDLSGYMNMWV RQAPGKGLEWVGIVIGINGATYYASWAKGRFTISRDN KTTVYLQMNSLRAEDTAVYFCARGDIWGQGTLLTVSS	SEQ ID NO: 202
Constant Region	ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPV TVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPS SLGTQTYICNVNHKPSNTKVDKRVKPKCDKTHTCPP CPAPPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVD VSHEDPEVKFNWYVDGVEVHNAKTKPREEQYASTYRV VSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKA KGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSD IAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVD KSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK	SEQ ID NO: 564
Alternative constant region	ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPV TVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPS SLGTQTYICNVNHKPSNTKVDKRVKPKCDKTHTCPP CPAPPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVD VSHEDPEVKFNWYVDGVEVHNAKTKPREEQYASTYRV VSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKA KGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSD IAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVD KSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPG	SEQ ID NO: 565
Full length Heavy Chain	EVQLVESGGGLVQPGGSLRLSCAVSGIDLSGYMNMWV RQAPGKGLEWVGIVIGINGATYYASWAKGRFTISRDN KTTVYLQMNSLRAEDTAVYFCARGDIWGQGTLLTVSS ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPV TVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPS SLGTQTYICNVNHKPSNTKVDARVEPKSCDKTHTCPP CPAPPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVD VSHEDPEVKFNWYVDGVEVHNAKTKPREEQYASTYRV VSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKA KGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSD IAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVD KSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK	SEQ ID NO: 201
Alternative Full length Heavy Chain	EVQLVESGGGLVQPGGSLRLSCAVSGIDLSGYMNMWV RQAPGKGLEWVGIVIGINGATYYASWAKGRFTISRDN KTTVYLQMNSLRAEDTAVYFCARGDIWGQGTLLTVSS ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPV TVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPS SLGTQTYICNVNHKPSNTKVDARVEPKSCDKTHTCPP CPAPPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVD VSHEDPEVKFNWYVDGVEVHNAKTKPREEQYASTYRV VSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKA KGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSD IAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVD KSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPG	SEQ ID NO: 566

Figure 2 - Light Chain Protein Sequence

Sequence Name: Ab6

FR1	QVLTQSPSSLSASVGDRVTINC	SEQ ID NO: 223
CDR1	QASQSVYHNTYLA	SEQ ID NO: 224
FR2	WYQQKPGKVPKQLIY	SEQ ID NO: 225
CDR2	DASTLAS	SEQ ID NO: 226
FR3	GVPSRFSGSGSGTDFTLTISSLQPEDVATYYC	SEQ ID NO: 227
CDR3	LGSYDCTNGDCFV	SEQ ID NO: 228
FR4	FGGGTKVEIKR	SEQ ID NO: 229
Variable Region	QVLTQSPSSLSASVGDRVTINCQASQSVYHNTYLAWY QQKPGKVPKQLIYDASTLASGVPSRFSGSGSGTDFTL TISSLQPEDVATYYCLGSYDCTNGDCFVFGGGTKVEI KR	SEQ ID NO: 222
Constant Region	TVAAPSVFIFPPSDEQLKSGTASVVCLLNFFYPREAK VQWKVDNALQSGNSQESVTEQDSKDYSLSTLTLS KADYEEKHKVYACEVTHQGLSSPVTKSFNRGEC	SEQ ID NO: 563
Full length light Chain	QVLTQSPSSLSASVGDRVTINCQASQSVYHNTYLAWY QQKPGKVPKQLIYDASTLASGVPSRFSGSGSGTDFTL TISSLQPEDVATYYCLGSYDCTNGDCFVFGGGTKVEI KRTVAAPSVFIFPPSDEQLKSGTASVVCLLNFFYPRE AKVQWKVDNALQSGNSQESVTEQDSKDYSLSTLT LSKADYEEKHKVYACEVTHQGLSSPVTKSFNRGEC	SEQ ID NO: 221

Figure 3A - Heavy Chain DNA Sequence

Sequence Name: Ab6

FR1	GAGGTGCAGCTTGTGGAGTCTGGGGGAGGCTTGGTCC AGCCTGGGGGGTCCCTGAGACTCTCCTGTGCAGTCTC TGGAAATCGACCTCAGT	SEQ ID NO: 213
CDR1	GGCTACTACATGAAC	SEQ ID NO: 214
FR2	TGGGTCCGTCAGGCTCCAGGGAAGGGGCTGGAGTGGG TCGGA	SEQ ID NO: 215
CDR2	GTCATTGGTATTAATGGTGCCACATACTACGCGAGCT GGGCGAAAGGC	SEQ ID NO: 216
FR3	CGATTACCATCTCCAGAGACAATTCCAAGACCACGG TGTATCTTCAAATGAACAGCCTGAGAGCTGAGGACAC TGCTGTGTATTTCTGTGCTAGA	SEQ ID NO: 217
CDR3	GGGGACATC	SEQ ID NO: 218
FR4	TGGGGCCAAGGGACCCTCGTCACCGTCTCGAGC	SEQ ID NO: 219
Variable Region	GAGGTGCAGCTTGTGGAGTCTGGGGGAGGCTTGGTCC AGCCTGGGGGGTCCCTGAGACTCTCCTGTGCAGTCTC TGGAAATCGACCTCAGTGGCTACTACATGAACTGGGTC CGTCAGGCTCCAGGGAAGGGGCTGGAGTGGGTCGGAG TCATTGGTATTAATGGTGCCACATACTACGCGAGCTG GGCGAAAGGCCGATTACCATCTCCAGAGACAATTCC AAGACCACGGTGTATCTTCAAATGAACAGCCTGAGAG CTGAGGACACTGCTGTGTATTTCTGTGCTAGAGGGGA CATCTGGGGCCAAGGGACCCTCGTCACCGTCTCGAGC	SEQ ID NO: 212
Full length Heavy Chain	GAGGTGCAGCTTGTGGAGTCTGGGGGAGGCTTGGTCC AGCCTGGGGGGTCCCTGAGACTCTCCTGTGCAGTCTC TGGAAATCGACCTCAGTGGCTACTACATGAACTGGGTC CGTCAGGCTCCAGGGAAGGGGCTGGAGTGGGTCGGAG TCATTGGTATTAATGGTGCCACATACTACGCGAGCTG GGCGAAAGGCCGATTACCATCTCCAGAGACAATTCC AAGACCACGGTGTATCTTCAAATGAACAGCCTGAGAG CTGAGGACACTGCTGTGTATTTCTGTGCTAGAGGGGA CATCTGGGGCCAAGGGACCCTCGTCACCGTCTCGAGC GCCTCCACCAAGGGCCCATCGGTCTTCCCCTGGGCAC CCTCCTCCAAGAGCACCTCTGGGGGCACAGCGGCCCT GGGCTGCCTGGTCAAGGACTACTTCCCCGAACCGGTG ACGGTGTCTGGAACTCAGGCGCCCTGACCAGCGGCG TGCACACCTTCCCGGCTGTCTACAGTCTCAGGACT CTACTCCCTCAGCAGCGTGGTGACCGTGCCCTCCAGC AGCTTGGGCACCCAGACCTACATCTGCAACGTGAATC ACAAGCCCAGCAACACCAAGGTGGACGCGAGAGTTGA GCCCCAATCTTGTGACAAAACCTCACACATGCCCCACG TGCCCAAGCACCTGAACTCCTGGGGGGACCGTCAGTCT TCCTCTTCCCCCAAAACCAAGGACACCCCTCATGAT CTCCCGGACCCCTGAGGTACATGCGTGGTGGTGGAC GTGAGCCACGAAGACCCTGAGGTCAAGTTCAACTGGT ACGTGGACGGCGTGGAGGTGCATAATGCCAAGACAAA GCCGCGGGAGGAGCAGTACGCCAGCACGTACCGTGTG GTCAGCGTCTCACCCTCCTGCACCAGGACTGGCTGA ATGGCAAGGAGTACAAGTGCAAGGTCTCCAACAAAGC CCTCCAGCCCCATCGAGAAAACCATCTCCAAGGCC AAAGGGCAGCCCCGAGAACCACAGGTGTACACCCTGC CCCCATCCCGGGAGGAGATGACCAAGAACCAGGTGAG CCTGACCTGCCTGGTCAAAGGCTTCTATCCAGCGAC	SEQ ID NO: 211

Figure 3B - Heavy Chain DNA Sequence

Sequence Name: Ab6

	ATCGCCGTGGAGTGGGAGAGCAATGGGCAGCCGGAGA ACAACTACAAGACCACGCCTCCCGTGCTGGACTCCGA CGGCTCCTTCTTCCTCTACAGCAAGCTCACCGTGGAC AAGAGCAGGTGGCAGCAGGGGAACGTCTTCTCATGCT CCGTGATGCATGAGGCTCTGCACAACCACTACACGCA GAAGAGCCTCTCCCTGTCTCCGGGTAAATGA	
Alternative Full length Heavy Chain	GAGGTGCAGCTTGTGGAGTCTGGGGGAGGCTTGGTCC AGCCTGGGGGGTCCCTGAGACTCTCCTGTGCAGTCTC TGGAAATCGACCTCAGTGGCTACTACATGAACTGGGTC CGTCAGGCTCCAGGGAAGGGGCTGGAGTGGGTCGGAG TCATTGGTATTAATGGTGCCACATACTACGCGAGCTG GGCGAAAGGCCGATTCAACATCTCCAGAGACAATTCC AAGACCACGGTGTATCTTCAAATGAACAGCCTGAGAG CTGAGGACACTGCTGTGTATTTCTGTGTAGAGGGGA CATCTGGGGCCAAGGGACCCTCGTCACCGTCTCGAGC GCCTCCACCAAGGGCCCATCGGTCTTCCCCCTGGCAC CCTCCTCCAAGAGCACCTCTGGGGGCACAGCGGCCCT GGGCTGCCTGGTCAAGGACTACTTCCCCGAACCGGTG ACGGTGTCTGTGGAACCTCAGGCGCCCTGACCAGCGGCG TGCACACCTTCCCGGCTGTCTACAGTCTCAGGACT CTACTCCCTCAGCAGCGTGGTGACCGTGCCCTCCAGC AGCTTGGGCACCCAGACCTACATCTGCAACGTGAATC ACAAGCCCAGCAACACCAAGGTGGACGCGAGAGTTGA GCCCCAAATCTTGTGACAAAACCTCACACATGCCACCG TGCCCAGCACCTGAACTCCTGGGGGGACCGTCAGTCT TCCTCTTCCCCCAAACCAAGGACACCCCTCATGAT CTCCCGGACCCCTGAGGTACATGCGTGGTGGTGGAC GTGAGCCACGAAGACCCTGAGGTCAAGTTCAACTGGT ACGTGGACGGCGTGGAGGTGCATAATGCCAAGACAAA GCCGCGGGAGGAGCAGTACGCCAGCACGTACCGTGTG GTCAGCGTCTCACCGTCTGCACCAGGACTGGCTGA ATGGCAAGGAGTACAAGTGCAAGGTCTCCAACAAAGC CCTCCCAGCCCCATCGAGAAAACCATCTCCAAGCC AAAGGGCAGCCCCGAGAACCACAGGTGTACACCCCTGC CCCCATCCCGGGAGGAGATGACCAAGAACCAGGTCAG CCTGACCTGCCTGGTCAAAGGCTTCTATCCCAGCGAC ATCGCCGTGGAGTGGGAGAGCAATGGGCAGCCGGAGA ACAACTACAAGACCACGCCTCCCGTGCTGGACTCCGA CGGCTCCTTCTTCCTCTACAGCAAGCTCACCGTGGAC AAGAGCAGGTGGCAGCAGGGGAACGTCTTCTCATGCT CCGTGATGCATGAGGCTCTGCACAACCACTACACGCA GAAGAGCCTCTCCCTGTCTCCGGGTGA	SEQ ID NO: 567

Figure 4 - Light Chain DNA Sequence

Sequence Name: Ab6

FR1	CAAGTGCTGACCCAGTCTCCATCCTCCCTGTCTGCAT CTGTAGGAGACAGAGTCACCATCAATTGC	SEQ ID NO: 233
CDR1	CAGGCCAGTCAGAGTGTTTATCATAACACCTACCTGG CC	SEQ ID NO: 234
FR2	TGGTATCAGCAGAAACCAGGGAAAGTTCCTAAGCAAC TGATCTAT	SEQ ID NO: 235
CDR2	GATGCATCCACTCTGGCATCT	SEQ ID NO: 236
FR3	GGGGTCCCATCTCGTTTCAGTGGCAGTGGATCTGGGA CAGATTTCACTCTCACCATCAGCAGCCTGCAGCCTGA AGATGTTGCAACTTATTACTGT	SEQ ID NO: 237
CDR3	CTGGGCAGTTATGATTGTACTAATGGTGATTGTTTTG TT	SEQ ID NO: 238
FR4	TTCGGCGGAGGAACCAAGGTGGAAATCAAACGT	SEQ ID NO: 239
Variable Region	CAAGTGCTGACCCAGTCTCCATCCTCCCTGTCTGCAT CTGTAGGAGACAGAGTCACCATCAATTGCCAGGCCAG TCAGAGTGTTTATCATAACACCTACCTGGCCTGGTAT CAGCAGAAACCAGGGAAAGTTCCTAAGCAACTGATCT ATGATGCATCCACTCTGGCATCTGGGGTCCCATCTCG TTTCAGTGGCAGTGGATCTGGGACAGATTTCACTCTC ACCATCAGCAGCCTGCAGCCTGAAGATGTTGCAACTT ATTACTGTCTGGGCAGTTATGATTGTACTAATGGTGA TTGTTTTGTTTTTCGGCGGAGGAACCAAGGTGGAAATC AAACGT	SEQ ID NO: 232
Full length light Chain	CAAGTGCTGACCCAGTCTCCATCCTCCCTGTCTGCAT CTGTAGGAGACAGAGTCACCATCAATTGCCAGGCCAG TCAGAGTGTTTATCATAACACCTACCTGGCCTGGTAT CAGCAGAAACCAGGGAAAGTTCCTAAGCAACTGATCT ATGATGCATCCACTCTGGCATCTGGGGTCCCATCTCG TTTCAGTGGCAGTGGATCTGGGACAGATTTCACTCTC ACCATCAGCAGCCTGCAGCCTGAAGATGTTGCAACTT ATTACTGTCTGGGCAGTTATGATTGTACTAATGGTGA TTGTTTTGTTTTTCGGCGGAGGAACCAAGGTGGAAATC AAACGTACGGTGGCTGCACCATCTGTCTTCATCTTCC CGCCATCTGATGAGCAGTTGAAATCTGGAACCTGCCTC TGTTGTGTGCCTGCTGAATAACTTCTATCCAGAGAG GCCAAAGTACAGTGGAAAGGTGGATAACGCCCTCCAAT CGGGTAACTCCCAGGAGAGTGTACAGAGCAGGACAG CAAGGACAGCACCTACAGCCTCAGCAGCACCTGACG CTGAGCAAAGCAGACTACGAGAAACACAAAGTCTACG CCTGCGAAGTCACCCATCAGGGCCTGAGCTCGCCCGT CACAAAGAGCTTCAACAGGGGAGAGTGTTAG	SEQ ID NO: 231

Figure 5

Heavy Chain Protein Sequence Features

Antibody	Variable Region Coordinates	SEQ ID NO:	CDR1 Coordinates	SEQ ID NO:	CDR2 Coordinates	SEQ ID NO:	CDR3 Coordinates	SEQ ID NO:
Ab6	1-111	202	31-35	204	50-65	206	98-100	208

Figure 6

Heavy Chain Protein Sequence Features

[illegible]

Figure 7

Light Chain Protein Sequence Features

Antibody	Variable Region Coordinates	SEQ ID NO:	CDR1 Coordinates	SEQ ID NO:	CDR2 Coordinates	SEQ ID NO:	CDR3 Coordinates	SEQ ID NO:
Ab6	1-113	222	23-35	224	51-57	226	90-102	228

Figure 8
Light Chain Protein Sequence Features

Antibo dy	FR1 Coordinat es	SEQ ID NO:	FR2 Coordi nates	SEQ ID NO:	FR3 Coordi nates	SEQ ID NO:	FR4 Coordin ates	SEQ ID NO:	Constant Region Coordinates	SEQ ID NO:
Ab6	1-22	223	36-50	225	58-89	227	103-113	229	114-219	563

Figure 9

Heavy Chain DNA Sequence Features

Antibody	Variable Region Coordinates	SEQ ID NO:	CDR1 Coordinates	SEQ ID NO:	CDR2 Coordinates	SEQ ID NO:	CDR3 Coordinates	SEQ ID NO:
Ab6	1-333	212	91-105	214	148-195	216	292-300	218

Figure 10

Heavy Chain DNA Sequence Features

Antibody	FR1 Coordin ates	SEQ ID NO:	FR2 Coordin ates	SEQ ID NO:	FR3 Coordin ates	SEQ ID NO:	FR4 Coordin ates	SEQ ID NO:
Ab6	1-90	213	106-147	215	196-291	217	301-333	219

Figure 11

Light Chain DNA Sequence Features

Antibody	Variable Region Coordinates	SEQ ID NO:	CDR1 Coordinates	SEQ ID NO:	CDR2 Coordinates	SEQ ID NO:	CDR3 Coordinates	SEQ ID NO:
Ab6	1-339	232	67-105	234	151-171	236	268-306	238

Figure 12

Light Chain DNA Sequence Features

Antibo dy	FR1 Coordi nates	SEQ ID NO:	FR2 Coordin ates	SEQ ID NO:	FR3 Coordin ates	SEQ ID NO:	FR4 Coordinat es	SEQ ID NO:
Ab6	1-66	233	106-150	235	172-267	237	307-339	239

Fig 13 - Study design of clinical trial of example 2

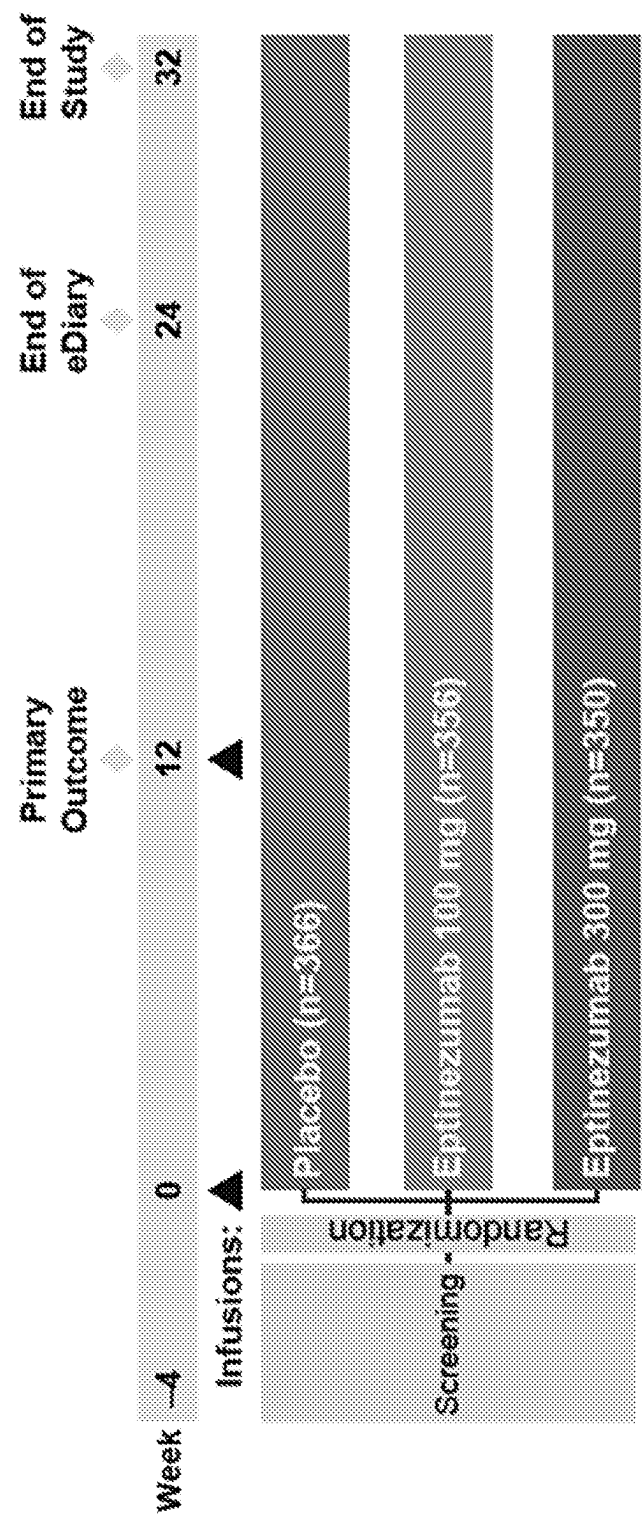


Fig 14 - Mean Monthly migraine days

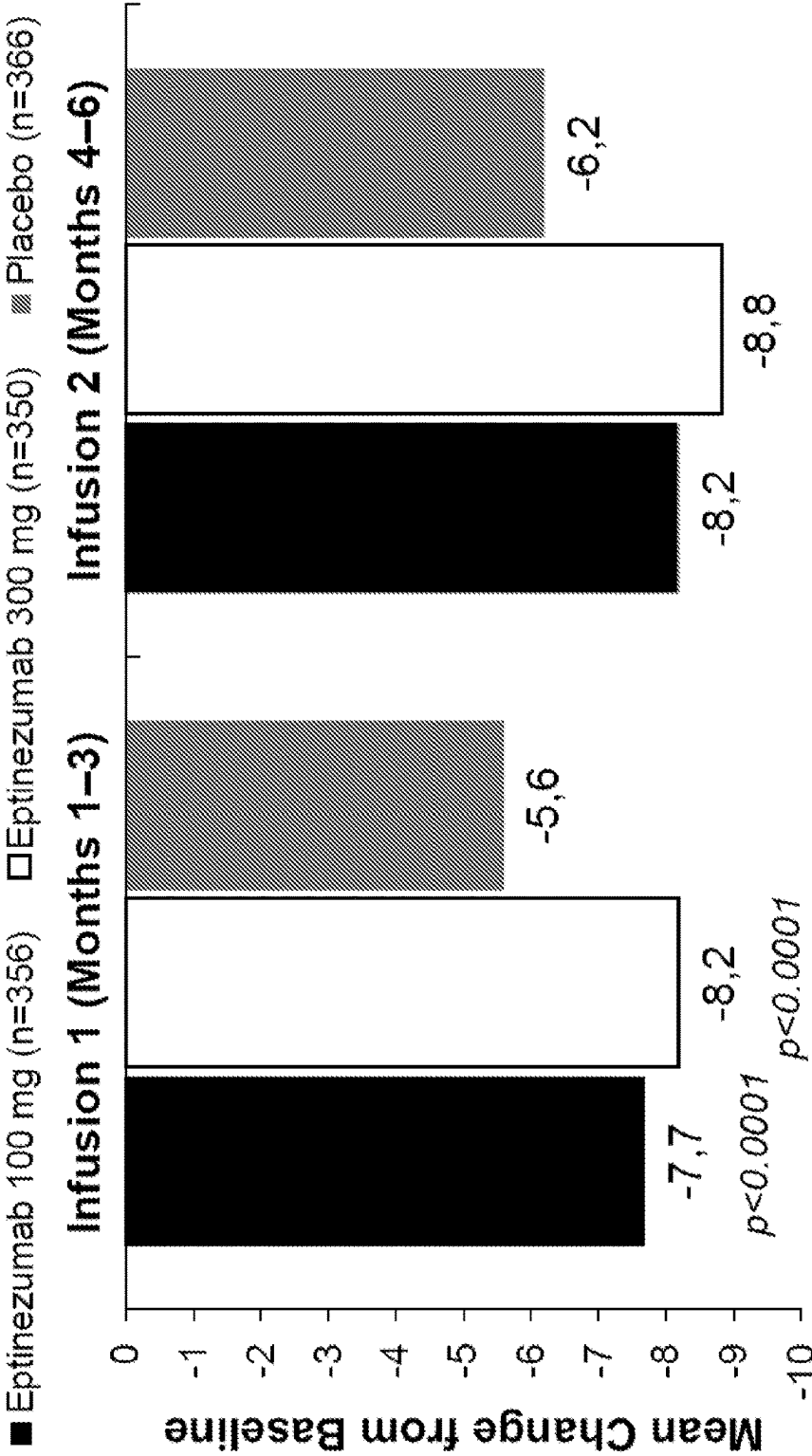


Fig. 15.—Most Bothersome Symptoms Across the Course of the Migraine

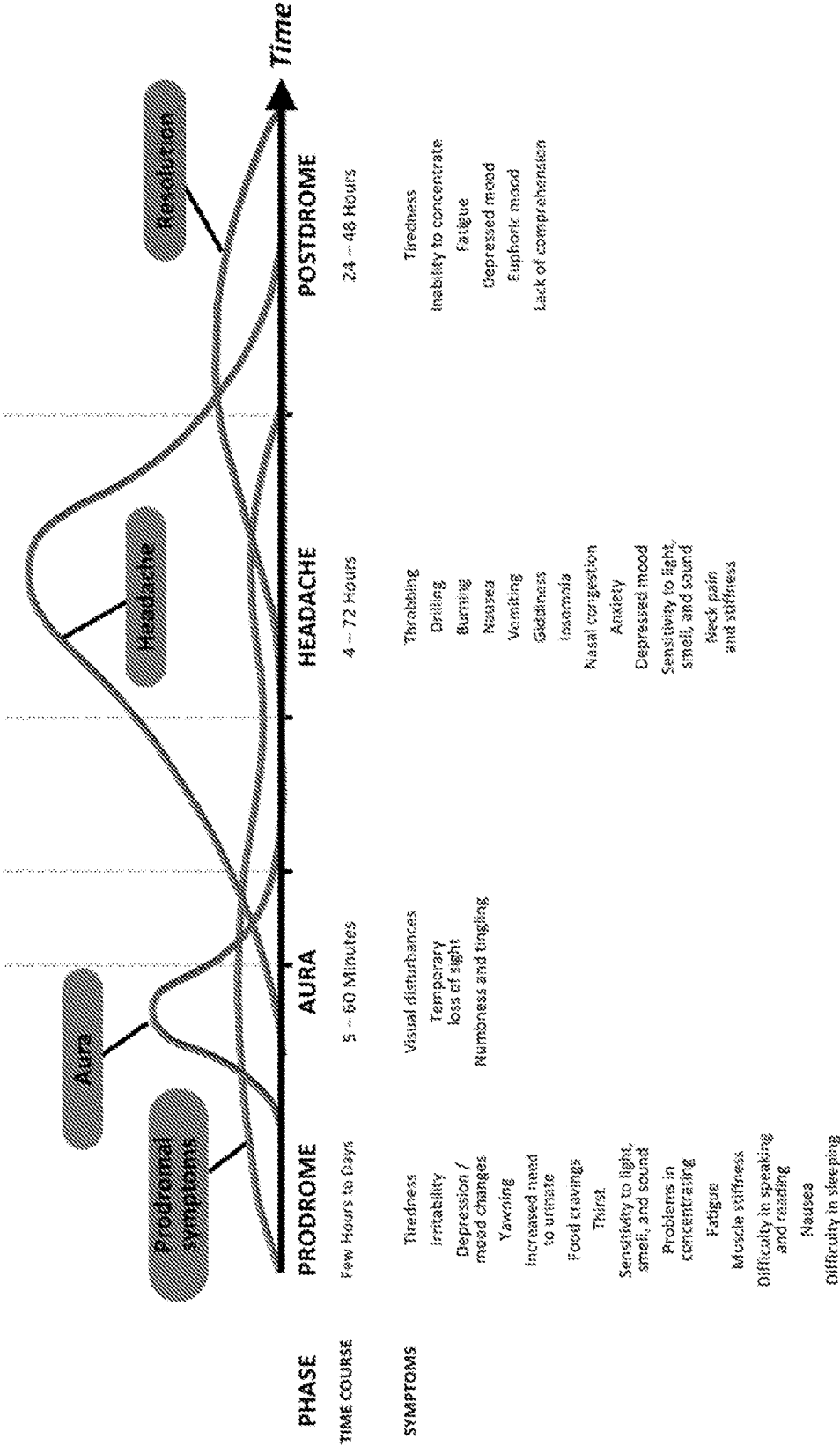


Fig 16 – MBS change from baseline during the 28 day screening period – i.e. before the first infusion of Ab6

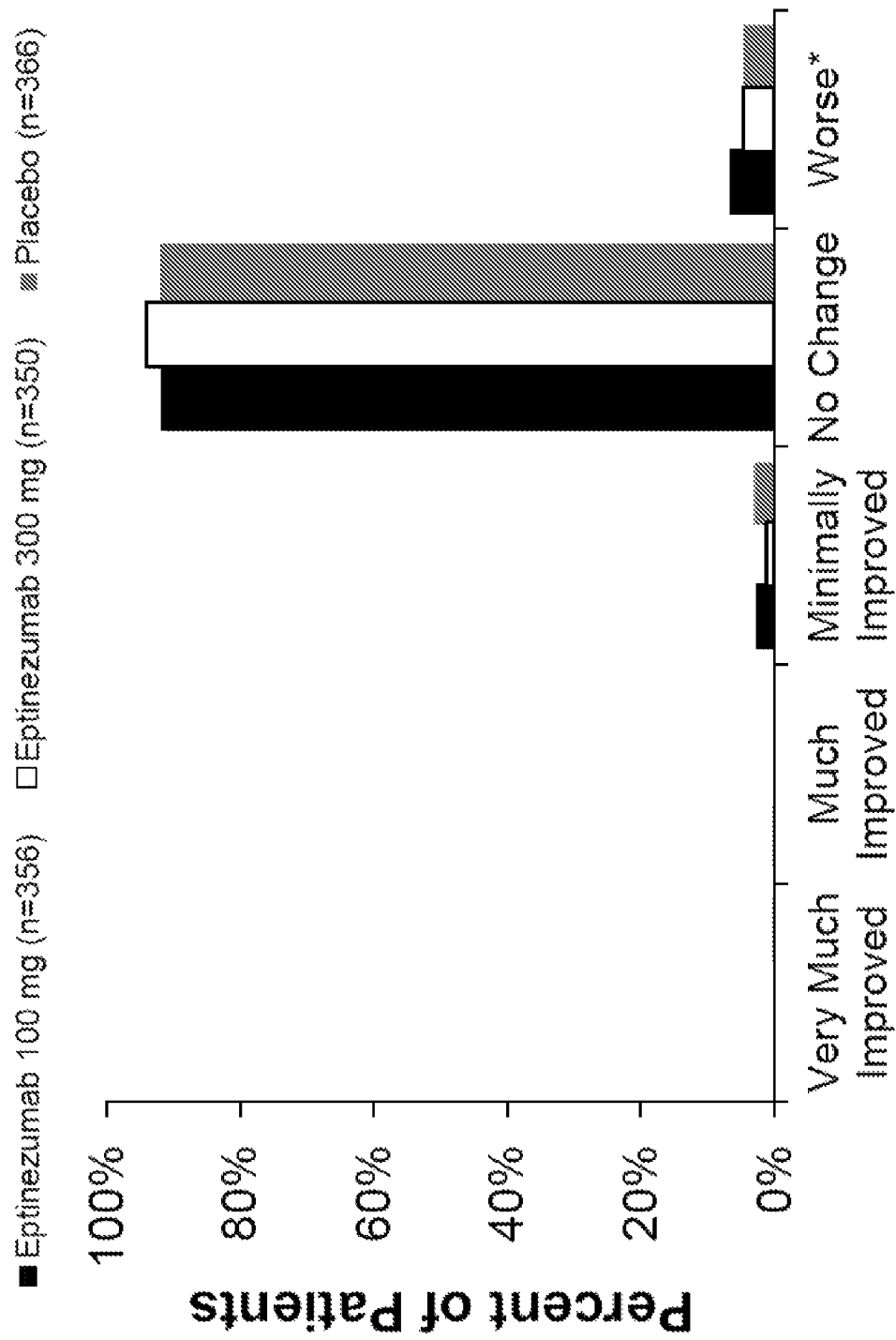


Fig 17 – MBS change from baseline 1 month after first infusion

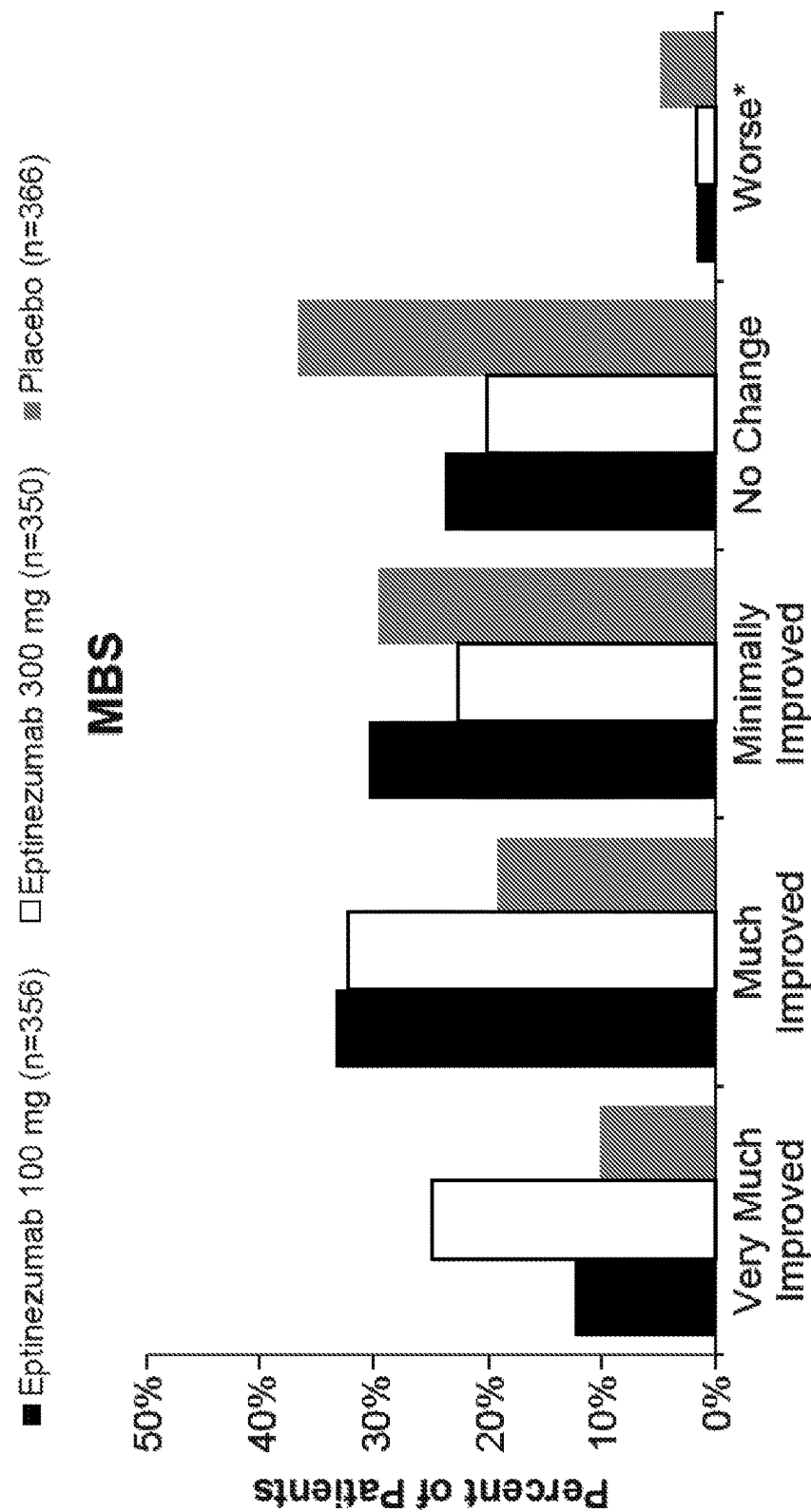


Fig 18 – PGIC from baseline 1 month after first infusion

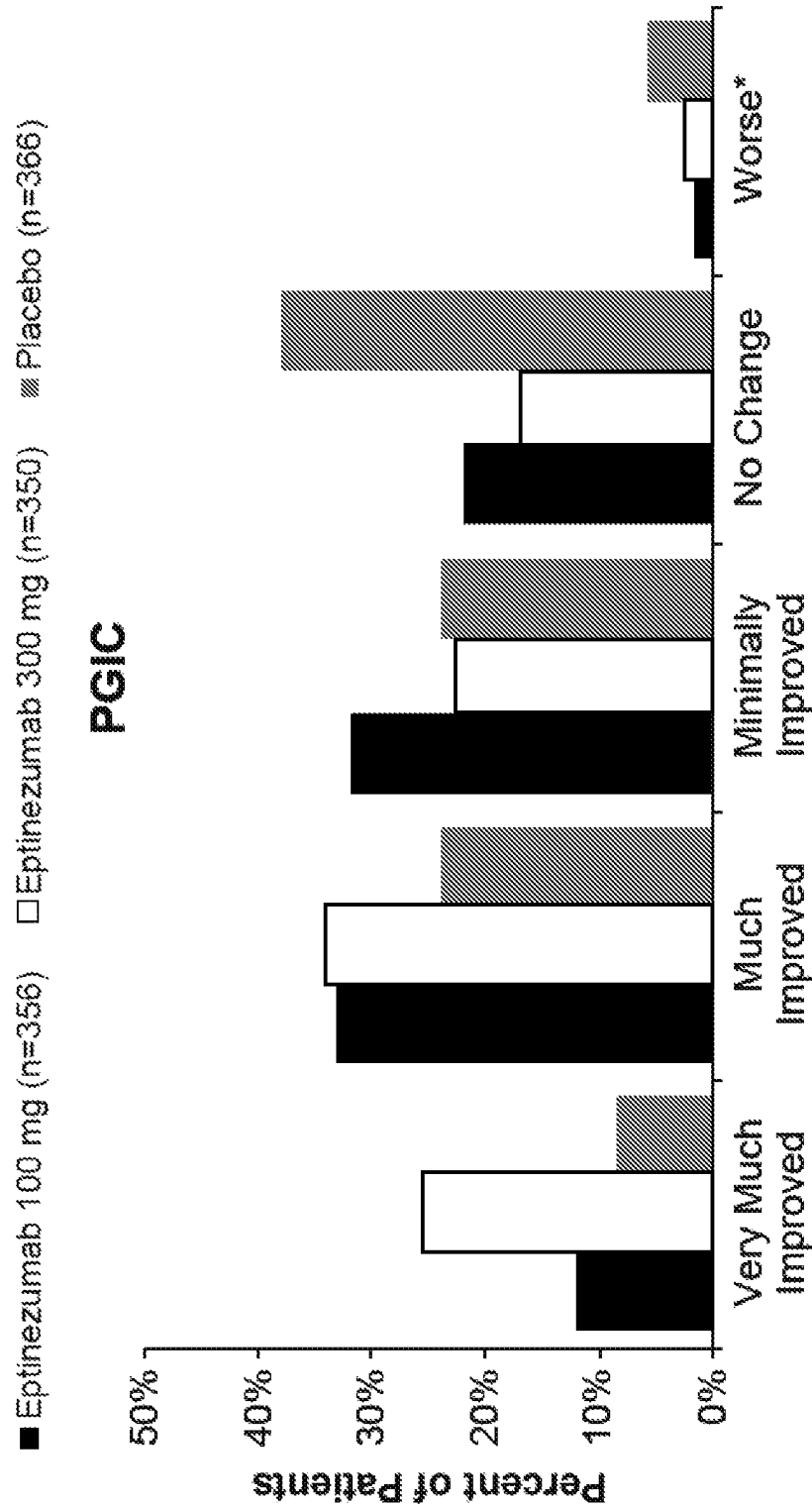


Fig 19 – MBS change from baseline 3 month after first infusion

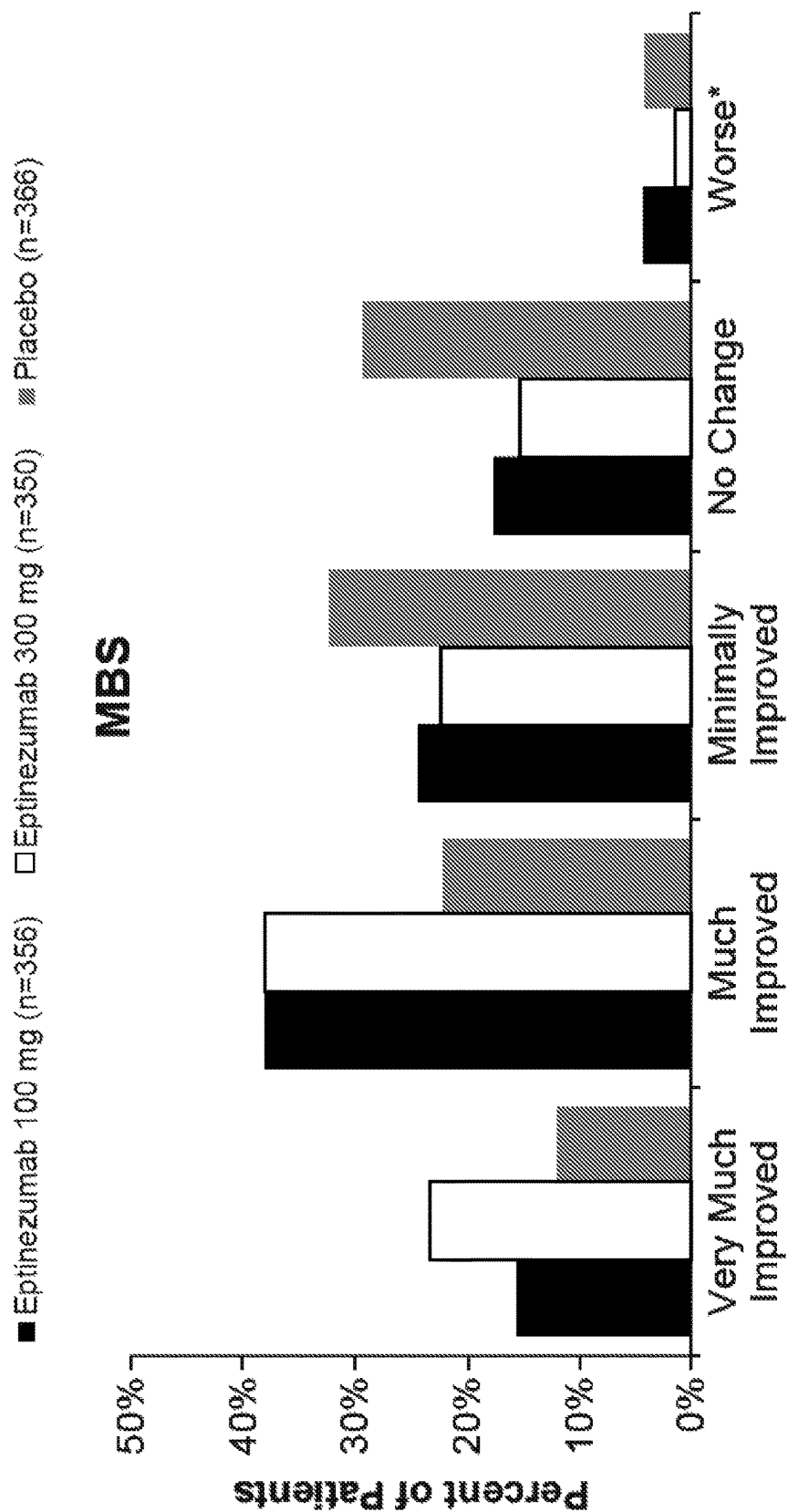


Fig 20 – PGIC from baseline 3 month after first infusion

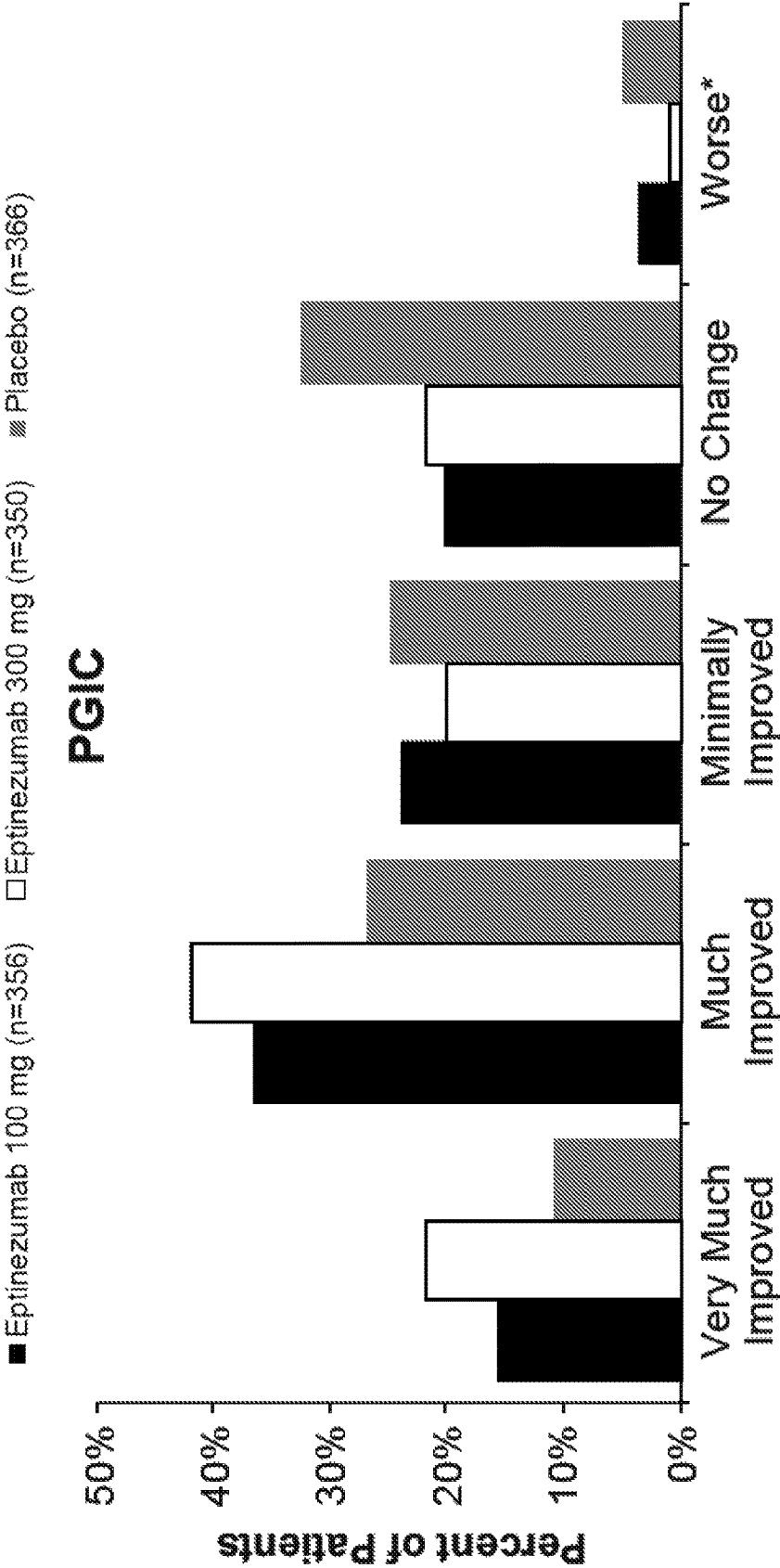


Fig 21 – MBS change from baseline 6 month after first infusion

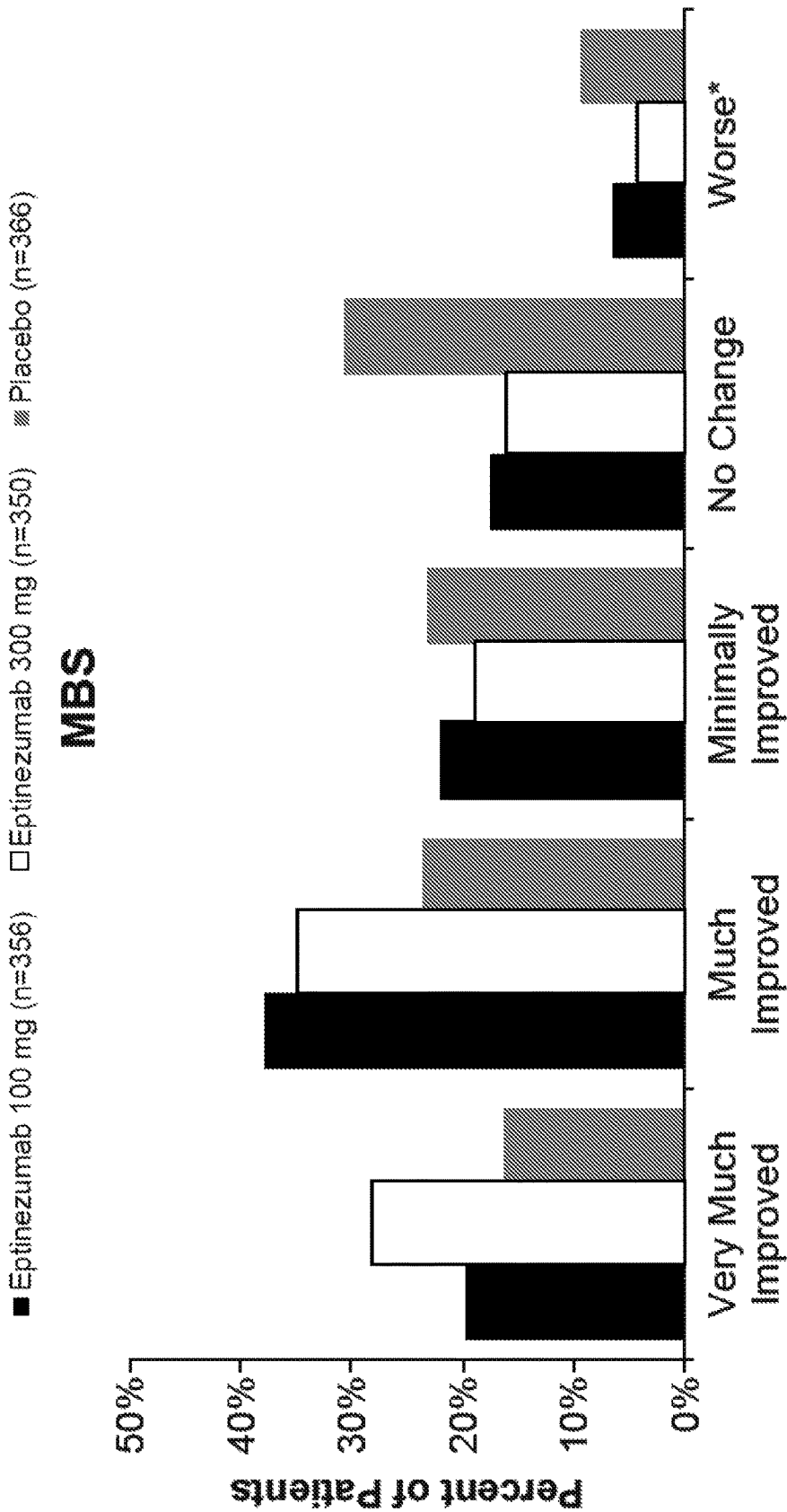
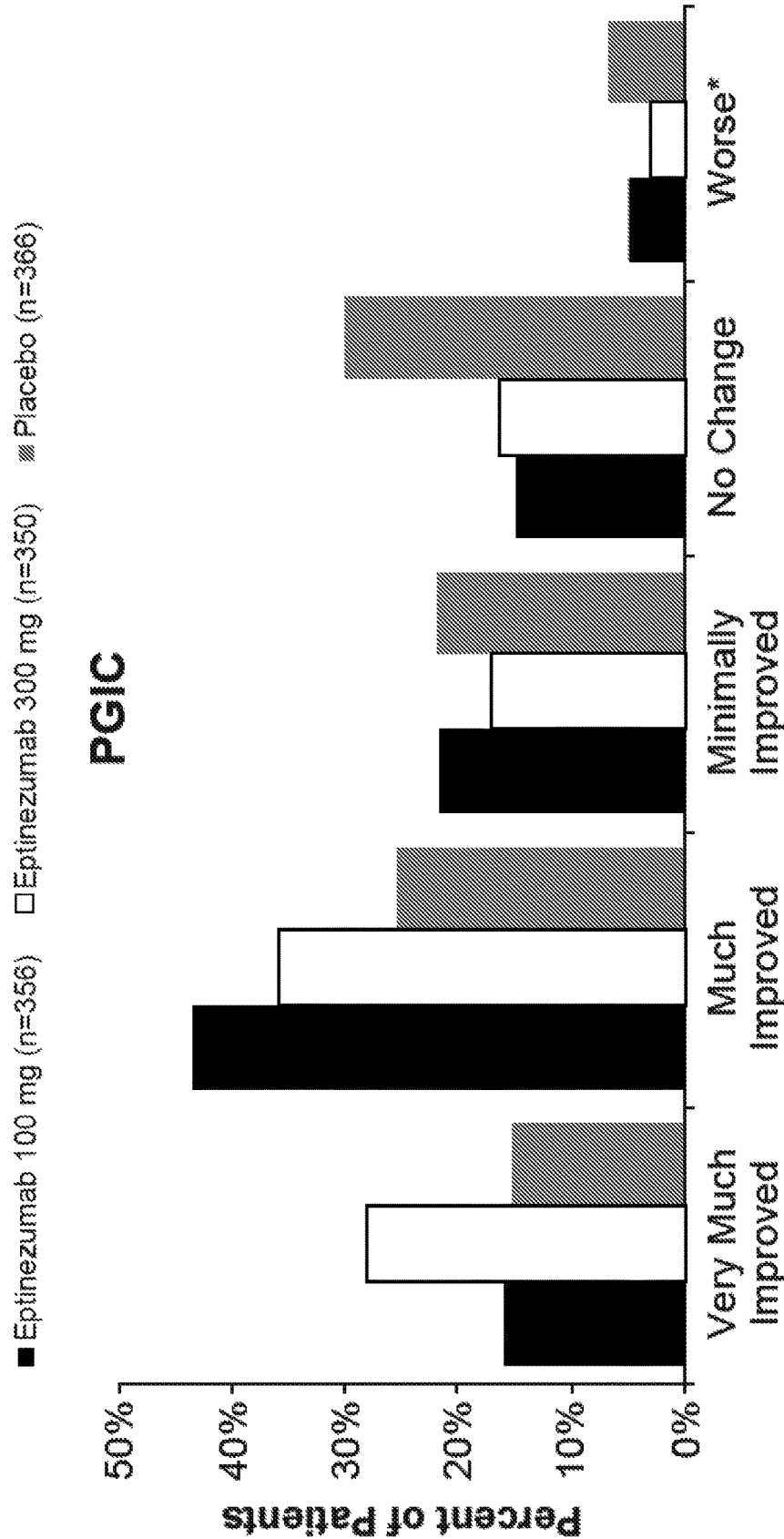


Fig 22 – PGIC from baseline 6 month after first infusion



1

TREATMENT OF MOST BOTHERSOME SYMPTOM (MBS) ASSOCIATED WITH MIGRAINE USING ANTI-CGRP ANTIBODIES

SEQUENCE LISTING DISCLOSURE

This application is a continuation of U.S. application Ser. No. 16/860,239 filed Apr. 28, 2020, which claims priority to U.S. Provisional Appl. No. 63/005,950, filed Apr. 6, 2020, which is hereby incorporated by reference in its entirety.

SEQUENCE LISTING DISCLOSURE

This application includes as part of its disclosure an electronic sequence listing text file named “1143257o009402.xml”, having a size of 771,211 bytes and created on Mar. 13, 2023, which is hereby incorporated by reference in its entirety.

SEQUENCES NOT PERMITTED TO BE ENTERED IN ST.26 XML FILE DUE TO SEQUENCE LENGTH

Table A below lists sequences present in the U.S. priority application Ser. No. 16/860,239 and 63/244,466 (identified above, which are both herein incorporated by reference in their entirety) but cannot be included in the “1143257o009402.xml” file submitted herewith due to the length of the sequences.

TABLE A

Sequence	Previous SEQ ID NO: #	Length	Type	Organism	Other Information
Gly Asp Ile	8	3	Protein	Artificial	Engineered antibody sequence
ggggacatc	18	9	DNA	Artificial	Engineered antibody sequence
Gly Asp Ile	48	3	Protein	Artificial	Engineered antibody sequence
ggggacatc	58	9	DNA	Artificial	Engineered antibody sequence
Gly Asp Ile	88	3	Protein	Artificial	Engineered antibody sequence
ggggacatc	98	9	DNA	Artificial	Engineered antibody sequence
Gly Asp Ile	128	3	Protein	Artificial	Engineered antibody sequence
ggggacatc	138	9	DNA	Artificial	Engineered antibody sequence
Gly Asp Ile	168	3	Protein	Artificial	Engineered antibody sequence
ggggacatc	178	9	DNA	Artificial	Engineered antibody sequence
Gly Asp Ile	208	3	Protein	Artificial	Engineered antibody sequence
ggggacatc	218	9	DNA	Artificial	Engineered antibody sequence
Gly Asp Ile	248	3	Protein	Artificial	Engineered antibody sequence
ggggacatc	258	9	DNA	Artificial	Engineered antibody sequence
Gly Asp Ile	288	3	Protein	Artificial	Engineered antibody sequence
ggggacatc	298	9	DNA	Artificial	Engineered antibody sequence
Gly Asp Ile	328	3	Protein	Artificial	Engineered antibody sequence
ggggacatc	338	9	DNA	Artificial	Engineered antibody sequence
Gly Asp Ile	368	3	Protein	Artificial	Engineered antibody sequence
ggggacatc	378	9	DNA	Artificial	Engineered antibody sequence
Gly Asp Ile	408	3	Protein	Artificial	Engineered antibody sequence
ggcgacatc	418	9	DNA	Artificial	Engineered antibody sequence
Gly Asp Ile	448	3	Protein	Artificial	Engineered antibody sequence
ggggacatc	458	9	DNA	Artificial	Engineered antibody sequence
Gly Asp Ile	528	3	Protein	Artificial	Engineered antibody sequence
ggggacatc	538	9	DNA	Artificial	Engineered antibody sequence

BACKGROUND OF THE INVENTION

Field of the Invention

This invention pertains to methods of treatment of most bothersome symptom associated with migraine, using antibodies and fragments thereof (including Fab fragments) that specifically bind to human Calcitonin Gene Related Peptide (hereinafter “CGRP”).

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Description of Related Art

Calcitonin Gene Related Peptide (CGRP) is produced as a multifunctional neuropeptide of 37 amino acids in length. Two forms of CGRP, the CGRP-alpha and CGRP-beta forms, exist in humans and have similar activities. CGRP-alpha and CGRP-beta differ by three amino acids in humans, and are derived from different genes. CGRP is released from numerous tissues such as trigeminal nerves, which when activated release neuropeptides within the meninges, mediating neurogenic inflammation that is characterized by vasodilation, vessel leakage, and mast-cell degradation. Durham, P. L., *New Eng. J. Med.*, 350 (11):1073-75 (2004). Biological effects of CGRP are mediated via the CGRP receptor (CGRP-R), which consists of a seven-transmembrane component, in conjunction with receptor-associated membrane protein (RAMP). CGRP-R further requires the activity of the receptor component protein (RCP), which is essential for an efficient coupling to adenylate cyclase through G proteins and the production of cAMP. Doods, H., *Curr. Op. Invest. Drugs*, 2(9):1261-68 (2001).

Migraines are neurovascular disorder affecting approximately 10% of the adult population in the U.S., and are typically accompanied by intense headaches. CGRP is believed to play a prominent role in the development of migraines. In fact, several companies, i.e., Amgen, Eli Lilly, Teva and Alder Biopharmaceuticals (recently acquired by Lundbeck A/S) have developed anti-CGRP and anti-

CGRP-R antibodies for use in treating or preventing migraine headaches. The present assignee has previously filed patent applications related to anti-CGRP antibodies and uses thereof including published PCT Application WO/2012/162243 filed May 21, 2012 entitled “ANTI-CGRP COMPOSITIONS AND USE THEREOF”, published PCT Application WO/2012/162257 filed May 21, 2012, entitled “USE OF ANTI-CGRP ANTIBODIES AND ANTIBODY FRAGMENTS TO PREVENT OR INHIBIT PHOTOPHOBIA OR LIGHT AVERSION IN SUBJECTS

IN NEED THEREOF, ESPECIALLY MIGRAINE SUFFERERS" published PCT Application WO/2012/162253, filed May 21, 2012, entitled "USE OF ANTI-CGRP OR ANTI-CGRP-R ANTIBODIES OR ANTIBODY FRAGMENTS TO TREAT OR PREVENT CHRONIC AND ACUTE FORMS OF DIARRHEA" and published PCT Application WO/2015/003122, filed Jul. 3, 2014, entitled "REGULATION OF GLUCOSE METABOLISM USING ANTI-CGRP ANTIBODIES" all of which applications are incorporated by reference in their entirety.

BRIEF SUMMARY

The present disclosure provides methods of treatment of most bothersome symptom (MBS) associated with migraine in patient suffering from chronic migraine, comprising administering to a patient in need an effective amount of at least one anti-CGRP antibody or antibody fragment thereof or an anti-CGRP-R antibody or antibody fragment thereof or one or more formulations comprising said antibody or antibody fragment as disclosed herein. Said antibody treatment may be initiated in the interictal period, i.e. in between migraine attacks or in the ictal phase, i.e. during the migraine episode. Said migraine may comprise e.g. chronic migraine or episodic migraine, in a specific aspect of the present invention the patient suffers from chronic migraine. In the present invention, said anti-CGRP antibody or antibody fragment is denoted Ab6. Ab6 is an anti-CGRP antibody or antibody fragment thereof having the light chain CDR 1, 2, and 3 polypeptide sequences of SEQ ID NO: 224; SEQ ID NO: 226; and SEQ ID NO: 228, respectively and the heavy chain CDR 1, 2, and 3 polypeptide sequences of SEQ ID NO: 204; SEQ ID NO: 206; and SEQ ID NO: 208; or having the light chain CDR 1, 2, and 3 polypeptide sequences encoded by SEQ ID NO: 234; SEQ ID NO: 236; and SEQ ID NO: 238, respectively and heavy chain CDR 1, 2, and 3 polypeptide sequences encoded by SEQ ID NO: 214; SEQ ID NO: 216; and SEQ ID NO: 218, respectively. Said anti-CGRP antibody may comprise the variable light chain polypeptide of SEQ ID NO: 222 and the variable heavy chain polypeptide of SEQ ID NO: 202. Said anti-CGRP antibody may comprise the variable light chain polypeptide encoded by SEQ ID NO: 232 and the variable heavy chain polypeptide encoded by SEQ ID NO: 212. Said anti-CGRP antibody may comprise the light chain polypeptide of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201 or SEQ ID NO: 566. Said anti-CGRP antibody may comprise the light chain polypeptide encoded by SEQ ID NO: 231 and the heavy chain polypeptide encoded by SEQ ID NO: 211 or SEQ ID NO: 567. Said anti-CGRP antibody may comprise the antibody expression product isolated from recombinant cells which express nucleic acid sequences encoding the variable light chain polypeptide of SEQ ID NO: 222 and the variable heavy chain polypeptide of SEQ ID NO: 202, which polypeptides optionally are respectively linked to human light and heavy constant region polypeptides, e.g., human IgG1, IgG2, IgG3 or IgG4 constant regions, which constant regions optionally may be modified to alter glycosylation or proteolysis, wherein said recombinant cells optionally comprise yeast or mammalian cells, e.g., *Pichia pastoris* or CHO cells. Said anti-CGRP antibody may comprise the antibody expression product isolated from recombinant cells which express nucleic acid sequences encoding the light chain of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201 or SEQ ID NO: 566, wherein said recombinant cells optionally comprise yeast or mammalian cells, e.g., *Pichia*

pastoris or CHO cells, wherein the constant regions thereof optionally may be modified to alter glycosylation or proteolysis or other effector functions. Any of the aforementioned anti-CGRP antibodies or antibody fragments, preferably Ab6, may be optionally comprised in a formulation as disclosed herein, e.g., comprising histidine (L-histidine), sorbitol, polysorbate 80, such as, per 1 mL volume, about 100 mg anti-CGRP antibody, about 3.1 mg L-Histidine, about 40.5 mg Sorbitol, and about 0.15 mg Polysorbate 80, having a pH of about 5.8. The administered dosage of said antibody may be between about 100 mg and about 300 mg, such as about 100 mg, about 300 mg, 100 mg, or 300 mg. The dosage may be administered by different means, e.g., intravenously, e.g., in a saline solution such as 0.9% sodium chloride in a suitable volume, such as 100 mL.

Said patient may exhibit less than 25 headache days per month, less than 20 headache days per month, less than 15 headache days per month, or less than 10 headache days per month. For example, said patient may exhibit less than 14 headache days, less than 13 headache days, less than 12 headache days, less than headache 11 days, less than 10 headache days, less than 9 headache days, less than 8 headache days, less than 7 headache days, or less than 6 headache days per month. Said patient may exhibit between 2-15 headache days, e.g., 3-14 headache days, 4-13 headache days, 5-12 headache days, 6-11 headache days, or 7-10 headache days/month.

Said patient may exhibit less than 10 migraines per month, such as between 1-9 migraines per month, such as between 2-8 migraines per month, between 3-7 migraine per month, between 4-6 migraine per month, or about 5 migraines per month. Said patient may exhibit fewer than 1 migraine per month on average, e.g., on average one migraine every 2 months, one every 3 months, one every 4 or 6 months, or intermediate values such as 2 every 3 months, etc. Said migraine may be diagnosed in accord with the ICHD-3 guidelines.

In addition to headache and associated symptoms as described in the diagnostic criteria of the International Classification of Headache Disorders (ICHD-3) for migraine with or without aura, migraine patients experience a variety of autonomic, cognitive, sensory and motor symptoms during migraine, these symptoms are experienced uniquely by individual patients. In the present invention, the patients were allowed to self-identify a specific symptom associated with chronic migraine that they considered to be most bothersome. In the present application these symptoms will be referred to at the most bothersome symptom (MBS) associated with migraine. In the present invention the patient could identify their MBS without limitation, which provides a unique patient-centered approach for identifying and measuring the efficacy of antibodies of the invention as treatment of these most bothersome migraine-associated symptoms and hence is expected to have a meaningful impact on the patients ability to function during migraine. Although nausea, vomiting, photophobia, and phonophobia are migraine-associated symptoms included in ICDH-3 diagnostic criteria, many other symptoms may be observed to occur prior to, after, and even between days with diagnosable migraine. Over the duration of a migraine attack, these can include cognitive symptoms (e.g. memory, executive function, attention deficit), affective symptoms (e.g. mood changes, depression, anxiety, irritability), other sensory symptoms (e.g. osmophobia, taste abnormalities), as well as blurry vision, nasal congestion, rhinorrhea, lacrimation, sweating, ptosis, yawning, polyuria, abdominal cramps, diarrhea, dizziness, and neck pain. The MBS associated with migraine

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reported by the patients enrolled in the clinical trial described in Example 2 is summarized in Table 1. Although nausea/vomiting, photophobia, and phonophobia were common in the patient population in Example 2, less than half of these patients named one of these 3 symptoms included in ICDH-3 diagnostic criteria as their patient-identified MBS.

Migraine is a complex disorder of the brain associated with multifaceted symptomatology yet expressed in a personalized unique manner. Often persisting over multiple days, the peri-ictal period of migraine can be classified into four distinct phases—prodrome/premonitory, preictal/aura, ictal/headache, postdrome/postictal—with overlapping symptoms occurring during each phase of migraine. The various types and timing of MBS across the course of the migraine is illustrated in FIG. 15. It is highly relevant to assess MBS in migraine patients during clinical trials, since it is recognized that headache pain alone is not considered sufficient to adequately eliminate the impact of migraine on the patients daily living and health status. The reduction in mean monthly migraine days (MMDs) or a similar endpoints in clinical trials do not fully capture the burden of migraine and the associated symptoms that are affected by therapeutic intervention. The inventors of the present invention found that in addition to reducing MMDs Ab6, an anti-CGRP antibody, was also effective in improving MBS in migraine patients. Improvements in these symptoms associated with treatment were correlated with improved patients' perception of disease status and indirectly with satisfaction with treatment response. It is known that migraine patients often continue to seek treatment for their migraine because of the burden of their MBS, thus supporting the clinical value of treating both the primary migraine pathology and the MBS associated with said migraine.

The present invention provides anti-CGRP antibodies or antibody fragments thereof, which are able to improve the MBS associated with migraine in patients suffering from migraine, such as chronic or episodic migraine. The MBS parameter rates the patient's assessment of change (improvement or worsening since the start of the study) in this symptom.

The present invention provides anti-CGRP antibodies or antibody fragments thereof, which are able to improve the patient global impression of change (PGIC) associated with migraine treatment in patients impacted by migraine, such as chronic or episodic migraine. The patient global impression of change (PGIC) associated with migraine parameter comprises a single question concerning the patient's impression of the overall change (improvement or worsening since the start of the study) in their disease status evaluated on a 7 point Likert scale anchored by very much improved and very much worse.

The present invention provides anti-CGRP antibodies or antibody fragments thereof, which are able to reduce MMDs as well as improve the patient's most bothersome symptom (MBS) associated with migraine in a manner that is highly correlated with positive change in the patient's global impression of change (PGIC) of migraine treatment. This dual action constitutes an improved treatment option for patient suffering from migraine, which goes beyond treating the migraine headache, and provides treatment for the collective migraine burden experienced by the patient comprising both migraine headache as well as MBS associated with migraine.

The present invention provides methods of improving most bothersome symptom (MBS) associated with migraine, comprising intravenously administering to a patient in need thereof between about 100 mg and about 300 mg of an

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anti-CGRP antibody, wherein said anti-CGRP antibody preferably comprises the light chain polypeptide of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201 or 566.

The present invention provides methods of improving patient global impression of change (PGIC), comprising intravenously administering to a patient in need thereof between about 100 mg and about 300 mg of an anti-CGRP antibody, wherein said anti-CGRP antibody preferably comprises the light chain polypeptide of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201 or 566.

In another aspect, the invention provides methods of improving most bothersome symptom (MBS) associated with migraine and simultaneously reduce the MMDs, comprising intravenously administering to a patient in need thereof between about 100 mg and about 300 mg of an anti-CGRP antibody, wherein said anti-CGRP antibody preferably comprises the light chain polypeptide of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201 or 566.

In another aspect, the invention provides methods of improving patient global impression of change (PGIC) associated with migraine and simultaneously reduce the MMDs, comprising intravenously administering to a patient in need thereof between about 100 mg and about 300 mg of an anti-CGRP antibody, wherein said anti-CGRP antibody preferably comprises the light chain polypeptide of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201 or 566.

In another aspect, the invention provides methods of improving most bothersome symptom (MBS) associated with migraine and patient global impression of change (PGIC) associated with migraine, comprising intravenously administering to a patient in need thereof between about 100 mg and about 300 mg of an anti-CGRP antibody, wherein said anti-CGRP antibody preferably comprises the light chain polypeptide of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201 or 566.

In another aspect, the invention provides methods of improving most bothersome symptom (MBS) associated with migraine and/or patient global impression of change (PGIC) associated with migraine and simultaneously reduce the MMDs, comprising intravenously administering to a patient in need thereof between about 100 mg and about 300 mg of an anti-CGRP antibody, wherein said anti-CGRP antibody preferably comprises the light chain polypeptide of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201 or 566.

In some exemplary embodiments the dosage of said anti-CGRP antibody may be 100 mg.

In other exemplary embodiments the dosage of said anti-CGRP antibody may be 300 mg.

The method may further comprise intravenously administering 100 mg of said anti-CGRP antibody every 10-14 weeks, preferably every 11-13 weeks, more preferably every 12 weeks.

The method may further comprise intravenously administering 300 mg of said anti-CGRP antibody every 10-14 weeks, preferably every 11-13 weeks, more preferably every 12 weeks.

The antibody may be provided or administered in a formulation as disclosed herein, e.g., comprising histidine (L-histidine), sorbitol, polysorbate 80, such as, per 1 mL volume, about 100 mg anti-CGRP antibody, about 3.1 mg L-Histidine, about 40.5 mg Sorbitol, and about 0.15 mg Polysorbate 80, having a pH of about 5.8.

Prior to first dosage, the patient may exhibit between about 10 and about 22 migraine days per month, such as between about 13 and about 19 migraine days per month, such as about 16 migraine days per month.

Prior to first dosage, the patient may exhibit between about 14 and about 27 headache days per month, such as between about 17 and about 24 headache days per month, such as about 20 or about 21 headache days per month.

Said patient may have been diagnosed with migraine at least 10 years prior to said first dosage, such as at least 15 years prior to said first dosage, such as at least 18 or at least 19 years prior to said first dosage.

Said patient may have been diagnosed with chronic migraine at least 5 years prior to said first dosage, such as at least 8 years prior to said first dosage, such as at least 11 or at least 12 years prior to said first dosage.

The patient may have a headache when administered the anti-CGRP antibody or fragments thereof of the invention.

The patient may have a migraine, such as a migraine with aura, when administered anti-CGRP antibody or fragments thereof of the invention.

Said patient may have a reduction in the number of migraine days by at least 50% in the one month period after being administered said first dose relative to the baseline number of migraine days experienced by that patient prior to said first dose.

Said patient may have a reduction in the number of migraine days by at least 75% in the one month period after being administered said first dose relative to the baseline number of migraine days experienced by that patient prior to said first dose.

Said patient may have a reduction in the number of migraine days by 100% in the one month period after being administered said first dose relative to the baseline number of migraine days experienced by that patient prior to said first dose.

Said patient may have a reduction in the number of migraine days by at least 50% in the 12 week period after being administered said first dose relative to the baseline number of migraine days experienced by that patient prior to said first dose.

Said patient may have a reduction in the number of migraine days by at least 75% in the 12 week period after being administered said first dose relative to the baseline number of migraine days experienced by that patient prior to said first dose.

Said patient may have a reduction in the number of migraine days by 100% in the 12 week period after being administered said first dose relative to the baseline number of migraine days experienced by that patient prior to said first dose.

Said patient may experience an improvement in their MBS associated with migraine in the one month period after being administered said first dose measured as the change from the baseline MBS.

Said patient may experience an improvement in their MBS associated with migraine in the 3 month period after being administered said first dose measured as the change from the baseline MBS.

Said patient may experience an improvement in their MBS associated with migraine in the 6 month period after being administered said first dose measured as the change from the baseline MBS.

Said patient may experience an improvement in their PGIC associated with migraine in the one month period after being administered said first dose measured as the change from the baseline.

Said patient may experience an improvement in their PGIC associated with migraine in the 3 month period after being administered said first dose measured as the change from the baseline.

Said patient may experience an improvement in their PGIC associated with migraine in the 6 month period after being administered said first dose measured as the change from the baseline.

The method may further comprise administering, e.g., intravenously, a second dose of an anti-CGRP antibody of the invention to said patient within about 10-14 weeks, preferably 11-13 weeks, more preferably about 12 weeks or about 3 months, after said first dose.

Said first dose may comprise about 100 mg, about 125 mg, about 150 mg, about 175 mg, about 200 mg, about 225 mg, about 250 mg, about 275 mg, or about 300 mg of said anti-CGRP antibody.

Prior to said administration, the patient may exhibit between about 15 and about 30 migraine days per month, such as between about 16 and about 28 migraine days per month, such as between about 17 and about 26 migraine days per month, such as about 16 migraine days per month.

Prior to said administration, the patient may exhibit between about 15 and about 27 headache days per month, such as between about 17 and about 24 headache days per month, such as about 20 or about 21 headache days per month.

Said patient may have been diagnosed with migraine at least 10 years prior to said administration, such as at least 15 years prior to said administration, such as at least 18 or at least 19 years prior to said administration.

Said patient may have been diagnosed with chronic migraine at least 5 years prior to said administration, such as at least 8 years prior to said administration, such as at least 11 or at least 12 years prior to said administration.

Said patient may have a reduction in the number of migraine days by at least 50% in the one month period after being administered said antibody relative to the baseline number of migraine days experienced by that patient prior to said administration.

Said patient may have a reduction in the number of migraine days by at least 75% in the one month period after being administered said antibody relative to the baseline number of migraine days experienced by that patient prior to said administration.

Said patient may have a reduction in the number of migraine days by 100% in the one month period after being administered said antibody relative to the baseline number of migraine days experienced by that patient prior to said administration.

Said patient may have a reduction in the number of migraine days by at least 50% in the 12 week period after being administered said antibody relative to the baseline number of migraine days experienced by that patient prior to said administration.

Said patient may have a reduction in the number of migraine days by at least 75% in the 12 week period after being administered said antibody relative to the baseline number of migraine days experienced by that patient prior to said administration.

Said patient may have a reduction in the number of migraine days by 100% in the 12 week period after being administered said antibody relative to the baseline number of migraine days experienced by that patient prior to said administration.

The method may further comprise administering, e.g., intravenously, a second dose of said anti-CGRP antibody to

said patient within about 10-14 weeks, preferably 11-13 weeks, more preferably about 12 weeks or about 3 months, after said administration.

Said administration may comprise about 100 mg, about 125 mg, about 150 mg, about 175 mg, about 200 mg, about 225 mg, about 250 mg, about 275 mg, or about 300 mg of said anti-CGRP antibody.

Said anti-CGRP antibody may be aglycosylated or if glycosylated only may contain only mannose residues.

Said anti-CGRP antibody may consist of the light chain polypeptide of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201 or SEQ ID NO: 566. Said anti-CGRP antibody may consist of the light chain polypeptide encoded by SEQ ID NO: 231 and the heavy chain polypeptide encoded by SEQ ID NO: 211 or SEQ ID NO: 567.

In some embodiments, said anti-human CGRP antibody or antibody fragment comprises the variable light chain of SEQ ID NO: 222 and/or the variable heavy chain of SEQ ID NO: 202. In some embodiments, said anti-human CGRP antibody or antibody fragment comprises the variable light chain encoded by SEQ ID NO: 232 and/or the variable heavy chain encoded by SEQ ID NO: 212.

In some embodiments, said anti-human CGRP antibody or antibody fragment comprises the light chain of SEQ ID NO: 221 and/or the heavy chain of SEQ ID NO: 201 or SEQ ID NO: 566. In some embodiments, said anti-human CGRP antibody or antibody fragment comprises the light chain encoded by SEQ ID NO: 231 and/or the heavy chain encoded by SEQ ID NO: 211 or SEQ ID NO: 567.

In some embodiments, said anti-CGRP antibody may comprise the antibody expression product isolated from recombinant cells which express nucleic acid sequences encoding the VL polypeptide of SEQ ID NO: 222 and the VH polypeptide of SEQ ID NO: 202, which polypeptides optionally are respectively linked to human light and heavy constant region polypeptides, e.g., human IgG1, IgG2, IgG3 or IgG4 constant regions, which constant regions optionally may be modified to alter glycosylation or proteolysis, wherein said recombinant cells optionally comprise yeast or mammalian cells, e.g., *Pichia pastoris* or CHO cells.

In some embodiments, said anti-CGRP antibody may comprise the antibody expression product isolated from recombinant cells which express nucleic acid sequences encoding the light chain of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201 or SEQ ID NO: 566, wherein said recombinant cells optionally comprise yeast or mammalian cells, e.g., *Pichia pastoris* or CHO cells, wherein the constant regions thereof optionally may be modified to alter glycosylation or proteolysis or other effector functions.

In some embodiments any of the aforementioned anti-CGRP antibodies or antibody fragments may be comprised in a formulation as disclosed herein, e.g., comprising histidine (L-histidine), sorbitol, polysorbate 80, such as, per 1 mL volume, about 100 mg anti-CGRP antibody, about 3.1 mg L-Histidine, about 40.5 mg Sorbitol, and about 0.15 mg Polysorbate 80, having a pH of about 5.8. The antibody or fragment may be administered by different means, e.g., intravenously, e.g., in a saline solution such as 0.9% sodium chloride in a suitable volume, such as 100 mL.

In some embodiments, about 100 mg, about 125 mg, about 150 mg, about 175 mg, about 200 mg, about 225 mg, about 250 mg, about 275 mg, or about 300 mg of said anti-CGRP antibody or antibody fragment is administered, e.g., intravenously.

In other embodiments, about 100 mg of said anti-CGRP antibody or antibody fragment is administered.

In other embodiments, about 300 mg of said anti-CGRP antibody or antibody fragment is administered, e.g., intravenously.

In exemplary embodiments, the anti-human CGRP antibody or antibody fragment is administered, e.g., intravenously at a frequency which is at most every 10-14 weeks, preferably every 11-13 weeks, more preferably every 3 months or every 12 weeks, wherein the antibody dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 10-14 weeks, preferably every 11-13 weeks, more preferably every 3 months or every 12 weeks. The phrase "the antibody dosage is administered in a single formulation or divided into different formulations" refers to the administration of the recited amount of antibody within a relatively short period of time, e.g., within a period of several hours, e.g., 1 to 8 hours, about one day, within about two days, or within about one week, which may be by the same or different routes (e.g., i.v., i.m., and/or s.c.), sites of administration. The term "different formulations" in this context refers to antibody dosages that are administered at different times and/or at different sites and/or different routes, irrespective of whether the dosages are the same or different with respect to the chemical composition of the pharmaceutical formulation in which each dosage is administered; for example, the concentration, excipients, carriers, pH, and the like may be the same or different between the different administered dosages.

In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 8 weeks or every 2 months.

In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 10-14 weeks, preferably every 11-13 weeks, more preferably every 12 weeks or every 3 months.

In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 16 weeks or every 4 months.

In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 20 weeks or every 5 months.

In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 24 weeks or every 6 months.

In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 28 weeks or every 7 months.

In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 32 weeks or every 8 months.

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In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 36 weeks or every 9 months.

In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 40 weeks or every 8 months.

In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 44 weeks or every 9 months.

In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 48 weeks or every 10 months.

In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 52 weeks or every 11 months.

In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 56 weeks or every 12 months.

In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 15-18 months.

In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 18-21 months.

In other exemplary embodiments, the anti-human CGRP antibody dosage or antibody fragment used in the aforementioned methods is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 2 years.

In other exemplary embodiments, the anti-human CGRP antibody used in the afore-mentioned methods is administered systemically.

In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment used in the afore-mentioned methods is administered by a mode of administration is selected from intravenous, intramuscular, intravenous, intrathecal, intracranial, topical, intranasal, and oral. In a preferred embodiment, the anti-human CGRP antibody or antibody fragment used in the afore-mentioned methods is administered intravenously.

In other exemplary embodiments, the anti-human CGRP antibody used in the afore-mentioned methods has an in vivo half-life of at least 10 days.

In other exemplary embodiments, the anti-human CGRP antibody has an in vivo half-life of at least 15 days.

In other exemplary embodiments, the anti-human CGRP antibody used in the afore-mentioned methods has an in vivo half-life of at least 20 days.

In other exemplary embodiments, the anti-human CGRP antibody used in the afore-mentioned methods has an in vivo half-life of at least 20-30 days.

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In other exemplary embodiments, the anti-human CGRP antibody is administered at a dosage of between about 100 mg and about 300 mg has an in vivo half-life of $\pm 20\%$ of at least about (284 $\pm$ 44 hours).

5 In other exemplary embodiments, the anti-human CGRP antibody used in the afore-mentioned methods binds to human α - and β -CGRP.

In other exemplary embodiments, the administered anti-human CGRP antibody dosage results in the inhibition of vasodilation induced by topically applied capsaicin at least 30 days after antibody administration.

10 In other exemplary embodiments, the administered anti-human CGRP antibody dosage results in the inhibition of vasodilation induced by topically applied capsaicin at least 60 days after antibody administration.

15 In other exemplary embodiments, the administered anti-human CGRP antibody dosage results in inhibition of vasodilation induced by topically applied capsaicin at least 90 days after antibody administration.

20 In other exemplary embodiments, the administered anti-human CGRP antibody dosage results in the inhibition of vasodilation induced by topically applied capsaicin at least 120 days after antibody administration.

25 In other exemplary embodiments, the administered anti-human CGRP antibody dosage results in the inhibition of vasodilation induced by topically applied capsaicin at least 150 days after antibody administration.

In other exemplary embodiments, the administered anti-human CGRP antibody dosage results in the inhibition of vasodilation induced by topically applied capsaicin at least 180 days after antibody administration.

30 In other exemplary embodiments, the administered anti-human CGRP antibody dosage results in the inhibition of vasodilation induced by topically applied capsaicin more than 180 days after antibody administration.

In other exemplary embodiments, the administered anti-human CGRP antibody dosage results in sustained pharmacodynamic (PK) activity, within 5% of the maximal response (I_{max}) (as compared to lower antibody doses).

35 In other exemplary embodiments, the administered anti-human CGRP antibody dosage results in sustained pharmacodynamic (PK) activity which is maintained for at least 2-3 months after antibody administration, wherein PK analysis of the anti-human CGRP antibody is derived from plasma concentrations.

In other exemplary embodiments, the administered anti-human CGRP antibody dosage is between about 100 mg and about 300 mg or more which is administered no more frequently than every 2 months.

40 The present invention is additionally directed to the use of specific antibodies and fragments thereof having binding specificity for CGRP, in particular antibodies having desired epitopic specificity, high affinity or avidity and/or functional properties. A preferred embodiment of the invention is directed to usage of chimeric or humanized antibodies and fragments thereof (including Fab fragments) capable of binding to CGRP and/or inhibiting the biological activities mediated by the binding of CGRP to the CGRP receptor ("CGRP-R") e.g., wherein such antibodies optionally are derived from recombinant cells engineered to express same, optionally yeast or mammalian cells, further optionally *Pichia pastoris* and CHO cells.

45 In another preferred embodiment of the invention, full length antibodies and Fab fragments thereof are contemplated that inhibit the CGRP- α -, CGRP- β -, and rat CGRP-driven production of cAMP. In a further preferred

embodiment of the invention, full length and Fab fragments thereof are contemplated that reduce vasodilation in a recipient following administration.

The invention also contemplates usage of conjugates of anti-CGRP antibodies and binding fragments thereof conjugated to one or more functional or detectable moieties. The invention also contemplates usage of chimeric or humanized anti-CGRP or anti-CGRP/CGRP-R complex antibodies and binding fragments thereof. In one embodiment, binding fragments include, but are not limited to, Fab, Fab', F(ab')₂, Fv, scFv fragments, SMIPs (small molecule immunopharmaceuticals), camelbodies, nanobodies, and IgNAR.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 provide the polypeptide sequences of the full-length heavy chain for antibody Ab6 with framework regions (FR), complementarity determining regions (CDRs), and constant region sequences delimited.

FIG. 2 provide the polypeptide sequences of the full-length light chain for antibody Ab6 with framework regions (FR), complementarity determining regions (CDRs), and constant region sequences delimited.

FIGS. 3A and 3B provide exemplary polynucleotide sequences encoding the full-length heavy chain for antibody Ab6 with framework regions (FR), complementarity determining regions (CDRs), and variable region coding sequences delimited.

FIG. 4 provide exemplary polynucleotide sequences encoding the full-length light chain for antibody Ab6 with their framework regions (FR), complementarity determining regions (CDRs), and variable region coding sequences delimited.

FIG. 5 provides the polypeptide sequence coordinates within the full-length heavy chain polypeptide sequences of antibodies Ab6 of sequence features including the variable region and complementarity determining regions (CDRs), and the SEQ ID NO of each individual feature.

FIG. 6 provides the polypeptide sequence coordinates within the full-length heavy chain polypeptide sequences of antibody Ab6 of sequence features including the framework regions (FRs) and constant region, and the SEQ ID NO of each individual feature.

FIG. 7 provides the polypeptide sequence coordinates within the full-length light chain polypeptide sequences of antibody Ab6 of sequence features including the variable region and complementarity determining regions (CDRs), and the SEQ ID NO of each individual feature.

FIG. 8 provides the polypeptide sequence coordinates within the full-length light chain polypeptide sequences of antibody Ab6 of sequence features including the framework regions (FRs) and constant region, and the SEQ ID NO of each individual feature.

FIG. 9 provides the polynucleotide sequence coordinates within the exemplary polynucleotide sequences encoding the full-length heavy chain polypeptide sequences of antibody Ab6 of sequence features including the variable region and complementarity determining regions (CDRs), and the SEQ ID NO of each individual feature.

FIG. 10 provides the polynucleotide sequence coordinates within the exemplary polynucleotide sequences encoding the full-length heavy chain polypeptide sequences of antibody Ab6 of sequence features including the framework regions (FRs) and constant region, and the SEQ ID NO of each individual feature.

FIG. 11 provides the polynucleotide sequence coordinates within the exemplary polynucleotide sequences encoding

the full-length light chain polypeptide sequences of antibody Ab6 of sequence features including the variable region and complementarity determining regions (CDRs), and the SEQ ID NO of each individual feature.

FIG. 12 provides the polynucleotide sequence coordinates within the exemplary polynucleotide sequences encoding the full-length light chain polypeptide sequences of antibody Ab6 of sequence features including the framework regions (FRs) and constant region, and the SEQ ID NO of each individual feature.

FIG. 13 Study design of the clinical trial protocol as summarized in Example 2.

FIG. 14 displays the efficacy of Ab6 on Mean Monthly Migraine Days (MMDs) in the clinical trial described in Example 2.

FIG. 15 Illustrates the types and timing of Most Bother-some Symptoms (MBS) across the course of the migraine

FIG. 16 Illustrates the MBS change from baseline during the 28 day screening period of the clinical trial described in Example 2—i.e. before the first infusion of Ab6.

FIG. 17 Illustrates the MBS change from baseline 1 month after the first infusion of Ab6 in the clinical trial described in Example 2.

FIG. 18 Illustrates the PGIC from baseline 1 month after the first infusion of Ab6 in the clinical trial described in Example 2.

FIG. 19 Illustrates the MBS change from baseline 3 month after the first infusion of Ab6 in the clinical trial described in Example 2.

FIG. 20 Illustrates the PGIC from baseline 3 month after the first infusion of Ab6 in the clinical trial described in Example 2.

FIG. 21 Illustrates the MBS change from baseline 6 month after the first infusion of Ab6 in the clinical trial described in Example 2.

FIG. 22 Illustrates the PGIC from baseline 6 month after the first infusion of Ab6 in the clinical trial described in Example 2.

DETAILED DESCRIPTION

Use of anti-CGRP antibodies for treatment of MBS and/or PGIC associated with migraine, such as chronic migraine or episodic migraine is described herein. Additionally, anti-CGRP antibodies are demonstrated herein to be effective for treatment of MMDs. The treatment efficacy on both MBS and PGIC are shown to be effective in providing relief of MBS and PGIC at 1 month, 3 months and 6 months following the first infusion of an anti-CGRP antibody or fragments thereof of the invention.

Definitions

It is to be understood that this invention is not limited to the particular methodology, protocols, cell lines, animal species or genera, and reagents described, as such may vary. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only, and is not intended to limit the scope of the present invention which will be limited only by the appended claims. As used herein the singular forms “a”, “and”, and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a cell” includes a plurality of such cells and reference to “the protein” includes reference to one or more proteins and equivalents thereof known to those skilled in the art, and so forth. All technical and scientific terms used herein have the same meaning as

commonly understood to one of ordinary skill in the art to which this invention belongs unless clearly indicated otherwise.

As used herein, the term “most bothersome symptom associated with migraine” refers to symptoms which is identified by an individual patient to be the most bothersome symptom they associate with their migraine. In the present invention the “most bothersome symptom associated with migraine” is specified in Table 1. The “most bothersome symptom associated with migraine” of the present invention described by the patient to the study investigator who assists in medical interpretation of the patients symptom. The investigator in the clinical study was able to with the patient consultation selected from the group of known migraine symptoms consisting of: Sensitivity to light (photophobia), Nausea/vomiting, Headache, Sensitivity to sound (phonophobia), Aura, Pain with activity, Pain, Throbbing/pulsation, Cognitive disruption, Fatigue, Mood changes, Sensitivity to smell (osmophobia or olfactophobia), Visual impact, Pressure/tightness, Pain (anatomical), Eye pain, Neck pain, Dizziness, Allodynia, Inactivity, Sensory disturbance, Sleep disturbance and Speech difficulty. A patient’s “most bothersome symptom associated with migraine” as used in the present invention refers to the self-identified “most bothersome symptom associated with migraine”, which may be one or more of the symptoms described herein above or may be classified as “other”

As used herein, the term “improvement of” or “improving” most bothersome symptom associated with migraine refers the change in the patient’s assessment of the MBS compared to baseline (i.e the MBS prior to the first dosing with anti-CGRP antibodies or fragments thereof of the invention). An improvement is characterized as ≥ 1 categorical change in the patients assesment of the MBS compared to baseline on the 7 point Likert scale described in Example 2.

As used herein, the term “improvement of” or “improving” patient global impression of change associated with migraine refers the change in the patient’s assessment of their disease status compared to baseline (i.e the disease status prior to the first dosing with anti-CGRP antibodies or fragments thereof of the invention). An improvement is characterized as ≥ 1 categorical change in the patients assesment of the PGIC compared to baseline on the 7-step scale described in Example 2.

As used herein, the term “chronic migraine” refers to a condition wherein a patient exhibits, on average, at least 15 headache per month with a subset of these headache days fulfilling the ICHD-3 criteria for migraine with or without aura. The term “episodic migraine” refers to a condition wherein a patient exhibits, on average, less than 15 day a month of headache with typically 4-15 being a migraine phenotype meeting the ICHD-3 definition of migraine with or without aura.

As used herein, the term “diagnosed with chronic migraine” refers to a patient meeting the clinical criteria for chronic migraine, whether or not a formal diagnosis of that patient was performed.

As used herein, the term “intravenously administering” refers to a mode of administration wherein a substance, e.g., an antibody, is introduced directly into the circulation of that patient, most typically into the venous circulation. The substance may be introduced in a carrier fluid, such as an aqueous solution, e.g., normal saline. The substance may be administered in a single formulation or in multiple formulations, as long as the administration is completed over a

short period of time (e.g., within 1 day, preferably within 12 hours, more preferably within 6 hours, and most preferably within 1-2 hours).

As used herein, the term “the baseline number of migraine days” refers to the number of migraine days exhibited by a patient in a specified time period, e.g., prior to treatment. For example, the baseline number of migraine days may be determined over a period of one month, or longer, e.g., by recording each day whether or not a migraine occurred.

As used herein, the term “migraine days per month” refers to the number of days per month on which a patient has a migraine, i.e., at any time during that day, the patient has symptoms that meet the clinical definition of migraine. The number of migraine days per month may be determined by recording each day whether or not a migraine occurred.

As used herein, the term “headache days per month” refers to the number of days per month on which a patient has a headache, i.e., at any time during that day, the patient has symptoms that meet the clinical definition of a headache. The number of headache days per month may be determined by recording each day whether or not a headache occurred.

Calcitonin Gene Related Peptide (CGRP): As used herein, CGRP encompasses not only the following *Homo sapiens* CGRP-alpha and *Homo sapiens* CGRP-beta amino acid sequences available from American Peptides (Sunnyvale CA) and Bachem (Torrance, CA):

CGRP-alpha: ACDTATCVTHRLAGLLSRSGGVVKN-NFVPTNVGSKAF-NH₂ (SEQ ID NO: 561), wherein the terminal phenylalanine is amidated;

CGRP-beta: ACNTATCVTHRLAGLLSRSGGMV-KSNFVPTNVGSKAF-NH₂ (SEQ ID NO: 562), wherein the terminal phenylalanine is amidated; but also any membrane-bound forms of these CGRP amino acid sequences, as well as mutants (mutiens), splice variants, isoforms, orthologs, homologues and variants of this sequence.

Expression Vector: These DNA vectors contain elements that facilitate manipulation for the expression of a foreign protein within the target host cell, e.g., a yeast or mammalian cell such as *Pichia pastoris* or CHO cells. Conveniently, manipulation of sequences and production of DNA for transformation is first performed in a bacterial host, e.g. *E. coli*, and usually vectors will include sequences to facilitate such manipulations, including a bacterial origin of replication and appropriate bacterial selection marker. Selection markers encode proteins necessary for the survival or growth of transformed host cells grown in a selective culture medium. Host cells not transformed with the vector containing the selection gene will not survive in the culture medium. Typical selection genes encode proteins that (a) confer resistance to antibiotics or other toxins, (b) complement auxotrophic deficiencies, or (c) supply critical nutrients not available from complex media. Exemplary vectors and methods for transformation of yeast are described, for example, in Burke, D., Dawson, D., & Steams, T. (2000). *Methods in yeast genetics: a Cold Spring Harbor Laboratory course manual*. Plainview, N.Y.: Cold Spring Harbor Laboratory Press.

Expression vectors for use in yeast or mammalian cells will generally further include yeast or mammalian specific sequences, including a selectable auxotrophic or drug marker for identifying transformed yeast strains or transformed mammalian cells. A drug marker may further be used to amplify copy number of the vector in the host cell.

The polypeptide coding sequence of interest is operably linked to transcriptional and translational regulatory sequences that provide for expression of the polypeptide in

host cells, e.g., *Pichia pastoris* or CHO cells. These vector components may include, but are not limited to, one or more of the following: an enhancer element, a promoter, and a transcription termination sequence. Sequences for the secretion of the polypeptide may also be included, e.g. a signal sequence, and the like. A yeast or mammalian origin of replication is optional, as expression vectors are often integrated into the host cell genome. In one embodiment of the invention, the polypeptide of interest is operably linked, or fused, to sequences providing for optimized secretion of the polypeptide from yeast diploid cells.

Nucleic acids are "operably linked" when placed into a functional relationship with another nucleic acid sequence. For example, DNA for a signal sequence is operably linked to DNA for a polypeptide if it is expressed as a preprotein that participates in the secretion of the polypeptide; a promoter or enhancer is operably linked to a coding sequence if it affects the transcription of the sequence. Generally, "operably linked" means that the DNA sequences being linked are contiguous, and, in the case of a secretory leader, contiguous and in reading frame. However, enhancers do not have to be contiguous. Linking is accomplished by ligation at convenient restriction sites or alternatively via a PCR/recombination method familiar to those skilled in the art (Gateway® Technology; Invitrogen, Carlsbad California). If such sites do not exist, the synthetic oligonucleotide adapters or linkers are used in accordance with conventional practice.

Promoters are untranslated sequences located upstream (5') to the start codon of a structural gene (generally within about 100 to 1000 bp) that control the transcription and translation of particular nucleic acid sequences to which they are operably linked. Such promoters fall into several classes: inducible, constitutive, and repressible promoters (that increase levels of transcription in response to absence of a repressor). Inducible promoters may initiate increased levels of transcription from DNA under their control in response to some change in culture conditions, e.g., the presence or absence of a nutrient or a change in temperature.

The promoter fragment may also serve as the site for homologous recombination and integration of the expression vector into the same site in the host genome; alternatively a selectable marker is used as the site for homologous recombination. Examples of suitable promoters from *Pichia* include the AOX1 and promoter (Cregg et al. (1989) *Mol. Cell. Biol.* 9:1316-1323); ICL1 promoter (Menendez et al. (2003) *Yeast* 20(13):1097-108); glyceraldehyde-3-phosphate dehydrogenase promoter (GAP) (Waterham et al. (1997) *Gene* 186(1):37-44); and FLD1 promoter (Shen et al. (1998) *Gene* 216(1):93-102). The GAP promoter is a strong constitutive promoter and the AOX and FLD1 promoters are inducible.

Other yeast promoters include ADH1, alcohol dehydrogenase II, GAL4, PHO3, PHO5, Pyk, and chimeric promoters derived therefrom. Additionally, non-yeast promoters may be used in the invention such as mammalian, insect, plant, reptile, amphibian, viral, and avian promoters. Most typically the promoter will comprise a mammalian promoter (potentially endogenous to the expressed genes) or will comprise a yeast or viral promoter that provides for efficient transcription in yeast systems.

Examples of mammalian promoters include cytomegalovirus (CMV) derived promoters, chicken 3-actin (CBM) derived promoters, adenomatous polyposis coli (APC) derived promoters, leucine-rich repeat containing G protein-coupled receptor 5 (LGR5) promoters, CAG promoter, Beta actin promoter, elongation factor-1 (EF1) promoter, early

growth response 1 (EGR-1) promoter, eukaryotic initiation factor 4A (EIF4A1) promoter, simian virus 40 (SV40) early promoter, mouse mammary tumor virus (MMTV), human immunodeficiency virus (HIV) long terminal repeat (LTR) promoter, MoMuLV promoter, an avian leukemia virus promoter, an Epstein-Barr virus immediate early promoter, a Rous sarcoma virus promoter, as well as human gene promoters such as, but not limited to, the actin promoter, the myosin promoter, the hemoglobin promoter, and the creatine kinase promoter, among others. Combinations of two or more of the foregoing promoters may also be used. Further, inducible promoters may be used. The use of an inducible promoter provides a molecular switch capable of turning on expression of the polynucleotide sequence which it is operatively linked when such expression is desired, or turning off the expression when expression is not desired. Examples of inducible promoters include, but are not limited to a metallothionein promoter, a glucocorticoid promoter, a progesterone promoter, and a tetracycline promoter.

The polypeptides of interest may be produced recombinantly not only directly, but also as a fusion polypeptide with a heterologous polypeptide, e.g. a signal sequence or other polypeptide having a specific cleavage site at the N-terminus of the mature protein or polypeptide. In general, the signal sequence may be a component of the vector, or it may be a part of the polypeptide coding sequence that is inserted into the vector. The heterologous signal sequence selected preferably is one that is recognized and processed through one of the standard pathways available within the host cell. The *S. cerevisiae* alpha factor pre-pro signal has proven effective in the secretion of a variety of recombinant proteins from *P. pastoris*. Other yeast signal sequences include the alpha mating factor signal sequence, the invertase signal sequence, and signal sequences derived from other secreted yeast polypeptides. Additionally, these signal peptide sequences may be engineered to provide for enhanced secretion in diploid yeast expression systems. Secretion signals for use in mammalian as well as yeast cells include mammalian signal sequences, which may be heterologous to the protein being secreted, or may be a native sequence for the protein being secreted. Signal sequences include pre-peptide sequences, and in some instances may include propeptide sequences. Many such signal sequences are known in the art, including the signal sequences found on immunoglobulin chains, e.g., K28 prepro toxin sequence, PHA-E, FACE, human MCP-1, human serum albumin signal sequences, human Ig heavy chain, human Ig light chain, and the like. For example, see Hashimoto et. al. *Protein Eng* 11(2) 75 (1998); and Kobayashi et. al. *Therapeutic Apheresis* 2(4) 257 (1998).

Transcription may be increased by inserting a transcriptional activator sequence into the vector. These activators are cis-acting elements of DNA, usually about from 10 to 300 bp, which act on a promoter to increase its transcription. Transcriptional enhancers are relatively orientation and position independent, having been found 5' and 3' to the transcription unit, within an intron, as well as within the coding sequence itself. The enhancer may be spliced into the expression vector at a position 5' or 3' to the coding sequence, but is preferably located at a site 5' from the promoter.

Expression vectors used in eukaryotic host cells may also contain sequences necessary for the termination of transcription and for stabilizing the mRNA. Such sequences are commonly available from 3' to the translation termination codon, in untranslated regions of eukaryotic or viral DNAs

or cDNAs. These regions contain nucleotide segments transcribed as polyadenylated fragments in the untranslated portion of the mRNA.

Construction of suitable vectors containing one or more of the above-listed components employs standard ligation techniques or PCR/recombination methods. Isolated plasmids or DNA fragments are cleaved, tailored, and re-ligated in the form desired to generate the plasmids required or via recombination methods. For analysis to confirm correct sequences in plasmids constructed, the ligation mixtures are used to transform host cells, and successful transformants selected by antibiotic resistance (e.g. ampicillin or Zeocin) where appropriate. Plasmids from the transformants are prepared, analyzed by restriction endonuclease digestion and/or sequenced.

As an alternative to restriction and ligation of fragments, recombination methods based on att sites and recombination enzymes may be used to insert DNA sequences into a vector. Such methods are described, for example, by Landy (1989) *Ann. Rev. Biochem.* 58:913-949; and are known to those of skill in the art. Such methods utilize intermolecular DNA recombination that is mediated by a mixture of lambda and *E. coli*-encoded recombination proteins. Recombination occurs between specific attachment (att) sites on the interacting DNA molecules. For a description of att sites see Weisberg and Landy (1983) Site-Specific Recombination in Phage Lambda, in *Lambda II*, Weisberg, ed. (Cold Spring Harbor, NY: Cold Spring Harbor Press), pp. 211-250. The DNA segments flanking the recombination sites are switched, such that after recombination, the att sites are hybrid sequences comprised of sequences donated by each parental vector. The recombination can occur between DNAs of any topology.

Att sites may be introduced into a sequence of interest by ligating the sequence of interest into an appropriate vector; generating a PCR product containing att B sites through the use of specific primers; generating a cDNA library cloned into an appropriate vector containing att sites; and the like.

Folding, as used herein, refers to the three-dimensional structure of polypeptides and proteins, where interactions between amino acid residues act to stabilize the structure. Proper folding is typically the arrangement of a polypeptide that results in optimal biological activity, and in the case of antibodies can conveniently be monitored by assays for activity, e.g. antigen binding.

The expression host may be further modified by the introduction of sequences encoding one or more enzymes that enhance folding and disulfide bond formation, i.e. foldases, chaperonins, etc. Such sequences may be constitutively or inducibly expressed in the yeast host cell, using vectors, markers, etc. as known in the art. Preferably the sequences, including transcriptional regulatory elements sufficient for the desired pattern of expression, are stably integrated in the yeast genome through a targeted methodology.

For example, the eukaryotic PDI is not only an efficient catalyst of protein cysteine oxidation and disulfide bond isomerization, but also exhibits chaperone activity. Co-expression of PDI can facilitate the production of active proteins having multiple disulfide bonds. Also of interest is the expression of BIP (immunoglobulin heavy chain binding protein); cyclophilin; and the like. In one embodiment of the invention, each of the haploid parental strains expresses a distinct folding enzyme, e.g. one strain may express BIP, and the other strain may express PDI or combinations thereof.

The terms "desired protein" or "desired antibody" are used interchangeably and refer generally to a parent anti-

body specific to a target, i.e., CGRP or a chimeric or humanized antibody or a binding portion thereof derived therefrom as described herein. The term "antibody" is intended to include any polypeptide chain-containing molecular structure with a specific shape that fits to and recognizes an epitope, where one or more non-covalent binding interactions stabilize the complex between the molecular structure and the epitope. The archetypal antibody molecule is the immunoglobulin, and in particular IgG etc., from all sources, e.g. human, rodent, rabbit, cow, sheep, pig, dog, other mammals, chicken, other avians, etc., are considered to be "antibodies." A preferred source for producing antibodies useful as starting material according to the invention is rabbits. Numerous antibody coding sequences have been described; and others may be raised by methods well-known in the art. Examples thereof include chimeric antibodies, human antibodies and other non-human mammalian antibodies, humanized antibodies, single chain antibodies (such as scFvs), camelbodies, nanobodies, IgNAR (single-chain antibodies derived from sharks), small-modular immunopharmaceuticals (SMIPs), and antibody fragments such as Fabs, Fab', F(ab')₂ and the like. See Streltsov V A, et al., Structure of a shark IgNAR antibody variable domain and modeling of an early-developmental isotype, *Protein Sci.* 2005 November; 14(11):2901-9. Epub 2005 Sep. 30; Greenberg A S, et al., A new antigen receptor gene family that undergoes rearrangement and extensive somatic diversification in sharks, *Nature*. 1995 Mar. 9; 374(6518): 168-73; Nuttall S D, et al., Isolation of the new antigen receptor from wobbegong sharks, and use as a scaffold for the display of protein loop libraries, *Mol Immunol.* 2001 August; 38(4):313-26; Hamers-Casterman C, et al., Naturally occurring antibodies devoid of light chains, *Nature*. 1993 Jun. 3; 363(6428):446-8; Gill D S, et al., Biopharmaceutical drug discovery using novel protein scaffolds, *Curr Opin Biotechnol.* 2006 December; 17(6):653-8. Epub 2006 Oct. 19.

For example, antibodies or antigen binding fragments may be produced by genetic engineering. In this technique, as with other methods, antibody-producing cells are sensitized to the desired antigen or immunogen. The messenger RNA isolated from antibody producing cells is used as a template to make cDNA using PCR amplification. A library of vectors, each containing one heavy chain gene and one light chain gene retaining the initial antigen specificity, is produced by insertion of appropriate sections of the amplified immunoglobulin cDNA into the expression vectors. A combinatorial library is constructed by combining the heavy chain gene library with the light chain gene library. This results in a library of clones which co-express a heavy and light chain (resembling the Fab fragment or antigen binding fragment of an antibody molecule). The vectors that carry these genes are co-transfected into a host cell. When antibody gene synthesis is induced in the transfected host, the heavy and light chain proteins self-assemble to produce active antibodies that can be detected by screening with the antigen or immunogen.

Antibody coding sequences of interest include those encoded by native sequences, as well as nucleic acids that, by virtue of the degeneracy of the genetic code, are not identical in sequence to the disclosed nucleic acids, and variants thereof. Variant polypeptides can include amino acid (aa) substitutions, additions or deletions. The amino acid substitutions can be conservative amino acid substitutions or substitutions to eliminate non-essential amino acids, such as to alter a glycosylation site, or to minimize misfolding by substitution or deletion of one or more cysteine

residues that are not necessary for function. Variants can be designed so as to retain or have enhanced biological activity of a particular region of the protein (e.g., a functional domain, catalytic amino acid residues, etc). Variants also include fragments of the polypeptides disclosed herein, particularly biologically active fragments and/or fragments corresponding to functional domains. Techniques for in vitro mutagenesis of cloned genes are known. Also included in the subject invention are polypeptides that have been modified using ordinary molecular biological techniques so as to improve their resistance to proteolytic degradation or to optimize solubility properties or to render them more suitable as a therapeutic agent.

Chimeric antibodies may be made by recombinant means by combining the variable light and heavy chain regions (V_L and V_H), obtained from antibody producing cells of one species with the constant light and heavy chain regions from another. Typically chimeric antibodies utilize rodent or rabbit variable regions and human constant regions, in order to produce an antibody with predominantly human domains. The production of such chimeric antibodies is well known in the art, and may be achieved by standard means (as described, e.g., in U.S. Pat. No. 5,624,659, incorporated herein by reference in its entirety). It is further contemplated that the human constant regions of chimeric antibodies of the invention may be selected from IgG1, IgG2, IgG3, and IgG4 constant regions.

Humanized antibodies are engineered to contain even more human-like immunoglobulin domains, and incorporate only the complementarity-determining regions of the animal-derived antibody. This is accomplished by carefully examining the sequence of the hyper-variable loops of the variable regions of the monoclonal antibody, and fitting them to the structure of the human antibody chains. Although facially complex, the process is straightforward in practice. See, e.g., U.S. Pat. No. 6,187,287, incorporated fully herein by reference.

In addition to entire immunoglobulins (or their recombinant counterparts), immunoglobulin fragments comprising the epitope binding site (e.g., Fab', F(ab')₂, or other fragments) may be synthesized. "Fragment," or minimal immunoglobulins may be designed utilizing recombinant immunoglobulin techniques. For instance "Fv" immunoglobulins for use in the present invention may be produced by synthesizing a fused variable light chain region and a variable heavy chain region. Combinations of antibodies are also of interest, e.g. diabodies, which comprise two distinct Fv specificities. In another embodiment of the invention, SMIPs (small molecule immunopharmaceuticals), camelbodies, nanobodies, and IgNAR are encompassed by immunoglobulin fragments.

Immunoglobulins and fragments thereof may be modified post-translationally, e.g. to add effector moieties such as chemical linkers, detectable moieties, such as fluorescent dyes, enzymes, toxins, substrates, bioluminescent materials, radioactive materials, chemiluminescent moieties and the like, or specific binding moieties, such as streptavidin, avidin, or biotin, and the like may be utilized in the methods and compositions of the present invention. Examples of additional effector molecules are provided infra.

A polynucleotide sequence "corresponds" to a polypeptide sequence if translation of the polynucleotide sequence in accordance with the genetic code yields the polypeptide sequence (i.e., the polynucleotide sequence "encodes" the polypeptide sequence), one polynucleotide sequence "corresponds" to another polynucleotide sequence if the two sequences encode the same polypeptide sequence.

A "heterologous" region or domain of a DNA construct is an identifiable segment of DNA within a larger DNA molecule that is not found in association with the larger molecule in nature. Thus, when the heterologous region encodes a mammalian gene, the gene will usually be flanked by DNA that does not flank the mammalian genomic DNA in the genome of the source organism. Another example of a heterologous region is a construct where the coding sequence itself is not found in nature (e.g., a cDNA where the genomic coding sequence contains introns, or synthetic sequences having codons different than the native gene). Allelic variations or naturally-occurring mutational events do not give rise to a heterologous region of DNA as defined herein.

A "coding sequence" is an in-frame sequence of codons that (in view of the genetic code) correspond to or encode a protein or peptide sequence. Two coding sequences correspond to each other if the sequences or their complementary sequences encode the same amino acid sequences. A coding sequence in association with appropriate regulatory sequences may be transcribed and translated into a polypeptide. A polyadenylation signal and transcription termination sequence will usually be located 3' to the coding sequence. A "promoter sequence" is a DNA regulatory region capable of binding RNA polymerase in a cell and initiating transcription of a downstream (3' direction) coding sequence. Promoter sequences typically contain additional sites for binding of regulatory molecules (e.g., transcription factors) which affect the transcription of the coding sequence. A coding sequence is "under the control" of the promoter sequence or "operatively linked" to the promoter when RNA polymerase binds the promoter sequence in a cell and transcribes the coding sequence into mRNA, which is then in turn translated into the protein encoded by the coding sequence.

Vectors are used to introduce a foreign substance, such as DNA, RNA or protein, into an organism or host cell. Typical vectors include recombinant viruses (for polynucleotides) and liposomes (for polypeptides). A "DNA vector" is a replicon, such as plasmid, phage or cosmid, to which another polynucleotide segment may be attached so as to bring about the replication of the attached segment. An "expression vector" is a DNA vector which contains regulatory sequences which will direct polypeptide synthesis by an appropriate host cell. This usually means a promoter to bind RNA polymerase and initiate transcription of mRNA, as well as ribosome binding sites and initiation signals to direct translation of the mRNA into a polypeptide(s). Incorporation of a polynucleotide sequence into an expression vector at the proper site and in correct reading frame, followed by transformation of an appropriate host cell by the vector, enables the production of a polypeptide encoded by said polynucleotide sequence.

"Amplification" of polynucleotide sequences is the in vitro production of multiple copies of a particular nucleic acid sequence. The amplified sequence is usually in the form of DNA. A variety of techniques for carrying out such amplification are described in a review article by Van Brunt (1990, *Bio Technol.*, 8(4):291-294). Polymerase chain reaction or PCR is a prototype of nucleic acid amplification, and use of PCR herein should be considered exemplary of other suitable amplification techniques.

The general structure of antibodies in vertebrates now is well understood (Edelman, G. M., *Ann. N.Y. Acad. Sci.*, 190: 5 (1971)). Antibodies consist of two identical light polypeptide chains of molecular weight approximately 23,000 daltons (the "light chain"), and two identical heavy chains of

molecular weight 53,000-70,000 (the "heavy chain"). The four chains are joined by disulfide bonds in a "Y" configuration wherein the light chains bracket the heavy chains starting at the mouth of the "Y" configuration. The "branch" portion of the "Y" configuration is designated the F_{ab} region; the stem portion of the "Y" configuration is designated the Fc region. The amino acid sequence orientation runs from the N-terminal end at the top of the "Y" configuration to the C-terminal end at the bottom of each chain. The N-terminal end possesses the variable region having specificity for the antigen that elicited it, and is approximately 100 amino acids in length, there being slight variations between light and heavy chain and from antibody to antibody.

The variable region is linked in each chain to a constant region that extends the remaining length of the chain and that within a particular class of antibody does not vary with the specificity of the antibody (i.e., the antigen eliciting it). There are five known major classes of constant regions that determine the class of the immunoglobulin molecule (IgG, IgM, IgA, IgD, and IgE corresponding to γ , μ , α , δ , and ϵ (gamma, mu, alpha, delta, or epsilon) heavy chain constant regions). The constant region or class determines subsequent effector function of the antibody, including activation of complement (Kabat, E. A., *Structural Concepts in Immunology and Immunochemistry*, 2nd Ed., p. 413-436, Holt, Rinehart, Winston (1976)), and other cellular responses (Andrews, D. W., et al., *Clinical Immunobiology*, pp 1-18, W. B. Sanders (1980); Kohl, S., et al., *Immunology*, 48: 187 (1983)); while the variable region determines the antigen with which it will react. Light chains are classified as either κ (kappa) or λ (lambda). Each heavy chain class can be prepared with either kappa or lambda light chain. The light and heavy chains are covalently bonded to each other, and the "tail" portions of the two heavy chains are bonded to each other by covalent disulfide linkages when the immunoglobulins are generated either by hybridomas or by B cells.

The expression "variable region" or "VR" refers to the domains within each pair of light and heavy chains in an antibody that are involved directly in binding the antibody to the antigen. Each heavy chain has at one end a variable domain (V_H) followed by a number of constant domains. Each light chain has a variable domain (V_L) at one end and a constant domain at its other end; the constant domain of the light chain is aligned with the first constant domain of the heavy chain, and the light chain variable domain is aligned with the variable domain of the heavy chain.

The expressions "complementarity determining region," "hypervariable region," or "CDR" refer to one or more of the hyper-variable or complementarity determining regions (CDRs) found in the variable regions of light or heavy chains of an antibody (See Kabat, E. A. et al., *Sequences of Proteins of Immunological Interest*, National Institutes of Health, Bethesda, Md., (1987)). These expressions include the hypervariable regions as defined by Kabat et al. ("Sequences of Proteins of Immunological Interest," Kabat E., et al., US Dept. of Health and Human Services, 1983) or the hypervariable loops in 3-dimensional structures of antibodies (Chothia and Lesk, *J Mol. Biol.* 196 901-917 (1987)). The CDRs in each chain are held in close proximity by framework regions and, with the CDRs from the other chain, contribute to the formation of the antigen binding site. Within the CDRs there are select amino acids that have been described as the selectivity determining regions (SDRs) which represent the critical contact residues used by the CDR in the antibody-antigen interaction (Kashmiri, S., *Methods*, 36:25-34 (2005)). In the present invention when

specific antibody amino acid or nucleic acid residues are referenced by number this generally refers to its position within a specified amino acid or nucleic acid sequence (i.e., particular sequence identifier) and/or in accordance with Kabat et al numbering.

The expressions "framework region" or "FR" refer to one or more of the framework regions within the variable regions of the light and heavy chains of an antibody (See Kabat, E. A. et al., *Sequences of Proteins of Immunological Interest*, National Institutes of Health, Bethesda, Md., (1987)). These expressions include those amino acid sequence regions interposed between the CDRs within the variable regions of the light and heavy chains of an antibody.

"Cmax" refers to the maximum (or peak) concentration that an antibody or other compound achieves in tested area (e.g., in the serum or another compartment such as cerebrospinal fluid) after the drug has been administered. For example, serum Cmax may be measured from serum, e.g., prepared by collecting a blood sample, allowing it to clot and separating solid components by centrifugation or other means to yield serum (blood containing neither blood cells nor clotting factors), and then detecting the concentration of the analyte in the serum by ELISA or other means known in the art.

"AUC" refers to the area under the concentration-time curve which is expressed in units of mg/mL*hr (or equivalently mg*hr/ml) unless otherwise specified. "AUC_{0-t}" refers to the area under the concentration-time curve from time=0 to last quantifiable concentration. "AUC_{0-inf}" refers to the area under the concentration-time curve from time=0 extrapolated to infinity.

"I_{max}" refers to the maximal pharmacodynamic response elicited by an anti-CGRP antibody dosage, preferably a dosage of 350 mg or more, more typically at least 750 or 1000 mg, as compared to the response elicited by a lower anti-CGRP antibody doses, e.g., wherein such response may be detected by the inhibition of vasodilation after topical application of capsaicin.

Anti-CGRP Antibodies and Binding Fragments Thereof Having Binding Specificity for CGRP

The invention specifically includes the use of Ab6, which is a specific anti-CGRP antibody or antibody fragment, which comprises or consists of the CDR, VL, VH, CL, CH polypeptides sequences identified in FIGS. 1-12. The polypeptides comprised in the anti-CGRP antibody, Ab6 is further described below.

Antibody Ab6

(SEQ ID NO: 222)

QVLTQSPSSLSASVGRVTINCAQASQSVYHNTYLAWYQQKPGKVPKQLI

YDASTLASGVPSRFSGSGSGTDFTLTISSLQPEDVATYYCLGSYDCTNG

DCFVFGGGTKVEIKR.

The invention also includes humanized antibodies having binding specificity to CGRP and possessing a light chain sequence comprising the sequence set forth below:

(SEQ ID NO: 221)

QVLTQSPSSLSASVGRVTINCAQASQSVYHNTYLAWYQQKPGKVPKQLI

YDASTLASGVPSRFSGSGSGTDFTLTISSLQPEDVATYYCLGSYDCTNG

DCFVFGGGTKVEIKRTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPR

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-continued

EAKVQWKVDNALQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKV
YACEVTHQGLSSPVTKSFNRGEC.

The invention further includes humanized antibodies having binding specificity to CGRP and possessing a variable heavy chain sequence comprising the sequence set forth below:

(SEQ ID NO: 202)
EVQLVESGGGLVQPGGSLRLSCAIVSGIDLSGYIMNWVRQAPGKGLEWVG
VIGINGATYYASWAKGRFTISRDNKSTTVYLMNSLRAEDTAVYFCARG
DIWGQGLTQVTVSS.

The invention also includes humanized antibodies having binding specificity to CGRP and possessing a heavy chain sequence comprising the sequence set forth below:

(SEQ ID NO: 201)
EVQLVESGGGLVQPGGSLRLSCAIVSGIDLSGYIMNWVRQAPGKGLEWVG
VIGINGATYYASWAKGRFTISRDNKSTTVYLMNSLRAEDTAVYFCARG
DIWGQGLTQVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEP
VTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNV
NHKPSNTKVDARVEPKSCDKTHTCPPCPAPPELLGGPSVFLFPPKPKDTL
MISRTPEVTCVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYASTY
RVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTPPV
LDSGDSFFLYSKLTVDKSRWQQGNVSCSVMHEALHNHYTQKSLSLSPGK.

Alternatively, the heavy chain of Ab6 may lack the C-terminal lysine of SEQ ID NO: 201, i.e., a heavy chain sequence comprising the sequence set forth below:

(SEQ ID NO: 566)
EVQLVESGGGLVQPGGSLRLSCAIVSGIDLSGYIMNWVRQAPGKGLEWVG
VIGINGATYYASWAKGRFTISRDNKSTTVYLMNSLRAEDTAVYFCARG
DIWGQGLTQVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEP
VTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNV
NHKPSNTKVDARVEPKSCDKTHTCPPCPAPPELLGGPSVFLFPPKPKDTL
MISRTPEVTCVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYASTY
RVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTPPV
LDSGDSFFLYSKLTVDKSRWQQGNVSCSVMHEALHNHYTQKSLSLSPG.

The invention further contemplates antibodies comprising one or more of the polypeptide sequences of SEQ ID NO: 224; SEQ ID NO: 226; and SEQ ID NO: 228 which correspond to the complementarity-determining regions (CDRs, or hypervariable regions) of the variable light chain sequence of SEQ ID NO: 222 or the light chain sequence of SEQ ID NO: 221, and/or one or more of the polypeptide sequences of SEQ ID NO: 204; SEQ ID NO: 206; and SEQ ID NO: 208 which correspond to the complementarity-determining regions (CDRs, or hypervariable regions) of the

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variable heavy chain sequence of SEQ ID NO: 202 or the heavy chain sequence of SEQ ID NO: 201 or SEQ ID NO: 566, or combinations of these polypeptide sequences. In another embodiment of the invention, the antibodies of the invention or fragments thereof comprise, or alternatively consist of, combinations of one or more of the CDRs, the variable heavy and variable light chain sequences, and the heavy and light chain sequences set forth above, including all of them.

The invention also contemplates fragments of the antibody having binding specificity to CGRP. In one embodiment of the invention, antibody fragments of the invention comprise, or alternatively consist of, the polypeptide sequence of SEQ ID NO: 222 or SEQ ID NO: 221. In another embodiment of the invention, antibody fragments of the invention comprise, or alternatively consist of, the polypeptide sequence of SEQ ID NO: 202 or SEQ ID NO: 201 or SEQ ID NO: 566.

In a further embodiment of the invention, fragments of the antibody having binding specificity to CGRP comprise, or alternatively consist of, one or more of the polypeptide sequences of SEQ ID NO: 224; SEQ ID NO: 226; and SEQ ID NO: 228 which correspond to the complementarity-determining regions (CDRs, or hypervariable regions) of the variable light chain sequence of SEQ ID NO: 222 or the light chain sequence of SEQ ID NO: 221.

In a further embodiment of the invention, fragments of the antibody having binding specificity to CGRP comprise, or alternatively consist of, one or more of the polypeptide sequences of SEQ ID NO: 204; SEQ ID NO: 206; and SEQ ID NO: 208 which correspond to the complementarity-determining regions (CDRs, or hypervariable regions) of the variable heavy chain sequence of SEQ ID NO: 202 or the heavy chain sequence of SEQ ID NO: 201 or SEQ ID NO: 566.

The invention also contemplates antibody fragments which include one or more of the antibody fragments described herein. In one embodiment of the invention, fragments of the antibodies having binding specificity to CGRP comprise, or alternatively consist of, one, two, three or more, including all of the following antibody fragments: the variable light chain region of SEQ ID NO: 222; the variable heavy chain region of SEQ ID NO: 202; the complementarity-determining regions (SEQ ID NO: 224; SEQ ID NO: 226; and SEQ ID NO: 228) of the variable light chain region of SEQ ID NO: 222; and the complementarity-determining regions (SEQ ID NO: 204; SEQ ID NO: 206; and SEQ ID NO: 208) of the variable heavy chain region of SEQ ID NO: 202.

In a particularly preferred embodiment of the invention, the humanized anti-CGRP antibody is Ab6, comprising, or alternatively consisting of, SEQ ID NO: 221 and SEQ ID NO: 201 or SEQ ID NO: 566, and having at least one of the biological activities set forth herein.

In a further particularly preferred embodiment of the invention, antibody fragments comprise, or alternatively consist of, Fab (fragment antigen binding) fragments having binding specificity for CGRP. With respect to antibody Ab6, the Fab fragment includes the variable light chain sequence of SEQ ID NO: 222 and the variable heavy chain sequence of SEQ ID NO: 202. This embodiment of the invention further contemplates additions, deletions, and variants of SEQ ID NO: 222 and/or SEQ ID NO: 202 in said Fab while retaining binding specificity for CGRP.

In another particularly preferred embodiment of the invention, said anti-CGRP antibody may comprise the antibody expression product isolated from recombinant cells

which express nucleic acid sequences encoding the variable light chain polypeptide of SEQ ID NO: 222 and the variable heavy chain polypeptide of SEQ ID NO: 202, which polypeptides optionally are respectively linked to human light and heavy constant region polypeptides, e.g., human IgG1, IgG2, IgG3 or IgG4 constant regions, which constant regions optionally may be modified to alter glycosylation or proteolysis, wherein said recombinant cells optionally comprise yeast or mammalian cells, e.g., *Pichia pastoris* or CHO cells.

In another particularly preferred embodiment of the invention, said anti-CGRP antibody may comprise the antibody expression product isolated from recombinant cells which express nucleic acid sequences encoding the light chain of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201 or SEQ ID NO: 566, wherein said recombinant cells optionally comprise yeast or mammalian cells, e.g., *Pichia pastoris* or CHO cells, wherein the constant regions thereof optionally may be modified to alter glycosylation or proteolysis or other effector functions.

In another particularly preferred embodiment of the invention, any of the aforementioned anti-CGRP antibodies or antibody fragments may be optionally comprised in a formulation as disclosed herein, e.g., comprising histidine (L-histidine), sorbitol, polysorbate 80, such as, per 1 mL volume, about 100 mg anti-CGRP antibody, about 3.1 mg L-Histidine, about 40.5 mg Sorbitol, and about 0.15 mg Polysorbate 80, having a pH of about 5.8.

In one embodiment of the invention described herein (infra), Fab fragments may be produced by enzymatic digestion (e.g., papain) of Ab6. In another embodiment of the invention, anti-CGRP antibodies such as Ab6 or Fab fragments thereof may be produced via expression in mammalian cells such as CHO, NSO or HEK 293 cells, fungal, insect, or microbial systems such as yeast cells (for example diploid yeast such as diploid *Pichia*) and other yeast strains. Suitable *Pichia* species include, but are not limited to, *Pichia pastoris*.

In another embodiment, antibody fragments may be present in one or more of the following non-limiting forms: Fab, Fab', F(ab')₂, Fv and single chain Fv antibody forms. In a preferred embodiment, the anti-CGRP antibodies described herein further comprises the kappa constant light chain sequence comprising the sequence set forth below:

(SEQ ID NO: 563)
TVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQSGG
NSQESVTEQDSKDSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVT
KSFNRGEC.

In another preferred embodiment, the anti-CGRP antibodies described herein further comprises the gamma-1 constant heavy chain polypeptide sequence comprising the sequence set forth below or the same sequence lacking the carboxy terminal lysine residue (SEQ ID NO: 564 and SEQ ID NO: 565, respectively):

(SEQ ID NO: 564)
ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPTVSWNSGALTSG
VHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRV
EPKSCDKTHTCPPCPAPELGGPSVFLFPPKPKDTLMISRTPEVTCVVV
DVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYASTYRVSVLTVLHQDW

-continued

LNGKEYCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQ
VSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFPLYSKLT
VDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK.

(SEQ ID NO: 565)
ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPTVSWNSGALTSG
VHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRV
EPKSCDKTHTCPPCPAPELGGPSVFLFPPKPKDTLMISRTPEVTCVVV
DVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYASTYRVSVLTVLHQDW
LNGKEYCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQ
VSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFPLYSKLT
VDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK.

For clarity, any antibody disclosed herein is intended to include any variant of the disclosed constant region variant sequences, e.g., Ab6 may comprise the constant region of SEQ ID NO: 564 containing the C-terminal lysine or may comprise the constant region of SEQ ID NO: 565 lacking the C-terminal lysine. Thus, every disclosure herein of the heavy chain of SEQ ID NO: 201 also includes a variant lacking the C-terminal lysine residue thereof, i.e., having the heavy chain variable region sequence of Ab6 (SEQ ID NO: 202) and the constant region sequence of SEQ ID NO: 565. For example, the sequence encoding an antibody comprising a C-terminal lysine in the heavy chain may, when expressed in cell lines such as CHO cells, produce an antibody lacking said C-terminal lysine due to proteolysis, or a mixture of heavy chains containing or lacking said C-terminal lysine.

In one embodiment of the invention, the antibodies or V_H or V_L polypeptides originate or are selected from one or more rabbit B cell populations prior to initiation of the humanization process referenced herein.

In another embodiment of the invention, the anti-CGRP antibodies and fragments thereof do not have binding specificity for CGRP-R. In a further embodiment of the invention, the anti-CGRP antibodies and fragments thereof inhibit the association of CGRP with CGRP-R. In another embodiment of the invention, the anti-CGRP antibodies and fragments thereof inhibit the association of CGRP with CGRP-R and/or additional proteins and/or multimers thereof, and/or antagonizes the biological effects thereof.

As stated herein, antibodies and fragments thereof may be modified post-translationally to add effector moieties such as chemical linkers, detectable moieties such as for example fluorescent dyes, enzymes, substrates, bioluminescent materials, radioactive materials, and chemiluminescent moieties, or functional moieties such as for example streptavidin, avidin, biotin, a cytotoxin, a cytotoxic agent, and radioactive materials.

Antibodies or fragments thereof may also be chemically modified to provide additional advantages such as increased solubility, stability and circulating time (in vivo half-life) of the polypeptide, or decreased immunogenicity (See U.S. Pat. No. 4,179,337). The chemical moieties for derivatization may be selected from water soluble polymers such as polyethylene glycol, ethylene glycol/propylene glycol copolymers, carboxymethylcellulose, dextran, polyvinyl alcohol and the like. The antibodies and fragments thereof may be modified at random positions within the molecule, or at predetermined positions within the molecule and may include one, two, three or more attached chemical moieties.

The polymer may be of any molecular weight, and may be branched or unbranched. For polyethylene glycol, the preferred molecular weight is between about 1 kDa and about 100 kDa (the term "about" indicating that in preparations of polyethylene glycol, some molecules will weigh more, some less, than the stated molecular weight) for ease in handling and manufacturing. Other sizes may be used, depending on the desired therapeutic profile (e.g., the duration of sustained release desired, the effects, if any on biological activity, the ease in handling, the degree or lack of antigenicity and other known effects of the polyethylene glycol to a therapeutic protein or analog). For example, the polyethylene glycol may have an average molecular weight of about 200, 500, 1000, 1500, 2000, 2500, 3000, 3500, 4000, 4500, 5000, 5500, 6000, 6500, 7000, 7500, 8000, 8500, 9000, 9500, 10,000, 10,500, 11,000, 11,500, 12,000, 12,500, 13,000, 13,500, 14,000, 14,500, 15,000, 15,500, 16,000, 16,500, 17,000, 17,500, 18,000, 18,500, 19,000, 19,500, 20,000, 25,000, 30,000, 35,000, 40,000, 50,000, 55,000, 60,000, 65,000, 70,000, 75,000, 80,000, 85,000, 90,000, 95,000, or 100,000 kDa. Branched polyethylene glycols are described, for example, in U.S. Pat. No. 5,643,575; Morpurgo et al., *Appl. Biochem. Biotechnol.* 56:59-72 (1996); Vorobjev et al., *Nucleosides Nucleotides* 18:2745-2750 (1999); and Caliceti et al., *Bioconjug. Chem.* 10:638-646 (1999), the disclosures of each of which are incorporated herein by reference.

There are a number of attachment methods available to those skilled in the art. See e.g., EP 0 401 384, herein incorporated by reference (coupling PEG to G-CSF). See also Malik et al., *Exp. Hematol.* 20:1028-1035 (1992) (reporting pegylation of GM-CSF using tresyl chloride). For example, polyethylene glycol may be covalently bound through amino acid residues via a reactive group, such as, a free amino or carboxyl group. Reactive groups are those to which an activated polyethylene glycol molecule may be bound. The amino acid residues having a free amino group may include lysine residues and the N-terminal amino acid residues; those having a free carboxyl group may include aspartic acid residues glutamic acid residues and the C-terminal amino acid residue. Sulfhydryl groups may also be used as a reactive group for attaching the polyethylene glycol molecules. Preferred for therapeutic purposes is attachment at an amino group, such as attachment at the N-terminus or lysine group.

As suggested above, polyethylene glycol may be attached to proteins via linkage to any of a number of amino acid residues. For example, polyethylene glycol can be linked to polypeptides via covalent bonds to lysine, histidine, aspartic acid, glutamic acid, or cysteine residues. One or more reaction chemistries may be employed to attach polyethylene glycol to specific amino acid residues (e.g., lysine, histidine, aspartic acid, glutamic acid, or cysteine) or to more than one type of amino acid residue (e.g., lysine, histidine, aspartic acid, glutamic acid, cysteine and combinations thereof).

Alternatively, antibodies or fragments thereof may have increased in vivo half-lives via fusion with albumin (including but not limited to recombinant human serum albumin or fragments or variants thereof (See, e.g., U.S. Pat. No. 5,876,969, issued Mar. 2, 1999, EP Patent 0 413 622, and U.S. Pat. No. 5,766,883, issued Jun. 16, 1998, herein incorporated by reference in their entirety)) or other circulating blood proteins such as transferrin or ferritin. In a preferred embodiment, polypeptides and/or antibodies of the present invention (including fragments or variants thereof) are fused with the mature form of human serum albumin (i.e., amino acids 1-585 of human serum albumin as shown in FIGS. 1

and 2 of EP Patent 0 322 094) which is herein incorporated by reference in its entirety. Polynucleotides encoding fusion proteins of the invention are also encompassed by the invention.

Regarding detectable moieties, further exemplary enzymes include, but are not limited to, horseradish peroxidase, acetylcholinesterase, alkaline phosphatase, beta-galactosidase and luciferase. Further exemplary fluorescent materials include, but are not limited to, rhodamine, fluorescein, fluorescein isothiocyanate, umbelliferone, dichlorotriazinylamine, phycoerythrin and dansyl chloride. Further exemplary chemiluminescent moieties include, but are not limited to, luminol. Further exemplary bioluminescent materials include, but are not limited to, luciferin and aequorin. Further exemplary radioactive materials include, but are not limited to, Iodine 125 (^{125}I), Carbon 14 (^{14}C), Sulfur 35 (^{35}S), Tritium (H) and Phosphorus 32 (^{32}P).

Regarding functional moieties, exemplary cytotoxic agents include, but are not limited to, methotrexate, aminopterin, 6-mercaptopurine, 6-thioguanine, cytarabine, 5-fluorouracil decarbazine; alkylating agents such as mechlorethamine, thioepa chlorambucil, melphalan, carmustine (BSNU), mitomycin C, lomustine (CCNU), 1-methylnitrosourea, cyclophosphamide, mechlorethamine, busulfan, dibromomannitol, streptozotocin, mitomycin C, cis-dichlorodiamine platinum (II) (DDP) cisplatin and carboplatin (paraplatin); anthracyclines include daunorubicin (formerly daunomycin), doxorubicin (adriamycin), doxorubicin, carminomycin, idarubicin, epirubicin, mitoxantrone and bisantrene; antibiotics include dactinomycin (actinomycin D), bleomycin, calicheamicin, mithramycin, and anthramycin (AMC); and antimetabolic agents such as the vinca alkaloids, vincristine and vinblastine. Other cytotoxic agents include paclitaxel (taxol), ricin, pseudomonas exotoxin, gemcitabine, cytochalasin B, gramicidin D, ethidium bromide, emetine, etoposide, teniposide, colchicin, dihydroxy anthracin dione, 1-dehydrotestosterone, glucocorticoids, procaine, tetracaine, lidocaine, propranolol, puromycin, procarbazine, hydroxyurea, asparaginase, corticosteroids, mytostane (O,P'-(DDD)), interferons, and mixtures of these cytotoxic agents.

Further cytotoxic agents include, but are not limited to, chemotherapeutic agents such as carboplatin, cisplatin, paclitaxel, gemcitabine, calicheamicin, doxorubicin, 5-fluorouracil, mitomycin C, actinomycin D, cyclophosphamide, vincristine and bleomycin. Toxic enzymes from plants and bacteria such as ricin, diphtheria toxin and *Pseudomonas* toxin may be conjugated to the humanized or chimeric antibodies, or binding fragments thereof, to generate cell-type-specific-killing reagents (Youle, et al., *Proc. Nat'l Acad. Sci. USA* 77:5483 (1980); Gilliland, et al., *Proc. Nat'l Acad. Sci. USA* 77:4539 (1980); Kroleick, et al., *Proc. Nat'l Acad. Sci. USA* 77:5419 (1980)).

Other cytotoxic agents include cytotoxic ribonucleases as described by Goldenberg in U.S. Pat. No. 6,653,104. Embodiments of the invention also relate to radioimmunoconjugates where a radionuclide that emits alpha or beta particles is stably coupled to the antibody, or binding fragments thereof, with or without the use of a complex-forming agent. Such radionuclides include beta-emitters such as Phosphorus-32 (^{32}P), Scandium-47 (^{47}Sc), Copper-67 (^{67}Cu), Gallium-67 (^{67}Ga), Yttrium-88 (^{88}Y), Yttrium-90 (^{90}Y), Iodine-125 (^{125}I), Iodine-131 (^{131}I), Samarium-153 (^{153}Sm), Lutetium-177 (^{177}Lu), Rhenium-186 (^{186}Re) or Rhenium-188 (^{188}Re), and alpha-emitters such as Astatine-211 (^{211}At), Lead-212 (^{212}Pb), Bismuth-212 (^{212}Bi) or -213 (^{213}Bi) or Actinium-225 (^{225}Ac).

Methods are known in the art for conjugating an antibody or binding fragment thereof to a detectable moiety and the like, such as for example those methods described by Hunter et al, *Nature* 144:945 (1962); David et al, *Biochemistry* 13:1014 (1974); Pain et al, *J Immunol. Meth.* 40:219 (1981); and Nygren, J., *Histochem. and Cytochem.* 30:407 (1982).

Embodiments described herein further include variants and equivalents that are substantially homologous to the antibodies, antibody fragments, diabodies, SMIPs, camelbodies, nanobodies, IgNAR, polypeptides, variable regions and CDRs set forth herein. These may contain, e.g., conservative substitution mutations, (i.e., the substitution of one or more amino acids by similar amino acids). For example, conservative substitution refers to the substitution of an amino acid with another within the same general class, e.g., one acidic amino acid with another acidic amino acid, one basic amino acid with another basic amino acid, or one neutral amino acid by another neutral amino acid. What is intended by a conservative amino acid substitution is well known in the art.

In another embodiment, the invention contemplates polypeptide sequences having at least 90% or greater sequence homology to any one or more of the polypeptide sequences of antibody fragments, variable regions and CDRs set forth herein. More preferably, the invention contemplates polypeptide sequences having at least 95% or greater sequence homology, even more preferably at least 98% or greater sequence homology, and still more preferably at least 99% or greater sequence homology to any one or more of the polypeptide sequences of antibody fragments, variable regions and CDRs set forth herein. Methods for determining homology between nucleic acid and amino acid sequences are well known to those of ordinary skill in the art.

In another embodiment, the invention further contemplates the above-recited polypeptide homologs of the antibody fragments, variable regions and CDRs set forth herein further having anti-CGRP activity. Non-limiting examples of anti-CGRP activity are set forth herein.

The invention further contemplates treatment methods wherein the one or more anti-human CGRP antibodies discussed above are aglycosylated or if glycosylated are only mannosylated; that contain an Fc region that has been modified to alter effector function, half-life, proteolysis, and/or glycosylation; are human, humanized, single chain or chimeric; and are a humanized antibody derived from a rabbit (parent) anti-human CGRP antibody. An exemplary mutation which impairs glycosylation comprises the mutation of the Asn residue at position 297 of an IgG heavy chain constant region such as IgG1 to another amino acid, such as Ala as described in U.S. Pat. No. 5,624,821, which is incorporated by reference in its entirety.

The invention further contemplates one or more anti-human CGRP antibodies wherein the framework regions (FRs) in the variable light region and the variable heavy regions of said antibody respectively are human FRs which are unmodified or which have been modified by the substitution of one or more human FR residues in the variable light or heavy chain region with the corresponding FR residues of the parent rabbit antibody, and wherein said human FRs have been derived from human variable heavy and light chain antibody sequences which have been selected from a library of human germline antibody sequences based on their high level of homology to the corresponding rabbit variable heavy or light chain regions relative to other human germline antibody sequences contained in the library.

The invention also contemplates that the treatment method may involve the administration of two or more

anti-CGRP antibodies or fragments thereof and disclosed herein. If more than one antibody is administered to the patient, the multiple antibodies may be administered simultaneously or concurrently, or may be staggered in their administration. The anti-CGRP activity of the anti-CGRP antibodies of the present invention, and fragments thereof having binding specificity to CGRP, may also be described by their strength of binding or their affinity for CGRP. In one embodiment of the invention, the anti-CGRP antibodies of the present invention, and fragments thereof having binding specificity to CGRP, bind to CGRP with a dissociation constant (K_D) of less than or equal to 5×10^{-7} M, 10^{-7} M, 5×10^{-8} M, 10^{-9} M, 5×10^{-9} M, 10^{-9} M, 5×10^{-10} M, 10^{-11} M, 5×10^{-11} M, 10^{-12} M, 5×10^{-12} M, 10^{-12} M, 5×10^{-13} M, or 10^{-13} M. Preferably, the anti-CGRP antibodies and fragments thereof bind CGRP with a dissociation constant of less than or equal to 10^{-1} M, 5×10^{-2} M, or 10^{-12} M. In a specific embodiment of the invention the anti-CGRP antibody is Ab6 having a dissociation constant of less than or equal to 10 pM, such as 2-8 pM, such as 3-6 pM, such as less than or equal to about 5 pM when measured using surface plasmon resonance (Misura, K et al, July 2019, Poster P220LB, AHS 61' annual scientific meeting). In another embodiment of the invention, the anti-CGRP antibodies of the present invention, and fragments thereof having binding specificity to CGRP, bind to a linear or conformational CGRP epitope.

In another embodiment of the invention, the anti-CGRP activity of the anti-CGRP antibodies of the present invention, and fragments thereof having binding specificity to CGRP, bind to CGRP with an off-rate of less than or equal to 10^{-4} S $^{-1}$, 5×10^{-5} S $^{-1}$, 10^{-5} S $^{-1}$, 5×10^{-6} S $^{-1}$, 10^{-6} S $^{-1}$, 5×10^{-7} S $^{-1}$, or 10^{-7} S $^{-1}$. In a specific embodiment of the invention the anti-CGRP antibody is Ab6 having an off-rate of less than or equal to 5×10^{-6} S $^{-1}$, such as less than or equal to 4×10^{-6} S $^{-1}$, such as less than or equal to 3×10^{-6} S $^{-1}$, such as less than or equal to 2×10^{-6} S $^{-1}$, such as less than or equal to 1×10^{-6} S $^{-1}$ when measured using surface plasmon resonance.

Polynucleotides Encoding Anti-CGRP Antibody Polypeptides

As aforementioned the invention specifically includes the use of specific anti-CGRP antibody or antibody fragment referred to herein as Ab6, which comprises or consists of the CDR, VL, VH, CL, and CH polypeptides having the sequences identified in FIGS. 1-12. The nucleic acid sequences encoding the foregoing VL, VH, CL, and CH polypeptides comprised in Ab6 are also comprised in FIGS. 1-12. The nucleic acid sequences which encode the CDR, VL, VH, CL, and CH polypeptides of an especially preferred anti-CGRP antibody, Ab6, are further described below.

Polynucleotides Encoding Antibody Ab6

The invention is further directed to polynucleotides encoding antibody polypeptides having binding specificity to CGRP. In one embodiment of the invention, polynucleotides of the invention comprise, or alternatively consist of, the following polynucleotide sequence encoding the variable light chain polypeptide sequence of SEQ ID NO: 222:

(SEQ ID NO: 232)

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CAAGTGCTGaccagctctccatccctcctgtctgcatctgtaggagaca
gagtcaccatcAATtgcCAGGCCAGTCAGAGTGTTATCATAACACCTA
CCTGGCCtggtatcagcagaaaccagggaagttcctaagCAActgatc
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tatGATGCATCCACTCTGGCATCTggggtcccatctcgtttcagtgga
gtggatctgggacagatttcaactctcaccatcagcagcctgcagcctga
agatgttgcaacttattactgtCTGGGCAGTTATGATTGTACTAATGGT
GATTGTTTTGTTtctcgcgaggaggaaccaaggtggaaatcaaacgt.

In one embodiment of the invention, polynucleotides of the invention comprise, or alternatively consist of, the following polynucleotide sequence encoding the light chain polypeptide sequence of SEQ ID NO: 221:

(SEQ ID NO: 231)

CAAGTGCTGacccagctctccatctccctgtctgcctctgtaggagaca
gagtcaccatcAATtgcCAGGCCAGTCAGAGTGTATTATATAACACCTA
CCTGGCCtggtatcagcagaaaccagggaagttcctaagCAActgac
tatGATGCATCCACTCTGGCATCTggggtcccatctcgtttcagtgga
gtggatctgggacagatttcaactctcaccatcagcagcctgcagcctga
agatgttgcaacttattactgtCTGGGCAGTTATGATTGTACTAATGGT
GATTGTTTTGTTtctcgcgaggaggaaccaaggtggaaatcaaacgtACGG
TGCGTGACCATCTGTCTTCTATCTCCCGCCATCTGATGAGCAGTTGAA
ATCTGGAAGTGCCTCTGTGTGTGCTGCTGAATAACTTCTATCCAGA
GAGGCCAAGTACAGTGAAGTGGATAACGCCCTCCAATCGGGTAACT
CCCAGGAGAGTGTACAGAGCAGGACAGCAAGGACAGCACCTACAGCCT
CAGCAGCACCTGACGCTGAGCAAAGCAGACTACGAGAAACACAAAGTC
TACGCTGCGAAGTACCCATCAGGGCCTGAGCTCGCCCGTCACAAAGA
GCTTCAACAGGGGAGAGTGTAG.

In another embodiment of the invention, polynucleotides of the invention comprise, or alternatively consist of, the following polynucleotide sequence encoding the variable heavy chain polypeptide sequence of SEQ ID NO: 202:

(SEQ ID NO: 212)

gaggtgcagctTgtggagctctggggaggcttggtccagcctggggggt
ccctgagactctcctgtgcaGTCtctggaATCGACCTCagtGGTACTA
CATGAACTgggtccgtcaggctccagggaaggggctggagtggtcGGA
GTCATTGGTATTAATGGTGCCACATACTACGCGAGCTGGGCGAAAGGCC
gattcaccatctccagagacaattccaagACCACGGTGTatcttcaaat
gaacagcctgagagctgaggacactgctgtgtatTTCTgtGCTAGAGGG
GACATCTggggccaagggaacctcgtcaccgtcTCGAGC.

In one embodiment of the invention, polynucleotides of the invention comprise, or alternatively consist of, the following polynucleotide sequence encoding the heavy chain polypeptide sequence of SEQ ID NO: 201:

(SEQ ID NO: 211)

gaggtgcagctTgtggagctctggggaggcttggtccagcctggggggt
ccctgagactctcctgtgcaGTCtctggaATCGACCTCagtGGTACTA
CATGAACTgggtccgtcaggctccagggaaggggctggagtggtcGGA

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GTCATTGGTATTAATGGTGCCACATACTACGCGAGCTGGGCGAAAGGCC
gattcaccatctccagagacaattccaagACCACGGTGTatcttcaaat
5 gaacagcctgagagctgaggacactgctgtgtatTTCTgtGCTAGAGGG
GACATCTggggccaagggaacctcgtcaccgtcTCGAGCGCTCCACCA
AGGGCCCATCGGTCTTCCCCCTGGCAcCCTCCTCCaAGAGCACCTCTGG
10 GGGCACAGCGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAACCG
GTGACGGTGTCTGGAACTCAGGCGCCCTGACCAGCGGCGTGCACACCT
TCCCGGCTGTCTACAGTCTCTCAGGACTCTACTCCCTCAGCAGCGTGGT
15 GACCGTGCCCTCCAGCAGCTTGGGCACCCAGACCTACATCTGCAACGTG
AATCACAAGCCCAGCAACACCAAGGTGGACGCGAGAGTTGAGCCCAAT
CTTGTGACAAAACCTACACATGCCACCGTGCCAGCACCTGAACCTCT
20 GGGGGGACCGTCAGTCTTCTCTTCCCCCAAAACCAAGGACACCCCTC
ATGaTCTCCCGGACCCCTGAGGTACATGCGTGGTGGTGGACGTGAGCC
ACGAAGACCCCTGAGGTCAAGTCAACTGGTACGTGGACGGCGTGGAGGT
GCATAATGCCAAGACAAAGCCGCGGGAGGAGCAGTACGCCAGCAGCTAC
25 CGTGTGGTCAGCGTCTCACCCTCTGACCAGGACTGGCTGAATGGCA
AGGAGTACAAGTGCAAGGTCTCCAACAAGCCCTCCAGCCCCCATCGA
GAAAACCATCTCCAAGCCAAAGGGCAGCCCCGAGAACCACAGGTGTAC
30 ACCCTGCCCCCATCCCGGGAGGAGATGACCAAGAACCAGGTGAGCTGA
CCTGCCTGGTCAAAGGCTTCTATCCAGCGACATCGCCGTGGATGGGA
GAGCAATGGGCAGCCGGAGAACAACTACAAGACCAGCCTCCCTGTCTG
35 GACTCCGACGGCTCTTCTTCTCTACAGCAAGCTCACCCTGGACAAGA
GCAGGTGGCAGCAGGGGAACGTCTTCTCATGCTCCGTGATGCATGAGGC
TCTGCACAACCACTACACGCAGAAGAGCCTCTCCCTGTCTCCGGGTAAA
40 TGA.

In one embodiment of the invention, polynucleotides of the invention comprise, or alternatively consist of, the following polynucleotide sequence encoding the heavy chain polypeptide sequence of SEQ ID NO: 566:

(SEQ ID NO: 567)

gaggtgcagctTgtggagctctggggaggcttggtccagcctggggggt
50 ccctgagactctcctgtgcaGTCtctggaATCGACCTCagtGGTACTA
CATGAACTgggtccgtcaggctccagggaaggggctggagtggtcGGA
GTCATTGGTATTAATGGTGCCACATACTACGCGAGCTGGGCGAAAGGCC
55 gattcaccatctccagagacaattccaagACCACGGTGTatcttcaaat
gaacagcctgagagctgaggacactgctgtgtatTTCTgtGCTAGAGGG
GACATCTggggccaagggaacctcgtcaccgtcTCGAGCGCTCCACCA
AGGGCCCATCGGTCTTCCCCCTGGCAcCCTCCTCCaAGAGCACCTCTGG
60 GGGCACAGCGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAACCG
GTGACGGTGTCTGGAACTCAGGCGCCCTGACCAGCGGCGTGCACACCT
TCCCGGCTGTCTACAGTCTCTCAGGACTCTACTCCCTCAGCAGCGTGGT
65 GACCGTGCCCTCCAGCAGCTTGGGCACCCAGACCTACATCTGCAACGTG

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AATCACAAAGCCAGCAACACCAAGGTGGACGCGAGAGTTGAGCCCAAAT
 CTTGTGACAAAACCTCACACATGCCACCGTGCCAGCACCTGAACCTCCT
 GGGGGGACCGTCAGTCTTCTCTTCCCCCAAACCAAGGACACCCCTC
 ATGaTCTCCCGGACCCCTGAGGTACATGCGTGGTGGAGCTGAGCC
 ACGAAGACCTGAGGTCAAGTTCAACTGGTACGTGGACGGCGTGGAGGT
 GCATAATGCCAAGACAAGCCGCGGAGGAGCAGTACGCCAGCACGTAC
 CGTGTGGTCAGCGTCTCACCCTCTGCACCAGGACTGGCTGAATGGCA
 AGGAGTACAAGTGCAAGGTCTCCAACAAAGCCCTCCAGCCCCCATCGA
 GAAAACCATCTCCAAAGCCAAGGGCAGCCCCGAGAACCACAGGTGTAC
 ACCCTGCCCCCATCCCGGAGGAGATGACCAAGAACCAGGTGAGCTGA
 CCTGCCTGGTCAAAGGCTTCTATCCAGCGACATCGCCGTGGAGTGGGA
 GAGCAATGGGCAGCCGAGAGAACAACTACAAGACCACGCTCCCGTGCTG
 GACTCCGACGGCTCCTTCTTCTCTACAGCAAGCTCACCGTGACAAGA
 GCAGGTGGCAGCAGGGGAACGTCTTCTCATGCTCCGTGATGCATGAGGC
 TCTGCACAACCACTACACGCAAGAGCCTCTCCCTGTCTCCGGTGA.

In a further embodiment of the invention, polynucleotides encoding antibody fragments having binding specificity to CGRP comprise, or alternatively consist of, one or more of the polynucleotide sequences of SEQ ID NO: 234; SEQ ID NO: 236; and SEQ ID NO: 238 which correspond to polynucleotides encoding the complementarity-determining regions (CDRs, or hypervariable regions) of the light chain variable sequence of SEQ ID NO: 222 or the light chain sequence of SEQ ID NO: 221.

In a further embodiment of the invention, polynucleotides encoding antibody fragments having binding specificity to CGRP comprise, or alternatively consist of, one or more of the polynucleotide sequences of SEQ ID NO: 214; SEQ ID NO: 216; and SEQ ID NO: 218 which correspond to polynucleotides encoding the complementarity-determining regions (CDRs, or hypervariable regions) of the heavy chain variable sequence of SEQ ID NO: 202 or the heavy chain sequence of SEQ ID NO: 201 or SEQ ID NO: 566.

The invention also contemplates polynucleotide sequences including one or more of the polynucleotide sequences encoding antibody fragments described herein. In one embodiment of the invention, polynucleotides encoding antibody fragments having binding specificity to CGRP comprise, or alternatively consist of, one, two, three or more, including all of the following polynucleotides encoding antibody fragments: the polynucleotide SEQ ID NO: 232 encoding the light chain variable sequence of SEQ ID NO: 222; the polynucleotide SEQ ID NO: 231 encoding the light chain sequence of SEQ ID NO: 221; the polynucleotide SEQ ID NO: 212 encoding the heavy chain variable sequence of SEQ ID NO: 202; the polynucleotide SEQ ID NO: 211 encoding the heavy chain sequence of SEQ ID NO: 201; the polynucleotide SEQ ID NO: 567 encoding the heavy chain sequence of SEQ ID NO: 566; polynucleotides encoding the complementarity-determining regions (SEQ ID NO: 234; SEQ ID NO: 236; and SEQ ID NO: 238) of the light chain variable sequence of SEQ ID NO: 222 or the light chain sequence of SEQ ID NO: 221; and polynucleotides encoding the complementarity-determining regions (SEQ ID NO: 214; SEQ ID NO: 216; and SEQ ID NO: 218) of the heavy

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chain variable sequence of SEQ ID NO: 202 or the heavy chain sequence of SEQ ID NO: 201 or SEQ ID NO: 566.

In a preferred embodiment of the invention, polynucleotides of the invention comprise, or alternatively consist of, polynucleotides encoding Fab (fragment antigen binding) fragments having binding specificity for CGRP. With respect to antibody Ab6, the polynucleotides encoding the full length Ab6 antibody comprise, or alternatively consist of, the polynucleotide SEQ ID NO: 231 encoding the light chain sequence of SEQ ID NO: 221 and the polynucleotide SEQ ID NO: 211 encoding the heavy chain sequence of SEQ ID NO: 201 or the polynucleotide SEQ ID NO: 567 encoding the heavy chain sequence of SEQ ID NO: 566.

Another embodiment of the invention contemplates these polynucleotides incorporated into an expression vector for expression in mammalian cells such as CHO, NSO, HEK-293, or in fungal, insect, or microbial systems such as yeast cells such as the yeast *Pichia*. Suitable *Pichia* species include, but are not limited to, *Pichia pastoris*. In one embodiment of the invention described herein (infra), Fab fragments may be produced by enzymatic digestion (e.g., papain) of Ab6 following expression of the full-length polynucleotides in a suitable host. In another embodiment of the invention, anti-CGRP antibodies such as Ab6 or Fab fragments thereof may be produced via expression of Ab6 polynucleotides in mammalian cells such as CHO, NSO or HEK 293 cells, fungal, insect, or microbial systems such as yeast cells (for example diploid yeast such as diploid *Pichia*) and other yeast strains. Suitable *Pichia* species include, but are not limited to, *Pichia pastoris*.

Host cells and vectors comprising said polynucleotides are also contemplated.

The invention further contemplates vectors comprising the polynucleotide sequences encoding the variable heavy and light chain polypeptide sequences, as well as the individual complementarity-determining regions (CDRs, or hypervariable regions), as set forth herein, as well as host cells comprising said vector sequences. In one embodiment of the invention, the host cell is a yeast cell. In another embodiment of the invention, the yeast host cell belongs to the genus *Pichia*.

Methods of Producing Antibodies and Fragments Thereof

In another embodiment, the present invention contemplates methods for producing anti-CGRP antibodies and fragments thereof. Methods for producing antibodies and fragments thereof secreted from polyploid, preferably diploid or tetraploid strains of mating competent yeast are taught, for example, in U.S. patent application publication no. US 2009/0022659 to Olson et al., and in U.S. Pat. No. 7,935,340 to Garcia-Martinez et al., the disclosures of each of which are herein incorporated by reference in their entireties. Methods for producing antibodies and fragments thereof in mammalian cells, e.g., CHO cells are further well known in the art.

Other methods of producing antibodies are also well known to those of ordinary skill in the art. For example, methods of producing chimeric antibodies are now well known in the art (See, for example, U.S. Pat. No. 4,816,567 to Cabilly et al.; Morrison et al., *P.N.A.S. USA*, 81:8651-55 (1984); Neuberger, M. S. et al., *Nature*, 314:268-270 (1985); Boulianne, G. L. et al., *Nature*, 312:643-46 (1984), the disclosures of each of which are herein incorporated by reference in their entireties).

Likewise, other methods of producing humanized antibodies are now well known in the art (See, for example, U.S. Pat. Nos. 5,530,101, 5,585,089, 5,693,762, and 6,180,370 to Queen et al; U.S. Pat. Nos. 5,225,539 and 6,548,640 to

Winter; U.S. Pat. Nos. 6,054,297, 6,407,213 and 6,639,055 to Carter et al; U.S. Pat. No. 6,632,927 to Adair; Jones, P. T. et al, *Nature*, 321:522-525 (1986); Reichmann, L., et al, *Nature*, 332:323-327 (1988); Verhoeyen, M., et al, *Science*, 239:1534-36 (1988), the disclosures of each of which are herein incorporated by reference in their entireties).

The present invention further includes the use of any of the pharmaceutical formulations disclosed herein in the manufacture of a medicament for the treatment, prevention and/or amelioration of most bothersome symptom associated with migraine.

Administration

In one embodiment of the invention, the anti-CGRP antibodies described herein, or CGRP binding fragments thereof, as well as combinations of said antibodies or antibody fragments, are administered to a subject at a concentration of between about 0.1 and 100.0 mg/kg of body weight of recipient subject. In a preferred embodiment of the invention, the anti-CGRP antibodies described herein, or CGRP binding fragments thereof, as well as combinations of said antibodies or antibody fragments, are administered to a subject at a concentration of about 0.4 mg/kg of body weight of recipient subject and/or at a dosage of 100 or 300 mg. In a preferred embodiment of the invention, the anti-CGRP antibodies described herein, or CGRP binding fragments thereof, as well as combinations of said antibodies or antibody fragments, are administered to a recipient subject with a frequency of once every twenty-six weeks or six months or less, such as once every sixteen weeks or four months or less, once every eight weeks or two months or less, once every four weeks or monthly or less, once every two weeks or bimonthly or less, once every week or less, or once daily or less. In general the administration of sequential doses may vary by plus or minus a few days from the aforementioned schedule, e.g., administration every 3 months or every 12 weeks includes administration of a dose varying from the schedule day by plus or minus 1, 2, 3, 4, 5, 5, or 7 days.

Fab fragments may be administered every two weeks or less, every week or less, once daily or less, multiple times per day, and/or every few hours. In one embodiment of the invention, a patient receives Fab fragments of 0.1 mg/kg to 40 mg/kg per day given in divided doses of 1 to 6 times a day, or in a sustained release form, effective to obtain desired results.

It is to be understood that the concentration of the antibody or Fab administered to a given patient may be greater or lower than the exemplary administration concentrations set forth above.

A person of skill in the art would be able to determine an effective dosage and frequency of administration through routine experimentation, for example guided by the disclosure herein and the teachings in Goodman, L. S., Gilman, A., Brunton, L. L., Lazo, J. S., & Parker, K. L. (2006). Goodman & Gilman's the pharmacological basis of therapeutics. New York: McGraw-Hill; Howland, R. D., Mycek, M. J., Harvey, R. A., Champe, P. C., & Mycek, M. J. (2006). Pharmacology. Lippincott's illustrated reviews. Philadelphia: Lippincott Williams & Wilkins; and Golan, D. E. (2008). Principles of pharmacology: the pathophysiologic basis of drug therapy. Philadelphia, Pa., [etc.]: Lippincott Williams & Wilkins.

In another embodiment of the invention, the anti-CGRP antibodies described herein, or CGRP binding fragments thereof, as well as combinations of said antibodies or antibody fragments, are administered to a subject in a pharmaceutical formulation.

A "pharmaceutical composition" refers to a chemical or biological composition suitable for administration to a mammal. Such compositions may be specifically formulated for administration via one or more of a number of routes, including but not limited to buccal, epicutaneous, epidural, inhalation, intraarterial, intracardial, intracerebroventricular, intradermal, intramuscular, intranasal, intraocular, intraperitoneal, intraspinal, intrathecal, intravenous, oral, parenteral, rectally via an enema or suppository, subcutaneous, subdermal, sublingual, transdermal, and transmucosal, preferably intravenous. In addition, administration can occur by means of injection, powder, liquid, gel, drops, or other means of administration.

A "pharmaceutical excipient" or a "pharmaceutically acceptable excipient" is a carrier, usually a liquid, in which an active therapeutic agent is formulated. In one embodiment of the invention, the active therapeutic agent is a humanized antibody described herein, or one or more fragments thereof. The excipient generally does not provide any pharmacological activity to the formulation, though it may provide chemical and/or biological stability, and release characteristics. Exemplary formulations can be found, for example, in Remington's Pharmaceutical Sciences, 19th Ed., Grennaro, A., Ed., 1995 which is incorporated by reference.

As used herein "pharmaceutically acceptable carrier" or "excipient" includes any and all solvents, dispersion media, coatings, antibacterial and antifungal agents, isotonic and absorption delaying agents that are physiologically compatible. In one embodiment, the carrier is suitable for parenteral administration. Alternatively, the carrier can be suitable for intravenous, intraperitoneal, intramuscular, or sublingual administration. Pharmaceutically acceptable carriers include sterile aqueous solutions or dispersions and sterile powders for the extemporaneous preparation of sterile injectable solutions or dispersions. The use of such media and agents for pharmaceutically active substances is well known in the art. Except insofar as any conventional media or agent is incompatible with the active compound, use thereof in the pharmaceutical compositions of the invention is contemplated. Supplementary active compounds can also be incorporated into the compositions.

Pharmaceutical compositions typically must be sterile and stable under the conditions of manufacture and storage. The invention contemplates that the pharmaceutical composition is present in lyophilized form. The composition can be formulated as a solution, microemulsion, liposome, or other ordered structure suitable to high drug concentration. The carrier can be a solvent or dispersion medium containing, for example, water, ethanol, polyol (for example, glycerol, propylene glycol, and liquid polyethylene glycol), and suitable mixtures thereof. The invention further contemplates the inclusion of a stabilizer in the pharmaceutical composition. The proper fluidity can be maintained, for example, by the maintenance of the required particle size in the case of dispersion and by the use of surfactants.

In many cases, it will be preferable to include isotonic agents, for example, sugars, polyalcohols such as mannitol, sorbitol, or sodium chloride in the composition. Prolonged absorption of the injectable compositions can be brought about by including in the composition an agent which delays absorption, for example, monostearate salts and gelatin. Moreover, the alkaline polypeptide can be formulated in a time release formulation, for example in a composition which includes a slow release polymer. The active compounds can be prepared with carriers that will protect the compound against rapid release, such as a controlled release formulation, including implants and microencapsulated

delivery systems. Biodegradable, biocompatible polymers can be used, such as ethylene vinyl acetate, polyanhydrides, polyglycolic acid, collagen, polyorthoesters, polylactic acid and polylactic, polyglycolic copolymers (PLG). Many methods for the preparation of such formulations are known to those skilled in the art.

An exemplary composition comprises, consists essentially of Ab6, an excipient such as histidine, an isotonic agent such as sorbitol, and a surfactant such as polysorbate 80 in an aqueous solution. For example, the composition may comprise, consist essentially of, or consist of histidine (L-histidine), sorbitol, polysorbate 80, such as, per 1 mL volume, about 100 mg Ab6, about 3.1 mg L-Histidine, about 40.5 mg Sorbitol, and about 0.15 mg Polysorbate 80, having a pH of about 5.8, or approximately that constitution, e.g., within 10% of those values, within 5% of those values, within 1% of those values, within 0.5% of those values, or within 0.1% of those values, and water. For example, the pH value may be within 10% of 5.8, i.e., between 5.22 and 6.38. The Ab6 antibody may comprise or consist of the variable light and heavy chain polypeptides of SEQ ID NO: 222 and SEQ ID NO: 202 respectively, or the light and heavy chain polypeptides of SEQ ID NO: 221 and SEQ ID NO: 201 respectively, or the light and heavy chain polypeptides of SEQ ID NO: 221 and SEQ ID NO: 566 respectively. The composition may be in the form of an aqueous solution, or a concentrate (e.g., lyophilized) which when reconstituted, e.g., by addition of water, yields the aforementioned constitution. An exemplary composition consists of, per mL, 100 mg of the light and heavy chain polypeptides of SEQ ID NO: 221 and SEQ ID NO: 201 respectively, about 3.1 mg L-Histidine, about 40.5 mg Sorbitol, and about 0.15 mg Polysorbate 80, and water Q.S. or approximately that constitution, e.g., within 10% of those quantities, within 5% of those quantities, within 1% of those quantities, within 0.5% of those quantities, or within 0.10% of those quantities. Another exemplary composition consists of, per mL, 100 mg of the light and heavy chain polypeptides of SEQ ID NO: 221 and SEQ ID NO: 566 respectively, about 3.1 mg L-Histidine, about 40.5 mg Sorbitol, and about 0.15 mg Polysorbate 80, and water Q.S. or approximately that constitution, e.g., within 10% of those quantities, within 5% of those quantities, within 1% of those quantities, within 0.5% of those quantities, or within 0.10% of those quantities. The composition may be suitable for intravenous or subcutaneous administration, preferably intravenous administration. For example, the composition may be suitable for mixing with an intravenous solution (such as 0.9% sodium chloride) at an amount of between about 100 mg and about 300 mg antibody added to 100 mL of intravenous solution. Preferably the composition may be shelf-stable for at least 1, 3, 6, 12, 18, or 24 months, e.g., showing formation of aggregates of no more than 5% or no more than 10% of the antibody or fragment after storage at room temperature or when refrigerated at 4° C. for the specified duration, or in an accelerated aging test that simulates storage for that duration.

For each of the recited embodiments, the compounds can be administered by a variety of dosage forms. Any biologically-acceptable dosage form known to persons of ordinary skill in the art, and combinations thereof, are contemplated. Examples of such dosage forms include, without limitation, reconstitutable powders, elixirs, liquids, solutions, suspensions, emulsions, powders, granules, particles, microparticles, dispersible granules, cachets, inhalants, aerosol inhalants, patches, particle inhalants, implants, depot implants,

injectables (including subcutaneous, intramuscular, intravenous, and intradermal, preferably intravenous), infusions, and combinations thereof.

The above description of various illustrated embodiments of the invention is not intended to be exhaustive or to limit the invention to the precise form disclosed. While specific embodiments of, and examples for, the invention are described herein for illustrative purposes, various equivalent modifications are possible within the scope of the invention, as those skilled in the relevant art will recognize. The teachings provided herein of the invention can be applied to other purposes, other than the examples described above.

These and other changes can be made to the invention in light of the above detailed description. In general, in the following claims, the terms used should not be construed to limit the invention to the specific embodiments disclosed in the specification and the claims. Accordingly, the invention is not limited by the disclosure, but instead the scope of the invention is to be determined entirely by the following claims.

The invention may be practiced in ways other than those particularly described in the foregoing description and examples. Numerous modifications and variations of the invention are possible in light of the above teachings and, therefore, are within the scope of the appended claims.

Certain CGRP antibody polynucleotides and polypeptides are disclosed in the sequence listing accompanying this patent application filing, and the disclosure of said sequence listing is herein incorporated by reference in its entirety.

The entire disclosure of each document cited (including patents, patent applications, journal articles, abstracts, manuals, books, or other disclosures) in the Background of the Invention, Detailed Description, and Examples is herein incorporated by reference in their entireties.

The following examples are put forth so as to provide those of ordinary skill in the art with a complete disclosure and description of how to make and use the subject invention, and are not intended to limit the scope of what is regarded as the invention. Efforts have been made to ensure accuracy with respect to the numbers used (e.g. amounts, temperature, concentrations, etc.) but some experimental errors and deviations should be allowed for. Unless otherwise indicated, parts are parts by weight, molecular weight is average molecular weight, temperature is in degrees centigrade; and pressure is at or near atmospheric.

Additional Exemplary Embodiments

Additional exemplary embodiments of the invention are provided as follows:

S1. Use of an anti-CGRP antibody for the manufacturing of a medicament for treating most bothersome symptom (MBS) associated with migraine, comprising administering to a migraine patient an anti-CGRP antibody.

S2. Use of an anti-CGRP antibody for the manufacturing of a medicament for treating most bothersome symptom (MBS) associated with migraine, comprising administering to a migraine patient an anti-CGRP antibody, wherein said migraine patient suffers from chronic migraine.

S3. Use of an anti-CGRP antibody for the manufacturing of a medicament for treating most bothersome symptom (MBS) associated with migraine, comprising administering to a migraine patient an anti-CGRP antibody, wherein said patient suffers from episodic migraine.

S4. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein the patients most bothersome symptom (MBS) associated with migraine is improved

at 1-12 hours post-completion of administration or infusion, such as 1-5 hours post-completion of administration or infusion, 1-2 hours post-completion of administration or infusion, or about 2 hours post-completion of administration or infusion.

S5. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein the patients most bothersome symptom (MBS) associated with migraine is improved within 1 month from the first dosing with said anti-CGRP antibody.

S6. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein the patients most bothersome symptom (MBS) associated with migraine is improved within 3 month from the first dosing with said anti-CGRP antibody.

S7. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein the patients most bothersome symptom (MBS) associated with migraine is improved within 6 month from the first dosing with said anti-CGRP antibody.

S8. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein the improvement is sustained for at least 3 months from the first dosing with said anti-CGRP antibody.

S9. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein the improvement is sustained for at least 6 months from the first dosing with said anti-CGRP antibody.

S10. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein the MBS is selected from the group consisting of: Sensitivity to light (photophobia), Nausea/vomiting, Headache, Sensitivity to sound (phonophobia), Aura, Pain with activity, Pain, Throbbing/pulsation, Cognitive disruption, Fatigue, Mood changes, Sensitivity to smell (osmophobia or olfactophobia), Visual impact, Pressure/tightness, Pain (anatomical), Eye pain, Neck pain, Dizziness, Allodynia, Inactivity, Sensory disturbance, Sleep disturbance and Speech difficulty.

S11. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein the MBS is selected from the group consisting of: Sensitivity to light (photophobia), Nausea/vomiting, Headache, Sensitivity to sound (phonophobia), Pain with activity, Pain, Throbbing/pulsation, Cognitive disruption, Fatigue, Mood changes and Sensitivity to smell (osmophobia or olfactophobia).

S12. Use of an anti-CGRP antibody for the manufacturing of a medicament for improving patient global impression of change (PGIC) associated with migraine, comprising administering to a migraine patient an anti-CGRP antibody.

S13. Use of an anti-CGRP antibody for the manufacturing of a medicament for improving patient global impression of change (PGIC) associated with migraine, comprising administering to a migraine patient an anti-CGRP antibody, wherein said migraine patient suffers from chronic migraine.

S14. Use of an anti-CGRP antibody for the manufacturing of a medicament for improving patient global impression of change (PGIC) associated with migraine, comprising administering to a migraine patient an anti-CGRP antibody, wherein said patient suffers from episodic migraine.

S15. Use of the anti-CGRP antibody of any one of embodiments S12-S14, wherein the administration of said medicament improves patient global impression of change (PGIC) associated with migraine within 1 month from the first dosing with said anti-CGRP antibody.

S16. Use of the anti-CGRP antibody of any one of embodiments S12-S14, wherein the administration of said medicament improves patient global impression of change

(PGIC) associated with migraine within 3 month from the first dosing with said anti-CGRP antibody.

S17. Use of the anti-CGRP antibody of any one of embodiments S12-S14, wherein the administration of said medicament improves patient global impression of change (PGIC) associated with migraine within 6 month from the first dosing with said anti-CGRP antibody.

S18. Use of the anti-CGRP antibody of any one of embodiments S12-S17, wherein the administration of said medicament improves patient global impression of change (PGIC) associated with migraine, and wherein the improvement is sustained for at least 3 months from the first dosing with said anti-CGRP antibody.

S19. Use of the anti-CGRP antibody of any one of embodiments S12-S18, wherein the administration of said medicament improves patient global impression of change (PGIC) associated with migraine, and wherein the improvement is sustained for at least 6 months from the first dosing with said anti-CGRP antibody.

S20. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said medicament is for intravenous or subcutaneous infusion.

S21. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said medicament is for intravenous infusion.

S22. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said patient is headache free 2 hours post-completion of administration or infusion.

S23. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises Ab6.

S24. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the light chain complementarity-determining region (CDR) 1, 2, and 3 polypeptide sequences of SEQ ID NO: 224; SEQ ID NO: 226; and SEQ ID NO: 228, respectively.

S25. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the light chain CDR 1, 2, and 3 polypeptide sequences encoded by SEQ ID NO: 234; SEQ ID NO: 236; and SEQ ID NO: 238, respectively.

S26. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the heavy chain CDR 1, 2, and 3 polypeptide sequences of SEQ ID NO: 204; SEQ ID NO: 206; and SEQ ID NO: 208, respectively.

S27. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the heavy chain CDR 1, 2, and 3 polypeptide sequences encoded by SEQ ID NO: 214; SEQ ID NO: 216; and SEQ ID NO: 218, respectively.

S28. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the light chain CDR 1, 2, and 3 polypeptide sequences of SEQ ID NO: 224; SEQ ID NO: 226; and SEQ ID NO: 228, respectively and heavy chain CDR 1, 2, and 3 polypeptide sequences of SEQ ID NO: 204; SEQ ID NO: 206; and SEQ ID NO: 208, respectively.

S29. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the light chain CDR 1, 2, and 3 polypeptide sequences encoded by SEQ ID NO: 234; SEQ ID NO: 236; and SEQ ID NO: 238, respectively and heavy chain CDR 1, 2, and 3 polypeptide sequences encoded by SEQ ID NO: 214; SEQ ID NO: 216; and SEQ ID NO: 218, respectively.

S30. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the variable light chain polypeptide of SEQ ID NO: 222.

S31. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the variable light chain polypeptide encoded by SEQ ID NO: 232.

S32. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the variable heavy chain polypeptide of SEQ ID NO: 202.

S33. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the variable heavy chain polypeptide encoded by SEQ ID NO: 212.

S34. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the variable light chain polypeptide of SEQ ID NO: 222 and the variable heavy chain polypeptide of SEQ ID NO: 202.

S35. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the variable light chain polypeptide encoded by SEQ ID NO: 232 and the variable heavy chain polypeptide encoded by SEQ ID NO: 212.

S36. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the light chain polypeptide of SEQ ID NO: 221.

S37. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the light chain polypeptide encoded by SEQ ID NO: 231.

S38. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the heavy chain polypeptide of SEQ ID NO: 201 or SEQ ID NO: 566.

S39. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the heavy chain polypeptide encoded by SEQ ID NO: 211 or SEQ ID NO: 567.

S40. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the light chain polypeptide of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201 or SEQ ID NO: 566.

S41. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the light chain polypeptide encoded by SEQ ID NO: 231 and the heavy chain polypeptide encoded by SEQ ID NO: 211 or SEQ ID NO: 567.

S42. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein the administered amount of said anti-CGRP antibody is between about 100 mg and about 300 mg, or is about 100 mg, or is about 300 mg.

S43. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein the administered amount of said anti-CGRP antibody is 100 mg.

S44. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said medicament is for intravenous administration in a dosage of 100 mg of said anti-CGRP antibody every 10-14 weeks, preferably every 11-13 weeks, more preferably every 12 weeks.

S45. Use of the anti-CGRP antibody of any one of the foregoing embodiments S1-S42, wherein said medicament is for intravenous administration in a dosage of 300 mg of said

anti-CGRP antibody every 10-14 weeks, preferably every 11-13 weeks, more preferably every 12 weeks.

S46. Use of the anti-CGRP antibody of any one of the foregoing embodiments wherein, prior to administration of said medicament, the patient exhibits between 1-10 migraine attacks per month in the month or in the 3 months prior to administration.

S47. Use of the anti-CGRP antibody of any one of the foregoing embodiments wherein, prior to administration of said medicament, the patient exhibits between 2-8 migraine attacks per month in the month or in the 3 months prior to administration.

S48. Use of the anti-CGRP antibody of any one of the foregoing embodiments wherein, prior to administration of said medicament, the patient exhibits between 3-7 migraine attacks per month in the month or in the 3 months prior to administration.

S49. Use of the anti-CGRP antibody of any one of the foregoing embodiments wherein, prior to administration of said medicament, the patient exhibits less than 25 headache days per month in the month or in the 3 months prior to administration.

S50. Use of the anti-CGRP antibody of any one of the foregoing embodiments wherein, prior to administration of said medicament, the patient exhibits less than 20 headache days per month in the month or in the 3 months prior to administration.

S51. Use of the anti-CGRP antibody of any one of the foregoing embodiments wherein, prior to administration of said medicament, the patient exhibits less than 15 headache days per month in the month or in the 3 months prior to administration.

S52. Use of the anti-CGRP antibody of any one of the foregoing embodiments wherein, prior to administration of said medicament, the patient exhibits less than 10 headache days per month in the month or in the 3 months prior to administration.

S53. Use of the anti-CGRP antibody of any one of the foregoing embodiments wherein said patient was diagnosed with migraine at least 10 years prior to administration of said medicament.

S54. Use of the anti-CGRP antibody of any one of the foregoing embodiments wherein said patient was diagnosed with migraine at least 15 years prior to administration of said medicament.

S55. Use of the anti-CGRP antibody of any one of the foregoing embodiments wherein said patient was diagnosed with migraine at least 18 or at least 19 years prior to administration of said medicament.

S56. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said patient further has a reduction in the number of migraine days by at least 50% in the one month period after being administered said antibody relative to the baseline number of migraine days experienced by that patient prior to administration of said medicament.

S57. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said patient further has a reduction in the number of migraine days by at least 75% in the one month period after being administered said antibody relative to the baseline number of migraine days experienced by that patient prior to administration of said medicament.

S58. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said patient further has a reduction in the number of migraine days by 100% in the one month period after being administered said antibody relative to the baseline number of migraine days experienced by that patient prior to administration of said medicament.

S59. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said patient further has a reduction in the number of migraine days by at least 50% in the 12 week period after being administered said antibody relative to the baseline number of migraine days experienced by that patient prior to administration of said medicament.

S60. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said patient further has a reduction in the number of migraine days by at least 75% in the 12 week period after being administered said antibody relative to the baseline number of migraine days experienced by that patient prior to administration of said medicament.

S61. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said patient further has a reduction in the number of migraine days by 100% in the 12 week period after being administered said antibody relative to the baseline number of migraine days experienced by that patient prior to administration of said medicament.

S62. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said medicament is further for administration in a second dose of said anti-CGRP antibody about 10-14 weeks, preferably 11-13 weeks, more preferably about 12 weeks or about 3 months after administration of said medicament.

S63. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said medicament comprises about 100 mg, about 125 mg, about 150 mg, about 175 mg, about 200 mg, about 225 mg, about 250 mg, about 275 mg, or about 300 mg of said anti-CGRP antibody.

S64. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody is aglycosylated or if glycosylated only contains only mannose residues.

S65. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody consists of the light chain polypeptide of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201 or SEQ ID NO: 566.

S66. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody consists of the light chain polypeptide encoded by SEQ ID NO: 231 and the heavy chain polypeptide encoded by SEQ ID NO: 211 or SEQ ID NO: 567.

S67. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said headache or said migraine is diagnosed according to the third edition of the International Classification of Headache Disorders.

S68. Use of the anti-CGRP antibody of any of any one of the foregoing embodiments, wherein said anti-CGRP antibody is expressed in or obtained by expression in *Pichia pastoris*.

S69. Use of the anti-CGRP antibody of any of any one of embodiments S1-S67, wherein said anti-CGRP antibody is expressed in or obtained by expression in CHO cells.

S70. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody or anti-CGRP antibody fragment is comprised in a formulation comprising or consisting of histidine (L-histidine), sorbitol, polysorbate 80, and water.

S71. Use of the anti-CGRP antibody of embodiment S70, wherein said formulation comprises or consists of, per 1 mL volume, 100 mg anti-CGRP antibody, 3.1 mg L-Histidine, 40.5 mg Sorbitol, and 0.15 mg Polysorbate 80, or having amounts of each constituent within $\pm 10\%$ of said values, and having a pH of 5.8 or within $\pm 10\%$ of said value.

S72. Use of the anti-CGRP antibody of embodiment S70, wherein said formulation comprises or consists of, per 1 mL

volume, 100 mg anti-CGRP antibody, 3.1 mg L-Histidine, 40.5 mg Sorbitol, and 0.15 mg Polysorbate 80, or having amounts of each constituent within $\pm 5\%$ of said values, and/or having a pH of 5.8 or within $\pm 5\%$ of said value.

S73. Use of the anti-CGRP antibody of embodiment S70, wherein said formulation comprises or consists of, per 1 mL volume, 100 mg anti-CGRP antibody, 3.1 mg L-Histidine, 40.5 mg Sorbitol, and 0.15 mg Polysorbate 80, or having amounts of each constituent within $\pm 1\%$ of said values, and/or having a pH of 5.8 or within $\pm 1\%$ of said value.

S74. Use of the anti-CGRP antibody of embodiment S70, wherein said formulation comprises or consists of, per 1 mL volume, 100 mg anti-CGRP antibody, 3.1 mg L-Histidine, 40.5 mg Sorbitol, and 0.15 mg Polysorbate 80, or having amounts of each constituent within $\pm 0.5\%$ of said values, and/or having a pH of 5.8 or within $\pm 0.5\%$ of said value.

S75. Use of the anti-CGRP antibody of embodiment S70, wherein said formulation comprises or consists of, per 1 mL volume, 100 mg anti-CGRP antibody, 3.1 mg L-Histidine, 40.5 mg Sorbitol, and 0.15 mg Polysorbate 80, or having amounts of each constituent within $\pm 0.1\%$ of said values, and/or having a pH of 5.8 or within $\pm 0.1\%$ of said value.

S76. Use of the anti-CGRP antibody of any of any one of the foregoing embodiments, wherein the anti-CGRP antibody has a dissociation constant of less than or equal to 10 pM, such as 2-8 pM, such as 3-6 pM, such as less than or equal to about 5 pM.

Further Exemplary Embodiments

Further exemplary embodiments of the invention are provided as follows:

E1. An anti-CGRP antibody for use in treating most bothersome symptom (MBS) associated with migraine in a patient suffering from migraine.

E2. The anti-CGRP antibody for use of embodiment E1, wherein the patient suffers from chronic migraine.

E3. The anti-CGRP antibody for use of embodiment E1, wherein the patient suffers from episodic migraine.

E4. The anti-CGRP antibody for use according to any of the foregoing embodiments, wherein the patients most bothersome symptom (MBS) associated with migraine is improved at 1-12 hours post-completion of administration or infusion, such as 1-5 hours post-completion of administration or infusion, 1-2 hours post-completion of administration or infusion, or about 2 hours post-completion of administration or infusion.

E5. The anti-CGRP antibody for use according to any of the foregoing embodiments, wherein the patients most bothersome symptom (MBS) associated with migraine is improved within 1 month from the first dosing with said anti-CGRP antibody.

E6. The anti-CGRP antibody for use according to any of the foregoing embodiments, wherein the patients most bothersome symptom (MBS) associated with migraine is improved within 3 month from the first dosing with said anti-CGRP antibody.

E7. The anti-CGRP antibody for use according to any of the foregoing embodiments, wherein the patients most bothersome symptom (MBS) associated with migraine is improved within 6 month from the first dosing with said anti-CGRP antibody.

E8. The anti-CGRP antibody for use according to any of the foregoing embodiments, wherein the improvement is sustained for at least 3 months from the first dosing with said anti-CGRP antibody.

E9. The anti-CGRP antibody for use according to any of the foregoing embodiments, wherein the improvement is sustained for at least 6 months from the first dosing with said anti-CGRP antibody.

E10. The anti-CGRP antibody for use according to any of the foregoing embodiments, wherein the MBS is selected from the group consisting of: Sensitivity to light (photophobia), Nausea/vomiting, Headache, Sensitivity to sound (phonophobia), Aura, Pain with activity, Pain, Throbbing/pulsation, Cognitive disruption, Fatigue, Mood changes, Sensitivity to smell (osmophobia or olfactophobia), Visual impact, Pressure/tightness, Pain (anatomical), Eye pain, Neck pain, Dizziness, Allodynia, Inactivity, Sensory disturbance, Sleep disturbance and Speech difficulty.

E11. The anti-CGRP antibody for use according to any of the foregoing embodiments, wherein the MBS is selected from the group consisting of: Sensitivity to light (photophobia), Nausea/vomiting, Headache, Sensitivity to sound (phonophobia), Aura, Pain with activity, Pain, Throbbing/pulsation, Cognitive disruption, Fatigue, Mood changes and Sensitivity to smell (osmophobia or olfactophobia).

E12. An anti-CGRP antibody for use in improving patient global impression of change (PGIC) associated with migraine in a patient suffering from migraine.

E13. The anti-CGRP antibody for use of embodiment E12, wherein the patient suffers from chronic migraine.

E14. The anti-CGRP antibody for use of embodiment E12, wherein the patient suffers from episodic migraine.

E15. The anti-CGRP antibody for use according to any of embodiments E12-E14, wherein the improvement of patient global impression of change (PGIC) associated with migraine is observed within 1 month from the first dosing with said anti-CGRP antibody.

E16. The anti-CGRP antibody for use according to any of embodiments E12-E14, wherein the improvement of patient global impression of change (PGIC) associated with migraine is observed within 3 month from the first dosing with said anti-CGRP antibody.

E17. The anti-CGRP antibody for use according to any of embodiments E12-E14, wherein the improvement of patient global impression of change (PGIC) associated with migraine is observed within 6 month from the first dosing with said anti-CGRP antibody.

E18. The anti-CGRP antibody for use according to any of embodiments E12-E17, wherein the improvement of patient global impression of change (PGIC) associated with migraine is sustained for 3 months from the first dosing with said anti-CGRP antibody.

E19. The anti-CGRP antibody for use according to any of embodiments E12-E18, wherein the improvement of patient global impression of change (PGIC) associated with migraine is sustained for 6 months from the first dosing with said anti-CGRP antibody.

E20. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody is for intravenous or subcutaneous infusion.

E21. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody is for intravenous infusion.

E22. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said patient is headache free 2 hours post-completion of administration or infusion.

E23. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises Ab6.

E24. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody

comprises the light chain complementarity-determining region (CDR) 1, 2, and 3 polypeptide sequences of SEQ ID NO: 224; SEQ ID NO: 226; and SEQ ID NO: 228, respectively.

E25. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the light chain CDR 1, 2, and 3 polypeptide sequences encoded by SEQ ID NO: 234; SEQ ID NO: 236; and SEQ ID NO: 238, respectively.

E26. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the heavy chain CDR 1, 2, and 3 polypeptide sequences of SEQ ID NO: 204; SEQ ID NO: 206; and SEQ ID NO: 208, respectively.

E27. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the heavy chain CDR 1, 2, and 3 polypeptide sequences encoded by SEQ ID NO: 214; SEQ ID NO: 216; and SEQ ID NO: 218, respectively.

E28. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the light chain CDR 1, 2, and 3 polypeptide sequences of SEQ ID NO: 224; SEQ ID NO: 226; and SEQ ID NO: 228, respectively and heavy chain CDR 1, 2, and 3 polypeptide sequences of SEQ ID NO: 204; SEQ ID NO: 206; and SEQ ID NO: 208, respectively.

E29. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the light chain CDR 1, 2, and 3 polypeptide sequences encoded by SEQ ID NO: 234; SEQ ID NO: 236; and SEQ ID NO: 238, respectively and heavy chain CDR 1, 2, and 3 polypeptide sequences encoded by SEQ ID NO: 214; SEQ ID NO: 216; and SEQ ID NO: 218, respectively.

E30. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the variable light chain polypeptide of SEQ ID NO: 222.

E31. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the variable light chain polypeptide encoded by SEQ ID NO: 232.

E32. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the variable heavy chain polypeptide of SEQ ID NO: 202.

E33. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the variable heavy chain polypeptide encoded by SEQ ID NO: 212.

E34. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the variable light chain polypeptide of SEQ ID NO: 222 and the variable heavy chain polypeptide of SEQ ID NO: 202.

E35. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the variable light chain polypeptide encoded by SEQ ID NO: 232 and the variable heavy chain polypeptide encoded by SEQ ID NO: 212.

E36. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the light chain polypeptide of SEQ ID NO: 221.

E37. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the light chain polypeptide encoded by SEQ ID NO: 231.

E38. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the heavy chain polypeptide of SEQ ID NO: 201 or SEQ ID NO: 566.

E39. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the heavy chain polypeptide encoded by SEQ ID NO: 211 or SEQ ID NO: 567.

E40. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the light chain polypeptide of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201 or SEQ ID NO: 566.

E41. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the light chain polypeptide encoded by SEQ ID NO: 231 and the heavy chain polypeptide encoded by SEQ ID NO: 211 or SEQ ID NO: 567.

E42. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein the administered amount of said anti-CGRP antibody is between about 100 mg and about 300 mg, or is about 100 mg, or is about 300 mg.

E43. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein the administered amount of said anti-CGRP antibody is 100 mg.

E44. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody is for intravenous administration in a dosage of 100 mg of said anti-CGRP antibody every 10-14 weeks, preferably every 11-13 weeks, more preferably every 12 weeks.

E45. The anti-CGRP antibody for use of any one of the foregoing embodiments E1-E42, wherein said anti-CGRP antibody is for intravenous administration in a dosage of administering 300 mg of said anti-CGRP antibody every 10-14 weeks, preferably every 11-13 weeks, more preferably every 12 weeks.

E46. The anti-CGRP antibody for use of any one of the foregoing embodiments wherein, prior to administration of said anti-CGRP antibody, the patient exhibits between 1-10 migraine attacks per month in the month or in the 3 months prior to administration.

E47. The anti-CGRP antibody for use of any one of the foregoing embodiments wherein, prior to administration of said anti-CGRP antibody, the patient exhibits between 2-8 migraine attacks per month in the month or in the 3 months prior to administration.

E48. The anti-CGRP antibody for use of any one of the foregoing embodiments wherein, prior to administration of said anti-CGRP antibody, the patient exhibits between 3-7 migraine attacks per month in the month or in the 3 months prior to administration.

E49. The anti-CGRP antibody for use of any one of the foregoing embodiments wherein, prior to administration of said anti-CGRP antibody, the patient exhibits less than 25 headache days per month in the month or in the 3 months prior to administration.

E50. The anti-CGRP antibody for use of any one of the foregoing embodiments wherein, prior to administration of said anti-CGRP antibody, the patient exhibits less than 20 headache days per month in the month or in the 3 months prior to administration.

E51. The anti-CGRP antibody for use of any one of the foregoing embodiments wherein, prior to administration of said anti-CGRP antibody, the patient exhibits less than 15 headache days per month in the month or in the 3 months prior to administration.

E52. The anti-CGRP antibody for use of any one of the foregoing embodiments wherein, prior to administration of said anti-CGRP antibody, the patient exhibits less than 10 headache days per month in the month or in the 3 months prior to administration.

E53. The anti-CGRP antibody for use of any one of the foregoing embodiments wherein said patient was diagnosed with migraine at least 10 years prior to the administration of said anti-CGRP antibody.

E54. The anti-CGRP antibody for use of any one of the foregoing embodiments wherein said patient was diagnosed with migraine at least 15 years prior to the administration of said anti-CGRP antibody.

E55. The anti-CGRP antibody for use of any one of the foregoing embodiments wherein said patient was diagnosed with migraine at least 18 or at least 19 years prior to the administration of said anti-CGRP antibody.

E56. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said patient has a reduction in the number of migraine days by at least 50% in the one month period after being administered said antibody relative to the baseline number of migraine days experienced by that patient prior to the administration of said anti-CGRP antibody.

E57. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said patient has a reduction in the number of migraine days by at least 75% in the one month period after being administered said antibody relative to the baseline number of migraine days experienced by that patient prior to the administration of said anti-CGRP antibody.

E58. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said patient has a reduction in the number of migraine days by 100% in the one month period after being administered said antibody relative to the baseline number of migraine days experienced by that patient prior to the administration of said anti-CGRP antibody.

E59. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said patient has a reduction in the number of migraine days by at least 50% in the 12 week period after being administered said antibody relative to the baseline number of migraine days experienced by that patient prior to the administration of said anti-CGRP antibody.

E60. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said patient has a reduction in the number of migraine days by at least 75% in the 12 week period after being administered said antibody relative to the baseline number of migraine days experienced by that patient prior to the administration of said anti-CGRP antibody.

E61. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said patient has a reduction in the number of migraine days by 100% in the 12 week period after being administered said antibody relative to the baseline number of migraine days experienced by that patient prior to the administration of said anti-CGRP antibody.

E62. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said use comprises administering a second dose of said anti-CGRP antibody to said patient about 10-14 weeks, preferably 11-13 weeks, more preferably about 12 weeks or about 3 months after the administration of said anti-CGRP antibody.

E63. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said use comprises admin-

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istering about 100 mg, about 125 mg, about 150 mg, about 175 mg, about 200 mg, about 225 mg, about 250 mg, about 275 mg, or about 300 mg of said anti-CGRP antibody.

E64. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody is aglycosylated or if glycosylated only contains only mannose residues.

E65. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody consists of the light chain polypeptide of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201 or SEQ ID NO: 566.

E66. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody consists of the light chain polypeptide encoded by SEQ ID NO: 231 and the heavy chain polypeptide encoded by SEQ ID NO: 211 or SEQ ID NO: 567.

E67. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said headache or said migraine is diagnosed according to the third edition of the International Classification of Headache Disorders.

E68. The anti-CGRP antibody for use of any of any one of the foregoing embodiments, wherein said anti-CGRP antibody is expressed in or obtained by expression in *Pichia pastoris*.

E69. The anti-CGRP antibody for use of any of any one of embodiments E1-E67, wherein said anti-CGRP antibody is expressed in or obtained by expression in CHO cells.

E70. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody or anti-CGRP antibody fragment is comprised in a formulation comprising or consisting of histidine (L-histidine), sorbitol, polysorbate 80, and water.

E71. The anti-CGRP antibody for use of embodiment E70, wherein said formulation comprises or consists of, per 1 mL volume, 100 mg anti-CGRP antibody, 3.1 mg L-Histidine, 40.5 mg Sorbitol, and 0.15 mg Polysorbate 80, or having amounts of each constituent within $\pm 10\%$ of said values, and having a pH of 5.8 or within $\pm 10\%$ of said value.

E72. The anti-CGRP antibody for use of embodiment E70, wherein said formulation comprises or consists of, per 1 mL volume, 100 mg anti-CGRP antibody, 3.1 mg L-Histidine, 40.5 mg Sorbitol, and 0.15 mg Polysorbate 80, or having amounts of each constituent within $\pm 5\%$ of said values, and/or having a pH of 5.8 or within $\pm 5\%$ of said value.

E73. The anti-CGRP antibody for use of embodiment E70, wherein said formulation comprises or consists of, per 1 mL volume, 100 mg anti-CGRP antibody, 3.1 mg L-Histidine, 40.5 mg Sorbitol, and 0.15 mg Polysorbate 80, or having amounts of each constituent within $\pm 1\%$ of said values, and/or having a pH of 5.8 or within $\pm 1\%$ of said value.

E74. The anti-CGRP antibody for use of embodiment E70, wherein said formulation comprises or consists of, per 1 mL volume, 100 mg anti-CGRP antibody, 3.1 mg L-Histidine, 40.5 mg Sorbitol, and 0.15 mg Polysorbate 80, or having amounts of each constituent within $\pm 0.5\%$ of said values, and/or having a pH of 5.8 or within $\pm 0.5\%$ of said value.

E75. The anti-CGRP antibody for use of embodiment E70, wherein said formulation comprises or consists of, per 1 mL volume, 100 mg anti-CGRP antibody, 3.1 mg L-Histidine, 40.5 mg Sorbitol, and 0.15 mg Polysorbate 80, or

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having amounts of each constituent within $\pm 0.1\%$ of said values, and/or having a pH of 5.8 or within $\pm 0.1\%$ of said value.

E76. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein the anti-CGRP antibody has a dissociation constant of less than or equal to 10 pM, such as 2-8 pM, such as 3-6 pM, such as less than or equal to about 5 pM.

EXAMPLES

The following examples are provided in order to illustrate the invention, but are not to be construed as limiting the scope of the claims in any way.

Example 1

Preparation of Antibodies that Bind CGRP

The preparation of exemplary anti-CGRP antibody Ab6 having the sequences in FIGS. 1-12 is disclosed in commonly owned PCT Application WO/2012/162243, published on Nov. 29, 2012, the contents of which are incorporated by reference herein. This application exemplifies synthesis of these antibodies in *Pichia pastoris* cells. The present Applicant further contemplates synthesis of anti-CGRP antibody Ab6 particularly in CHO cells.

Example 2

Human Clinical Study Evaluating the Safety and Efficacy of an Anti-CGRP Antibody in Chronic Migraine Patients

This example describes a randomized, double-blind, placebo-controlled clinical trial evaluating the safety and efficacy of Ab6 for chronic migraine prevention. In the study, 1,072 patients were randomized to receive Ab6 (300 mg or 100 mg), or placebo administered by infusion once every 12 weeks. The study design is depicted in FIG. 13. To be eligible for the trial, patients must have experienced at least 15 headache days per month, of which at least eight met criteria for migraine. Patients that participated in the trial had an average of 16.1 migraine days per month at baseline. The primary endpoint of the present study was the change from baseline in mean monthly migraine days (MMDs) over weeks 1-12 following the first infusion of Ab6. The change from baseline in mean monthly migraine days (MMDs) following the second infusion at week 12 was also assessed and the results are shown in FIG. 14.

Study endpoints further included patient-identified MBS as part of the predefined key secondary endpoints. At screening, patients verbally identified the MBS associated with their migraine, which was pooled across treatment arms for this analysis. The change from baseline of these symptoms were then rated by the patient every month of the study beginning from Day 0.

In the present study, patients verbally identified the most bothersome symptom (MBS) associated with their migraine at screening. The MBS associated with their migraine was then categorized by the investigator into a predefined list of 8 symptoms or an "other" option. The predefined list included the terms nausea, vomiting, sensitivity to light, sensitivity to sound, mental cloudiness, fatigue, pain with activity, and mood changes. The "other" option provided investigators the opportunity to identify any migraine-associated symptom without limitation described by the patient as most bothersome but did not easily fit into the check list of symptoms included in the work study checklist. For those patients who selected the "other" category for their MBS,

their “write-in” responses were re-coded post hoc and re-classified to the predefined list or to new symptom classes. At subsequent visits, patients were asked to rate the change from the screening visit in their self-reported MBS on a 7-point scale, which is shown below:

Very Much Improved	Much Improved	Minimally Improved	No Change	Minimally Worse	Much Worse	Very Much Worse
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In addition to MBS, the patients were also requested to evaluate the efficacy of the treatment on patient global impression of change (PGIC), which is a parameter comprising a single question assessing the patient’s own impression of the overall change in their disease status since the start of the study. This parameter was also rated by the patients at a 7-point scale identical to the one used to assess change in MBS as displayed above and at the same time points in the study. In FIGS. 16-22 the “worse” category includes “minimally worse”, “much worse”, and “very much worse”.

At the screening visit in, patients indicated a wide range of symptoms as their MBS, with the “other” category being the most frequent response (40%-42% across the 3 treatment groups). The patients who selected the “other” category generally provided more details and/or had more than 1 symptom as their MBS, allowing for these symptoms to be recorded. The overall list of MBS is summarized in Table 1 below.

TABLE 1

Summary of patient-identified MBS in the present study as described in Example 2				
Symptom, n (%)	Eptinezumab 100 mg (n = 356)	Eptinezumab 300 mg (n=350)	Placebo (n = 366)	Total (N = 1072)
ICHD-3 Symptoms				
Sensitivity to light	67 (18.8)	64 (18.3)	69 (18.9)	200 (18.7)
Nausea/vomiting	55 (15.4)	46 (13.1)	61 (16.7)	162 (15.1)
Headache	45 (12.6)	43 (12.3)	32 (8.7)	120 (11.2)
Sensitivity to sound	22 (6.2)	28 (8.0)	28 (7.7)	78 (7.3)
Aura	4 (1.1)	1 (<1)	2 (<1)	7 (0.7)
Additional Symptoms				
Pain with activity	53 (14.9)	45 (12.9)	49 (13.4)	147 (13.7)
Pain	35 (9.8)	45 (12.9)	53 (14.5)	133 (12.4)
Throbbing/pulsation	18 (5.1)	17 (4.9)	15 (4.1)	50 (4.7)
Cognitive disruption	17 (4.8)	14 (4.0)	13 (3.6)	44 (4.1)
Fatigue	7 (2.0)	11 (3.1)	8 (2.2)	26 (2.4)
Mood changes	8 (2.2)	4 (1.1)	4 (1.1)	16 (1.5)
Sensitivity to smell	1 (<1)	1 (<1)	8 (2.2)	10 (0.9)
Visual impact	2 (<1)	3 (<1)	3 (<1)	8 (0.7)
Pressure/tightness	2 (<1)	2 (<1)	3 (<1)	7 (0.7)
Pain, anatomical	3 (<1)	3 (<1)	0	6 (0.6)
Eye pain	4 (1.1)	1 (<1)	1 (<1)	6 (0.6)
Neck pain	1 (<1)	1 (<1)	3 (<1)	5 (0.5)
Dizziness	2 (<1)	2 (<1)	1 (<1)	5 (0.5)
Allodynia	1 (<1)	1 (<1)	1 (<1)	3 (0.3)
Inactivity	0	1 (<1)	1 (<1)	2 (0.2)
Sensory disturbance	1 (<1)	0	0	1 (0.1)
Sleep disturbance	0	0	1 (<1)	1 (0.1)
Speech difficulty	0	0	1 (<1)	1 (0.1)
Multiple*	7 (2.0)	12 (3.4)	8 (2.2)	27 (2.5)
Other	1 (<1)	5 (1.4)	1 (<1)	7 (0.7)

*Patient’s most bothersome symptom included more than 1 symptom type.
ICHD-3 = International Classification of Headache Disorders 3rd edition.

The most commonly reported symptoms were light sensitivity, nausea/vomiting, pain with activity, pain, headache,

sound sensitivity, throbbing/pulsation, cognitive disruption, fatigue, mood changes, and sensitivity to smell, with each category having at least 10 patients reporting these events as their MBS. At the end of the 28-day screening period (i.e., before dosing at the baseline visit), patients were asked to rate the change in their identified MBS from very much worse to very much improved, with >90% reporting no change in their MBS, which is illustrated in FIG. 16. This suggests that the bothersomeness of patient-identified MBS was quite stable among this cohort with chronic migraine during the screening period.

Infusion of Ab6 in doses of 100 mg and 300 mg provided significantly reduction in mean MMDs across months 1-3 of the study, with further reduction after an additional infusion at week 12 of the study. This effect is shown in FIG. 14.

The efficacy of Ab6 on the MBS was demonstrated at 1 month (FIG. 17), 3 months (FIG. 19), and 6 months (FIG. 21), following the first infusion of Ab6 in doses of 100 mg and 300 mg. The efficacy of Ab6 on the PGIC was demonstrated at 1 month (FIG. 18), 3 months (FIG. 20), and 6 months (FIG. 22), following the first infusion of Ab6 in doses of 100 mg and 300 mg. The efficacy on these parameters were sustained or increased through 2 doses of Ab6 over 6 months; at Month 1, 75-82% of Ab6-treated patients indicated some level of improvement compared to 56-59% for the placebo-treated patients; at Month 3 ratings of improvement were similar to those of month 1; at Month 6, ~80% of Ab6-treated patients indicated ≥ 1 categorical level of improvement in MBS and PGIC. The distribution of ratings for MBS improvement and PGIC were similar across time points, suggesting that the 2 identically rated measures in patients with chronic migraine move in parallel. These data suggest that improvements in patient-identified most bothersome migraine-associated symptoms are highly correlated with the patient’s perception of an improved disease status in patients with chronic migraine.

The administered antibody, Ab6, is an anti-CGRP antibody consisting of the light chain polypeptide of SEQ ID NO: 221 and heavy chain polypeptide of SEQ ID NO: 201.

Patient characteristics are summarized in Table 2 below, with separate columns for patients receiving placebo, 100 mg of the antibody, or 300 mg of the antibody. Patients had a mean number of years from migraine diagnosis of between 17.0 and 19.0 years, a mean duration of suffering from chronic migraine of between 11.5 and 12.4 years, and between 44.3% and 45.2% of patients utilized at least one prophylactic medication. In addition, patients with a dual diagnosis of chronic migraine and medications overuse excluding opioid and butalbital over were included in this study. At baseline, in both antibody treatment groups the mean number of migraine days per month was 16.1, while for the placebo group, the mean number of migraine days per month was 16.2.

TABLE 2

Summarizes the characteristics of patients in each treatment group in the clinical trials described in Example 2.			
	Eptinezumab 100 mg n = 356	Eptinezumab 300 mg n = 350	Placebo n = 366
Age (years), mean (SD)	41.0 (11.72)	41.0 (10.36)	39.6 (11.28)
Sex, n (%)			
Male	49 (13.8%)	36 (10.3%)	41 (11.2%)
Female	307 (86.2%)	314 (89.7%)	325 (88.8%)

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TABLE 2-continued

Summarizes the characteristics of patients in each treatment group in the clinical trials described in Example 2.			
	Eptinezumab 100 mg n = 356	Eptinezumab 100 mg n = 350	Placebo n = 366
Race, n (%)			
White	332 (93.3%)	322 (92.0%)	321 (87.7%)
Black or African American	21 (5.9%)	23 (6.6%)	38 (10.4%)
Other*	3 (0.8%)	5 (1.4%)	7 (1.9%)
BMI (kg/m ²), mean (SD)	26.4 (4.98)	26.3 (7.14)	27.0 (5.56)
Age at migraine diagnosis (years), mean (SD)	22.8 (10.64)	22.0 (9.30)	22.6 (9.98)
Duration of migraine diagnosis (years), mean (SD)	18.3 (12.22)	19.0 (11.50)	17.0 (11.63)

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TABLE 2-continued

Summarizes the characteristics of patients in each treatment group in the clinical trials described in Example 2.			
	Eptinezumab 100 mg n = 356	Eptinezumab 100 mg n = 350	Placebo n = 366
Duration of chronic migraines (years), mean (SD)	11.6 (11.72)	12.3 (11.15)	11.6 (10.90)
Number of migraine days, mean (SD) †	16.1 (4.61)	16.1 (4.77)	16.2 (4.55)
Medication-overuse headache diagnosis, n (%)§	139 (39.0%)	147 (42.0%)	145 (39.6%)

BMI, body mass index;
SD, standard deviation,
*Other includes Asian, American Indian or Alaska Native, Native Hawaiian or Other Pacific Islander, multiple races, and other.
† As reported by the eDiary in the 28-day screening period.
§Based on 3rd edition of the International Classification of Headache Disorders (beta).

SEQUENCE LISTING

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source           1..439
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                 organism = synthetic construct

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WAKGRFTISR ASSTTVDLKM TSLTTEDTAT YFCARGDIWG PGTLVTVSSA STKGPSVFPL 120
APSSKSTSGG TAALGCLVKD YFPEPVTVSW NSGALTSGVH TFPAVLQSSG LYSLSVTV 180
PSSSLGTQTY ICNVNHKPSN TKVDKRVEPK SCDKTHTCPP CPAPELLGGP SVFLPPPKPK 240
DTLMISRTPV VTCVVVDVSH EDPEVKFNWY VDGVEVHNAK TKPREEQYAS TYRVVSVLTV 300
LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM TKNQVSLTCL 360
VKGPYPSPDIA VEWESNGQPE NNYKTTTPPVL DSDGSFFLYS KLTVDKSRWQ QGNVPSCSVM 420
HEALHNHYTQ KSLSLSPGK 439

SEQ ID NO: 2      moltype = AA  length = 109
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source           1..109
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                 organism = synthetic construct

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WAKGRFTISR ASSTTVDLKM TSLTTEDTAT YFCARGDIWG PGTLVTVSS 109

SEQ ID NO: 3      moltype = AA  length = 29
FEATURE          Location/Qualifiers
REGION           1..29
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source           1..29
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 3
QSLEESGGRL VTPGTPLTLT CTVSGLDLSS 29

SEQ ID NO: 4      moltype = AA  length = 5
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                 organism = synthetic construct

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SEQ ID NO: 5      moltype = AA  length = 14
FEATURE          Location/Qualifiers

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	organism = synthetic construct	
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SEQ ID NO: 6	moltype = AA length = 16	
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REGION	1..16	
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source	1..16	
	mol_type = protein	
	organism = synthetic construct	
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VIGINDNTYY ASWAKG		16
SEQ ID NO: 7	moltype = AA length = 31	
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REGION	1..31	
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source	1..31	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 7		
RFTISRASST TVDLKMTSLT TEDTATYFCA R		31
SEQ ID NO: 8	moltype = length =	
SEQUENCE: 8		
000		
SEQ ID NO: 9	moltype = AA length = 11	
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source	1..11	
	mol_type = protein	
	organism = synthetic construct	
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WGPGLVTVS S		11
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source	1..330	
	mol_type = protein	
	organism = synthetic construct	
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ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS		60
GLYSLSSVVT VPSSSLGTQT YICNVNHNKPS NTKVDRKVEP KSCDKHTTCP PCPAPELLGG		120
PSVFLFPPPK KDTLMISRTP EVTCVVVDVS HEDPEVKFNW YVDGVEVHNA KTKPREEQYA		180
STYRVVSVLT VLNQDNLNGK EYKCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSREE		240
MTKNQVSLTC LVKGFYPSDI AVEWESNGQP ENNYKTTTPV LDSDGSFFLY SKLTVDKSRW		300
QQGNVFSCSV MHEALHNHYT QKSLSLSPGK		330
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source	1..1320	
	mol_type = other DNA	
	organism = synthetic construct	
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tgcacagtct ctggactcga cctcagtagc tactacatgc aatgggtccg ccagggtcca		120
gggaaggggc tggaaatggat cggagtcatt ggtattaatg ataacacata ctacgcgagc		180
tgggcgaaag gccgattcac catctccaga gcctcgtcga ccacggtgga tctgaaaatg		240
accagtctga caaccgagga caccggccacc tatctctgtg ccagagggga catctggggc		300
ccaggcaccg tcgtcaccgt ctcgagcgcc tccaccaagg gcccatcggt cttccccctg		360
gcaccctcct ccaagagcac ctctgggggc acagcggccc tgggctgcct ggtcaaggac		420
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accttccccg ctgtcctaca gtctcagga ctctactccc tcagcagcgt ggtgaccgtg		540
ccctccagca gcttggggc cccagacctac atctgcaacg tgaatcacaa gccagccaac		600
accaaggttg acaagagagt tgagcccaaa tcttgtgaca aaactcacac atgcccaccg		660
tgcccagcac ctgaactcct ggggggaccg tcagtcttcc tcttcccccc aaaacccaag		720
gacaccctca tgatctcccc gaccctgag gtcacatgcg tgggtggtgga cgtgagccac		780
gaagaccctg aggtcaagtt caactggtac gtggacggcg tggaggtgca taatgccaa		840

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acaaagccgc gggaggagca gtaacccagc acgtaccgtg tggtcagcgt cctcaccgtc 900
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ccagcccccac tcgagaaaac catctccaaa gccaaagggc agccccgaga accacagggtg 1020
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aagctcaccg tggacaagag caggtggcag caggggaacg tcttctcatg ctccgtgatg 1260
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SEQ ID NO: 12      moltype = DNA length = 327
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SEQUENCE: 12
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gggaaggggc tggaatggat cggagtcatt ggtattaatg ataacacata ctacgcgagc 180
tgggcgaaag gccgattcac catctccaga gcctcgctga ccacgggtga tctgaaaatg 240
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SEQUENCE: 13
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SEQUENCE: 14
agctactaca tgcaa 15

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SEQ ID NO: 15      moltype = DNA length = 42
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SEQ ID NO: 18      moltype = length =
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source            1..219
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SRFKGSGSGT QFTLTISDLE CADAATYYCL GSYDCSSGDC FVFGGGTEVV VKRTVAAPSV 120
FIFPPSDEQL KSGTASVVCL LNNFYFPREK VQWKVDNALQ SGNSQESVTE QDSKDYSTSL 180
SSTLTLSKAD YEKHKVYACE VTHQLSSPV TKSFNRGEC 219

SEQ ID NO: 22      moltype = AA length = 113
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REGION            1..113
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source            1..113
                   mol_type = protein
                   organism = synthetic construct

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SRFKGSGSGT QFTLTISDLE CADAATYYCL GSYDCSSGDC FVFGGGTEVV VKR 113

SEQ ID NO: 23      moltype = AA length = 22
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REGION            1..22
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source            1..22
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 23
QVLTQTASPV SAAVGSTVTI NC 22

SEQ ID NO: 24      moltype = AA length = 13
FEATURE           Location/Qualifiers
REGION            1..13
                   note = Engineered antibody sequence
source            1..13
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 24

```

-continued

QASQSVYDNN YLA	13
SEQ ID NO: 25	moltype = AA length = 15
FEATURE	Location/Qualifiers
REGION	1..15
source	note = Engineered antibody sequence 1..15 mol_type = protein organism = synthetic construct
SEQUENCE: 25	
WYQQKPGQPP KQLIY	15
SEQ ID NO: 26	moltype = AA length = 7
FEATURE	Location/Qualifiers
REGION	1..7
source	note = Engineered antibody sequence 1..7 mol_type = protein organism = synthetic construct
SEQUENCE: 26	
STSTLAS	7
SEQ ID NO: 27	moltype = AA length = 32
FEATURE	Location/Qualifiers
REGION	1..32
source	note = Engineered antibody sequence 1..32 mol_type = protein organism = synthetic construct
SEQUENCE: 27	
GVSSRFKSGS SGTQFTLTIS DLECADAATY YC	32
SEQ ID NO: 28	moltype = AA length = 13
FEATURE	Location/Qualifiers
REGION	1..13
source	note = Engineered antibody sequence 1..13 mol_type = protein organism = synthetic construct
SEQUENCE: 28	
LGSYDCSSGD CFV	13
SEQ ID NO: 29	moltype = AA length = 11
FEATURE	Location/Qualifiers
REGION	1..11
source	note = Engineered antibody sequence 1..11 mol_type = protein organism = synthetic construct
SEQUENCE: 29	
FGGGTEVVVK R	11
SEQ ID NO: 30	moltype = AA length = 106
FEATURE	Location/Qualifiers
REGION	1..106
source	note = Engineered antibody sequence 1..106 mol_type = protein organism = synthetic construct
SEQUENCE: 30	
TVAAPSVFIF PPSDEQLKSG TASVVCLLNN FYPREAKVQW KVDNALQSGN SQESVTEQDS	60
KDSTYLSLST LTLKADYK HKVYACEVTH QGLSSPVTKS FNRGEC	106
SEQ ID NO: 31	moltype = DNA length = 660
FEATURE	Location/Qualifiers
misc_feature	1..660
source	note = Engineered antibody sequence 1..660 mol_type = other DNA organism = synthetic construct
SEQUENCE: 31	
caagtgtcga cccagactgc atcccccgctg tctgcagctg tgggaagcac agtcaccatc	60
aattgccagg ccagtcagag tgtttatgat aacaactacc tagcctggta tcagcagaaa	120
ccagggcagc ctcccaagca actgactctat tctacatcca ctctggcatc tgggggtctca	180
tcgcggttca aaggcagtg atctgggaca cagttcactc tcaccatcag cgacctggag	240
tgtgccgatg ctgccactta ctactgtcta ggcagttatg attgtagtag tggtgattgt	300
tttgttttcg gcggaggac cgaggtggtg gtcaaacgta cggtggctgc accatctgtc	360
ttcatcttcc cgccatctga tgagcagttg aaatctggaa ctgcctctgt tgtgtgctg	420
ctgaataact tctatcccag agaggccaaa gtacagtggg aggtggataa cgccctccaa	480

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tcgggtaact ccaggagag tgccacagag caggacagca aggacagcac ctacagcctc 540
agcagcaccg tgacgctgag caaagcagac tacgagaaac acaaagtcta cgctgcgaa 600
gtcaccatcg agggcctgag ctgcgccgtc acaaagagct tcaacagggg agagtgttag 660

```

```

SEQ ID NO: 32      moltype = DNA length = 339
FEATURE           Location/Qualifiers
misc_feature       1..339
                   note = Engineered antibody sequence
source            1..339
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 32
caagtgtga ccagactgc atccccctg tctgcagctg tgggaagcac agtcaccatc 60
aattgccagg ccagtcagag tgtttatgat aacaactacc tagcctggta tcagcagaaa 120
ccagggcagc ctcccaagca actgatctat tctacatcca ctctggcatc tggggtctca 180
tcgcggttca aaggcagtg atctgggaca cagttcactc tcaccatcag cgacctggag 240
tgtgccgatg ctgccactta ctactgtcta ggcagttatg attgtagtag tggtgattgt 300
ttgtttttcg gcggaggagc cgaggtggtg gtcaaacgt 339

```

```

SEQ ID NO: 33      moltype = DNA length = 66
FEATURE           Location/Qualifiers
misc_feature       1..66
                   note = Engineered antibody sequence
source            1..66
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 33
caagtgtga ccagactgc atccccctg tctgcagctg tgggaagcac agtcaccatc 60
aattgc 66

```

```

SEQ ID NO: 34      moltype = DNA length = 39
FEATURE           Location/Qualifiers
misc_feature       1..39
                   note = Engineered antibody sequence
source            1..39
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 34
caggccagtc agagtgttta tgataacaac tacctagcc 39

```

```

SEQ ID NO: 35      moltype = DNA length = 45
FEATURE           Location/Qualifiers
misc_feature       1..45
                   note = Engineered antibody sequence
source            1..45
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 35
tggtatcagc agaaaccagg gcagcctccc aagcaactga tctat 45

```

```

SEQ ID NO: 36      moltype = DNA length = 21
FEATURE           Location/Qualifiers
misc_feature       1..21
                   note = Engineered antibody sequence
source            1..21
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 36
tctacatcca ctctggcatc t 21

```

```

SEQ ID NO: 37      moltype = DNA length = 96
FEATURE           Location/Qualifiers
misc_feature       1..96
                   note = Engineered antibody sequence
source            1..96
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 37
gggggtctcat cgcggttcaa aggcagtgga tctgggacac agttcactct caccatcagc 60
gacctggagt gtgccgatgc tgccacttac tactgt 96

```

```

SEQ ID NO: 38      moltype = DNA length = 39
FEATURE           Location/Qualifiers
misc_feature       1..39
                   note = Engineered antibody sequence
source            1..39
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 38

```


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```

source          1..14
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 45
WVRQAPGKGL EWVG                                     14

SEQ ID NO: 46      moltype = AA  length = 16
FEATURE           Location/Qualifiers
REGION            1..16
                 note = Engineered antibody sequence
source            1..16
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 46
VIGINDNTYY ASWAKG                                     16

SEQ ID NO: 47      moltype = AA  length = 32
FEATURE           Location/Qualifiers
REGION            1..32
                 note = Engineered antibody sequence
source            1..32
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 47
RFTISRDN SK TTVYLQMN SL RAEDTAVYFC AR                 32

SEQ ID NO: 48      moltype =      length =
SEQUENCE: 48
000

SEQ ID NO: 49      moltype = AA  length = 11
FEATURE           Location/Qualifiers
REGION            1..11
                 note = Engineered antibody sequence
source            1..11
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 49
WGQGTLLVTVS S                                         11

SEQ ID NO: 50      moltype = AA  length = 330
FEATURE           Location/Qualifiers
REGION            1..330
                 note = Engineered antibody sequence
source            1..330
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 50
ASTKGPSVF LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 60
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKRV EP KSCDKTHTCP PCPAPELLGG 120
PSVFLFPPKP KDTLMISRTP EVTCVVVDVS HEDPEVKFNW YVDGVEVHNA KTKPREEQYA 180
STYRVVSVLT VLNQDWLNGK EYKCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSREE 240
MTKNQVSLTC LVKGFYPSDI AVEWESNGQP ENNYKTTTPV LDSDGSFFLY SKLTVDKSRW 300
QQGNVFSCSV MHEALHNHYT QKSLSLSPGK                    330

SEQ ID NO: 51      moltype = DNA  length = 1326
FEATURE           Location/Qualifiers
misc_feature      1..1326
                 note = Engineered antibody sequence
source            1..1326
                 mol_type = other DNA
                 organism = synthetic construct

SEQUENCE: 51
gaggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctgggggggc cctgagactc 60
tcctgtgcag tctctggact cgacctcagt agctactaca tgcaatgggt ccgtcaggct 120
ccagggaag ggtcggagtg ggtcggagtc attggtatca atgataacac atactacgcg 180
agctgggcga aaggccgatt caccatctcc agagacaatt ccaagaccac ggtgtatctt 240
caaatgaaca gcctgagagc tgaggacact gctgtgtatt tctgtgctag aggggacatc 300
tggggccaag ggaccctcgt caccgtctcg agcgctcca ccaagggccc atcggtcttc 360
cccctggcac cctcctccaa gagcacctct gggggcacag cggccctggg ctgcctggtc 420
aaggactact tcccgaacc ggtgacggtg tcgtggaact caggcgccct gaccagcggc 480
gtgcacacct tcccggctgt cctacagtcc tcaggactct actccctcag cagcgtggtg 540
accgtgccct ccagcagctt gggcacccag acctacatct gcaacgtgaa tcacaagccc 600
agcaaacacca aggtggacaa gagagttgag cccaatctt gtgacaaaac tcacacatgc 660
ccaccgtgcc cagcacctga actcctgggg ggaccgtcag tcttctctct cccccaaaaa 720
cccaaggaca cctcatgat ctcccgacc cctgaggtca catgcgtggt ggtggacgtg 780
agccacgaag acctgaggt caagttcaac tggtagctgg acggcgtgga ggtgcataat 840
gccaagacaa agccgcggga ggagcagtag gccagcacgt accgtgtggt cagcgtcctc 900
accgtcctgc accaggactg gctgaatggc aaggagtaca agtgcaaggt ctccaacaaa 960

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```

gcctctccag ccccatcgaa gaaaaccatc tccaaagcca aagggcagcc ccgagaacca 1020
caggtgtaca cctgcccc atcccgagg gagatgacca agaaccagg cagcctgacc 1080
tgctgtgtca aaggttcta tcccagcgac atcgccgtgg agtgggagag caatgggcag 1140
ccggagaaca actacaagac cagcctccc gtgctggact ccgacggctc cttcttctc 1200
tacagcaagc tcacgtgga caagagcagg tggcagcagg ggaacgtct ctcagtctcc 1260
gtgatgcatg aggctctgca caaccactac acgcagaaga gcctctccct gtctccgggt 1320
aatga 1326

```

```

SEQ ID NO: 52      moltype = DNA length = 333
FEATURE
misc_feature      Location/Qualifiers
                  1..333
                  note = Engineered antibody sequence
source            1..333
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 52
gaggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctgggggggtc cctgagactc 60
tcctgtgcag tctctggact cgacctcagt agctactaca tgcaatgggt ccgtcaggct 120
ccagggaagg ggctggagtg ggtcggagtc attggtatca atgataacac atactacgcg 180
agctgggcga aaggccgatt caccatctcc agagacaatt ccaagaccac ggtgtatctt 240
caaatgaaca gcctgagagc tgaggacact gctgtgtatt tctgtgctag aggggacatc 300
tggggccaag ggacctcgt caccgtctcg agc 333

```

```

SEQ ID NO: 53      moltype = DNA length = 90
FEATURE
misc_feature      Location/Qualifiers
                  1..90
                  note = Engineered antibody sequence
source            1..90
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 53
gaggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctgggggggtc cctgagactc 60
tcctgtgcag tctctggact cgacctcagt 90

```

```

SEQ ID NO: 54      moltype = DNA length = 15
FEATURE
misc_feature      Location/Qualifiers
                  1..15
                  note = Engineered antibody sequence
source            1..15
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 54
agctactaca tgcaa 15

```

```

SEQ ID NO: 55      moltype = DNA length = 42
FEATURE
misc_feature      Location/Qualifiers
                  1..42
                  note = Engineered antibody sequence
source            1..42
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 55
tgggtccgtc aggtccagg gaaggggctg gagtgggtcg ga 42

```

```

SEQ ID NO: 56      moltype = DNA length = 48
FEATURE
misc_feature      Location/Qualifiers
                  1..48
                  note = Engineered antibody sequence
source            1..48
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 56
gtcattggtg tcaatgataa cacatactac gcgagctggg cgaaaggc 48

```

```

SEQ ID NO: 57      moltype = DNA length = 96
FEATURE
misc_feature      Location/Qualifiers
                  1..96
                  note = Engineered antibody sequence
source            1..96
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 57
cgattcacca tctccagaga caattccaag accacgggtg atcttcaaat gaacagcctg 60
agagctgagg aactgctgtg gtatttctgt gctaga 96

```

```

SEQ ID NO: 58      moltype = length =
SEQUENCE: 58
000

```

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```

SEQ ID NO: 59      moltype = DNA length = 33
FEATURE           Location/Qualifiers
misc_feature      1..33
                  note = Engineered antibody sequence
source           1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 59
tggggccaag ggaccctcgt caccgtctcg agc                               33

SEQ ID NO: 60      moltype = DNA length = 993
FEATURE           Location/Qualifiers
misc_feature      1..993
                  note = Engineered antibody sequence
source           1..993
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 60
gcctccacca agggcccatc ggtcttcccc ctggcaccct cctccaagag cacctctggg 60
ggcacagcgg ccttgggctg cctgggtcaag gactacttcc ccgaaccggt gacgggtgtcg 120
tggaaactcag gcgccctgac cagcggcgctg cacaccttcc cggctgtcct acagtctctca 180
ggactctact cctcagcagc cgtgggtgacc gtgccctcca gcagcttggg caccagacc 240
tacatctgca acgtgaatca caagcccagc aacaccaagg tggacaagag agttgagccc 300
aaatcttgtg acaaaactca cacatgcccc ccggtgccca cactgaact cctgggggga 360
ccgtcagttc tcctcttccc cccaaaaccc aaggacaccc tcatgatctc ccggaccctc 420
gaggtcacat gcgtgggtgt ggacgtgagc cacgaagacc ctgaggtcaa gttcaactgg 480
tacgtggacg gcgtggaggt gcataatgcc aagacaaagc cgcggggagga gcagtacgcc 540
agcacgtacc gtgtgggtcag cgtcctcacc gtctcgcacc aggactgggt gaatggcaag 600
gagtacaagt gcaagggtct caacaaagcc ctcccagccc ccacgagaaa aaccatctcc 660
aaagcccaag ggcagccccc agaaccacag gtgtacaccc tgcccccatc ccgggaggag 720
atgaccaaga accagggtcag cctgacctgc ctggtcaaa gcttctatcc cagcgacatc 780
gccgtggagt gggagagcaa tgggcagccg gagaacaact acaagaccac gcctcccggt 840
ctggactccg acggtctctt ctctctctac agcaagctca ccgtggacaa gagcagggtg 900
cagcagggga acgtcttctc atgtcccggt atgcatgagg ctctgcacaa ccactacacg 960
cagaagagcc tctccctgtc tccgggtaaa tga                               993

SEQ ID NO: 61      moltype = AA length = 219
FEATURE           Location/Qualifiers
REGION           1..219
                  note = Engineered antibody sequence
source           1..219
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 61
QVLTQSPSSL SASVGDRVTI NCQASQSVYD NNYLAWYQQK PGKVPKQLIY STSTLASGVP 60
SRFSGSGSGT DFTLTISLQ PEDVATYYCL GSYDCSSGDC FVFGGGTKVE IKRTVAAPSV 120
FIFPPSDEQL KSGTASVCL LNNFYPREAK VQWKVDNALQ SGNSQESVTE QDSKSTYSL 180
SSTLTLSKAD YEKHKVYACE VTHQGLSSPV TKSFNRGEC                               219

SEQ ID NO: 62      moltype = AA length = 113
FEATURE           Location/Qualifiers
REGION           1..113
                  note = Engineered antibody sequence
source           1..113
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 62
QVLTQSPSSL SASVGDRVTI NCQASQSVYD NNYLAWYQQK PGKVPKQLIY STSTLASGVP 60
SRFSGSGSGT DFTLTISLQ PEDVATYYCL GSYDCSSGDC FVFGGGTKVE IKR          113

SEQ ID NO: 63      moltype = AA length = 22
FEATURE           Location/Qualifiers
REGION           1..22
                  note = Engineered antibody sequence
source           1..22
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 63
QVLTQSPSSL SASVGDRVTI NC                               22

SEQ ID NO: 64      moltype = AA length = 13
FEATURE           Location/Qualifiers
REGION           1..13
                  note = Engineered antibody sequence
source           1..13
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 64
QASQSVYDNN YLA                               13

```

SEQ ID NO: 65	moltype = AA	length = 15	
FEATURE	Location/Qualifiers		
REGION	1..15		
source	note = Engineered antibody sequence		
	1..15		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 65			
WYQKPKGKVP KQLIY			15
SEQ ID NO: 66	moltype = AA	length = 7	
FEATURE	Location/Qualifiers		
REGION	1..7		
source	note = Engineered antibody sequence		
	1..7		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 66			
STSTLAS			7
SEQ ID NO: 67	moltype = AA	length = 32	
FEATURE	Location/Qualifiers		
REGION	1..32		
source	note = Engineered antibody sequence		
	1..32		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 67			
GVPSRFRSGSG SGTDFTLTIS SLQPEDVATY YC			32
SEQ ID NO: 68	moltype = AA	length = 13	
FEATURE	Location/Qualifiers		
REGION	1..13		
source	note = Engineered antibody sequence		
	1..13		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 68			
LGSYDCSSGD CFV			13
SEQ ID NO: 69	moltype = AA	length = 11	
FEATURE	Location/Qualifiers		
REGION	1..11		
source	note = Engineered antibody sequence		
	1..11		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 69			
FGGGTKVEIK R			11
SEQ ID NO: 70	moltype = AA	length = 106	
FEATURE	Location/Qualifiers		
REGION	1..106		
source	note = Engineered antibody sequence		
	1..106		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 70			
TVAAPSVFIF PPSDEQLKSG TASVCLLN FYPREAKVQW KVDNALQSGN SQESVTEQDS			60
KDSTYSLSST LTLKADYEEK HKVYACEVTH QGLSSPVTKS FNRGEC			106
SEQ ID NO: 71	moltype = DNA	length = 660	
FEATURE	Location/Qualifiers		
misc_feature	1..660		
source	note = Engineered antibody sequence		
	1..660		
	mol_type = other DNA		
	organism = synthetic construct		
SEQUENCE: 71			
caagtgcctga cccagctctcc atcctcctctg tctgcatctg taggagacag agtcaccatc			60
aattgccagg ccagtcagag tgtttatgat aacaactacc tagcctggta tcagcagaaa			120
ccaggggaaag ttctctaagca actgatctat tctacatcca ctctggcatc tgggggtccca			180
tctcgtttca gtggcagtggt atctggggaca gatttcaactc tcaccatcag cagcctgcag			240
cttgaagatg ttgcaactta ttactgtcta ggcagttatg attgtagtag tggtgattgt			300
ttgtttttcg gcggaggaac caaggttgaa atcaaacgta cggtggctgc accatctgtc			360
ttcatcttcc cgccatctga tgagcagttg aaatctggaa ctgcctctgt tgtgtgcctg			420
ctgaataact tctatcccag agaggccaaa gtacagtgga aggtggataa cgccctccaa			480
tcqqqtaact cccaqqagaa ttctacacaq caqqacagca aqqacagcac ctacaqcctc			540

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```

agcagcacc c tgacgctgag caaagcagac tacgagaaac acaaagtcta cgcctgcgaa 600
gtcaccatc agggcctgag ctgcccgtc acaaagagct tcaacagggg agagtgttag 660

```

```

SEQ ID NO: 72      moltype = DNA length = 339
FEATURE           Location/Qualifiers
misc_feature       1..339
                   note = Engineered antibody sequence
source            1..339
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 72
caagtgtga cccagtctcc atcctccctg tctgcatctg taggagacag agtcaccatc 60
aattgccagg ccagtcagag tgtttatgat aacaactacc tagcctggta tcagcagaaa 120
ccagggaag ttcctaagca actgatctat tctacatcca ctctggcatc tgggggtcca 180
tctcgtttca gtggcagtg atctgggaca gatttcactc tcaccatcag cagcctgcag 240
cctgaagatg ttgcaactta ttactgtcta ggcagttatg attgtagtag tgggtattgt 300
ttgttttcg gcggaggaac caaggaggaa atcaaacgt 339

```

```

SEQ ID NO: 73      moltype = DNA length = 66
FEATURE           Location/Qualifiers
misc_feature       1..66
                   note = Engineered antibody sequence
source            1..66
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 73
caagtgtga cccagtctcc atcctccctg tctgcatctg taggagacag agtcaccatc 60
aattgc 66

```

```

SEQ ID NO: 74      moltype = DNA length = 39
FEATURE           Location/Qualifiers
misc_feature       1..39
                   note = Engineered antibody sequence
source            1..39
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 74
caggccagtc agagtgttta tgataacaac tacctagcc 39

```

```

SEQ ID NO: 75      moltype = DNA length = 45
FEATURE           Location/Qualifiers
misc_feature       1..45
                   note = Engineered antibody sequence
source            1..45
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 75
tggatcagc agaaaccagg gaaagttcct aagcaactga tctat 45

```

```

SEQ ID NO: 76      moltype = DNA length = 21
FEATURE           Location/Qualifiers
misc_feature       1..21
                   note = Engineered antibody sequence
source            1..21
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 76
tctacatcca ctctggcatc t 21

```

```

SEQ ID NO: 77      moltype = DNA length = 96
FEATURE           Location/Qualifiers
misc_feature       1..96
                   note = Engineered antibody sequence
source            1..96
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 77
gggggtccat ctctgttcag tggcagtga tctgggacag atttcaactct caccatcagc 60
agcctgcagc ctgaagatgt tgcaacttat tactgt 96

```

```

SEQ ID NO: 78      moltype = DNA length = 39
FEATURE           Location/Qualifiers
misc_feature       1..39
                   note = Engineered antibody sequence
source            1..39
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 78
ctaggcagtt atgattgtag tagtggtgat tgttttgtt 39

```

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```

SEQ ID NO: 79      moltype = DNA length = 33
FEATURE           Location/Qualifiers
misc_feature      1..33
                  note = Engineered antibody sequence
source           1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 79
ttcggcggag gaaccaaggt ggaaatcaaa cgt                               33

SEQ ID NO: 80      moltype = DNA length = 321
FEATURE           Location/Qualifiers
misc_feature      1..321
                  note = Engineered antibody sequence
source           1..321
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 80
acgggtggctg caccatctgt cttcatcttc cgcctatctg atgagcagtt gaaatctgga 60
actgcctctg ttgtgtgcct gctgaataac ttctatccca gagaggccaa agtacagtgg 120
aaggtggata acgcctcca atcgggtaac tccaggaga gtgtcacaga gcaggacagc 180
aaggacagca cctacagcct cagcagcacc ctgacgctga gcaaagcaga ctacgagaaa 240
cacaagctct acgcctgcga agtcacccat cagggcctga gctcgcccg cacaagagc 300
ttcaacaggg gagagtgtta g                                           321

SEQ ID NO: 81      moltype = AA length = 441
FEATURE           Location/Qualifiers
REGION           1..441
                  note = Engineered antibody sequence
source           1..441
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 81
EVQLVESGGG LVQPGGSLRL SCAVSGLDLS SYMQWVRQA PGKGLEWVGV IGINDNTYYA 60
SWAKGRFTIS RDNSKTTVYL QMNSLRAEDT AVYFCARGDI WQGGLVTVS SASTKGPSVF 120
PLAPSSKSTS GGTAALGCLV KDYFPEPTV SWNSGALTSG VHTFPAVLQS SGLYSLSSV 180
TVPSSSLGTQ TYICNVNHPK SNTKVDARVE PKSCDKHTHC PPCPAPELLG GPSVFLFPPK 240
PKDTLMISRT PEVTCVVVDV SHEDPEVKFN WYVDGVEVHN AKTKPREEQY ASTYRVVSVL 300
TVLHQDWLNG KEYKCKVSNK ALPAPIEKTI SKAKGQPREP QVYTLPPSRE EMTKNQVSLT 360
CLVKGFPYPSD IAEWESNGQ PENNYKTPP VLDSDGSFPL YSKLTVDKSR WQGNVFSCS 420
VMHEALHNYH TQKSLSLSPG K                                           441

SEQ ID NO: 82      moltype = AA length = 111
FEATURE           Location/Qualifiers
REGION           1..111
                  note = Engineered antibody sequence
source           1..111
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 82
EVQLVESGGG LVQPGGSLRL SCAVSGLDLS SYMQWVRQA PGKGLEWVGV IGINDNTYYA 60
SWAKGRFTIS RDNSKTTVYL QMNSLRAEDT AVYFCARGDI WQGGLVTVS S          111

SEQ ID NO: 83      moltype = AA length = 30
FEATURE           Location/Qualifiers
REGION           1..30
                  note = Engineered antibody sequence
source           1..30
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 83
EVQLVESGGG LVQPGGSLRL SCAVSGLDLS                               30

SEQ ID NO: 84      moltype = AA length = 5
FEATURE           Location/Qualifiers
REGION           1..5
                  note = Engineered antibody sequence
source           1..5
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 84
SYMQ                                                                5

SEQ ID NO: 85      moltype = AA length = 14
FEATURE           Location/Qualifiers
REGION           1..14
                  note = Engineered antibody sequence
source           1..14

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mol_type = protein
organism = synthetic construct

SEQUENCE: 85
WVRQAPGKGL EWVG 14

SEQ ID NO: 86
FEATURE
REGION
source
moltype = AA length = 16
Location/Qualifiers
1..16
note = Engineered antibody sequence
1..16
mol_type = protein
organism = synthetic construct

SEQUENCE: 86
VIGINDNTYY ASWAKG 16

SEQ ID NO: 87
FEATURE
REGION
source
moltype = AA length = 32
Location/Qualifiers
1..32
note = Engineered antibody sequence
1..32
mol_type = protein
organism = synthetic construct

SEQUENCE: 87
RFTISRDNK TTVYLQMNSL RAEDTAVYFC AR 32

SEQ ID NO: 88
SEQUENCE: 88
000

SEQ ID NO: 89
FEATURE
REGION
source
moltype = AA length = 11
Location/Qualifiers
1..11
note = Engineered antibody sequence
1..11
mol_type = protein
organism = synthetic construct

SEQUENCE: 89
WGQGLTVTVS S 11

SEQ ID NO: 90
FEATURE
REGION
source
moltype = AA length = 330
Location/Qualifiers
1..330
note = Engineered antibody sequence
1..330
mol_type = protein
organism = synthetic construct

SEQUENCE: 90
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 60
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDARVEP KSCDKHTTCP PCPAPELLGG 120
PSVFLFPPPK KDTLMISRTPEVTCVVVDVS HEDPEVKFNW YVDGVEVHNA KTKPREEQYA 180
STYRVSVLT VLHQDWLNGK EYKCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSREE 240
MTKNQVSLTC LVKGFYPSDI AVEWESNGQP ENNYKTTTPV LDSGGSFFLY SKLTVDKSRW 300
QQGNVFSCSV MHEALHNHYT QKSLSLSPGK 330

SEQ ID NO: 91
FEATURE
misc_feature
source
moltype = DNA length = 1326
Location/Qualifiers
1..1326
note = Engineered antibody sequence
1..1326
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 91
gaggtgcagc ttgtggagtc tgggggaggc ttgggtccagc ctgggggggc cctgagactc 60
tcctgtgcag tcctctggact cgacctcagt agctactaca tgcaatgggt ccgtcaggct 120
ccagggaagg ggctggagtg ggtcggagtc attggtatca atgataacac atactacgcg 180
agctgggcga aaggccgatt caccatctcc agagacaatt ccaagaccac ggtgtatctt 240
caaatgaaca gcctgagagc tgaggacact gctgtgtatt tctgtgctag aggggacatc 300
tggggccaag ggaccctcgt caccgtctcg agcgctcca ccaagggccc atcggtcttc 360
cccctggcac cctcctccaa gaccacctct gggggcacag cggccctggg ctgcctggtc 420
aaggactact tccccgaacc ggtgacggtg tcgtggaact caggcgccct gaccagcggc 480
gtgcacacct tcccggctgt cctacagtcc tcaggactct actccctcag cagcgtggtg 540
accgtgcctt ccagcagctt gggcaccacg acctacatct gcaacgtgaa tcacaagccc 600
agcaacacca aggtggagcg gagagttgag cccaaatctt gtgacaaaac tcacacatgc 660
ccaccgtgcc cagcacctga actcctgggg ggaccgtcag tcttctctct cccccaaaa 720
cccaaggaca cctcatgat cteccggacc cctgagggtc catgcgtggt ggtggacgtg 780
agccacgaag accctgaggt caagttcaac tggtagctgg acggcgtgga ggtgcataat 840
gccaagacaa agccgcggga ggagcagtag gccagcacgt accgtgtggt cagcgtcttc 900
accgtcctgc accaggactg gctgaatggc aaggagtaca agtgcaaggt ctccaacaaa 960
gcctcccccag ccccatcgaa gaaaaccatc tccaaagcca aagggcagcc ccgagaacca 1020

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caggtgtaca cectgcccc atccccggag gagatgacca agaaccaggt cagcctgacc 1080
tgccctggta aaggcttcta tcccagcgac atcgccgtgg agtgggagag caatgggcag 1140
ccggagaaca actacaagac cagcctccc gtgctggact ccgacggctc cttcttctc 1200
tacagcaagc tcaccgtgga caagagcagg tggcagcagg ggaacgtctt ctcagtctcc 1260
gtgatgcatg aggctctgca caaccactac acgcagaaga gcctctccct gtctccgggt 1320
aatga                                           1326

```

```

SEQ ID NO: 92      moltype = DNA length = 333
FEATURE           Location/Qualifiers
misc_feature       1..333
                   note = Engineered antibody sequence
source            1..333
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 92
gaggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctgggggggtc cctgagactc 60
tcctgtgcag tctctggact cgacctcagt agctactaca tgcaatgggt ccgtcaggct 120
ccagggaagg ggctggagtg ggtcggagtc attggtatca atgataacac atactacgcg 180
agctgggcga aaggccgatt caccatctcc agagacaatt ccaagaccac ggtgtatctt 240
caaatgaaca gcctgagagc tgaggacact gctgtgtatt tctgtgctag aggggacatc 300
tggggccaag ggaccctcgt caccgtctcg agc                               333

```

```

SEQ ID NO: 93      moltype = DNA length = 90
FEATURE           Location/Qualifiers
misc_feature       1..90
                   note = Engineered antibody sequence
source            1..90
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 93
gaggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctgggggggtc cctgagactc 60
tcctgtgcag tctctggact cgacctcagt                               90

```

```

SEQ ID NO: 94      moltype = DNA length = 15
FEATURE           Location/Qualifiers
misc_feature       1..15
                   note = Engineered antibody sequence
source            1..15
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 94
agctactaca tgcaa                                           15

```

```

SEQ ID NO: 95      moltype = DNA length = 42
FEATURE           Location/Qualifiers
misc_feature       1..42
                   note = Engineered antibody sequence
source            1..42
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 95
tgggtccgtc aggctccagg gaaggggctg gagtgggtcg ga           42

```

```

SEQ ID NO: 96      moltype = DNA length = 48
FEATURE           Location/Qualifiers
misc_feature       1..48
                   note = Engineered antibody sequence
source            1..48
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 96
gtcattggta tcaatgataa cacatactac gcgagctggg cgaaaggc 48

```

```

SEQ ID NO: 97      moltype = DNA length = 96
FEATURE           Location/Qualifiers
misc_feature       1..96
                   note = Engineered antibody sequence
source            1..96
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 97
cgattcacca tctccagaga caattccaag accacggtgt atcttcaaat gaacagcctg 60
agagctgagg aactgtctgt gtatttctgt gctaga                    96

```

```

SEQ ID NO: 98      moltype = length =
SEQUENCE: 98
000

```

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SEQ ID NO: 99      moltype = DNA length = 33

```

-continued

FEATURE Location/Qualifiers
 misc_feature 1..33
 note = Engineered antibody sequence
 source 1..33
 mol_type = other DNA
 organism = synthetic construct
 SEQUENCE: 99
 tggggccaag ggaccctcgt caccgtctcg agc 33

SEQ ID NO: 100 moltype = DNA length = 993
 FEATURE Location/Qualifiers
 misc_feature 1..993
 note = Engineered antibody sequence
 source 1..993
 mol_type = other DNA
 organism = synthetic construct
 SEQUENCE: 100
 gcctccacca agggcccac cgtcttcccc ctggcaccct cctccaagag cacctctggg 60
 ggacacagcg ccctgggctg cctgggtcaag gactacttcc ccgaaccggg gacgggtgctg 120
 tggaaactcag gcgccttgac cagcggcgctg cacacdtccc cggtgtgctt acagtctctca 180
 ggactctact ccctcagcag cgtgggtgacc gtgccctcca gcagcttggg caccagacc 240
 tacatctgca acgtgaatca caagcccagc aacaccaagg tggacgctg agttgagccc 300
 aaatcttgtg acaaaaactca cacatgcccc ccgtgcccag cacctgaact cctggggggga 360
 ccgtcagctt tcctcttccc cccaaaaccc aaggacaccc tcatgatctc ccggacccct 420
 gaggtcacat gcgtgggtgt ggacgtgagc cagcaagacc ctgaggtcaa gttcaactgg 480
 tacgtggagc gcgtggaggt gcataatgcc aagacaaaag cgcgaggaga gcagtacgcc 540
 agcacgtacc gtgtgggtcag cgtcctcacc gtccctgcacc aggactggct gaatggcaag 600
 gagtacaagt gcaagggtctc caacaaagcc ctcccagccc ccatcgagaa aacctcttc 660
 aaagccaaag ggcagccccc agaaccacag gtgtacaccc tgccccatc ccgggaggag 720
 atgaccaaga accagggtcag cctgacctgc ctgggtcaaa gcttctatcc cagcgacatc 780
 gccgtggagt gggagagcaa tgggcagccg gagaacaact acaagaccac gcctcccggt 840
 ctggactccg acggtctctt ctctctctac agcaagctca ccgtggacaa gagcagggtg 900
 cagcagggga acgtctcttc atgctccgtg atgcatgagg ctctgcacaa ccactacacg 960
 cagaagagcc tctccctgtc tccgggtaaa tga 993

SEQ ID NO: 101 moltype = AA length = 219
 FEATURE Location/Qualifiers
 REGION 1..219
 note = Engineered antibody sequence
 source 1..219
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 101
 QVLTQSPSSL SASVGDRVTI NCQASQSVYD NNYLAWYQQK PGKVPKQLIY STSTLASGVP 60
 SRFSGSGSGT DFTLTISLQ PEDVATYYCL GSYDCSSGDC FVFGGKVE IKRTVAAPSV 120
 FIFPPSDEQL KSGTASVVL LNNFYPREAK VQWKVDNALQ SGNSQESVTE QDSKSTYSL 180
 SSTLTLSKAD YEKHKVYACE VTHQGLSSPV TKSFPNRGEC 219

SEQ ID NO: 102 moltype = AA length = 113
 FEATURE Location/Qualifiers
 REGION 1..113
 note = Engineered antibody sequence
 source 1..113
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 102
 QVLTQSPSSL SASVGDRVTI NCQASQSVYD NNYLAWYQQK PGKVPKQLIY STSTLASGVP 60
 SRFSGSGSGT DFTLTISLQ PEDVATYYCL GSYDCSSGDC FVFGGKVE IKR 113

SEQ ID NO: 103 moltype = AA length = 22
 FEATURE Location/Qualifiers
 REGION 1..22
 note = Engineered antibody sequence
 source 1..22
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 103
 QVLTQSPSSL SASVGDRVTI NC 22

SEQ ID NO: 104 moltype = AA length = 13
 FEATURE Location/Qualifiers
 REGION 1..13
 note = Engineered antibody sequence
 source 1..13
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 104
 QASQSVYDNN YLA 13

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gtcaccatc agggcctgag ctgcgccgtc acaaagagct tcaacagggg agagtgttag 660

SEQ ID NO: 112 moltype = DNA length = 339
 FEATURE Location/Qualifiers
 misc_feature 1..339
 note = Engineered antibody sequence
 source 1..339
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 112
 caagtgtga cccagtctcc atcctcctg tctgcatctg taggagacag agtcaccatc 60
 aattgccagg ccagtcagag tgtttatgat aacaactacc tagcctggta tcagcagaaa 120
 ccagggaag ttctaagca actgatctat tctacatcca ctctggcatc tgggggtcca 180
 tctcgtttca gtggcagtg atctgggaca gatttcaact tcaccatcag cagcctgcag 240
 cctgaagatg ttgcaactta ttactgtcta ggcagttatg attgtagtag tgggtattgt 300
 ttgttttcg gcggaggaac caaggtggaa atcaaacgt 339

SEQ ID NO: 113 moltype = DNA length = 66
 FEATURE Location/Qualifiers
 misc_feature 1..66
 note = Engineered antibody sequence
 source 1..66
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 113
 caagtgtga cccagtctcc atcctcctg tctgcatctg taggagacag agtcaccatc 60
 aattgc 66

SEQ ID NO: 114 moltype = DNA length = 39
 FEATURE Location/Qualifiers
 misc_feature 1..39
 note = Engineered antibody sequence
 source 1..39
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 114
 caggccagtc agagtgttta tgataacaac tacctagcc 39

SEQ ID NO: 115 moltype = DNA length = 45
 FEATURE Location/Qualifiers
 misc_feature 1..45
 note = Engineered antibody sequence
 source 1..45
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 115
 tggatcagc agaaaccagg gaaagttcct aagcaactga tctat 45

SEQ ID NO: 116 moltype = DNA length = 21
 FEATURE Location/Qualifiers
 misc_feature 1..21
 note = Engineered antibody sequence
 source 1..21
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 116
 tctacatcca ctctggcatc t 21

SEQ ID NO: 117 moltype = DNA length = 96
 FEATURE Location/Qualifiers
 misc_feature 1..96
 note = Engineered antibody sequence
 source 1..96
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 117
 ggggtcccat ctcgtttcag tggcagtgga tctgggacag atttcaactct caccatcagc 60
 agcctgcagc ctgaagatgt tgcaacttat tactgt 96

SEQ ID NO: 118 moltype = DNA length = 39
 FEATURE Location/Qualifiers
 misc_feature 1..39
 note = Engineered antibody sequence
 source 1..39
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 118
 ctaggcagtt atgattgtag tagtggtgat tgttttgtt 39

-continued

SEQ ID NO: 119 moltype = DNA length = 33
 FEATURE Location/Qualifiers
 misc_feature 1..33
 note = Engineered antibody sequence
 source 1..33
 mol_type = other DNA
 organism = synthetic construct

 SEQUENCE: 119
 ttcggcggag gaaccaaggt ggaatcaaaa cgt 33

SEQ ID NO: 120 moltype = DNA length = 321
 FEATURE Location/Qualifiers
 misc_feature 1..321
 note = Engineered antibody sequence
 source 1..321
 mol_type = other DNA
 organism = synthetic construct

 SEQUENCE: 120
 acggtggctg caccatctgt cttcatcttc cggccatctg atgagcagtt gaaatctgga 60
 actgcctctg ttgtgtgctt gctgaataac ttctatccca gagaggccaa agtacagtgg 120
 aaggtggata acgcccctcca atcgggtaac tcccaggaga gtgtcacaga gcaggacagc 180
 aaggacagca cctacagcct cagcagcacc ctgacgctga gcaaagcaga ctacagagaaa 240
 cacaaggtct acgcctgcca agtcacccat cagggcctga gctcgccgt cacaagagc 300
 ttcaacaggg gagagtgtta g 321

SEQ ID NO: 121 moltype = AA length = 439
 FEATURE Location/Qualifiers
 REGION 1..439
 note = Engineered antibody sequence
 source 1..439
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 121
 QSLEESGGRL VTPGTPLTLT CSVSGIDLSG YYMNWVRQAP GKGLEWIGVI GINGATYYAS 60
 WAKGRFTISK TSSTTVDLKM TSLTTEDTAT YFCARGDIWG PGTLVTVSSA STKGPSVFPPL 120
 APSSKSTSGG TAALGCLVKD YFPEPVTWSW NSGALTSGVH TFPVAVLQSSG LYSLSVVTV 180
 PSSSLGTQTY ICNVNHKPSN TKVDRVEPK SCDKTHTCP CPAPELLGGP SVFLFPPKPK 240
 DTLMISRTPE VTCVVVDVSH EDPEVKFNWY VDGVEVHNAK TKPREEQYAS TYRVVSVLTV 300
 LHQDWLNGKE YKCKVSNKAL PAPIETISK AKGQPREPQV YTLPPSREEM TKNQVSLTCL 360
 VKGFYPSDIA VEWESNGQPE NNYKTPPVVL DSDGSFFLYS KLTVDKSRWQ QGNVFSCSVM 420
 HEALHNYHQ KSLSLSPGK 439

SEQ ID NO: 122 moltype = AA length = 109
 FEATURE Location/Qualifiers
 REGION 1..109
 note = Engineered antibody sequence
 source 1..109
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 122
 QSLEESGGRL VTPGTPLTLT CSVSGIDLSG YYMNWVRQAP GKGLEWIGVI GINGATYYAS 60
 WAKGRFTISK TSSTTVDLKM TSLTTEDTAT YFCARGDIWG PGTLVTVSS 109

SEQ ID NO: 123 moltype = AA length = 29
 FEATURE Location/Qualifiers
 REGION 1..29
 note = Engineered antibody sequence
 source 1..29
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 123
 QSLEESGGRL VTPGTPLTLT CSVSGIDLS 29

SEQ ID NO: 124 moltype = AA length = 5
 FEATURE Location/Qualifiers
 REGION 1..5
 note = Engineered antibody sequence
 source 1..5
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 124
 GYYMN 5

SEQ ID NO: 125 moltype = AA length = 14
 FEATURE Location/Qualifiers
 REGION 1..14
 note = Engineered antibody sequence
 source 1..14
 mol_type = protein

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organism = synthetic construct
 SEQUENCE: 125
 WVRQAPGKGL EWIG 14

SEQ ID NO: 126
 FEATURE
 REGION
 source
 moltype = AA length = 16
 Location/Qualifiers
 1..16
 note = Engineered antibody sequence
 1..16
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 126
 VIGINGATYY ASWAKG 16

SEQ ID NO: 127
 FEATURE
 REGION
 source
 moltype = AA length = 31
 Location/Qualifiers
 1..31
 note = Engineered antibody sequence
 1..31
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 127
 RFTISKTSST TVDLKMTSLT TEDTATYFCA R 31

SEQ ID NO: 128
 SEQUENCE: 128
 000
 moltype = length =

SEQ ID NO: 129
 FEATURE
 REGION
 source
 moltype = AA length = 11
 Location/Qualifiers
 1..11
 note = Engineered antibody sequence
 1..11
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 129
 WGPGLVTVS S 11

SEQ ID NO: 130
 FEATURE
 REGION
 source
 moltype = AA length = 330
 Location/Qualifiers
 1..330
 note = Engineered antibody sequence
 1..330
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 130
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTPPAVLQSS 60
 GLYSLSSVVT VPSSSLGTQT YICNVNHNKPS NTKVDKRVEP KSCDKHTTCP PCPAPELLGG 120
 PSVFLFPPKP KDTLMISRTPEVTCVVVDVS HEDPEVKFNW YVDGVEVHNA KTKPREEQYA 180
 STYRVVSVLT VLHQDWLNGK EYKCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSREE 240
 MTKNQVSLTCLVKGFYPSDI AVEWESNGQP ENNYKTTTPV LDSGGSFFLY SKLTVDKSRW 300
 QQGNVFSCSV MHEALHNHYT QKSLSLSPGK 330

SEQ ID NO: 131
 FEATURE
 misc_feature
 source
 moltype = DNA length = 1320
 Location/Qualifiers
 1..1320
 note = Engineered antibody sequence
 1..1320
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 131
 cagtcgctgg aggagtcagg gggctgcctg gtcacgcctg ggacacccct gacactcacc 60
 tgttcctgtc ctggcatcga cctcagtggt tactacatga actgggtccg ccagggtcca 120
 gggaaggggc tggaaatggat cggagtcatt ggtattaatg gtgccacata ctacgcgagc 180
 tgggcgaaaag gccgattcac catctccaaa acctcgctga ccacggtgga tctgaaaatg 240
 accagtctga caaccgagga cacggccacc tattctgtg ccagagggga catctggggc 300
 ccgggcaccc tcgtcacctg ctgcagcgcc tccaccaagg gcccatcggt ctteccctcg 360
 gcaccctcct ccaagagcac ctctgggggc acagcggccc tgggtgcct ggccaaggac 420
 tacttccccg aaccggtgac ggtgtcgtgg aactcaggcg ccctgaccag cggcgtgcac 480
 accttccccg ctgtcctaca gtctcagga ctctactccc tcagcagcgt ggtgaccgtg 540
 cctccagca gcttgggcac ccagacctac atctgcaacg tgaatcaca gccacgcaac 600
 accaaggtgg acaagagagt tgagcccaaa tcttgtgaca aaactcacac atgccaccg 660
 tgccagcac ctgaactcct ggggggaccg tcagtcttcc tcttcccccc aaaacccaag 720
 gacaccctca tgatctccc gaccctgag gtcacatgag tgggtggtgga cgtgagccac 780
 gaagaccctg aggtcaagtt caactggtac gtggacggcg tggaggtgca taatgccaa 840
 acaaagccgc gggaggagca gtacgccagc acgtaccgtg tggtcagcgt cctcaccgtc 900
 ctgcaccagg actggctgaa tggcaaggag tacaagtga aggtctccaa caaagccctc 960
 ccagcccca tcgagaaaac catctccaaa gccaaaggcg agccccgaga accacagggt 1020
 tacaccctgc ccccatcccg ggaggagatg accaagaacc aggtcagcct gacctgctg 1080

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gtcaaaggct tctatcccag cgacatcgcc gtggagtggg agagcaatgg gcagccggag 1140
aacaactaca agaccacgcc tcccgtgctg gactcgcagc gctccttctt cctctacagc 1200
aagctcaccg tggacaagag caggtggcag caggggaacg tcttctcatg ctccgtgatg 1260
catgaggctc tgcacaacca ctacacgcag aagagcctct cctgtgtctc gggtaaatga 1320

```

```

SEQ ID NO: 132      moltype = DNA length = 327
FEATURE
misc_feature       1..327
                    note = Engineered antibody sequence
source             1..327
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 132
cagtcgctgg aggagtccgg gggtcgctg gtcacgcctg ggacaccct gacactcacc 60
tgttccgtct ctggcatcga cctcagtggc tactacatga actgggtccg ccagggtcca 120
gggaaggggc tgggaatggat cggagtcatt ggtattaatg gtgccacata ctacgcgagc 180
tgggcgaaag gccgattcac catctccaaa acctcgtcga ccacgggtga tctgaaaatg 240
accagtctga caaccgagga cacggccacc tatttctgtg ccagagggga catctggggc 300
ccgggcaccc tcgtcaccgt ctcgagc                                     327

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```

SEQ ID NO: 133      moltype = DNA length = 87
FEATURE
misc_feature       1..87
                    note = Engineered antibody sequence
source             1..87
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 133
cagtcgctgg aggagtccgg gggtcgctg gtcacgcctg ggacaccct gacactcacc 60
tgttccgtct ctggcatcga cctcagtc                                     87

```

```

SEQ ID NO: 134      moltype = DNA length = 15
FEATURE
misc_feature       1..15
                    note = Engineered antibody sequence
source             1..15
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 134
ggctactaca tgaac                                             15

```

```

SEQ ID NO: 135      moltype = DNA length = 42
FEATURE
misc_feature       1..42
                    note = Engineered antibody sequence
source             1..42
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 135
tgggtccgcc aggtccagg gaaggggctg gaatggatcg ga                                     42

```

```

SEQ ID NO: 136      moltype = DNA length = 48
FEATURE
misc_feature       1..48
                    note = Engineered antibody sequence
source             1..48
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 136
gtcattggtg ttaatgggtc cacatactac gcgagctggg cgaaaggc                                     48

```

```

SEQ ID NO: 137      moltype = DNA length = 93
FEATURE
misc_feature       1..93
                    note = Engineered antibody sequence
source             1..93
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 137
cgattcacca tctccaaaac ctgctcgacc acggtggatc tgaaaatgac cagtctgaca 60
accgaggaca cggccaccta tttctgtgcc aga                                     93

```

```

SEQ ID NO: 138      moltype = length =
SEQUENCE: 138
000

```

```

SEQ ID NO: 139      moltype = DNA length = 33
FEATURE
misc_feature       1..33

```

-continued

```

source          note = Engineered antibody sequence
                1..33
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 139
tggggcccgg gcaccctcgt caccgtctcg agc                               33

SEQ ID NO: 140      moltype = DNA length = 993
FEATURE            Location/Qualifiers
misc_feature        1..993
                    note = Engineered antibody sequence
source              1..993
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 140
gcctccacca agggcccacg ggtcttcccc ctggcacccct cctccaagag caccctctggg 60
ggcacagcgg ccttggggtg cctgggtcaag gactacttcc ccgaaccggg gacgggtgtcg 120
tggaactcag gcgccttgac cagcggcgctg cacaccttcc cggctgtcct acagtctcctca 180
ggactctact ccttcagcag cgtgggtgacc gtgccctcca gcagcttggg caccagacc 240
tacatctgca acgtgaatca caagcccagc aacaccaagg tggacaagag agttgagccc 300
aaatcttgtg acaaaactca cacatgcccc ccgtgcccag cactgaact cctgggggga 360
ccgtcagtc tctcttcccc ccctaaaccc aaggacaccc tcatgatctc ccggaccct 420
gaggtcacat gcgtgggtgg ggacgtgagc cacgaagacc ctgaggtcaa gttcaactgg 480
tacgtggacg gcgtggaggt gcataatgcc aagacaaagg cgcgggagga gcagtacgcc 540
agcacgtacc gtgtgggtcag cgtctcacc gtctgcacc aggactggct gaatggcaag 600
gagtacaagt gcaagggtct caacaaagcc ctcccagccc ccacgagaa aacctctctc 660
aaagccaaag ggcagccccc agaaccacag gtgtacaccc tgcccccatc ccgggaggag 720
atgaccaaga accagggtcag cctgacctgc ctggtcaaag gcttctatcc cagcgacatc 780
gccgtggagt gggagagcaa tgggcagccg gagaacaact acaagaccac gcctcccgtg 840
ctggactccg acggctcctt ctctctctac agcaagctca ccgtggacaa gagcagggtg 900
cagcagggga agctctcttc atgctccgtg atgcatgagg ctctgcacaa ccactacacg 960
cagaagagcc tctccctgtc tccgggtaaa tga                               993

SEQ ID NO: 141      moltype = AA length = 219
FEATURE            Location/Qualifiers
REGION            1..219
                    note = Engineered antibody sequence
source            1..219
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 141
QVLTQTPSPV SAAVGSTVTI NCQASQSVYH NTYLAWYQQK PGQPPKQLIY DASTLASGVP 60
SRFSGSGSGT QFTLTISGVQ CNDAAAYYCL GSYDCTNGDC FVFGGTEVV VKRTVAAPSV 120
FIFPPSDEQL KSGTASVCL LNNFYPREAK VQWKVDNALQ SGNSQESVTE QDSKSTYSL 180
SSTLTLSKAD YEKHKVYACE VTHQGLSSPV TKSFNRGEC                               219

SEQ ID NO: 142      moltype = AA length = 113
FEATURE            Location/Qualifiers
REGION            1..113
                    note = Engineered antibody sequence
source            1..113
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 142
QVLTQTPSPV SAAVGSTVTI NCQASQSVYH NTYLAWYQQK PGQPPKQLIY DASTLASGVP 60
SRFSGSGSGT QFTLTISGVQ CNDAAAYYCL GSYDCTNGDC FVFGGTEVV VKR                               113

SEQ ID NO: 143      moltype = AA length = 22
FEATURE            Location/Qualifiers
REGION            1..22
                    note = Engineered antibody sequence
source            1..22
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 143
QVLTQTPSPV SAAVGSTVTI NC                               22

SEQ ID NO: 144      moltype = AA length = 13
FEATURE            Location/Qualifiers
REGION            1..13
                    note = Engineered antibody sequence
source            1..13
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 144
QASQSVYHNT YLA                               13

SEQ ID NO: 145      moltype = AA length = 15
FEATURE            Location/Qualifiers

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REGION	1..15	
	note = Engineered antibody sequence	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 145		
WYQKPGQPP KQLIY		15
SEQ ID NO: 146	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = Engineered antibody sequence	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 146		
DASTLAS		7
SEQ ID NO: 147	moltype = AA length = 32	
FEATURE	Location/Qualifiers	
REGION	1..32	
	note = Engineered antibody sequence	
source	1..32	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 147		
GVPSRFSGSG SGTQFTLTIS GVQCNDAAAY YC		32
SEQ ID NO: 148	moltype = AA length = 13	
FEATURE	Location/Qualifiers	
REGION	1..13	
	note = Engineered antibody sequence	
source	1..13	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 148		
LGSYDCTNGD CFV		13
SEQ ID NO: 149	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Engineered antibody sequence	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 149		
FGGGTEVVVK R		11
SEQ ID NO: 150	moltype = AA length = 106	
FEATURE	Location/Qualifiers	
REGION	1..106	
	note = Engineered antibody sequence	
source	1..106	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 150		
TVAAPSVFIF PPSDEQLKSG TASVVCLLNN FYPREAKVQW KVDNALQSGN SQESVTEQDS		60
KDSTYLSST LTLSKADYK HKVYACEVTH QGLSPVTKS FNRGEC		106
SEQ ID NO: 151	moltype = DNA length = 660	
FEATURE	Location/Qualifiers	
misc_feature	1..660	
	note = Engineered antibody sequence	
source	1..660	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 151		
caagtgtcta cccagactcc atcccccggtg tctgcagctg tgggaagcac agtcaccatc		60
aattgccagg ccagtcagag tgtttatcat aacacctacc tggcctggta tcagcagaaa		120
ccagggcagc ctcccaaaaca actgatctat gatgcattca ctctggcgctc tgggggtccca		180
tcgcgggttca gcggcagtggt atctgggaca cagttcactc tcaccatcag cggcgtgcag		240
tgtaacgatg ctgccgctta ctactgtctg ggcagttatg attgtactaa tgggtgattgt		300
tttgttttct gcggagggac cgaggtgggtg gtcaaacgta cggtggcgctc accatctgtc		360
ttcatcttcc cgcatctga tgagcagttg aaatctggaa ctgcctctgt tgtgtgacctg		420
ctgaataact tctatccag agaggccaaa gtacagtggg aggtgggataa cgccctccaa		480
tcgggtaact cccaggagag tgtcacagag caggacagca aggacagcac ctacagcctc		540
agcagcaccg tgacgctgag caaagcagac tacgagaaac acaaagtcta cgccctgcgaa		600
gtcaccatcc agggcctgag ctgcgccgtc acaaagagct tcaacagggg agagtgttag		660

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SEQ ID NO: 152      moltype = DNA length = 339
FEATURE            Location/Qualifiers
misc_feature        1..339
                    note = Engineered antibody sequence
source              1..339
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 152
caagtgctga cccagactcc atcccccggtg tctgcagctg tgggaagcac agtcaccatc 60
aattgccagg ccagtcagag tgtttatcat aacacctacc tggcctggta tcagcagaaa 120
ccagggcagc ctcccaaaca actgatctat gatgcattcca ctctggcgtc tgggggtcca 180
tcgcggttca gcgcagctgg atctgggaca cagttcactc tcaccatcag cggcgtgcag 240
tgtaacgatg ctgccgctta ctactgtctg ggcagttatg attgtactaa tgggtattgt 300
ttgtttttcg gcggagggac cgagggtggtg gtcaaactg 339

SEQ ID NO: 153      moltype = DNA length = 66
FEATURE            Location/Qualifiers
misc_feature        1..66
                    note = Engineered antibody sequence
source              1..66
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 153
caagtgctga cccagactcc atcccccggtg tctgcagctg tgggaagcac agtcaccatc 60
aattgc                                           66

SEQ ID NO: 154      moltype = DNA length = 39
FEATURE            Location/Qualifiers
misc_feature        1..39
                    note = Engineered antibody sequence
source              1..39
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 154
caggccagtc agagtgttta tcataacacc tacctggcc 39

SEQ ID NO: 155      moltype = DNA length = 45
FEATURE            Location/Qualifiers
misc_feature        1..45
                    note = Engineered antibody sequence
source              1..45
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 155
tggatcagc agaaaccagg gcagcctccc aaacaactga tctat 45

SEQ ID NO: 156      moltype = DNA length = 21
FEATURE            Location/Qualifiers
misc_feature        1..21
                    note = Engineered antibody sequence
source              1..21
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 156
gatgcattcca ctctggcgtc t 21

SEQ ID NO: 157      moltype = DNA length = 96
FEATURE            Location/Qualifiers
misc_feature        1..96
                    note = Engineered antibody sequence
source              1..96
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 157
gggggtcccat cgcggttcag cggcagtgga tctgggacac agttcactct caccatcagc 60
ggcgtgcagt gtaacgatgc tgccgcttac tactgt 96

SEQ ID NO: 158      moltype = DNA length = 39
FEATURE            Location/Qualifiers
misc_feature        1..39
                    note = Engineered antibody sequence
source              1..39
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 158
ctgggcagtt atgattgtac taatgggtgat tgttttgtt 39

SEQ ID NO: 159      moltype = DNA length = 33
FEATURE            Location/Qualifiers

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misc_feature      1..33
                  note = Engineered antibody sequence
source            1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 159
ttcggcggag ggaccgaggt ggtggtcaaa cgt                               33

SEQ ID NO: 160      moltype = DNA length = 321
FEATURE            Location/Qualifiers
misc_feature        1..321
                  note = Engineered antibody sequence
source              1..321
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 160
acggtggtctg caccatctgt cttcatcttc cgcctatctg atgagcagtt gaaatctgga 60
actgcctctg  ttgtgtgctt gctgaataac ttctatccca gagaggccaa agtacagtgg 120
aaggtggata  acgcccctcca atcgggtaac tcccaggaga gtgtcacaga gcaggacagc 180
aaggacagca  cctacagcct  cagcagcacc ctgacgctga gcaaagcaga ctacgagaaa 240
cacaaagtct  acgcctgcga  agtcacccat cagggcctga gtcgcccgt cacaaagagc 300
ttcaacaggg  gagagtgtta  g                               321

SEQ ID NO: 161      moltype = AA length = 441
FEATURE            Location/Qualifiers
REGION              1..441
                  note = Engineered antibody sequence
source              1..441
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 161
EVQLVESGGG LVQPGGSLRL SCAVSGIDLS GYYMNWVRQA PGKGLEWVGV IGINGATYYA 60
SWAKGRFTIS RDNSKTTVYL QMNSLRAEDT AVYFCARGDI WGQGTLVTVS SASTKGPSVF 120
PLAPSSKSTS GGTAALGCLV KDYFPEPVTV SWNSGALTSG VHTFPAVLQS SGLYSLSSV 180
TVPSSSLGTQ TYICNVNHPK SNTKVDKRVK PKSCDKHTHC PPCPAPELLG GPSVFLFPPK 240
PKDTLMISRT PEVTCVVVDV SHEDPEVKFN WYVDGVEVHN AKTKPREEQY ASTYRVVSVL 300
TVLHQDWLNG KEYKCKVSNK ALPAPIEKTI SKAKGQPREP QVYTLPPSRE EMTKNQVSLT 360
CLVKGFPYPSD IAVEWESNGQ PENNYKTPPP VLDSDGSFPL YSKLTVDKSR WQQGNVFSCS 420
VMHEALHNNHY TQKSLSLSPG K                               441

SEQ ID NO: 162      moltype = AA length = 111
FEATURE            Location/Qualifiers
REGION              1..111
                  note = Engineered antibody sequence
source              1..111
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 162
EVQLVESGGG LVQPGGSLRL SCAVSGIDLS GYYMNWVRQA PGKGLEWVGV IGINGATYYA 60
SWAKGRFTIS RDNSKTTVYL QMNSLRAEDT AVYFCARGDI WGQGTLVTVS S          111

SEQ ID NO: 163      moltype = AA length = 30
FEATURE            Location/Qualifiers
REGION              1..30
                  note = Engineered antibody sequence
source              1..30
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 163
EVQLVESGGG LVQPGGSLRL SCAVSGIDLS                               30

SEQ ID NO: 164      moltype = AA length = 5
FEATURE            Location/Qualifiers
REGION              1..5
                  note = Engineered antibody sequence
source              1..5
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 164
GYMN                               5

SEQ ID NO: 165      moltype = AA length = 14
FEATURE            Location/Qualifiers
REGION              1..14
                  note = Engineered antibody sequence
source              1..14
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 165

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WVRQAPGKGL EWVG	14
SEQ ID NO: 166	moltype = AA length = 16
FEATURE	Location/Qualifiers
REGION	1..16
	note = Engineered antibody sequence
source	1..16
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 166	
VIGINGATYY ASWAKG	16
SEQ ID NO: 167	moltype = AA length = 32
FEATURE	Location/Qualifiers
REGION	1..32
	note = Engineered antibody sequence
source	1..32
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 167	
RFTISRDNK TTVYLQMNSL RAEDTAVYFC AR	32
SEQ ID NO: 168	moltype = length =
SEQUENCE: 168	
000	
SEQ ID NO: 169	moltype = AA length = 11
FEATURE	Location/Qualifiers
REGION	1..11
	note = Engineered antibody sequence
source	1..11
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 169	
WGQGLTVTVS S	11
SEQ ID NO: 170	moltype = AA length = 330
FEATURE	Location/Qualifiers
REGION	1..330
	note = Engineered antibody sequence
source	1..330
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 170	
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS	60
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKRVPE KSCDKHTTCP PCPAPELLGG	120
PSVFLFPPPK KDTLMISRTP EVTCVVVDVS HEDPEVKFNW YVDGVEVHNA KTKPREEQYA	180
STYRVVSVLT VLNQDNLNGK EYKCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSREE	240
MTKNQVSLTC LVKGFYPSDI AVEWESNGQP ENNYKTTTPV LDSDGSFFLY SKLTVDKSRW	300
QQGNVFCSCV MHEALHNHYT QKSLSLSPGK	330
SEQ ID NO: 171	moltype = DNA length = 1326
FEATURE	Location/Qualifiers
misc_feature	1..1326
	note = Engineered antibody sequence
source	1..1326
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 171	
gaggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctgggggggc cctgagactc	60
tcctgtgcag tctctggaat cgacctcagt ggctactaca tgaactgggt ccgtcaggct	120
ccagggaagg ggctggagtg ggtcggagtc attggtatta atggtgccac atactacgcg	180
agctgggcga aaggccgatt caccatctcc agagacaatt ccaagaccac ggtgtatctt	240
caaatgaaca gcctgagagc tgaggacact gctgtgtatt tctgtgctag aggggacatc	300
tggggccaag ggaccctcgt caccgtctcg agcgctcca ccaagggccc atcggtcttc	360
cccctggcac cctcctccaa gagcacctct gggggcacag cggccctggg ctgcctggtc	420
aaggactact tccccgaacc ggtgacggtg tcgtggaact caggcgccct gaccagcggc	480
gtgcacacct tcccggctgt cctacagtcc tcaggactct actccctcag cagcgtgggtg	540
accgtgccct ccagcagctt gggcaccacg acctacatct gcaacgtgaa tcacaagccc	600
agcaaaccca aggtggacaa gagagttgag ccccaatctt gtgacaaaac tcacacatgc	660
ccaccgtgcc cagcacctga actcctgggg ggaccgtcag tcttctctt cccccaaaa	720
cccaaggaca cctcatgat ctcccgagc cctgaggtca catgcgtggt ggtggacgtg	780
agccacgaag acctgaggt caagttcaac tggtagctgg acggcgtgga ggtgcataat	840
gccaagacaa agccgcggga ggagcagtag gccagcacgt accgtgtggt cagcgtctc	900
accgtcctgc accaggtg gctgaatggc aaggagtaca agtgaagggt ctccaacaaa	960
gccctccag ccccatcga gaaaaccatc tccaaagcca aagggcagcc ccgagaacca	1020
caggtgtaca ccttgcctcc atcccgagg gagatgacca agaaccaggt cagcctgacc	1080
tgcttggtca aaggcttcta tcccagcgac atcgccgtgg agtgggagag caatgggcag	1140
ccggagaaca actacaagac cagcctccc gtgctggact ccgacggctc cttcttctc	1200

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source          1..33
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 179
tggggccaag ggaccctcgt caccgtctcg agc                               33

SEQ ID NO: 180      moltype = DNA length = 993
FEATURE            Location/Qualifiers
misc_feature       1..993
                  note = Engineered antibody sequence
source            1..993
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 180
gcctccacca agggcccac cctcttcccc ctggcaccct cctccaagag cacctctggg 60
ggcacagcgg cccctgggctg cctgggtcaag gactacttcc ccgaaccggg gacgggtgtcg 120
tggaaactcag gcgccttgac cagcggcgtg cacaccttcc cggtgtgctt acagtccctca 180
ggactctact ccctcagcag cgtgggtgacc gtgccctcca gcagcttggg caccagacc 240
tacatctgca acgtgaatca caagcccagc aacaccaagg tggacaagag agttgagccc 300
aaatcttgtg acaaaaactca cacatgcccc ccgtgcccag cacctgaact cctgggggga 360
ccgtcagctc tcctcttccc cccaaaaccc aaggacaccc tcatgatctc ccggaccct 420
gaggtccatc gcgtgggtgt ggacgtgagc cagcaagacc ctgaggtcaa gttcaactgg 480
tacgtggacg gcgtggaggt gcataatgcc aagacaaagc cgcggggagga gcagtacgcc 540
agcacgtacc gtgtgggtcag cgtcctcacc gtccctgcacc aggactggct gaatggcaag 600
gagtacaagt gcaagggtctc caacaaagcc ctcccagccc ccatcgagaa aaccatctcc 660
aaagccaaag ggcagccccg agaaccacag gtgtacaccc tgcccccatc ccgggaggag 720
atgaccaaga accaggtcag cctgacctgc ctgggtcaaag gcttctatcc cagcgacatc 780
gccgtggagt gggagagcaa tgggcagccg gagaacaact acaagaccac gcctcccggt 840
ctggactccg acggctcctt ctctctctac agcaagctca ccgtggacaa gagcagggtg 900
cagcagggga acgtcttctc atgctccgtg atgcatgagg ctctgcacaa ccactacacg 960
cagaagagcc tctccctgtc tccgggtaaa tga                               993

SEQ ID NO: 181      moltype = AA length = 219
FEATURE            Location/Qualifiers
REGION            1..219
                  note = Engineered antibody sequence
source            1..219
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 181
QVLTQSPSSL SASVGDRVIT NCQASQSVYH NTYLAWYQQK PGKVPKQLIY DASTLASGVP 60
SRFSGSGSGT DFTLTISLQ PEDVATYYCL GSYDCTNGDC FVFGGGTKVE IKRTVAAPSV 120
FIFPPSDEQL KSGTASVVCL LNNFYPREAK VQWKVDNALQ SGNSQESVTE QDSKDSSTYSL 180
SSTLTLSKAD YEKHKVYACE VTHQGLSSPV TKSFNRGEC'                               219

SEQ ID NO: 182      moltype = AA length = 113
FEATURE            Location/Qualifiers
REGION            1..113
                  note = Engineered antibody sequence
source            1..113
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 182
QVLTQSPSSL SASVGDRVIT NCQASQSVYH NTYLAWYQQK PGKVPKQLIY DASTLASGVP 60
SRFSGSGSGT DFTLTISLQ PEDVATYYCL GSYDCTNGDC FVFGGGTKVE IKR                               113

SEQ ID NO: 183      moltype = AA length = 22
FEATURE            Location/Qualifiers
REGION            1..22
                  note = Engineered antibody sequence
source            1..22
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 183
QVLTQSPSSL SASVGDRVIT NC                               22

SEQ ID NO: 184      moltype = AA length = 13
FEATURE            Location/Qualifiers
REGION            1..13
                  note = Engineered antibody sequence
source            1..13
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 184
QASQSVYHNT YLA                               13

SEQ ID NO: 185      moltype = AA length = 15
FEATURE            Location/Qualifiers
REGION            1..15

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source	note = Engineered antibody sequence 1..15 mol_type = protein organism = synthetic construct	
SEQUENCE: 185 WYQQKPGKVP KQLIY		15
SEQ ID NO: 186 FEATURE REGION	moltype = AA length = 7 Location/Qualifiers 1..7 note = Engineered antibody sequence	
source	1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 186 DASTLAS		7
SEQ ID NO: 187 FEATURE REGION	moltype = AA length = 32 Location/Qualifiers 1..32 note = Engineered antibody sequence	
source	1..32 mol_type = protein organism = synthetic construct	
SEQUENCE: 187 GVPSRFGSGG SGTDFTLTIS SLQPEDVATY YC		32
SEQ ID NO: 188 FEATURE REGION	moltype = AA length = 13 Location/Qualifiers 1..13 note = Engineered antibody sequence	
source	1..13 mol_type = protein organism = synthetic construct	
SEQUENCE: 188 LGSYDCTNGD CFV		13
SEQ ID NO: 189 FEATURE REGION	moltype = AA length = 11 Location/Qualifiers 1..11 note = Engineered antibody sequence	
source	1..11 mol_type = protein organism = synthetic construct	
SEQUENCE: 189 FGGKTKVEIK R		11
SEQ ID NO: 190 FEATURE REGION	moltype = AA length = 106 Location/Qualifiers 1..106 note = Engineered antibody sequence	
source	1..106 mol_type = protein organism = synthetic construct	
SEQUENCE: 190 TVAAPSVFIF PPSDEQLKSG TASVVCLLNN FYPREKQVQW KVDNALQSGN SQESVTEQDS KDSTYLSST LTLSKADYK HKVYACEVTH QGLSSPVTKS FNRGEC		60 106
SEQ ID NO: 191 FEATURE misc_feature	moltype = DNA length = 660 Location/Qualifiers 1..660 note = Engineered antibody sequence	
source	1..660 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 191 caagtgcgta cccagtcctc atcctccctg tctgcatctg taggagacag agtcaccatc aattgccagg ccagtcagag tgtttatcat aacacctacc tggcctggta tcagcagaaa ccagggaag ttcctaagca actgatctat gatgcatcca ctctggcatc tgggggccca tctcgtttca gtggcagtg atctgggaca gatttcactc tcaccatcag cagcctgcag cctgaagatg ttgcaactta ttaactgtctg ggcagttatg attgtactaa tggtgattgt tttgttttcg gcggaggaac caaggtggaa atcaaacgta cggtggctgc accatctgtc ttcatcttcc cgccatctga tgagcagttg aaatctggaa ctgcctctgt tgtgtgcctg ctgaataact tctatcccag agaggccaaa gtacagtggg aggtggataa cgccctccaa tcgggtaact cccaggagag tgtcacagag caggacagca aggacagcac ctacagcctc agcagcaccg tgacgctgag caaagcagac tacgagaaac acaaagtcta cgctgcgaa gtcaccatc agggcctgag ctgcgccgtc acaaagagct tcaacagggg agagtgttag	60 120 180 240 300 360 420 480 540 600 660	
SEQ ID NO: 192	moltype = DNA length = 339	

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FEATURE	Location/Qualifiers	
misc_feature	1..339	
	note = Engineered antibody sequence	
source	1..339	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 192		
caagtgtga cccagtctcc atcctccctg tctgcatctg taggagacag agtcaccatc		60
aattgccagg ccagtcagag tgtttatcat aacacctacc tggcctggta tcagcagaaa		120
ccagggaag ttcctaagca actgatctat gatgcatcca ctctggcatc tgggggtcca		180
tctcgtttca gtggcagtggt atctgggaca gatttcactc tcaccatcag cagcctgcag		240
cctgaagatg ttgcaactta ttactgtctg ggcagttatg attgtactaa tggtgattgt		300
tttgttttcg gcggaggaac caaggtggaa atcaaacgt		339
SEQ ID NO: 193	moltype = DNA length = 66	
FEATURE	Location/Qualifiers	
misc_feature	1..66	
	note = Engineered antibody sequence	
source	1..66	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 193		
caagtgtga cccagtctcc atcctccctg tctgcatctg taggagacag agtcaccatc		60
aattgc		66
SEQ ID NO: 194	moltype = DNA length = 39	
FEATURE	Location/Qualifiers	
misc_feature	1..39	
	note = Engineered antibody sequence	
source	1..39	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 194		
caggccagtc agagtgttta tcataacacc tacctggcc		39
SEQ ID NO: 195	moltype = DNA length = 45	
FEATURE	Location/Qualifiers	
misc_feature	1..45	
	note = Engineered antibody sequence	
source	1..45	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 195		
tggatcagc agaaaccagg gaaagttcct aagcaactga tctat		45
SEQ ID NO: 196	moltype = DNA length = 21	
FEATURE	Location/Qualifiers	
misc_feature	1..21	
	note = Engineered antibody sequence	
source	1..21	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 196		
gatgcatcca ctctggcatc t		21
SEQ ID NO: 197	moltype = DNA length = 96	
FEATURE	Location/Qualifiers	
misc_feature	1..96	
	note = Engineered antibody sequence	
source	1..96	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 197		
gggggtccat ctcgttttcag tggcagtgga tctgggacag atttcaactct caccatcagc		60
agcctgcagc ctgaagatgt tgcaacttat tactgt		96
SEQ ID NO: 198	moltype = DNA length = 39	
FEATURE	Location/Qualifiers	
misc_feature	1..39	
	note = Engineered antibody sequence	
source	1..39	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 198		
ctgggcagtt atgattgtac taatgggtgat tgttttgtt		39
SEQ ID NO: 199	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	

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source          note = Engineered antibody sequence
                1..33
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 199
ttcggcggag gaaccaaggt ggaaatcaaa cgt                               33

SEQ ID NO: 200      moltype = DNA length = 321
FEATURE            Location/Qualifiers
misc_feature        1..321
                    note = Engineered antibody sequence
source             1..321
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 200
acggtggctg caccatctgt cttcatcttc cggccatctg atgagcagtt gaaatctgga   60
actgcctctg ttgtgtgctt gctgaataac ttctatccca gagaggccaa agtacagtgg   120
aaggtggata acgcccctcca atcgggtaac tcccaggaga gtgtcacaga gcaggacagc   180
aaggacagca cctacagcct cagcagcacc ctgacgctga gcaaagcaga ctacgagaaa   240
cacaaagtct acgcctgcga agtcacccat cagggcctga gctcgcccg cacaagagc    300
ttcaacaggg gagagtgtta g                                           321

SEQ ID NO: 201      moltype = AA length = 441
FEATURE            Location/Qualifiers
REGION             1..441
                    note = Engineered antibody sequence
source             1..441
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 201
EVQLVESGGG LVQPGGSLRL SCAVSGIDLS GYYMNWVRQA PGKGLEWVG V IGINGATYYA   60
SWAKGRFTIS RDNSKTTVYL QMNSLRAEDT AVYFCARGDI WGQGT LVTVS SASTKGPSVF   120
PLAPSSKSTS GGTAALGCLV KDYPPEPVTV SWNSGALTSG VHTFPAVLQS SGLYSLSSVV   180
TVPSSSLGTQ TYICNVNHKP SNTKVDARVE PKSCDKHTC PPCPAPELLG GPSVFLFPPK   240
PKDTLMISRT PEVTCVVVDV SHEDPEVKFN WYVDGVEVHN AKTKPREEQY ASTYRVVSVL   300
TVLHQDWLNG KEYKCKVSNK ALPAPIEKTI SKAKGQPREP QVYTLPPSRE EMTKNQVSLT   360
CLVKGFPYPSD IAVEWESNGQ PENNYKTPPP VLDSDGSFPL YSKLTVDKSR WQQGNVFS   420
VMHEALHNHY TQKSLSLSPG K                                           441

SEQ ID NO: 202      moltype = AA length = 111
FEATURE            Location/Qualifiers
REGION             1..111
                    note = Engineered antibody sequence
source             1..111
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 202
EVQLVESGGG LVQPGGSLRL SCAVSGIDLS GYYMNWVRQA PGKGLEWVG V IGINGATYYA   60
SWAKGRFTIS RDNSKTTVYL QMNSLRAEDT AVYFCARGDI WGQGT LVTVS S          111

SEQ ID NO: 203      moltype = AA length = 30
FEATURE            Location/Qualifiers
REGION             1..30
                    note = Engineered antibody sequence
source             1..30
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 203
EVQLVESGGG LVQPGGSLRL SCAVSGIDLS                               30

SEQ ID NO: 204      moltype = AA length = 5
FEATURE            Location/Qualifiers
REGION             1..5
                    note = Engineered antibody sequence
source             1..5
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 204
GYMN                                                                5

SEQ ID NO: 205      moltype = AA length = 14
FEATURE            Location/Qualifiers
REGION             1..14
                    note = Engineered antibody sequence
source             1..14
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 205
WVRQAPGKGL EWVG                                                  14

```

SEQ ID NO: 206	moltype = AA length = 16		
FEATURE	Location/Qualifiers		
REGION	1..16		
	note = Engineered antibody sequence		
source	1..16		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 206			
VIGINGATYY ASWAKG			16
SEQ ID NO: 207	moltype = AA length = 32		
FEATURE	Location/Qualifiers		
REGION	1..32		
	note = Engineered antibody sequence		
source	1..32		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 207			
RFTISRDN SK TTVY LQMNSL RAEDTAVYFC AR			32
SEQ ID NO: 208	moltype = length =		
SEQUENCE: 208			
000			
SEQ ID NO: 209	moltype = AA length = 11		
FEATURE	Location/Qualifiers		
REGION	1..11		
	note = Engineered antibody sequence		
source	1..11		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 209			
WGQGLTVTVS S			11
SEQ ID NO: 210	moltype = AA length = 330		
FEATURE	Location/Qualifiers		
REGION	1..330		
	note = Engineered antibody sequence		
source	1..330		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 210			
ASTKGPSVFP LAPSSKSTSG	GTAALGCLVK	DYFPEPVTVS	WNSGALTSGV HTFPAVLQSS 60
GLYSLSSVVT VPSSSLGTQT	YICNVNHKPS	NTKVDARVEP	KSCDKHTHTCP PCPAPELLGG 120
PSVFLFPPPK KDTLMISRTP	EVTCTVVVDVS	HEDPEVKFNW	YVDGVEVHNA KTKPREEQYA 180
STYRVVSVLT VLHQDWLNGK	EYKCKVSNKA	LPAPIEKTIS	KAKGQPREPQ VYTLPPSREE 240
MTKNQVSLTC LVKGFYPSDI	AVEWESNGQP	ENNYKTTPPV	LDSDGSFFLY SKLTVDKSRW 300
QQGNVFSCSV MHEALHNHYT	QKSLSLSPGK		330
SEQ ID NO: 211	moltype = DNA length = 1326		
FEATURE	Location/Qualifiers		
misc_feature	1..1326		
	note = Engineered antibody sequence		
source	1..1326		
	mol_type = other DNA		
	organism = synthetic construct		
SEQUENCE: 211			
gggtgtgcagc ttgtggagtc	tggggggaggc	ttggtccagc	ctggggggggtc cctgagactc 60
tctgtgtgcag tctctggaat	cgacctcagt	ggctactaca	tgaactgggt ccgtcaggct 120
ccagggaaagg ggtctggagt	ggtcggagtc	attggtatta	atggtgccac atactacgcg 180
agctggggcga aaggccgatt	caccatctcc	agagacaatt	ccaagaccac ggtgtatctt 240
caaatgaaca gcttgagagc	tgaggacact	gctgtgtatt	tctgtgtcag aggggacatc 300
tgggggcgaag ggaccctcgt	caccgtctcg	agcgctcca	ccaagggccc atcggtcttc 360
ccccctggcc cctcctccaa	gagcacctct	ggggggcaac	cgggcctggg ctgcctggtc 420
agcgactact tccccgaacc	gttgagcagt	tcgtggaact	caggcgccct caggcaggcg 480
gtgcacacct tccccgctgt	cctacagtc	tcaggactct	actccctcag cagcgtggtg 540
accgtgcctc ccagcagctt	ggggaccagc	acctacatct	gcaactgtga tcacaagccc 600
agcaacacca aggttgagcg	gagagtgtag	cccaaatctt	gtgacaaaac tcacacatgc 660
ccaccgtgcc cagcacctga	actcctgggg	ggaccgtcag	tcttctctct ccccccaaaa 720
cccaaggaca cctctcatgat	ctccccgacc	cctgagtgca	catgcgtggt ggtggacgtg 780
agccacgaag acctgaggt	caagttcaac	tgtgactgtg	acggcgtgga ggtgcataat 840
gccaaagacaa agccggggga	gcagcagtag	gccagcagct	accgtgtggt cagcgctctc 900
accgtctctgc accaggactg	gctgaatggc	aaggagtaca	agtgcagggt ctccaacaaa 960
gcctctcccg cccccatcga	gaaaaccatc	tcctaaagcca	aaggggcagcc ccgagaacca 1020
caggtgttaca cctcgcccc	atccccggag	gagatgacca	agaaccaggt cagcctgacc 1080
tgctgtgttca aaggcttcta	tcccagcgac	atcgccgtgg	agtgggagag caatggggcag 1140
ccggagaaaca actacaagac	cagccctccc	gtgctggagt	ccgacggctc cttctctctc 1200
tacaqcaaac tcaccgtqqa	caacgaacgt	tggcaqcaag	ggaacgtctt ctcatgctcc 1260

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```
gtgatgcatg aggctctgca caaccactac acgcagaaga gcctctccct gtctccgggt 1320
aatga 1326
```

```
SEQ ID NO: 212      moltype = DNA length = 333
FEATURE            Location/Qualifiers
misc_feature        1..333
                    note = Engineered antibody sequence
source              1..333
                    mol_type = other DNA
                    organism = synthetic construct
```

```
SEQUENCE: 212
gaggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctgggggggtc cctgagactc 60
tcctgtgcag tctctggaat cgacctcagt ggctactaca tgaactgggt ccgtcaggct 120
ccagggaagg ggctggagtg ggtcggagtc attggtatta atggtgccac atactacgcg 180
agctgggcga aaggccgatt caccatctcc agagacaatt ccaagaccac ggtgtatctt 240
caaatgaaca gcctgagagc ttagggacact gctgtgtatt tctgtgctag agggggacatc 300
tggggccaag ggacctcgt caccgtctcg agc 333
```

```
SEQ ID NO: 213      moltype = DNA length = 90
FEATURE            Location/Qualifiers
misc_feature        1..90
                    note = Engineered antibody sequence
source              1..90
                    mol_type = other DNA
                    organism = synthetic construct
```

```
SEQUENCE: 213
gaggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctgggggggtc cctgagactc 60
tcctgtgcag tctctggaat cgacctcagt 90
```

```
SEQ ID NO: 214      moltype = DNA length = 15
FEATURE            Location/Qualifiers
misc_feature        1..15
                    note = Engineered antibody sequence
source              1..15
                    mol_type = other DNA
                    organism = synthetic construct
```

```
SEQUENCE: 214
ggctactaca tgaac 15
```

```
SEQ ID NO: 215      moltype = DNA length = 42
FEATURE            Location/Qualifiers
misc_feature        1..42
                    note = Engineered antibody sequence
source              1..42
                    mol_type = other DNA
                    organism = synthetic construct
```

```
SEQUENCE: 215
tgggtccgtc aggctccagg gaaggggctg gagtgggtcg ga 42
```

```
SEQ ID NO: 216      moltype = DNA length = 48
FEATURE            Location/Qualifiers
misc_feature        1..48
                    note = Engineered antibody sequence
source              1..48
                    mol_type = other DNA
                    organism = synthetic construct
```

```
SEQUENCE: 216
gtcattggtg ttaatggtgc cacatactac gcgagctggg cgaaaggc 48
```

```
SEQ ID NO: 217      moltype = DNA length = 96
FEATURE            Location/Qualifiers
misc_feature        1..96
                    note = Engineered antibody sequence
source              1..96
                    mol_type = other DNA
                    organism = synthetic construct
```

```
SEQUENCE: 217
cgattcacca tctccagaga caattccaag accacgggtg atcttcaaat gaacagcctg 60
agagctgagg aactgtgtgt gtatttctgt gctaga 96
```

```
SEQ ID NO: 218      moltype = length =
SEQUENCE: 218
000
```

```
SEQ ID NO: 219      moltype = DNA length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = Engineered antibody sequence
source              1..33
```

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		mol_type = other DNA	
		organism = synthetic construct	
SEQUENCE: 219			
tgggggccaag	ggaccctcgt	caccgtctcgt	agc
			33
SEQ ID NO: 220	moltype = DNA length = 993		
FEATURE	Location/Qualifiers		
misc_feature	1..993		
	note = Engineered antibody sequence		
source	1..993		
	mol_type = other DNA		
	organism = synthetic construct		
SEQUENCE: 220			
gcctccacca	agggcccatc	ggtcttcccc	ctggcacacct
ggcacagcgg	cctctgggctg	ccttggtcaag	gactactctc
tggaactcag	gcgcacctgac	cagcggcgctg	cacacctctc
ggactctact	cctctcagcag	cgtgggtgacc	gtgccctcca
tacatctgca	acgtgaatca	caagcccagc	aacaccaagg
aaatcttggt	acaaaaatca	cacatgccca	ccgtgcccag
ccgtcagttc	tctctctccc	cccaaaaacc	aaggacacct
gaggtcacat	gcgtgggtgt	ggacgtgagc	cacgaagacc
tacgtggacg	gcgtggaggt	gcataatgcc	aagacaaagc
agcacgtacc	gtgtgggtcag	cgctcctcacc	gtcctgcacc
gagtcacaag	gcaaggctctc	gcacaaagcc	ctcccagccc
aaagccaaag	ggcagccccg	agaaccacag	gtgtacacct
atgaccaaga	accagggtcag	cctgacctgc	ctgggtcaag
gcggtggagt	gggagagcaa	tgggcagccg	gagaacaact
ctggactccg	acggctcctt	cttcctctac	agcaaagtca
cagcagggga	acgtcttctc	atgctccgtg	atgcatgagg
cagaagagcc	tctccctgtc	tccgggttaa	tga
			993
SEQ ID NO: 221	moltype = AA length = 219		
FEATURE	Location/Qualifiers		
REGION	1..219		
	note = Engineered antibody sequence		
source	1..219		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 221			
QVLTSQSSSL	SASVGDRVTI	NCQASQSVYH	NTYLAWYQQK
SRFSGSGSGT	DFTLTISSLQ	PEDVATYIYCL	GSYDCTNGDC
FIFPPSDEQL	KSGTASVVLQ	LNNFYPREAK	VQWKVDNALQ
SSTLTLSKAD	YEKHKVYACE	VTHQGLSSPV	TKSFNRGEC
			60
			120
			180
			219
SEQ ID NO: 222	moltype = AA length = 113		
FEATURE	Location/Qualifiers		
REGION	1..113		
	note = Engineered antibody sequence		
source	1..113		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 222			
QVLTSQSSSL	SASVGDRVTI	NCQASQSVYH	NTYLAWYQQK
SRFSGSGSGT	DFTLTISSLQ	PEDVATYIYCL	GSYDCTNGDC
			60
			113
SEQ ID NO: 223	moltype = AA length = 22		
FEATURE	Location/Qualifiers		
REGION	1..22		
	note = Engineered antibody sequence		
source	1..22		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 223			
QVLTSQSSSL	SASVGDRVTI	NC	
			22
SEQ ID NO: 224	moltype = AA length = 13		
FEATURE	Location/Qualifiers		
REGION	1..13		
	note = Engineered antibody sequence		
source	1..13		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 224			
QASQSVYHNT	YLA		
			13
SEQ ID NO: 225	moltype = AA length = 15		
FEATURE	Location/Qualifiers		
REGION	1..15		
	note = Engineered antibody sequence		

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source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 225		
WYQQKPGKVP KQLIY		15
SEQ ID NO: 226	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = Engineered antibody sequence	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 226		
DASTLAS		7
SEQ ID NO: 227	moltype = AA length = 32	
FEATURE	Location/Qualifiers	
REGION	1..32	
	note = Engineered antibody sequence	
source	1..32	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 227		
GVPSRFSGSG SGTDFTLTIS SLQPEDVATY YC		32
SEQ ID NO: 228	moltype = AA length = 13	
FEATURE	Location/Qualifiers	
REGION	1..13	
	note = Engineered antibody sequence	
source	1..13	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 228		
LGSYDCTNGD CFV		13
SEQ ID NO: 229	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Engineered antibody sequence	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 229		
FGGGTKVEIK R		11
SEQ ID NO: 230	moltype = AA length = 106	
FEATURE	Location/Qualifiers	
REGION	1..106	
	note = Engineered antibody sequence	
source	1..106	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 230		
TVAAPSVFIF PPSDEQLKSG TASVVCLLNN FYPREAKVQW KVDNALQSGN SQESVTEQDS		60
KDSTYLSLST LTLKADYK HKVYACEVTH QGLSSPVTKS FNRGEC		106
SEQ ID NO: 231	moltype = DNA length = 660	
FEATURE	Location/Qualifiers	
misc_feature	1..660	
	note = Engineered antibody sequence	
source	1..660	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 231		
caagtgcctga cccagctctcc atcctccctg tctgcatctg taggagacag agtcaccatc		60
aattgccagg ccagtcagag tggttatcat aacacctacc tggcctggta tcagcagaaa		120
ccagggaag ttcctaagca actgatctat gatgcatcca ctctggcatc tggggtccca		180
tctcgtttca gtggcagtg atctgggaca gatttcaact tcaccatcag cagcctgcag		240
cctgaagatg ttgcaactta ttactgtctg ggcagttatg attgtactaa tgggtgattgt		300
ttgtttttcg gcggaggaac caaggtggaa atcaaacgta cggtggctgc accatctgtc		360
ttcatcttcc cgccatctga tgagcagttg aaatctggaa ctgcctctgt tgtgtgctctg		420
ctgaataact tctatccag agaggccaaa gtacagtggg aggtggataa cgccctccaa		480
ctgggtaact ccaggagag tgtcacagag caggacagca aggacagcac ctacagcctc		540
agcagcacc tcacgctgag caaagcagac tacgagaaac acaaagtcta cgctgcgaa		600
gtcaccatc agggcctgag ctgcgccgtc acaaagagct tcaacagggg agagtgttag		660
SEQ ID NO: 232	moltype = DNA length = 339	
FEATURE	Location/Qualifiers	

misc_feature	1..339	
	note = Engineered antibody sequence	
source	1..339	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 232		
caagtgtga cccagtcctc	atctctccctg tctgcatctg taggagacag agtcaccatc	60
aattgccagg ccagtcagag	tgtttatcat aacacctacc tggcctggta tcagcagaaa	120
ccagggaaag ttctaagca	actgatctat gatgcatcca ctctggcatc tgggggtccca	180
tctcgtttca gtggcagtg	atctgggaca gatttcactc tcaccatcag cagcctgcag	240
cctgaagatg ttgcaactta	ttactgtctg ggcagttatg attgtactaa tgggtattgt	300
tttgttttcg gcggaggaac	caaggtggaa atcaaacgt	339
SEQ ID NO: 233	moltype = DNA length = 66	
FEATURE	Location/Qualifiers	
misc_feature	1..66	
	note = Engineered antibody sequence	
source	1..66	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 233		
caagtgtga cccagtcctc	atctctccctg tctgcatctg taggagacag agtcaccatc	60
aattgc		66
SEQ ID NO: 234	moltype = DNA length = 39	
FEATURE	Location/Qualifiers	
misc_feature	1..39	
	note = Engineered antibody sequence	
source	1..39	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 234		
caggccagtc agagtgttta	tcataacacc tacctggcc	39
SEQ ID NO: 235	moltype = DNA length = 45	
FEATURE	Location/Qualifiers	
misc_feature	1..45	
	note = Engineered antibody sequence	
source	1..45	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 235		
tggtatcagc agaaaccagg	gaaagttcct aagcaactga tctat	45
SEQ ID NO: 236	moltype = DNA length = 21	
FEATURE	Location/Qualifiers	
misc_feature	1..21	
	note = Engineered antibody sequence	
source	1..21	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 236		
gatgcatcca ctctggcatc	t	21
SEQ ID NO: 237	moltype = DNA length = 96	
FEATURE	Location/Qualifiers	
misc_feature	1..96	
	note = Engineered antibody sequence	
source	1..96	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 237		
ggggtcctat ctcgtttcag	tggcagtgga tctgggacag atttcactct caccatcagc	60
agctgcagc ctgaagatgt	tgcaacttat tactgt	96
SEQ ID NO: 238	moltype = DNA length = 39	
FEATURE	Location/Qualifiers	
misc_feature	1..39	
	note = Engineered antibody sequence	
source	1..39	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 238		
ctgggcagtt atgattgtac	taatggtgat tgttttgtt	39
SEQ ID NO: 239	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Engineered antibody sequence	

source	1..33 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 239		
ttcggcgggag gaaccaaggt ggaatatcaaa cgt		33
SEQ ID NO: 240	moltype = DNA length = 321	
FEATURE	Location/Qualifiers	
misc_feature	1..321 note = Engineered antibody sequence	
source	1..321 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 240		
acgggtggctg caccatctgt cttcatcttc cgcacctctg atgagcagtt gaaatctgga		60
actgcctctg ttgtgtgctt gctgaataac ttctatccca gagaggccaa agtacagtgg		120
aagggtgata acgcccctcca atcgggtaac tcccaggaga gtgtcacaga gcaggacagc		180
aaggacagca cctacagcct cagcagcacc ctgacgctga gcaaaagcaga ctacgagaaa		240
cacaaagtct acgctctgca agtcacccat cagggcctga gctcgcccgt cacaaagagc		300
ttcaacaggg gagagtgtta g		321
SEQ ID NO: 241	moltype = AA length = 440	
FEATURE	Location/Qualifiers	
REGION	1..440 note = Engineered antibody sequence	
source	1..440 mol_type = protein organism = synthetic construct	
SEQUENCE: 241		
QEQLKESGGR LVTPGTSLT TCTVSGIDL NHYMQWRQA PGKGLEWIGV VGINGRTYYA		60
SWAKGRFTIS RTSSTTVDLK MTRLTTEDTA TYFCARGDIW GPGTLVTVSS ASTKGPSVFP		120
LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS GLYSLSSVVT		180
VPSSSLGTQT YICNVNHKPS NTKVDKRVEP KSCDKTHTCP PCPAPELLGG PSVFLFPPPK		240
KDTLMISRTP EVTCVVVDVS HEDPEVKFNN YVDGVEVHNA KTKPREEQYA STYRVVSVLT		300
VLHQDWLNGK EYKCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSREE MTKNQVSLTC		360
LVHGFYPDSI AVEWESNGQP ENNYKTTTPV LDSGDSFPLY SKLTVDKSRW QQGNVFSCSV		420
MHEALHNHYT QKSLSLSPGK		440
SEQ ID NO: 242	moltype = AA length = 110	
FEATURE	Location/Qualifiers	
REGION	1..110 note = Engineered antibody sequence	
source	1..110 mol_type = protein organism = synthetic construct	
SEQUENCE: 242		
QEQLKESGGR LVTPGTSLT TCTVSGIDL NHYMQWRQA PGKGLEWIGV VGINGRTYYA		60
SWAKGRFTIS RTSSTTVDLK MTRLTTEDTA TYFCARGDIW GPGTLVTVSS		110
SEQ ID NO: 243	moltype = AA length = 30	
FEATURE	Location/Qualifiers	
REGION	1..30 note = Engineered antibody sequence	
source	1..30 mol_type = protein organism = synthetic construct	
SEQUENCE: 243		
QEQLKESGGR LVTPGTSLT TCTVSGIDL		30
SEQ ID NO: 244	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
REGION	1..5 note = Engineered antibody sequence	
source	1..5 mol_type = protein organism = synthetic construct	
SEQUENCE: 244		
NHYMQ		5
SEQ ID NO: 245	moltype = AA length = 14	
FEATURE	Location/Qualifiers	
REGION	1..14 note = Engineered antibody sequence	
source	1..14 mol_type = protein organism = synthetic construct	
SEQUENCE: 245		
WVROAPGKGL EWIG		14

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tga 1323

SEQ ID NO: 252      moltype = DNA  length = 330
FEATURE            Location/Qualifiers
misc_feature        1..330
                    note = Engineered antibody sequence
source              1..330
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 252
caggagcagc tgaaggagtc cgggggtcgc ctggtcacgc ctgggacatc cctgacactc 60
acctgcaccg tctctggaat cgacctcagt aaccactaca tgcaatgggt ccgccaggct 120
ccagggaagg ggctggagtg gatcggagtc gttggtatta atggtcgcac atactacgcg 180
agctgggcga aaggccgatt caccatctcc agaacctcgt cgaccacggt ggatctgaaa 240
atgaccaggc tgacaaccga ggacacggcc acctatttct gtgccagagg ggacatctgg 300
ggcccaggca ccttggtcac cgtctcgagc 330

SEQ ID NO: 253      moltype = DNA  length = 90
FEATURE            Location/Qualifiers
misc_feature        1..90
                    note = Engineered antibody sequence
source              1..90
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 253
caggagcagc tgaaggagtc cgggggtcgc ctggtcacgc ctgggacatc cctgacactc 60
acctgcaccg tctctggaat cgacctcagt 90

SEQ ID NO: 254      moltype = DNA  length = 15
FEATURE            Location/Qualifiers
misc_feature        1..15
                    note = Engineered antibody sequence
source              1..15
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 254
aaccactaca tgcaa 15

SEQ ID NO: 255      moltype = DNA  length = 42
FEATURE            Location/Qualifiers
misc_feature        1..42
                    note = Engineered antibody sequence
source              1..42
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 255
tgggtccgcc aggtccagg gaaggggctg gagtggatcg ga 42

SEQ ID NO: 256      moltype = DNA  length = 48
FEATURE            Location/Qualifiers
misc_feature        1..48
                    note = Engineered antibody sequence
source              1..48
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 256
gtcgttggtt ttaatggtcg cacatactac gcgagctggg cgaaaggc 48

SEQ ID NO: 257      moltype = DNA  length = 93
FEATURE            Location/Qualifiers
misc_feature        1..93
                    note = Engineered antibody sequence
source              1..93
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 257
cgattcacca tctccagaac ctcgctgacc acggtggatc tgaaaatgac caggctgaca 60
accgaggaca cggccaccta ttctgtgcc aga 93

SEQ ID NO: 258      moltype = length =
SEQUENCE: 258
000

SEQ ID NO: 259      moltype = DNA  length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = Engineered antibody sequence
source              1..33
                    mol_type = other DNA

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                organism = synthetic construct
SEQUENCE: 259
tggggcccag gcaccctggt caccgtctcg agc                               33

SEQ ID NO: 260                moltype = DNA length = 993
FEATURE                      Location/Qualifiers
misc_feature                  1..993
                              note = Engineered antibody sequence
source                       1..993
                              mol_type = other DNA
                              organism = synthetic construct

SEQUENCE: 260
gcctccacca agggcccacac ggtcttcccc ctggcaccct cctccaagag caccctctggg 60
ggcacagcgg ccttgggctg cctgggtcaag gactacttcc ccgaaccggt gacgggtgtcg 120
tggaaactcag gcgccctgac cagcggcgctg cacaccttcc cggctgtcct acagtccctca 180
ggactctact ccttcagcag cgtgggtgacc gtgccctcca gcagcttggg caccagacc 240
tacatctgca acgtgaatca caagcccagc aacaccaagg tggacaagag agttgagccc 300
aaatcttgtg acaaaactca cacatgcccc ccggtgccca cactgaact cctgggggga 360
ccgtcagttc tctcttcccc cccaaaaccc aaggacaccc tcatgatctc ccggaccct 420
gaggtcacat gcgtgggtgt ggacgtgagc cacgaagacc ctgaggtcaa gttcaactgg 480
tacgtggacg gcgtggaggt gcataatgcc aagacaaagc cgcgaggagga gcagtacgcc 540
agcacgtacc gtgtgggtcag cgtcctcacc gtctctgacc aggactgggt gaatggcaag 600
gagtacaagt gcaagggtct caacaaagcc ctcccagccc ccacgagaaa aaccatctcc 660
aaagccaaag ggcagccccc agaaccacag gtgtacaccc tgcccccatc ccgggaggag 720
atgaccaaga accagggtcag cctgacctgc ctggtcaaa gcttctatcc cagcgacatc 780
gccgtggagt gggagagcaa tgggcagccg gagaacaact acaagaccac gcctcccggt 840
ctggactccg acggtctcct ctctctctac agcaagctca ccgtggacaa gacgaggtgg 900
cagcagggga acgtcttctc atgctccgtg atgcatgagg ctctgcacaa ccactacacg 960
cagaagagcc tctccctgtc tccgggtaaa tga                               993

SEQ ID NO: 261                moltype = AA length = 219
FEATURE                      Location/Qualifiers
REGION                      1..219
                              note = Engineered antibody sequence
source                       1..219
                              mol_type = protein
                              organism = synthetic construct

SEQUENCE: 261
QVLTQTASPV SAAVGSTVTI NCQASQSVYN YNYLAWYQQK PGQPPKQLIY STSTLASGVS 60
SRFKGSGSGT QFTLTISDVQ CDDAATYYCL GSYDCSTGDC FVFGGTEVV VKRTVAAPSV 120
FIFPPSDEQL KSGTASVVCL LNNFYPREAK VQWKVDNALQ SGNSQESVTE QDSKDYSTSL 180
SSTLTLSKAD YEKHKVYACE VTHQGLSSPV TKSFNRGEC                               219

SEQ ID NO: 262                moltype = AA length = 113
FEATURE                      Location/Qualifiers
REGION                      1..113
                              note = Engineered antibody sequence
source                       1..113
                              mol_type = protein
                              organism = synthetic construct

SEQUENCE: 262
QVLTQTASPV SAAVGSTVTI NCQASQSVYN YNYLAWYQQK PGQPPKQLIY STSTLASGVS 60
SRFKGSGSGT QFTLTISDVQ CDDAATYYCL GSYDCSTGDC FVFGGTEVV VKR                               113

SEQ ID NO: 263                moltype = AA length = 22
FEATURE                      Location/Qualifiers
REGION                      1..22
                              note = Engineered antibody sequence
source                       1..22
                              mol_type = protein
                              organism = synthetic construct

SEQUENCE: 263
QVLTQTASPV SAAVGSTVTI NC                               22

SEQ ID NO: 264                moltype = AA length = 13
FEATURE                      Location/Qualifiers
REGION                      1..13
                              note = Engineered antibody sequence
source                       1..13
                              mol_type = protein
                              organism = synthetic construct

SEQUENCE: 264
QASQSVYNYN YLA                               13

SEQ ID NO: 265                moltype = AA length = 15
FEATURE                      Location/Qualifiers
REGION                      1..15
                              note = Engineered antibody sequence
source                       1..15

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SEQUENCE: 265	mol_type = protein	
WYQKPGQPP KQLIY	organism = synthetic construct	15
SEQ ID NO: 266	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
source	note = Engineered antibody sequence	
	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 266		7
STSTLAS		
SEQ ID NO: 267	moltype = AA length = 32	
FEATURE	Location/Qualifiers	
REGION	1..32	
source	note = Engineered antibody sequence	
	1..32	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 267		32
GVSSRFKSGS SGTQFTLTIS DVQCDDAATY YC		
SEQ ID NO: 268	moltype = AA length = 13	
FEATURE	Location/Qualifiers	
REGION	1..13	
source	note = Engineered antibody sequence	
	1..13	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 268		13
LGSYDCSTGD CFV		
SEQ ID NO: 269	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
source	note = Engineered antibody sequence	
	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 269		11
FGGGTEVVVK R		
SEQ ID NO: 270	moltype = AA length = 106	
FEATURE	Location/Qualifiers	
REGION	1..106	
source	note = Engineered antibody sequence	
	1..106	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 270		60
TVAAPSVFIF PPSDEQLKSG TASVVCCLNN FYPREAKVQW KVDNALQSGN SQESVTEQDS		
KDSTYLSST LTLKADYK HKVYACEVTH QGLSPVTKS FNRGEC		106
SEQ ID NO: 271	moltype = DNA length = 660	
FEATURE	Location/Qualifiers	
misc_feature	1..660	
source	note = Engineered antibody sequence	
	1..660	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 271		
caagtgtgta cccagactgc atcccccggtg tctgcagctg tgggaagcac agtcaccatc	60	
aattgccagg ccagtcagag tgtttataat tacaactacc ttgcctggta tcagcagaaa	120	
ccagggcagc ctcccaagca actgatctat tctacatcca ctctggcatc tgggggtctca	180	
tcgcgattca aaggcagtggt atctgggaca cagttcactc tcaccatcag cgacgtgcag	240	
tgtgacgatg ctgccactta ctactgtcta ggcagttatg actgtagtac tggtgattgt	300	
tttgttttcg gcgaggggac cgaggtggtg gtcaaacgta cggtgggtgc accatctgtc	360	
ttcatcttcc cgccatctga tgagcagttg aaatctggaa ctgcctctgt tgtgtgctg	420	
ctgaataact tctatcccag agaggccaaa gtacagtggg aggtggataa cgccctccaa	480	
tcgggtaact cccaggagag tgtcacagag caggacagca aggacagcac ctacagcctc	540	
agcagcacc tgacgctgag caaagcagac tacgagaaac acaaagtcta cgcctgcgaa	600	
gtcaccatcc agggcctgag ctgcgccgtc acaaagagct tcaacagggg agagtgttag	660	
SEQ ID NO: 272	moltype = DNA length = 339	
FEATURE	Location/Qualifiers	
misc_feature	1..339	

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source          note = Engineered antibody sequence
                1..339
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 272
caagtgtctga cccagactgc atcccccgctg tctgcagctg tgggaagcac agtcaccatc 60
aattgccagg ccagtcagag tgtttataat tacaactacc ttgcctggta tcagcagaaa 120
ccagggcagc ctcccaagca actgatctat tctacatcca ctctggcatc tggggtotca 180
tcgcgattca aaggcagtg atctgggaca cagttcactc tcaccatcag cgacgtgcag 240
tgtgacgatg ctgccactta ctactgtcta ggcagttatg actgtagtac tggtgattgt 300
tttgttttcg gcggagggac cgagggtggg gtcaaacgt 339

SEQ ID NO: 273      moltype = DNA length = 66
FEATURE            Location/Qualifiers
misc_feature        1..66
                    note = Engineered antibody sequence
source              1..66
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 273
caagtgtctga cccagactgc atcccccgctg tctgcagctg tgggaagcac agtcaccatc 60
aattgc 66

SEQ ID NO: 274      moltype = DNA length = 39
FEATURE            Location/Qualifiers
misc_feature        1..39
                    note = Engineered antibody sequence
source              1..39
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 274
caggccagtc agagtgttta taattacaac taccttgcc 39

SEQ ID NO: 275      moltype = DNA length = 45
FEATURE            Location/Qualifiers
misc_feature        1..45
                    note = Engineered antibody sequence
source              1..45
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 275
tggatcagc agaaaccagg gcagcctccc aagcaactga tctat 45

SEQ ID NO: 276      moltype = DNA length = 21
FEATURE            Location/Qualifiers
misc_feature        1..21
                    note = Engineered antibody sequence
source              1..21
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 276
tctacatcca ctctggcatc t 21

SEQ ID NO: 277      moltype = DNA length = 96
FEATURE            Location/Qualifiers
misc_feature        1..96
                    note = Engineered antibody sequence
source              1..96
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 277
ggggtctcat cgcgattcaa aggcagtga tctgggacac agttcactct caccatcagc 60
gacgtgcagt gtgacgatgc tgccacttac tactgt 96

SEQ ID NO: 278      moltype = DNA length = 39
FEATURE            Location/Qualifiers
misc_feature        1..39
                    note = Engineered antibody sequence
source              1..39
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 278
ctaggcagtt atgactgtag tactgggtgat tgttttgtt 39

SEQ ID NO: 279      moltype = DNA length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = Engineered antibody sequence
source              1..33

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mol_type = other DNA
organism = synthetic construct

SEQUENCE: 279
ttcggcggag ggaccgaggt ggtggtcaaa cgt 33

SEQ ID NO: 280      moltype = DNA length = 321
FEATURE            Location/Qualifiers
misc_feature        1..321
                    note = Engineered antibody sequence
source              1..321
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 280
acgggtggctg caccatctgt cttcatcttc cggccatctg atgagcagtt gaaatctgga 60
actgcctctg ttgtgtgcct gctgaataac ttctatccca gagaggccaa agtacagtgg 120
aagggtggata acgcccctcca atcgggtaac tcccaggaga gtgtcacaga gcaggacagc 180
aaggacagca cctacagcct cagcagcacc ctgacgctga gcaaagcaga ctacgagaaa 240
cacaagctct acgcctgcga agtcacccat cagggcctga gctcgcccg cacaagagc 300
ttcaacaggg gagagtgtta g 321

SEQ ID NO: 281      moltype = AA length = 441
FEATURE            Location/Qualifiers
REGION              1..441
                    note = Engineered antibody sequence
source              1..441
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 281
EVQLVESGGG LVQPGGSLRL SCAVSGIDLS NHYMQWVRQA PGKGLEWVG VINGRITYYA 60
SWAKGRFTIS RDNSKTTVYL QMNSLRAEDT AVYFCARGDI WGQGTLVTVS SASTKGPSVF 120
PLAPSSKSTS GGTAALGCLV KDYPPEPVTV SWNSGALTSG VHTFPAVLQS SGLYSLSSV 180
TVPSSSLGTQ TYICNVNHPK SNTKVDKRVK PKSCDKHTHC PPCPAPELLG GPSVFLFPPK 240
PKDTLMISRT PEVTCVVVDV SHEDPEVKFN WYVDGVEVHN AKTKPREEQY ASTYRVVSVL 300
TVLHQDWLNG KEYKCKVSNK ALPAPIEKTI SKAKGQPREP QVYTLPPSRE EMTKNQVSLT 360
CLVKGFPYPSD IAVEWESNGQ PENNYKTPP VLDSDGSFPL YSKLTVDKSR WQQGNVFS 420
VMHEALHNYH TQKSLSLSPG K 441

SEQ ID NO: 282      moltype = AA length = 111
FEATURE            Location/Qualifiers
REGION              1..111
                    note = Engineered antibody sequence
source              1..111
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 282
EVQLVESGGG LVQPGGSLRL SCAVSGIDLS NHYMQWVRQA PGKGLEWVG VINGRITYYA 60
SWAKGRFTIS RDNSKTTVYL QMNSLRAEDT AVYFCARGDI WGQGTLVTVS S 111

SEQ ID NO: 283      moltype = AA length = 30
FEATURE            Location/Qualifiers
REGION              1..30
                    note = Engineered antibody sequence
source              1..30
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 283
EVQLVESGGG LVQPGGSLRL SCAVSGIDLS 30

SEQ ID NO: 284      moltype = AA length = 5
FEATURE            Location/Qualifiers
REGION              1..5
                    note = Engineered antibody sequence
source              1..5
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 284
NHYMQ 5

SEQ ID NO: 285      moltype = AA length = 14
FEATURE            Location/Qualifiers
REGION              1..14
                    note = Engineered antibody sequence
source              1..14
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 285
WVRQAPGKGL EWVG 14

SEQ ID NO: 286      moltype = AA length = 16

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FEATURE	Location/Qualifiers	
REGION	1..16	
	note = Engineered antibody sequence	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 286		
VVGINGRTYY ASWAKG		16
SEQ ID NO: 287	moltype = AA length = 32	
FEATURE	Location/Qualifiers	
REGION	1..32	
	note = Engineered antibody sequence	
source	1..32	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 287		
RFTISRDNK TTVYLQMNLS RAEDTAVYFC AR		32
SEQ ID NO: 288	moltype = length =	
SEQUENCE: 288		
000		
SEQ ID NO: 289	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Engineered antibody sequence	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 289		
WGQGLVTVS S		11
SEQ ID NO: 290	moltype = AA length = 330	
FEATURE	Location/Qualifiers	
REGION	1..330	
	note = Engineered antibody sequence	
source	1..330	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 290		
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS		60
GLYSLSVVVT VPSSSLGTQT YICNVNHKPS NTKVDKRVPE KSCDKHTTCP PCPAPELLGG		120
PSVFLFPPPK KDTLMISRTPEVTCVVVDVS HEDPEVKFNW YVDGVEVHNA KTKPREEQYA		180
STYRVVSVLT VLHQDWLNGK EYKCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSREE		240
MTKNQVSLTC LVKGFYPSDI AVEWESNGQP ENNYKTTTPPV LDSDGSFFLY SKLTVDKSRW		300
QQGNVFSCSV MHEALHNHYT QKSLSLSPGK		330
SEQ ID NO: 291	moltype = DNA length = 1326	
FEATURE	Location/Qualifiers	
misc_feature	1..1326	
	note = Engineered antibody sequence	
source	1..1326	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 291		
gaggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctgggggggc cctgagactc		60
tcctgtgcag tctctggaat cgacctcagt aaccactaca tgcaatgggt ccgtcaggct		120
ccagggaagg ggtcggagtg ggtcggagtc gttggtatca atggtcgcac atactacgcg		180
agctgggcga aaggccgatt caccatctcc agagacaatt ccaagaccac ggtgatctt		240
caaatgaaca gctgagagc tgaggacact gctgtgtatt tctgtgctag aggggacatc		300
tggggccaag ggaccctcgt caccgtctcg agcgctcca ccaagggccc atcggtcttc		360
cccctggcac cctcctccaa gagcacctct gggggcacag cggccctggg ctgcctggtc		420
aaggactact tcccgaacc ggtgacgggt tctgtggaact caggcgccct gaccagcggc		480
gtgcacacct tcccggctgt cctacagtcc tcaggactct actccctcag cagcgtggtg		540
accgtgccc cagcagctt gggcacccag acctacatct gcaacgtgaa tcacaagccc		600
agcaacacca aggtggacaa gagagttgag cccaaatctt gtgacaaaac tcacacatgc		660
ccacgtgcc cagcacctga actcctgggg ggaccgtcag tcttctctct cccccaaaa		720
cccaaggaca cctcatgat ctcccgacc cctgaggtca catgcgtggt ggtggacgtg		780
agccacgaag acctgaggt caagttcaac tggtagctgg acggcgtgga ggtgcataat		840
gccaagacaa agccgcggga ggagcagtag gccagcacgt accgtgtggt cagcgtcctc		900
accgtcctgc accaggactg gctgaatggc aaggagtaca agtgcaaggt ctccaacaaa		960
gcccctccag ccccatcga gaaaaccatc tccaaagcca aagggcagcc ccgagaacca		1020
caggtgtaca cctgtcccc atccccggag gagatgacca agaaccaggt cagcctgacc		1080
tgccctggta aagcttctc tcccagcgac atcgccgtgg agtgggagag caatgggcag		1140
ccggagaaca actacaagac cagcctccc gtgctggact ccgacggctc cttcttctc		1200
tacagcaagc tcaccgtgga caagagcagg tggcagcagg ggaacgtctt ctcagtctc		1260
gtgatgcatg aggtctgcga caaccactac acgcagaaga gcctctccct gtctccgggt		1320
aatga		1326

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SEQ ID NO: 292      moltype = DNA length = 333
FEATURE            Location/Qualifiers
misc_feature        1..333
                    note = Engineered antibody sequence
source              1..333
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 292
gaggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctgggggggc cctgagactc 60
tcctgtgcag tctctggaat cgacctcagt aaccactaca tgcaatgggt ccgtcaggct 120
ccagggaagg ggctggagtg ggtcggagtc gttggtatca atggtcgcac atactacgcg 180
agctgggcga aaggccgatt caccatctcc agagacaatt ccaagaccac ggtgtatctt 240
caaatgaaca gcctgagagc tgaggacact gctgtgtatt tctgtgctag aggggacatc 300
tggggccaag ggaccctcgt caccgtctcg agc 333

SEQ ID NO: 293      moltype = DNA length = 90
FEATURE            Location/Qualifiers
misc_feature        1..90
                    note = Engineered antibody sequence
source              1..90
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 293
gaggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctgggggggc cctgagactc 60
tcctgtgcag tctctggaat cgacctcagt 90

SEQ ID NO: 294      moltype = DNA length = 15
FEATURE            Location/Qualifiers
misc_feature        1..15
                    note = Engineered antibody sequence
source              1..15
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 294
aaccactaca tgcaa 15

SEQ ID NO: 295      moltype = DNA length = 42
FEATURE            Location/Qualifiers
misc_feature        1..42
                    note = Engineered antibody sequence
source              1..42
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 295
tgggtccgtc aggctccagg gaaggggctg gagtgggtcg ga 42

SEQ ID NO: 296      moltype = DNA length = 48
FEATURE            Location/Qualifiers
misc_feature        1..48
                    note = Engineered antibody sequence
source              1..48
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 296
gtcgttggtg tcaatggtcg cacatactac gcgagctggg cgaaaggc 48

SEQ ID NO: 297      moltype = DNA length = 96
FEATURE            Location/Qualifiers
misc_feature        1..96
                    note = Engineered antibody sequence
source              1..96
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 297
cgattcacca tctccagaga caattccaag accacgggtg atcttcaaat gaacagcctg 60
agagctgagg aactgctgtg gtatttctgt gctaga 96

SEQ ID NO: 298      moltype = length =
SEQUENCE: 298
000

SEQ ID NO: 299      moltype = DNA length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = Engineered antibody sequence
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 299
 tggggccaag ggaccctcgt caccgtctcg agc 33

SEQ ID NO: 300 moltype = DNA length = 993
 FEATURE Location/Qualifiers
 misc_feature 1..993
 note = Engineered antibody sequence
 source 1..993
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 300
 gcctccacca agggcccac cgtcttcccc ctggcaccct cctccaagag cacctctggg 60
 ggacacagcg ccttgggctg cctgggtcaag gactacttcc ccgaaccggg gacgggtgtcg 120
 tgggaactcag gcgccttgac cagcggcgctg cacaccttcc cggctgtcct acagtctctca 180
 ggactctact ccctcagcag cgtgggtgacc gtgccctcca gcagcttggg caccagacc 240
 tacatctgca acgtgaatca caagccagc aacaccaagg tggacaagag agttgagccc 300
 aaatcttgtg acaaaaactca cacatgccc cegtgccag cacctgaact cctgggggga 360
 ccgtcagctc tcctcttccc cccaaaaccc aaggacaccc tcatgatctc ccggaccct 420
 gaggtcacat gcgtgggtgt ggacgtgagc cagcaagacc ctgaggtcaa gttcaactgg 480
 tacgtggacg gcgtggaggt gcataatgcc aagacaaagc cgcgggagga gcagtacgcc 540
 agcacgtacc gtgtgggtcag cgtcctcacc gtctcgcacc aggactggct gaatggcaag 600
 gagtacaagt gcaagggtctc caacaaagcc ctcccagccc ccatcgagaa aacctctcc 660
 aaagccaaag ggcagccccg agaaccacag gtgtacaccc tgccccatc ccgggaggag 720
 atgaccaaga accaggtcag cctgacctgc ctgggtcaaag gcttctatcc cagcgacatc 780
 gccgtggagt gggagagcaa tgggcagccg gagaacaact acaagaccac gcctccgtg 840
 ctggactcgc acggtctctt ctctctctac agcaagctca ccgtggacaa gagcaggtgg 900
 cagcagggga acgtctcttc atgctccgtg atgcatgagg ctctgcacaa ccactacacg 960
 cagaagagcc tctccctgtc tccgggtaaa tga 993

SEQ ID NO: 301 moltype = AA length = 219
 FEATURE Location/Qualifiers
 REGION 1..219
 note = Engineered antibody sequence
 source 1..219
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 301
 QVLTQSPSSL SASVGDRVIT NCQASQSVYN YNYLAWYQQK PGKVPKQLIY STSTLASGVP 60
 SRFSGSGSGT DFTLTISLQ PEDVATYYCL GSYDCSTGDC FVFGGGTKVE IKRTVAAPSV 120
 FIFPPSDEQL KSGTASVCL LNNFYPREAK VQWKVDNALQ SGNSQESVTE QDSKSTYSL 180
 SSTLTLSKAD YEKHKVYACE VTHQGLSSPV TKSFNRGEC 219

SEQ ID NO: 302 moltype = AA length = 113
 FEATURE Location/Qualifiers
 REGION 1..113
 note = Engineered antibody sequence
 source 1..113
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 302
 QVLTQSPSSL SASVGDRVIT NCQASQSVYN YNYLAWYQQK PGKVPKQLIY STSTLASGVP 60
 SRFSGSGSGT DFTLTISLQ PEDVATYYCL GSYDCSTGDC FVFGGGTKVE IKR 113

SEQ ID NO: 303 moltype = AA length = 22
 FEATURE Location/Qualifiers
 REGION 1..22
 note = Engineered antibody sequence
 source 1..22
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 303
 QVLTQSPSSL SASVGDRVIT NC 22

SEQ ID NO: 304 moltype = AA length = 13
 FEATURE Location/Qualifiers
 REGION 1..13
 note = Engineered antibody sequence
 source 1..13
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 304
 QASQSVYNYN YLA 13

SEQ ID NO: 305 moltype = AA length = 15
 FEATURE Location/Qualifiers
 REGION 1..15
 note = Engineered antibody sequence
 source 1..15
 mol_type = protein

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organism = synthetic construct

SEQUENCE: 305
WYQQKPGKVP KQLIY 15

SEQ ID NO: 306
FEATURE
REGION
source
moltype = AA length = 7
Location/Qualifiers
1..7
note = Engineered antibody sequence
mol_type = protein
organism = synthetic construct

SEQUENCE: 306
STSTLAS 7

SEQ ID NO: 307
FEATURE
REGION
source
moltype = AA length = 32
Location/Qualifiers
1..32
note = Engineered antibody sequence
mol_type = protein
organism = synthetic construct

SEQUENCE: 307
GVPSRFSGSG SGTDFTLTIS SLQPEDVATY YC 32

SEQ ID NO: 308
FEATURE
REGION
source
moltype = AA length = 13
Location/Qualifiers
1..13
note = Engineered antibody sequence
mol_type = protein
organism = synthetic construct

SEQUENCE: 308
LGSYDCSTGD CFV 13

SEQ ID NO: 309
FEATURE
REGION
source
moltype = AA length = 11
Location/Qualifiers
1..11
note = Engineered antibody sequence
mol_type = protein
organism = synthetic construct

SEQUENCE: 309
FGGGTKVEIK R 11

SEQ ID NO: 310
FEATURE
REGION
source
moltype = AA length = 106
Location/Qualifiers
1..106
note = Engineered antibody sequence
mol_type = protein
organism = synthetic construct

SEQUENCE: 310
TVAAPSVFIF PPSDEQLKSG TASVVCLLNN FYPREKVQW KVDNALQSGN SQESVTEQDS 60
KDSTYLSST LTLSKADY EK HKVYACEVTH QGLSSPVTKS FNRGEC 106

SEQ ID NO: 311
FEATURE
misc_feature
source
moltype = DNA length = 660
Location/Qualifiers
1..660
note = Engineered antibody sequence
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 311
caagtgtga cccagctctc atcctccctg tctgcatctg taggagacag agtcaccatc 60
aattgccagg ccagtcagag tgtttacaat tacaactacc ttgcctggta tcagcagaaa 120
ccagggaag ttcctaagca actgatctat tctacatcca ctctggcatc tggggtecca 180
tctcgtttca gtggcagtg atctgggaca gatttcactc tcaccatcag cagcctgcag 240
cctgaagatg ttgcaactta ttaactgtctg ggcagttatg attgtagtac tggtgattgt 300
tttgttttcg gccgagggaac caaggtggaa atcaaacgta cggtggctgc accatctgtc 360
ttcatcttcc cgccatctga tgagcagttg aaatctggaa ctgcctctgt tgtgtgctg 420
ctgaataact tctatcccag agaggccaaa gtacagtgga aggtggataa cgccctccaa 480
tcgggtaact cccaggagag tgtcacagag caggacagca aggacagcac ctacagcctc 540
agcagcacc tgacgctgag caaagcagac tacgagaaac acaaagtcta cgctgcgaa 600
gtcaccatc agggcctgag ctgcgccgtc acaaagagct tcaacagggg agagtgttag 660

SEQ ID NO: 312
FEATURE
misc_feature
moltype = DNA length = 339
Location/Qualifiers
1..339
note = Engineered antibody sequence

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source                1..339
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 312
caagtgctga cccagtctcc atcctccttg tctgcatctg taggagacag agtcaccatc 60
aattgccagg ccagtcagag tgtttacaat tacaactacc ttgcctggta tcagcagaaa 120
ccagggaag  ttctaagca actgatctat tctacatcca ctctggcatc tggggtccca 180
tctcgtttca gtggcagtg  atctgggaca gatttcactc tcaccatcag cagcctgcag 240
cctgaagatg ttgcaactta ttactgtctg ggcagttatg attgtagtac tggtgattgt 300
tttgttttcg gcgagggaac caaggtggaa atcaaacgt 339

SEQ ID NO: 313        moltype = DNA length = 66
FEATURE              Location/Qualifiers
misc_feature          1..66
                      note = Engineered antibody sequence
source                1..66
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 313
caagtgctga cccagtctcc atcctccttg tctgcatctg taggagacag agtcaccatc 60
aattgc 66

SEQ ID NO: 314        moltype = DNA length = 39
FEATURE              Location/Qualifiers
misc_feature          1..39
                      note = Engineered antibody sequence
source                1..39
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 314
caggccagtc agagtgttta caattacaac taccttgcc 39

SEQ ID NO: 315        moltype = DNA length = 45
FEATURE              Location/Qualifiers
misc_feature          1..45
                      note = Engineered antibody sequence
source                1..45
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 315
tggatcacgc agaaaccagg gaaagttcct aagcaactga tctat 45

SEQ ID NO: 316        moltype = DNA length = 21
FEATURE              Location/Qualifiers
misc_feature          1..21
                      note = Engineered antibody sequence
source                1..21
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 316
tctacatcca ctctggcatc t 21

SEQ ID NO: 317        moltype = DNA length = 96
FEATURE              Location/Qualifiers
misc_feature          1..96
                      note = Engineered antibody sequence
source                1..96
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 317
ggggtcccat ctcgttttcag tggcagtgga tctgggacag atttcaactct caccatcagc 60
agcctgcagc ctgaagatgt tgcaacttat tactgt 96

SEQ ID NO: 318        moltype = DNA length = 39
FEATURE              Location/Qualifiers
misc_feature          1..39
                      note = Engineered antibody sequence
source                1..39
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 318
ctgggcagtt atgattgtag tactgggtgat tgttttgtt 39

SEQ ID NO: 319        moltype = DNA length = 33
FEATURE              Location/Qualifiers
misc_feature          1..33
                      note = Engineered antibody sequence
source                1..33
                      mol_type = other DNA

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                                organism = synthetic construct
SEQUENCE: 319
ttcggcggag gaaccaaggt ggaaatcaaa cgt                                     33

SEQ ID NO: 320                  moltype = DNA   length = 321
FEATURE                        Location/Qualifiers
misc_feature                   1..321
                                note = Engineered antibody sequence
source                        1..321
                                mol_type = other DNA
                                organism = synthetic construct

SEQUENCE: 320
acggtggctg caccatctgt cttcatcttc cggccatctg atgagcagtt gaaatctgga   60
actgcctctg ttgtgtgctt gctgaataac ttctatccca gagaggccaa agtacagtgg   120
aagggtggata acgcccctcca atcgggtaac tcccaggaga gtgtcacaga gcaggacagc   180
aaggacagca cctacagctc cagcagcacc ctgacgctga gcaaagcaga ctacgagaaa   240
cacaaggtct acgctgtgca agtcacccat cagggcctga gctcgcccg cacaagagc    300
ttcaacaggg gagagtgtta g                                              321

SEQ ID NO: 321                  moltype = AA   length = 439
FEATURE                        Location/Qualifiers
REGION                        1..439
                                note = Engineered antibody sequence
source                        1..439
                                mol_type = protein
                                organism = synthetic construct

SEQUENCE: 321
QSLEESGGRL VTPGTPLTLT CTVSGIGLSS YMQWVRQSP GRGLEWIGVI GSDGKTYIAT   60
WAKGRFTISK TSSTTVDLRM ASLTTEDTAT YFCTRGDIWG PGTLVTVSSA STKGPSVFPPL  120
APSSKSTSGG TAALGCLVKD YFPEPVTWSW NSGALTSGVH TFPAVLQSSG LYSLSVVTV   180
PSSSLGTQTY ICNVNHPKPSN TKVDRKVEPK SCDKTHTCPP CPAPELLGGP SVFLFPPKPK  240
DTLMISRTPV VTCVVVDVSH EDPEVKFNWY VDGVEVHNAK TKPREEQYAS TYRVVSVLTV   300
LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM TKNQVSLTCL  360
VKGFYPSDIA VEWESNGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWQ QGNVFSCSVM  420
HEALHNHYTQ KSLSLSPGK                                              439

SEQ ID NO: 322                  moltype = AA   length = 109
FEATURE                        Location/Qualifiers
REGION                        1..109
                                note = Engineered antibody sequence
source                        1..109
                                mol_type = protein
                                organism = synthetic construct

SEQUENCE: 322
QSLEESGGRL VTPGTPLTLT CTVSGIGLSS YMQWVRQSP GRGLEWIGVI GSDGKTYIAT   60
WAKGRFTISK TSSTTVDLRM ASLTTEDTAT YFCTRGDIWG PGTLVTVSS              109

SEQ ID NO: 323                  moltype = AA   length = 29
FEATURE                        Location/Qualifiers
REGION                        1..29
                                note = Engineered antibody sequence
source                        1..29
                                mol_type = protein
                                organism = synthetic construct

SEQUENCE: 323
QSLEESGGRL VTPGTPLTLT CTVSGIGLS                                     29

SEQ ID NO: 324                  moltype = AA   length = 5
FEATURE                        Location/Qualifiers
REGION                        1..5
                                note = Engineered antibody sequence
source                        1..5
                                mol_type = protein
                                organism = synthetic construct

SEQUENCE: 324
SYMQ                                                                    5

SEQ ID NO: 325                  moltype = AA   length = 14
FEATURE                        Location/Qualifiers
REGION                        1..14
                                note = Engineered antibody sequence
source                        1..14
                                mol_type = protein
                                organism = synthetic construct

SEQUENCE: 325
WVRQSPGRGL EWIG                                                       14

SEQ ID NO: 326                  moltype = AA   length = 16
FEATURE                        Location/Qualifiers

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REGION 1..16
 note = Engineered antibody sequence
 source 1..16
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 326
 VIGSDGKTTY ATWAKG 16

SEQ ID NO: 327
 FEATURE Location/Qualifiers
 REGION 1..31
 note = Engineered antibody sequence
 source 1..31
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 327
 RFTISKTSST TVDLRMASLT TEDTATYFCT R 31

SEQ ID NO: 328
 SEQUENCE: 328
 000
 moltype = length =

SEQ ID NO: 329
 FEATURE Location/Qualifiers
 REGION 1..11
 note = Engineered antibody sequence
 source 1..11
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 329
 WGPGLTVTVS S 11

SEQ ID NO: 330
 FEATURE Location/Qualifiers
 REGION 1..330
 note = Engineered antibody sequence
 source 1..330
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 330
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTPPAVLQSS 60
 GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKRVPE KSCDKTHTCP PCPAPELLGG 120
 PSVFLFPPKP KDTLMISRTPEVTCVVVDVSHEDPEVKFNW YVDGVEVHNA KTKPREEQYA 180
 STYRVVSVLT VHLQDWLNGK EYKCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSREE 240
 MTKNQVSLTCLVKGFYPSDI AVEWESNGQP ENNYKTTTPV LDSGGSFFLY SKLTVDKSRW 300
 QQGNVFSCSV MHEALHNHYT QKSLSLSPGK 330

SEQ ID NO: 331
 FEATURE Location/Qualifiers
 misc_feature 1..1320
 note = Engineered antibody sequence
 source 1..1320
 mol_type = other DNA
 organism = synthetic construct
 SEQUENCE: 331
 cagtcgctgg aggagtcagg gggtcgctg gtcacgcctg ggacacccct gacactcacc 60
 tgcacagctc ctggaatcgg cctcagtagc tactacatgc agtgggtccg ccagtcctca 120
 gggaggggggc tgggaatggat cggagtcatt ggtagtgatg gtaagacata ctacgcgacc 180
 tgggcgaaaag gccgattcac catctccaag acctcgctga ccacgggtgga tctgagaatg 240
 gccagtcctga caaccgagga caccggccacc tattctctga ccagagggga catctggggc 300
 ccgggggaccc tcgtcaccgt ctcgagcgcg tccaccaagg gcccatcggt cttccccctg 360
 gcaccctcct ccaagagcac ctctgggggc acagcggccc tgggctgcct ggtcaaggac 420
 tacttccccg aaccgggtgac ggtgtcgtgg aactcaggcg ccctgaccag cggcgtgcac 480
 acccttcccg ctgtcctaca gtctcagga ctctactccc tcagcagcgt ggtgaccgtg 540
 ccctccagca gcttgggcac ccagacctac atctgcaacg tgaatcacia gccccagcaac 600
 accaagggtg acaagagagt tgagcccaaa tcttgtagaca aaactcacac atgccccaccg 660
 tgcccagcac ctgaactcct ggggggaccg tcagtcctcc tcttcccccc aaaacccaag 720
 gacacccca ctgctctccc gaccctgag gtcacatgcg tgggtggtgga cgtgagccac 780
 gaagaccctg aggtcaagtt caactggtac gtggacggcg tggaggtgca taatgccaaag 840
 acaaagccgc gggaggagca gtacgcccag acgtaccgtg tggtcagcgt cctcaccgtc 900
 ctgcaccagg actggctgaa tggcaaggag tacaagtgcg aggtctccaa caaagccctc 960
 ccagccccca tcgagaaaac catctccaaa gccaaaggcg agccccgaga accacagggtg 1020
 tacacccctg ccccatcccc ggaggagatg accaagaacc aggtcagcct gacctgcctg 1080
 gtcaaaaggct tctatccccg gcacatcgcc gtggagtggg agagcaatgg gcagccggag 1140
 aacaactaca agaccacgcc tcccgctgtg gactccgacg gctccttctt cctctacagc 1200
 aagctcaccg tggacaagag cagggtggcag caggggaaacg tcttctcatg ctccgtgatg 1260
 catgaggctc tgcacaacca ctacacgcag aagagcctct cctgtctctc gggtaaatga 1320

SEQ ID NO: 332
 moltype = DNA length = 327

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FEATURE	Location/Qualifiers	
misc_feature	1..327	
	note = Engineered antibody sequence	
source	1..327	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 332		
cagtcgctgg aggagtcggt gggtcgctg gtcacgcctg ggacacccct gacactcacc		60
tgcacagtct ctggaatcgg cctcagtagc tactacatgc agtgggtccg ccagtctcca		120
gggagggggc tgggaatggat cggagtcatt ggtagtgatg gtaagacata ctacgcgacc		180
tgggcgaaag gccgattcac catctccaag acctcgctga ccacggtgga tctgagaatg		240
gccagtctga caaccgagga caccggccacc tatttctgta ccagagggga catctggggc		300
ccggggagccc tcgtcacctg ctcgagc		327
SEQ ID NO: 333	moltype = DNA length = 87	
FEATURE	Location/Qualifiers	
misc_feature	1..87	
	note = Engineered antibody sequence	
source	1..87	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 333		
cagtcgctgg aggagtcggt gggtcgctg gtcacgcctg ggacacccct gacactcacc		60
tgcacagtct ctggaatcgg cctcagtagc		87
SEQ ID NO: 334	moltype = DNA length = 15	
FEATURE	Location/Qualifiers	
misc_feature	1..15	
	note = Engineered antibody sequence	
source	1..15	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 334		
agctactaca tgcag		15
SEQ ID NO: 335	moltype = DNA length = 42	
FEATURE	Location/Qualifiers	
misc_feature	1..42	
	note = Engineered antibody sequence	
source	1..42	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 335		
tgggtccgcc agtctccagg gagggggctg gaatggatcg ga		42
SEQ ID NO: 336	moltype = DNA length = 48	
FEATURE	Location/Qualifiers	
misc_feature	1..48	
	note = Engineered antibody sequence	
source	1..48	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 336		
gtcattggta gtgatggtaa gacatactac gcgacctggg cgaaaggc		48
SEQ ID NO: 337	moltype = DNA length = 93	
FEATURE	Location/Qualifiers	
misc_feature	1..93	
	note = Engineered antibody sequence	
source	1..93	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 337		
cgattcacca tctccaagac ctgcgtcgacc acggtggatc tgagaatggc cagtctgaca		60
accgaggaca cggccaccta tttctgtacc aga		93
SEQ ID NO: 338	moltype = length =	
SEQUENCE: 338		
000		
SEQ ID NO: 339	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Engineered antibody sequence	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 339		
tggggcccg ggaccctcgt caccgtctcg agc		33

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SEQ ID NO: 340      moltype = DNA length = 993
FEATURE            Location/Qualifiers
misc_feature        1..993
                    note = Engineered antibody sequence
source              1..993
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 340
gcctccacca agggcccatc ggtcttcccc ctggcaccct cctccaagag cacctctggg 60
ggcacagcgg ccttgggctg cctgggtcaag gactacttcc ccgaaccggg gacgggtgtcg 120
tggaaactcag gcgccttgac cagcggcgtg cacaccttcc cggctgtcct acagtctctca 180
ggactctact ccttcagcag cgtgggtgacc gtgccctcca gcagcttggg caccagacc 240
tacatctgca acgtgaatca caagcccagc aacaccaagg tggacaagag agttgagccc 300
aaatcttgtg acaaaactca cacatgcccc ccggtcccag cacctgaact cctgggggga 360
ccgtcagtc tctcttcccc cccaaaaccc aaggacacc tcctgatctc ccggaccct 420
gaggtcacat gcgtgggtgt ggacgtgagc cacgaagacc ctgaggtcaa gttcaactgg 480
tacgtggacg gcgtggaggt gcataatgcc aagacaaagc cgcgggagga gcagtacgcc 540
agcacgtacc gtgtgggtcag cgtcctcacc gtctgcacc aggactgggt gaatggcaag 600
gagtacaaat gcaagggtct caacaaagcc ctcccagccc ccacgcagaa aacctctctc 660
aaagccaaag ggcagccccc agaaccacag gtgtacaccc tgcccccatc ccgggaggag 720
atgaccaaga accagggtcag cctgacctgc ctgggtcaag gcttctatcc cagcgacatc 780
gccgtggagt ggggaggtcaa tgggcagccc gagaaacaact acaagaccac gcctcccggt 840
ctggactccg accgtctctt ctctctctac agcaagctca ccgtggacaa gagcagggtg 900
cagcagggga acgtctcttc atgctccgtg atgcatgagg ctctgcacaa ccactacacg 960
cagaagagcc tctccctgtc tccgggtaaa tga 993

SEQ ID NO: 341      moltype = AA length = 219
FEATURE            Location/Qualifiers
REGION              1..219
                    note = Engineered antibody sequence
source              1..219
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 341
QVLTQTPSPV SAAVGSTVTI NCQASQNVYN NNYLAWYQQK PGQPPKQLIY STSTLASGVS 60
SRFRGSGSGT QFTLTISDVQ CDDAATYYCL GSYDCSRGDC FVFGGTEVV VKRTVAAPSV 120
FIFPPSDEQL KSGTASVVCL LNNFYPREAK VQWKVDNALQ SGNSQESVTE QDSKDYSTYSL 180
SSTLTLSKAD YEKHKVYACE VTHQGLSSPV TKSFNRGEC 219

SEQ ID NO: 342      moltype = AA length = 113
FEATURE            Location/Qualifiers
REGION              1..113
                    note = Engineered antibody sequence
source              1..113
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 342
QVLTQTPSPV SAAVGSTVTI NCQASQNVYN NNYLAWYQQK PGQPPKQLIY STSTLASGVS 60
SRFRGSGSGT QFTLTISDVQ CDDAATYYCL GSYDCSRGDC FVFGGTEVV VKR 113

SEQ ID NO: 343      moltype = AA length = 22
FEATURE            Location/Qualifiers
REGION              1..22
                    note = Engineered antibody sequence
source              1..22
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 343
QVLTQTPSPV SAAVGSTVTI NC 22

SEQ ID NO: 344      moltype = AA length = 13
FEATURE            Location/Qualifiers
REGION              1..13
                    note = Engineered antibody sequence
source              1..13
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 344
QASQNVYNNN YLA 13

SEQ ID NO: 345      moltype = AA length = 15
FEATURE            Location/Qualifiers
REGION              1..15
                    note = Engineered antibody sequence
source              1..15
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 345

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WYQQKPGQPP KQLIY		15
SEQ ID NO: 346	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = Engineered antibody sequence	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 346		
STSTLAS		7
SEQ ID NO: 347	moltype = AA length = 32	
FEATURE	Location/Qualifiers	
REGION	1..32	
	note = Engineered antibody sequence	
source	1..32	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 347		
GVSSRFRGSG SGTQFTLTIS DVQCDDAATY YC		32
SEQ ID NO: 348	moltype = AA length = 13	
FEATURE	Location/Qualifiers	
REGION	1..13	
	note = Engineered antibody sequence	
source	1..13	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 348		
LGSYDCSRGD CFV		13
SEQ ID NO: 349	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Engineered antibody sequence	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 349		
FGGGTEVVVK R		11
SEQ ID NO: 350	moltype = AA length = 106	
FEATURE	Location/Qualifiers	
REGION	1..106	
	note = Engineered antibody sequence	
source	1..106	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 350		
TVAAPSVFIF PPSDEQLKSG TASVVCLLNN FYPREAKVQW KVDNALQSGN SQESVTEQDS		60
KDSTYLSST LTLSKADY EK HKVYACEVTH QGLSSPVT KS FNRGEC		106
SEQ ID NO: 351	moltype = DNA length = 660	
FEATURE	Location/Qualifiers	
misc_feature	1..660	
	note = Engineered antibody sequence	
source	1..660	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 351		
caagtgtcga cccagactcc atcccccgctg tctgcagctg tgggaagcac agtcaccatc		60
aattgccagg ccagtcagaa tgtttataat aacaactacc tagcctggta tcagcagaaa		120
ccagggcagc ctcccaagca actgatctat tctacgtcca ctctggcatc tgggggtctca		180
tcgcgattca gaggcagtggt atctgggaca cagttcactc tcaccatcag cgacgtgcag		240
tgtgacgatg ctgccactta ctactgtcta ggcagttatg attgtagtcg tgggtgattgt		300
tttgttttcg gcggaggggac cgaggtgggtg gtcaaacgta cggtgggtgc accatctgtc		360
ttcatcttcc cgccatctga tgagcagttg aaatctggaa ctgcctctgt tgtgtgacctg		420
ctgaataact tctatcccag agaggccaaa gtacagtggga aggtggataa cgccctccaa		480
tcgggtaact cccaggagag tgtcacagag caggacagca aggacagcac ctacagcctc		540
agcagcaccg tgacgctgag caaagcagac tacgagaac acaaagtcta cgccctgcgaa		600
gtcaccatc agggcctgag ctcgcccgtc acaaagagct tcaacagggg agagtgttag		660
SEQ ID NO: 352	moltype = DNA length = 339	
FEATURE	Location/Qualifiers	
misc_feature	1..339	
	note = Engineered antibody sequence	
source	1..339	
	mol_type = other DNA	

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                                organism = synthetic construct
SEQUENCE: 352
caagtgtctga cccagactcc atcccccggtg tctgcagctg tgggaagcac agtcaccatc 60
aattgccagg ccagtcagaa tgtttataat aacaactacc tagcctggta tcagcagaaa 120
ccagggcagc ctcccaagca actgatctat tctacgtcca ctctggcatc tgggggtotca 180
tcgcgattca gaggcagtg atctgggaca cagttcactc tcaccatcag cgacgtgcag 240
tgtgacgatg ctgccactta ctactgtcta ggcagttatg attgtagtcg tggtgattgt 300
tttgttttcg gcggaggggac cgagggtggtg gtcaaacgt 339

SEQ ID NO: 353      moltype = DNA length = 66
FEATURE            Location/Qualifiers
misc_feature       1..66
                    note = Engineered antibody sequence
source            1..66
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 353
caagtgtctga cccagactcc atcccccggtg tctgcagctg tgggaagcac agtcaccatc 60
aattgc                                           66

SEQ ID NO: 354      moltype = DNA length = 39
FEATURE            Location/Qualifiers
misc_feature       1..39
                    note = Engineered antibody sequence
source            1..39
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 354
caggccagtc agaattgttta taataacaac tacctagcc 39

SEQ ID NO: 355      moltype = DNA length = 45
FEATURE            Location/Qualifiers
misc_feature       1..45
                    note = Engineered antibody sequence
source            1..45
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 355
tggtatcagc agaaaccagg gcagcctccc aagcaactga tctat 45

SEQ ID NO: 356      moltype = DNA length = 21
FEATURE            Location/Qualifiers
misc_feature       1..21
                    note = Engineered antibody sequence
source            1..21
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 356
tctacgtcca ctctggcatc t 21

SEQ ID NO: 357      moltype = DNA length = 96
FEATURE            Location/Qualifiers
misc_feature       1..96
                    note = Engineered antibody sequence
source            1..96
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 357
gggggtctcat cgcgattcag aggcagtgga tctgggacac agttcactct caccatcagc 60
gacgtgcagt gtgacgatgc tgccacttac tactgt 96

SEQ ID NO: 358      moltype = DNA length = 39
FEATURE            Location/Qualifiers
misc_feature       1..39
                    note = Engineered antibody sequence
source            1..39
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 358
ctaggcagtt atgattgtag tctggtgat tgttttgtt 39

SEQ ID NO: 359      moltype = DNA length = 33
FEATURE            Location/Qualifiers
misc_feature       1..33
                    note = Engineered antibody sequence
source            1..33
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 359

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ttcggcgagg ggaccgaggt ggtgggtcaaa cgt		33
SEQ ID NO: 360	moltype = DNA length = 321	
FEATURE	Location/Qualifiers	
misc_feature	1..321	
	note = Engineered antibody sequence	
source	1..321	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 360		
acggtggctg caccatctgt	cttcattctt cgcgcattcg atgagcagtt gaaatctgga	60
actgcctctg ttgtgtgcct	gctgaataac ttctatccca gagaggccaa agtacagtgg	120
aaggtggata acgccctcca	atcgggtaac tcccaggaga gtgtcacaga gcaggacagc	180
aaggacagca cctacagcct	cagcagcacc ctgacgctga gcaaagcaga ctacgagaaa	240
cacaaagtct acgcctgcga	agtcaccact cagggcctga gctcgccctg cacaaagagc	300
ttcaacaggg gagagtgtta	g	321
SEQ ID NO: 361	moltype = AA length = 441	
FEATURE	Location/Qualifiers	
REGION	1..441	
	note = Engineered antibody sequence	
source	1..441	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 361		
EVQLVESGGG LVQPGGSLRL	SCAVSGIGLS SYIMQWVRQA PGKGLEWVGV IGSDGKITYYA	60
TWAKGRFTIS RDNSKTTYVL	QMNSLRAEDT AVYFCTRGLD WGQGTTLVTVS SASTKGPSVF	120
PLAPSSKSTS GGTAAALGCLV	KDYFPPEPVT SWNSGALTSG VHTFPAVLQS SGLYSLSSVV	180
TVPSSSLGTQ TYICNVNHKP	SNTKVDKRVK PKSCDKHTC PPCPAPELLG GPSVFLFPPK	240
PKDTLMISRT PEVTCVVVDV	SHEDPEVKFN WYVDGVEVHN AKTKPREEQY ASTYRVVSVL	300
TVLHQDWLNG KEYCKVSNK	ALPAPIEKTI SKAKQQPREP QVYTLPPSRE EMTKNQVSLT	360
CLVKGFYPSD IAEWESNGQ	PENNYKTTTP VLDSDGSFFL YSKLTVDKSR WQQGNVFS	420
VMHEALHNNH TQKLSLSLSPG	K	441
SEQ ID NO: 362	moltype = AA length = 111	
FEATURE	Location/Qualifiers	
REGION	1..111	
	note = Engineered antibody sequence	
source	1..111	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 362		
EVQLVESGGG LVQPGGSLRL	SCAVSGIGLS SYIMQWVRQA PGKGLEWVGV IGSDGKITYYA	60
TWAKGRFTIS RDNSKTTYVL	QMNSLRAEDT AVYFCTRGLD WGQGTTLVTVS S	111
SEQ ID NO: 363	moltype = AA length = 30	
FEATURE	Location/Qualifiers	
REGION	1..30	
	note = Engineered antibody sequence	
source	1..30	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 363		
EVQLVESGGG LVQPGGSLRL	SCAVSGIGLS	30
SEQ ID NO: 364	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
REGION	1..5	
	note = Engineered antibody sequence	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 364		
SYMQ		5
SEQ ID NO: 365	moltype = AA length = 14	
FEATURE	Location/Qualifiers	
REGION	1..14	
	note = Engineered antibody sequence	
source	1..14	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 365		
WVRQAPGKGL EWVG		14
SEQ ID NO: 366	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
REGION	1..16	
	note = Engineered antibody sequence	

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source 1..16
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 366
 VIGSDGKTY Y ATWAKG 16

SEQ ID NO: 367 moltype = AA length = 32
 FEATURE Location/Qualifiers
 REGION 1..32
 note = Engineered antibody sequence

source 1..32
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 367
 RFTISRDN SK TTVYLQMNSL RAEDTAVYFC TR 32

SEQ ID NO: 368 moltype = length =
 SEQUENCE: 368
 000

SEQ ID NO: 369 moltype = AA length = 11
 FEATURE Location/Qualifiers
 REGION 1..11
 note = Engineered antibody sequence

source 1..11
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 369
 WGGGTLVTVS S 11

SEQ ID NO: 370 moltype = AA length = 330
 FEATURE Location/Qualifiers
 REGION 1..330
 note = Engineered antibody sequence

source 1..330
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 370
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 60
 GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKRV EP KSCDKTHTCP PCPAPELLGG 120
 PSVFLFPPPK KDTLMISRTPEVTCVVVDVSHEDPEVKFNW YVDGVEVHNA KTKPREEQYA 180
 STYRVVSVLT VLNQDNLNGK EYKCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSREE 240
 MTKNQVSLTLC LVKGFPYSDI AVEWESNGQP ENNYKTTTPPV LDSDGSFFLY SKLTVDKSRW 300
 QQGNVFSCSV MHEALHNHYT QKSLSLSPGK 330

SEQ ID NO: 371 moltype = DNA length = 1326
 FEATURE Location/Qualifiers
 misc_feature 1..1326
 note = Engineered antibody sequence

source 1..1326
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 371
 gaggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctgggggggc cctgagactc 60
 tcctgtgcag tctctggaat cggcctcagc agctactaca tgcaatgggt ccgtcaggct 120
 ccagggaagg ggctggagtg ggtcggagtc attggtatgt atggtaagac atactacgcg 180
 acctgggcga aaggccgatt caccatctcc agagacaatt ccaagaccac ggtgtatctt 240
 caaatgaaca gctgagagc tgaggacact gctgtgtatt tctgtaccag aggggacatc 300
 tggggccaag ggaccctcgt caccgtctcg agcgctcca ccaagggccc atcggtcttc 360
 cccctggcac cctcctccaa gagcacctct gggggcacag cggccctggg ctgcctggtc 420
 aaggactact tcccgaacc ggtgacgggt tcgtggaact caggcgccct gaccagcggc 480
 gtgcacacct tcccggtgt cctacagtc tcaggactct actccctcag cagcgtggtg 540
 accgtgcccc ccagcagctt gggcacccag acctacatct gcaacgtgaa tcacaagccc 600
 agcaacacca agcgcgagg gagagttgag cccaaatctt gtgacaaaac tcacacatgc 660
 ccaccgtgcc cagcacctga actcctgggg ggaccgtcag tcttctctct ccccccaaaa 720
 cccaaggaca cctcatgat ctcccgagc cctgaggtca catgcgtggt ggtggacgtg 780
 agccacgaag acctgaggt caagtcaac tggtagctgg acggcgtgga ggtgcataat 840
 gccaaagcaa agcgcgagg gagagcagtc gccagcacgt accgtgtggt cagcgtcctc 900
 accgtcctgc accaggactg gctgaatggc aaggagtaca agtgcaaggc ctccaacaaa 960
 gccctccag ccccatcgaa gaaaaccatc tccaaagcca aagggcagcc ccgagaacca 1020
 cagggtgtaca cctgcccccc atcccgagg gagatgacca agaaccaggc cagcctgacc 1080
 tgctgtgtca aaggcttcta tcccagcgac atcgccgtgg agtgggagag caatgggcag 1140
 ccggagaaca actacaagac cagcgtctcc gtgctggact ccgacggctc cttctctctc 1200
 tacagcaagc tcaccgtgga caagagcagg tggcagcagg ggaacgtctt ctcagctccc 1260
 gtgatgcatg aggcctctgca caaccactac acgcagaaga gcctctccct gtctccgggt 1320
 aaatga 1326

SEQ ID NO: 372 moltype = DNA length = 333
 FEATURE Location/Qualifiers

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misc_feature      1..333
                  note = Engineered antibody sequence
source            1..333
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 372
gaggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctgggggggtc cctgagactc 60
tcctgtgcag tctctggaat cggcctcagt agctactaca tgcaatgggt ccgtcaggct 120
ccagggaagg ggctggagtg ggtcggagtc attggtagtg atggtaagac atactacgcg 180
acctgggcga aaggccgatt caccatctcc agagacaatt ccaagaccac ggtgtatctt 240
caaatgaaca gcctgagagc tgaggacact gctgtgtatt tctgtaccag agggggacatc 300
tggggccaag ggaccctcgt caccgtctcg agc 333

SEQ ID NO: 373    moltype = DNA length = 90
FEATURE          Location/Qualifiers
misc_feature      1..90
                  note = Engineered antibody sequence
source            1..90
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 373
gaggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctgggggggtc cctgagactc 60
tcctgtgcag tctctggaat cggcctcagt 90

SEQ ID NO: 374    moltype = DNA length = 15
FEATURE          Location/Qualifiers
misc_feature      1..15
                  note = Engineered antibody sequence
source            1..15
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 374
agctactaca tgcaa 15

SEQ ID NO: 375    moltype = DNA length = 42
FEATURE          Location/Qualifiers
misc_feature      1..42
                  note = Engineered antibody sequence
source            1..42
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 375
tgggtccgtc aggctccagg gaaggggctg gagtgggtcg ga 42

SEQ ID NO: 376    moltype = DNA length = 48
FEATURE          Location/Qualifiers
misc_feature      1..48
                  note = Engineered antibody sequence
source            1..48
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 376
gtcattggta gtgatggtaa gacatactac gcgacctggg cgaaaggc 48

SEQ ID NO: 377    moltype = DNA length = 96
FEATURE          Location/Qualifiers
misc_feature      1..96
                  note = Engineered antibody sequence
source            1..96
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 377
cgattcacca tctccagaga caattccaag accacgggtg atcttcaaat gaacagcctg 60
agagctgagg acaactgctgt gtatttctgt accaga 96

SEQ ID NO: 378    moltype = length =
SEQUENCE: 378
000

SEQ ID NO: 379    moltype = DNA length = 33
FEATURE          Location/Qualifiers
misc_feature      1..33
                  note = Engineered antibody sequence
source            1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 379
tggggccaag ggaccctcgt caccgtctcg agc 33

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SEQ ID NO: 380      moltype = DNA length = 993
FEATURE            Location/Qualifiers
misc_feature       1..993
                   note = Engineered antibody sequence
source            1..993
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 380
gctccacca agggcccatc ggtcttcccc ctggcaccct cctccaagag cacctctggg 60
ggcacagcgg ccttgggctg cctgggtcaag gactacttcc ccgaaccggg gacgggtgtcg 120
tggaactcag gcgccttgac cagcggcggtg cacacdtccc cggtgtgctt acagtctctca 180
ggactctact ccttcagcag cgtgggtgacc gtgccctcca gcagcttggg caccagacc 240
tacatctgca acgtgaatca caagcccagc aacaccaagg tggacaagag agttgagccc 300
aaatcttgtg acaaaaactca cacatgcccc ccgtgcccag cacctgaact cctgggggga 360
ccgtcagtcct tcctcttccc cccaaaaccc aaggaccccc tcatgatctc ccggaccct 420
gaggtcacat gcgtgggtgg ggacgtgagc cacgaagacc ctgaggtcaa gttcaactgg 480
tacgtggacg gcgtggaggt gcataatgcc aagacaaagc cgcgggagga gcagtacgcc 540
agcacgtacc gtgtgggtcag cgtcctcacc gtctctgacc aggactggct gaatggcaag 600
gagtacaagt gcaaggtctc caacaaagcc ctcccagccc ccatcgagaa aacctctcc 660
aaagccaaag ggcagccccc agaaccacag gtgtacaccc tgcccccatc ccgggaggag 720
atgaccaaga accaggtcag cctgacctgc ctgggtcaaag gcttctatcc cagcgacatc 780
gccgtggagt gggagagcaa tgggcagccg gagaacaact acaagaccac gcctccgtg 840
ctggactccg acggtctctt ctctctctac agcaagctca ccgtggacaa gagcagggtg 900
cagcagggga acgtctcttc atgctccgtg atgcatgagg ctctgcacaa ccactacacg 960
cagaagagcc tctccctgtc tccgggtaaa tga 993

SEQ ID NO: 381      moltype = AA length = 219
FEATURE            Location/Qualifiers
REGION            1..219
                   note = Engineered antibody sequence
source            1..219
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 381
QVLTQSPSSL SASVGDRVTI NCQASQNVYN NNYLAWYQQK PGKVPKQLIY STSTLASGVP 60
SRFSGSGSGT DFTLTISLQ PEDVATYYCL GSYDCSRGDC FVFGGGTKVE IKRTVAAPSV 120
FIFFPSDEQL KSGTASVVCL LNNFYPREAK VQWKVDNALQ SGNSQESVTE QDSKSTYSL 180
SSTLTLSKAD YEKHKVYACE VTHQGLSSPV TKSPNRGEC 219

SEQ ID NO: 382      moltype = AA length = 113
FEATURE            Location/Qualifiers
REGION            1..113
                   note = Engineered antibody sequence
source            1..113
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 382
QVLTQSPSSL SASVGDRVTI NCQASQNVYN NNYLAWYQQK PGKVPKQLIY STSTLASGVP 60
SRFSGSGSGT DFTLTISLQ PEDVATYYCL GSYDCSRGDC FVFGGGTKVE IKR 113

SEQ ID NO: 383      moltype = AA length = 22
FEATURE            Location/Qualifiers
REGION            1..22
                   note = Engineered antibody sequence
source            1..22
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 383
QVLTQSPSSL SASVGDRVTI NC 22

SEQ ID NO: 384      moltype = AA length = 13
FEATURE            Location/Qualifiers
REGION            1..13
                   note = Engineered antibody sequence
source            1..13
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 384
QASQNVYNNN YLA 13

SEQ ID NO: 385      moltype = AA length = 15
FEATURE            Location/Qualifiers
REGION            1..15
                   note = Engineered antibody sequence
source            1..15
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 385
WYQQKPGKVP KQLIY 15

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SEQ ID NO: 386	moltype = AA	length = 7	
FEATURE	Location/Qualifiers		
REGION	1..7		
source	note = Engineered antibody sequence 1..7 mol_type = protein organism = synthetic construct		
SEQUENCE: 386			
STSTLAS			7
SEQ ID NO: 387	moltype = AA	length = 32	
FEATURE	Location/Qualifiers		
REGION	1..32		
source	note = Engineered antibody sequence 1..32 mol_type = protein organism = synthetic construct		
SEQUENCE: 387			
GVPSRFGSGG SGTDFLTIS	SLQPEDVATY	YC	32
SEQ ID NO: 388	moltype = AA	length = 13	
FEATURE	Location/Qualifiers		
REGION	1..13		
source	note = Engineered antibody sequence 1..13 mol_type = protein organism = synthetic construct		
SEQUENCE: 388			
LGSYDCSRGD CFV			13
SEQ ID NO: 389	moltype = AA	length = 11	
FEATURE	Location/Qualifiers		
REGION	1..11		
source	note = Engineered antibody sequence 1..11 mol_type = protein organism = synthetic construct		
SEQUENCE: 389			
FGGGTKVEIK R			11
SEQ ID NO: 390	moltype = AA	length = 106	
FEATURE	Location/Qualifiers		
REGION	1..106		
source	note = Engineered antibody sequence 1..106 mol_type = protein organism = synthetic construct		
SEQUENCE: 390			
TVAAPSVFIF PPSDEQLKSG	TASVCLLN	FYPREAKVQW	KVDNALQSGN
KDSTYLSST LTLKADYEK	HKVYACEVTH	QGLSSPVTKS	FNRGEC
			60
			106
SEQ ID NO: 391	moltype = DNA	length = 660	
FEATURE	Location/Qualifiers		
misc_feature	1..660		
source	note = Engineered antibody sequence 1..660 mol_type = other DNA organism = synthetic construct		
SEQUENCE: 391			
caagtgtctga cccagctctcc	atctctccctg	tctgcatctg	taggagacag
aattgccagg ccagtcagaa	tgtttacaat	aacaactacc	tagccttgga
ccagggaag ttcttaagca	actgatctat	tctacatcca	ctctggcatc
tctcgtttca gtggcagtg	atctgggaca	gatttcactc	tcaccatcag
cttgaagatg ttgcaactta	ttaactgtctg	ggcagttatg	attgtagtctg
tttgttttcg gcggaggaac	caaggtgtgaa	atccaaagta	cggtggctgc
ttcatcttcc cgccatctga	tgagcagttg	aaatctggaa	ctgcctctgt
ctgaataact tctatccag	agaggccaaa	gtacagtgga	agggtggataa
tcgggtaact cccaggagag	tgtcacagag	caggacagca	aggacagcac
agcagcaccc tgacgctgag	caaagcagac	tacgagaaac	acaaagtcta
gtcaccatc agggcctgag	ctcgcccgctc	acaaagagct	tcaacagggg
			60
			120
			180
			240
			300
			360
			420
			480
			540
			600
			660
SEQ ID NO: 392	moltype = DNA	length = 339	
FEATURE	Location/Qualifiers		
misc_feature	1..339		
source	note = Engineered antibody sequence 1..339 mol_type = other DNA organism = synthetic construct		

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SEQUENCE: 392
caagtgtga cccagtcctc atcctcctg tctgcatctg taggagacag agtcaccatc 60
aattgccagg ccagtcagaa tgtttacaat aacaactacc tagcctggta tcagcagaaa 120
ccagggaag ttctaagca actgatctat tctacatcca ctctggcatc tgggggtcca 180
tctcgtttca gtggcagtg atctgggaca gatttcactc tcaccatcag cagcctgcag 240
cctgaagatg ttgcaactta ttactgtctg ggcagttatg attgtagtcg tggtgattgt 300
ttgttttcg gcggaggaac caaggtggaa atcaaacgt 339

SEQ ID NO: 393 moltype = DNA length = 66
FEATURE Location/Qualifiers
misc_feature 1..66
 note = Engineered antibody sequence
source 1..66
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 393
caagtgtga cccagtcctc atcctcctg tctgcatctg taggagacag agtcaccatc 60
aattgc 66

SEQ ID NO: 394 moltype = DNA length = 39
FEATURE Location/Qualifiers
misc_feature 1..39
 note = Engineered antibody sequence
source 1..39
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 394
caggccagtc agaattgtta caataacaac tacctagcc 39

SEQ ID NO: 395 moltype = DNA length = 45
FEATURE Location/Qualifiers
misc_feature 1..45
 note = Engineered antibody sequence
source 1..45
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 395
tggatcagc agaaaccagg gaaagtctct aagcaactga tctat 45

SEQ ID NO: 396 moltype = DNA length = 21
FEATURE Location/Qualifiers
misc_feature 1..21
 note = Engineered antibody sequence
source 1..21
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 396
tctacatcca ctctggcatc t 21

SEQ ID NO: 397 moltype = DNA length = 96
FEATURE Location/Qualifiers
misc_feature 1..96
 note = Engineered antibody sequence
source 1..96
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 397
gggggtccat ctcgtttcag tggcagtgga tctgggacag atttcaactct caccatcagc 60
agcctgcagc ctgaagatgt tgcaacttat tactgt 96

SEQ ID NO: 398 moltype = DNA length = 39
FEATURE Location/Qualifiers
misc_feature 1..39
 note = Engineered antibody sequence
source 1..39
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 398
ctgggcagtt atgattgtag tctgtgtgat tgttttgtt 39

SEQ ID NO: 399 moltype = DNA length = 33
FEATURE Location/Qualifiers
misc_feature 1..33
 note = Engineered antibody sequence
source 1..33
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 399
ttcggcggag gaaccaaggt ggaaatcaaa cgt 33

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SEQ ID NO: 400      moltype = DNA  length = 321
FEATURE            Location/Qualifiers
misc_feature        1..321
                    note = Engineered antibody sequence
source              1..321
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 400
acggtggctg caccatctgt cttcatcttc ccgccatctg atgagcagtt gaaatctgga   60
actgcctctg ttgtgtgect gctgaataac ttctatccca gagaggccaa agtacagtgg   120
aaggtggata acgcctctca atcgggtaac tcccaggaga gtgtcacaga gcaggacagc   180
aaggacagca cctacagcct cagcagcacc ctgacgctga gcaaagcaga ctacgagaaa   240
cacaaagtct acgcctgcga agtcacccat cagggcctga gctcgcccg cacaagagc    300
ttcaacaggg gagagtgtta g                                     321

SEQ ID NO: 401      moltype = AA  length = 439
FEATURE            Location/Qualifiers
REGION              1..439
                    note = Engineered antibody sequence
source              1..439
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 401
QSLEESGGRL VTPGGSLLTL CTVSGIDVTN YYMQWVRQAP GKGLEWIGVI GVNGKRYIAS   60
WAKGRFTISK TSSTTVDLKM TSLTTEDTAT YFCARGDIWG PGTLVTVSSA STKGPSVFPL   120
APSSKSTSGG TAALGCLVKD YFPEPVTWSW NSGALTSGVH TFPVAVLQSSG LYSLSVVTV   180
PSSSLGTQTY ICNVNHKPSN TKVDRVEPK  SCDKTHTCP CPAPELLGGP SVFLFPPKPK   240
DTLMISRTP ETCVVVDVSH EDPEVKFNWY VDGVEVHNAK TKPREEQYAS TYRVVSVLTV   300
LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM TKNQVSLTCL   360
VKGFYPSDIA VEWESNGQPE NNYKTPPVVL DSDGSFFLYS KLTVDKSRWQ QGNVFSCSVM   420
HEALHNYHQ KSLSLSPGK                                     439

SEQ ID NO: 402      moltype = AA  length = 109
FEATURE            Location/Qualifiers
REGION              1..109
                    note = Engineered antibody sequence
source              1..109
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 402
QSLEESGGRL VTPGGSLLTL CTVSGIDVTN YYMQWVRQAP GKGLEWIGVI GVNGKRYIAS   60
WAKGRFTISK TSSTTVDLKM TSLTTEDTAT YFCARGDIWG PGTLVTVSS              109

SEQ ID NO: 403      moltype = AA  length = 29
FEATURE            Location/Qualifiers
REGION              1..29
                    note = Engineered antibody sequence
source              1..29
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 403
QSLEESGGRL VTPGGSLLTL CTVSGIDVTN                                     29

SEQ ID NO: 404      moltype = AA  length = 5
FEATURE            Location/Qualifiers
REGION              1..5
                    note = Engineered antibody sequence
source              1..5
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 404
NYMQ                                                       5

SEQ ID NO: 405      moltype = AA  length = 14
FEATURE            Location/Qualifiers
REGION              1..14
                    note = Engineered antibody sequence
source              1..14
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 405
WVRQAPGKGL EWIG                                           14

SEQ ID NO: 406      moltype = AA  length = 16
FEATURE            Location/Qualifiers
REGION              1..16
                    note = Engineered antibody sequence
source              1..16

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mol_type = protein
organism = synthetic construct

SEQUENCE: 406
VIGVNGKRY Y ASWAKG 16

SEQ ID NO: 407
FEATURE
REGION
note = Engineered antibody sequence
1..31
source
mol_type = protein
organism = synthetic construct

SEQUENCE: 407
RFTISKTSST TVDLKMTSLT TEDTATYFCA R 31

SEQ ID NO: 408
SEQUENCE: 408
000

SEQ ID NO: 409
FEATURE
REGION
note = Engineered antibody sequence
1..11
source
mol_type = protein
organism = synthetic construct

SEQUENCE: 409
WGPGLTVTVS S 11

SEQ ID NO: 410
FEATURE
REGION
note = Engineered antibody sequence
1..330
source
mol_type = protein
organism = synthetic construct

SEQUENCE: 410
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTPPAVLQSS 60
GLYSLSSVVT VPSSSLGTQT YICNVNHNKPS NTKVDKRVPE KSCDKTHTCP PCPAPELLGG 120
PSVFLFPPKP KDTLMISRTPEVTCVVVDVSHEDPEVKFNW YVDGVEVHNA KTKPREEQYA 180
STYRVVSVLT VHLQDWLNGK EYKCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSREE 240
MTKNQVSLTC LVKGFYPSDI AVEWESNGQP ENNYKTPPV LDSDGSFFLY SKLTVDKSRW 300
QQGNVFSCSV MHEALHNHYT QKLSLSLSPGK 330

SEQ ID NO: 411
FEATURE
misc_feature
note = Engineered antibody sequence
1..1320
source
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 411
cagtcgctgg agggagtcgg gggctgcctg gtcacgcctg gaggatccct gacactcacc 60
tgcacagctc ctggaatcga cgtcactaac tactatatgc aatgggtccg ccagggtcca 120
gggaaggggc tgggaatggat cggagtcatt ggtgtgaatg gtaagagata ctacgcgagc 180
tgggcgaaag gccgattcac catctccaaa acctcgtcga ccacgggtgga tctgaaaaatg 240
accagtcctga caaccgagga cacggccacc tatttctgtg ccagaggcga catctggggc 300
ccggggaccc tcgtcaccgt ctcgagcgcc tccaccaagg gcccatcggt ctccccctg 360
gcacctcct ccaagagcac ctctgggggc acagcgccc tgggctgcct ggtcaaggac 420
tacttccccg aaccgggtgac ggtgtcgtgg aactcaggcg ccctgaccag cggcgtgcac 480
accttccccg ctgtcctaca gtctcagga ctctactccc tcagcagcgt ggtgaccgtg 540
cctccagca gcttgggcac ccagacctac atctgcaacg tgaatcaca gccacgcaac 600
accaagggtg acaagagagt tgagcccaaa tcttgtgaca aaactcacac atgccccaccg 660
tgccagcac ctgaactcct ggggggacgc tcagtccttc tcttcccccc aaaacccaag 720
gacacctca tgatctcccc gacccctgag gtcacatgcg tgggtggtgga cgtgagccac 780
gaagaccctg aggtcaagtt caactggtac gtggacggcg tggaggtgca taatgccaaag 840
acaaagccgc gggaggagca gtacgcccag acgtaccgtg tggtcagcgt cctcaccgtc 900
ctgcaccagg actggctgaa tggcaaggag tacaagtgca aggtctccaa caaagccctc 960
ccagccccca tcgagaaaac catctccaaa gccaaagggc agccccgaga accacagggtg 1020
tacacctgc ccccatcccg ggaggagatg accaagaacc aggtcagcct gacctgcctg 1080
gtcaaaggct tctatcccg cgacatcgcc gtggagtggg agagcaatgg gcagccggag 1140
aacaactaca agaccacgcc tcccgctgct gactccgacg gctccttctt cctctacagc 1200
aagctcaccg tggacaagag caggtggcag caggggaacg tcttctcatg ctccgtgatg 1260
catgaggctc tgcacaacca ctacacgcag aagagcctct cctgtctctc gggtaaatga 1320

SEQ ID NO: 412
FEATURE
misc_feature
note = Engineered antibody sequence
1..327

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source                1..327
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 412
cagtcgctgg aggagtcagg gggtcgcttg gtcacgcctg gaggatccct gacactcacc 60
tgcacagtct ctggaatcga cgtcactaac tactatatgc aatgggtccg ccaggctcca 120
gggaaggggc tggaatggat cggagtcatt ggtgtgaatg gtaagagata ctacgcgagc 180
tgggcgaaag gccgattcac catctccaaa acctcgtcga ccacgggtga tctgaaaatg 240
accagtctga caaccgagga caccggccacc tattctctgt ccagaggcga catctggggc 300
ccggggaccc tcgtcacctg ctcgagc                                     327

SEQ ID NO: 413        moltype = DNA length = 87
FEATURE              Location/Qualifiers
misc_feature          1..87
                      note = Engineered antibody sequence
source                1..87
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 413
cagtcgctgg aggagtcagg gggtcgcttg gtcacgcctg gaggatccct gacactcacc 60
tgcacagtct ctggaatcga cgtcact                                     87

SEQ ID NO: 414        moltype = DNA length = 15
FEATURE              Location/Qualifiers
misc_feature          1..15
                      note = Engineered antibody sequence
source                1..15
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 414
aactactata tgcaa                                             15

SEQ ID NO: 415        moltype = DNA length = 42
FEATURE              Location/Qualifiers
misc_feature          1..42
                      note = Engineered antibody sequence
source                1..42
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 415
tgggtccgcc aggtccagg gaaggggctg gaatggatcg ga                                     42

SEQ ID NO: 416        moltype = DNA length = 48
FEATURE              Location/Qualifiers
misc_feature          1..48
                      note = Engineered antibody sequence
source                1..48
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 416
gtcattggtg tgaatggtaa gagatactac gcgagctggg cgaaaggc      48

SEQ ID NO: 417        moltype = DNA length = 93
FEATURE              Location/Qualifiers
misc_feature          1..93
                      note = Engineered antibody sequence
source                1..93
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 417
cgattcacca tctccaaaac ctcgtcgacc acgggtggatc tgaaaatgac cagtctgaca 60
accgaggaca cggccaccta ttctgtgcc aga                               93

SEQ ID NO: 418        moltype = length =
SEQUENCE: 418
000

SEQ ID NO: 419        moltype = DNA length = 33
FEATURE              Location/Qualifiers
misc_feature          1..33
                      note = Engineered antibody sequence
source                1..33
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 419
tggggcccg ggaccctcgt caccgtctcg agc                               33

SEQ ID NO: 420        moltype = DNA length = 993
FEATURE              Location/Qualifiers

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misc_feature      1..993
                  note = Engineered antibody sequence
source            1..993
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 420
gcctccacca agggcccac ggtcttcccc ctggcaccct cctccaagag cacctctggg 60
ggcacagcgg ccttgggctg cctgggtcaag gactacttcc ccgaaccggg gacgggtgctg 120
tggaactcag gcgccctgac cagcggcgctg cacaccttcc cggctgtcct acagtccctca 180
ggactctact ccttcagcag cgtgggtgacc gtgccctcca gcagcttggg caccagacc 240
tacatctgca acgtgaatca caagcccagc aacaccaagg tggacaagag agttgagccc 300
aaatcttggt acaaaactca cacatgccca ccgtgccca cactgaact cctgggggga 360
ccgtcagtc tctcttccc cccaaaaccc aaggacacc tcctgatctc ccggaccct 420
gaggtcacat gcgtgggtgg ggacgtgagc cacgaagacc ctgaggtcaa gttcaactgg 480
tacgtggacg gcgtggaggt gcataatgcc aagacaaagc cgcgggagga gcagtacgcc 540
agcacgtacc gtgtgggtcag cgtcctcacc gtctgcacc aggactgggt gaatggcaag 600
gagtacaagt gcaagggtcc caacaaagcc ctcccagccc ccctcgagaa aacctctctc 660
aaagccaaag ggcagccccc agaaccacag gtgtacaccc tgcccccatc ccgggaggag 720
atgaccaaga accagggtcag cctgacctgc ctgggtcaag gcttctatcc cagcgacatc 780
gccgtggagt gggagagcaa tgggcagccg gagaacaact acaagaccac gcctcccgtg 840
ctggactccg acggtctcct ctctctctac agcaagctca ccgtggacaa gacgagggtg 900
cagcagggga acgtcttctc atgctccgtg atgcatgagg ctctgcacaa ccactacacg 960
cagaagagcc tctccctgtc tccgggtaaa tga 993

SEQ ID NO: 421      moltype = AA length = 219
FEATURE             Location/Qualifiers
REGION              1..219
                    note = Engineered antibody sequence
source              1..219
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 421
QVLTQTASPV SPAVGSTVTI NCRASQSVYY NNYLAWYQQK PGQPPKQLIY STSTLASGVS 60
SRFKGSGSGT QFTLTISDVQ CDDAATYYCL GSYDCSNGDC FVFGGTEVV VKRTVAAPSV 120
FIFPPSDEQL KSGTASVVCL LNNFYPREAK VQWKVDNALQ SGNSQESVTE QDSKDYSTYSL 180
SSTLTLSKAD YEKHKVYACE VTHQGLSSPV TKSFNRGEC 219

SEQ ID NO: 422      moltype = AA length = 113
FEATURE             Location/Qualifiers
REGION              1..113
                    note = Engineered antibody sequence
source              1..113
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 422
QVLTQTASPV SPAVGSTVTI NCRASQSVYY NNYLAWYQQK PGQPPKQLIY STSTLASGVS 60
SRFKGSGSGT QFTLTISDVQ CDDAATYYCL GSYDCSNGDC FVFGGTEVV VKR 113

SEQ ID NO: 423      moltype = AA length = 22
FEATURE             Location/Qualifiers
REGION              1..22
                    note = Engineered antibody sequence
source              1..22
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 423
QVLTQTASPV SPAVGSTVTI NC 22

SEQ ID NO: 424      moltype = AA length = 13
FEATURE             Location/Qualifiers
REGION              1..13
                    note = Engineered antibody sequence
source              1..13
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 424
RASQSVYYNN YLA 13

SEQ ID NO: 425      moltype = AA length = 15
FEATURE             Location/Qualifiers
REGION              1..15
                    note = Engineered antibody sequence
source              1..15
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 425
WYQQKPGQPP KQLIY 15

SEQ ID NO: 426      moltype = AA length = 7

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FEATURE	Location/Qualifiers	
REGION	1..7	
	note = Engineered antibody sequence	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 426		
STSTLAS		7
SEQ ID NO: 427	moltype = AA length = 32	
FEATURE	Location/Qualifiers	
REGION	1..32	
	note = Engineered antibody sequence	
source	1..32	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 427		
GVSSRFKGS SGTQFTLTIS DVQCDDAATY YC		32
SEQ ID NO: 428	moltype = AA length = 13	
FEATURE	Location/Qualifiers	
REGION	1..13	
	note = Engineered antibody sequence	
source	1..13	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 428		
LGSYDCSNGD CFV		13
SEQ ID NO: 429	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Engineered antibody sequence	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 429		
FGGGTEVVVK R		11
SEQ ID NO: 430	moltype = AA length = 106	
FEATURE	Location/Qualifiers	
REGION	1..106	
	note = Engineered antibody sequence	
source	1..106	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 430		
TVAAPSVFIF PPSDEQLKSG TASVVCLLNN FYPREAKVQW KVDNALQSGN SQESVTEQDS		60
KDSTYLSLST LTLSKADYK HKVYACEVTH QGLSSPVTKS FNRGEC		106
SEQ ID NO: 431	moltype = DNA length = 660	
FEATURE	Location/Qualifiers	
misc_feature	1..660	
	note = Engineered antibody sequence	
source	1..660	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 431		
caggtgctga cccagactgc atcccccgctg tctccagctg tgggaagcac agtcaccatc		60
aattgccggg ccagtcagag tgtttattat aacaactacc tagcctggta tcagcagaaa		120
ccagggcagc ctcccaagca actgatctat tctacatcca ctctggcatc tgggggtctca		180
tcgcgggtca aaggcagtg atctgggaca cagttcactc tcaccatcag cgacgtgcag		240
tgtgacgatg ctgccactta ctactgtcta ggcagttatg attgtagtaa tgggtgattgt		300
tttgttttcg gcggaggagc cgagggtggtg gtcaaacgta cggtggctgc accatctgtc		360
ttcatcttcc cgccatctga tgagcagttg aaatctggaa ctgcctctgt tgtgtgctctg		420
ctgaataact tctatcccag agaggccaaa gtacagtgga aggtggataa cgccctccaa		480
tcgggtaact cccaggagag tgtcacagag caggacagca aggacagcac ctacagcctc		540
agcagcacc tgacgctgag caaagcagac tacgagaac acaaagtcta cgctgcgaa		600
gtcaccatc agggcctgag ctgcgccgtc acaaagagct tcaacagggg agagtgttag		660
SEQ ID NO: 432	moltype = DNA length = 339	
FEATURE	Location/Qualifiers	
misc_feature	1..339	
	note = Engineered antibody sequence	
source	1..339	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 432		
caggtgctga cccagactgc atcccccgctg tctccagctg tgggaagcac agtcaccatc		60

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aattgccggg ccagtcagag tgtttattat aacaactacc tagcctggta tcagcagaaa 120
ccagggcagc ctcccaagca actgatctat tctacatcca ctctggcatc tggggtotca 180
tcgcggttca aaggcagtg atctgggaca cagttcactc tcaccatcag cgacgtgcag 240
tgtgacgatg ctgccactta ctactgtcta ggcagttatg attgtagtaa tggtgattgt 300
tttgttttcg gcggagggac cgagggtggg gtcaaacgt 339

SEQ ID NO: 433      moltype = DNA length = 66
FEATURE            Location/Qualifiers
misc_feature        1..66
                    note = Engineered antibody sequence
source              1..66
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 433
cagggtgctga cccagactgc atcccccggtg tctccagctg tgggaagcac agtcaccatc 60
aattgc                                           66

SEQ ID NO: 434      moltype = DNA length = 39
FEATURE            Location/Qualifiers
misc_feature        1..39
                    note = Engineered antibody sequence
source              1..39
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 434
cgggccagtc agagtgttta ttataacaac tacctagcc 39

SEQ ID NO: 435      moltype = DNA length = 45
FEATURE            Location/Qualifiers
misc_feature        1..45
                    note = Engineered antibody sequence
source              1..45
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 435
tggatcagc agaaaccagg gcagcctccc aagcaactga tctat 45

SEQ ID NO: 436      moltype = DNA length = 21
FEATURE            Location/Qualifiers
misc_feature        1..21
                    note = Engineered antibody sequence
source              1..21
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 436
tctacatcca ctctggcatc t 21

SEQ ID NO: 437      moltype = DNA length = 96
FEATURE            Location/Qualifiers
misc_feature        1..96
                    note = Engineered antibody sequence
source              1..96
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 437
ggggtctcat cgcggttcaa aggcagtgga tctgggacac agttcactct caccatcagc 60
gacgtgcagt gtgacgatgc tgccacttac tactgt 96

SEQ ID NO: 438      moltype = DNA length = 39
FEATURE            Location/Qualifiers
misc_feature        1..39
                    note = Engineered antibody sequence
source              1..39
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 438
ctaggcagtt atgattgtag taatgggtgat tgttttgtt 39

SEQ ID NO: 439      moltype = DNA length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = Engineered antibody sequence
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 439
ttcggcggag ggaccgaggt ggtggtcaaa cgt 33

SEQ ID NO: 440      moltype = DNA length = 321

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FEATURE	Location/Qualifiers		
misc_feature	1..321	note = Engineered antibody sequence	
source	1..321	mol_type = other DNA	
	organism = synthetic construct		
SEQUENCE: 440			
acggtggctg	caccatctgt	cttcattcttc	ccgccatctg
atgagcagtt	gaaatctgga	60	
actgcctctg	ttgtgtgcct	gctgaataac	ttctatccca
gagaggccaa	agtacagtgg	120	
aaggtggata	acgccctcca	atcgggtaac	ttcccaggaga
gtgtcacaga	gcaggacagc	180	
aaggacagca	cctacagcct	cagcagcacc	ctgacgctga
gcaaagcaga	ctacagagaaa	240	
cacaaagtct	acgcctgcga	agtcacccat	cagggcctga
gctcgcccg	cacaagagc	300	
ttcaacaggg	gagagtgtta	g	321
SEQ ID NO: 441	moltype = AA length = 441		
FEATURE	Location/Qualifiers		
REGION	1..441	note = Engineered antibody sequence	
source	1..441	mol_type = protein	
	organism = synthetic construct		
SEQUENCE: 441			
EVQLVESGGG	LVPQGGSLRL	SCAVSGIDVT	NYMQWVRQA
PGKLEWVG	IGVNGKRYIA	60	
SWAKGRFTIS	RDNSKTTVYL	QMNSLRAEDT	AVYFCARGDI
WGQGLTVTVS	SASTKGPSVF	120	
PLAPSSKSTS	GGTAALGCLV	KDYFPEPVTV	SWNSGALTSG
VHTFPAVLQS	SGLYSLSSV	180	
TVPSSSLGTQ	TYICNVNHKP	SENTKVDKRV	E
PKDTLMISRT	PEVTCVVVDV	SHEDPEVKFN	WYVDGVEVHN
AKTKPREEQY	ASTYRVVSVL	300	
TVLHQDWLNG	KEYKCKVSNK	ALPAPIEIKTI	SKAKGQPREP
QVYTLPPSRE	EMTKNQVSLT	360	
CLVKGFYPSD	IAVEWESNQ	PENNYKTTTP	VLDSDGSFFL
YSKLTVDKSR	WQQGNVPSCS	420	
VMHEALHNNH	TQKSLSLSPG	K	441
SEQ ID NO: 442	moltype = AA length = 111		
FEATURE	Location/Qualifiers		
REGION	1..111	note = Engineered antibody sequence	
source	1..111	mol_type = protein	
	organism = synthetic construct		
SEQUENCE: 442			
EVQLVESGGG	LVPQGGSLRL	SCAVSGIDVT	NYMQWVRQA
PGKLEWVG	IGVNGKRYIA	60	
SWAKGRFTIS	RDNSKTTVYL	QMNSLRAEDT	AVYFCARGDI
WGQGLTVTVS	S	111	
SEQ ID NO: 443	moltype = AA length = 30		
FEATURE	Location/Qualifiers		
REGION	1..30	note = Engineered antibody sequence	
source	1..30	mol_type = protein	
	organism = synthetic construct		
SEQUENCE: 443			
EVQLVESGGG	LVPQGGSLRL	SCAVSGIDVT	30
SEQ ID NO: 444	moltype = AA length = 5		
FEATURE	Location/Qualifiers		
REGION	1..5	note = Engineered antibody sequence	
source	1..5	mol_type = protein	
	organism = synthetic construct		
SEQUENCE: 444			
NYMQ			5
SEQ ID NO: 445	moltype = AA length = 14		
FEATURE	Location/Qualifiers		
REGION	1..14	note = Engineered antibody sequence	
source	1..14	mol_type = protein	
	organism = synthetic construct		
SEQUENCE: 445			
WVRQAPGKGL	EWVG		14
SEQ ID NO: 446	moltype = AA length = 16		
FEATURE	Location/Qualifiers		
REGION	1..16	note = Engineered antibody sequence	
source	1..16	mol_type = protein	
	organism = synthetic construct		

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SEQUENCE: 446
VIGVNGKRY Y ASWAKG 16

SEQ ID NO: 447 moltype = AA length = 32
FEATURE Location/Qualifiers
REGION 1..32
note = Engineered antibody sequence
source 1..32
mol_type = protein
organism = synthetic construct

SEQUENCE: 447
RFTISRDN SK TTVYLQMN SL RAEDTAVYFC AR 32

SEQ ID NO: 448 moltype = length =
SEQUENCE: 448
000

SEQ ID NO: 449 moltype = AA length = 11
FEATURE Location/Qualifiers
REGION 1..11
note = Engineered antibody sequence
source 1..11
mol_type = protein
organism = synthetic construct

SEQUENCE: 449
WGQGLTVTVS S 11

SEQ ID NO: 450 moltype = AA length = 330
FEATURE Location/Qualifiers
REGION 1..330
note = Engineered antibody sequence
source 1..330
mol_type = protein
organism = synthetic construct

SEQUENCE: 450
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 60
GLYSLSSVVT VPSSSLGTQT YICNVN HKPS NTKV DKRVEP KSCDKTHTCP PCPAPELLGG 120
PSVFLFPPPKP KDTLMISRTPEVTCVVVDVS HEDPEVKFNW YVDGVEVHNA KTKPREEQYA 180
STYRVVSVLT VLNQDWN LGK EYKCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSREE 240
MTKNQVSLTC LVKGFYPSDI AVEWESNGQP ENNYKTTTPV LDSGSGFFLY SKLTVDKSRW 300
QQGNVFSCSV MHEALHNHYT QKSLSLSPGK 330

SEQ ID NO: 451 moltype = DNA length = 1326
FEATURE Location/Qualifiers
misc_feature 1..1326
note = Engineered antibody sequence
source 1..1326
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 451
gaggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctgggggggc cctgagactc 60
tcctgtgcag tctctggaat cgacgtcact aactactaca tgcaatgggt ccgtcaggct 120
ccagggaagg ggctggagtg ggtcggagtc attggtgtga atggtgaag atactacgcg 180
agctgggcga aaggccgatt caccatctcc agagacaatt ccaagaccac ggtgatctt 240
caaatgaaca gctgagagc tgaggacact gctgtgtatt tctgtgccag aggggacatc 300
tggggccaag ggaccctcgt caccgtctcg agcgcctcca ccaagggccc atcggtcttc 360
cccctggcac cctcctccaa gagcacctct gggggcacag cggccctggg ctgcctggtc 420
aaggactact tcccgaacc ggtgacggtg tctgtggaact caggcgccct gaccagcggc 480
gtgcacacct tcccggctgt cctacagtc tccaggactct actccctcag cagcgtggtg 540
accgtgccct ccagcagctt gggcacccag acctacatct gcaacgtgaa tcacaagccc 600
agcaacacca aggtggacaa gagagttgag cccaaatctt gtgacaaaac tcacacatgc 660
ccaccgtgcc cagcacctga actcctgggg ggaccgtcag tcttctcttt cccccaaaa 720
cccaaggaca cctcatgat ctcccgacc cctgaggta catgcgtggt ggtggacgtg 780
agccacgaag cctgaggt caagtcaac tggtagctgg acggcgtgga ggtgcataat 840
gccaagacaa agccgcggga ggagcagtag gccagcacgt accgtgtggt cagcgtcttc 900
accgtcctgc accaggactg gctgaatggc aaggagtaca agtgcaaggt ctccaacaaa 960
gcccctccag ccccatcga gaaaaccatc tccaaagcca aagggcagcc ccgagaacca 1020
caggtgtaca cctgcctccc atcccgggag gagatgacca agaaccaggt cagcctgacc 1080
tgccctggtc aaggcttcta tcccagcgac atcgccgtgg agtgggagag caatgggcag 1140
cgggagaaca actacaagac cagcctccc gtgctggact ccgacggctc cttcttcttc 1200
tacagcaagc tcaccgtgga caagagcagg tggcagcagg ggaacgtctt ctcagctcc 1260
gtgatgcatg aggtctctga caaccactac acgcagaaga gcctctccct gtctccgggt 1320
aatga 1326

SEQ ID NO: 452 moltype = DNA length = 333
FEATURE Location/Qualifiers
misc_feature 1..333
note = Engineered antibody sequence
source 1..333

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mol_type = other DNA
organism = synthetic construct

SEQUENCE: 452
gaggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctgggggggc cctgagactc 60
tcctgtgcag tctctggaat cgacgtcact aactactaca tgcaatgggt ccgtcaggct 120
ccagggaagg ggctggagtg ggtcggagtc attggtgtga atggtaagag atactacgcg 180
agctgggcga aaggccgatt caccatctcc agagacaatt ccaagaccac ggtgtatctt 240
caaatgaaca gcctgagagc tgaggacact gctgtgtatt tctgtgccag aggggacatc 300
tggggccaag ggaccctcgt caccgtctcg agc 333

SEQ ID NO: 453      moltype = DNA length = 90
FEATURE            Location/Qualifiers
misc_feature        1..90
                    note = Engineered antibody sequence
source              1..90
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 453
gaggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctgggggggc cctgagactc 60
tcctgtgcag tctctggaat cgacgtcact 90

SEQ ID NO: 454      moltype = DNA length = 15
FEATURE            Location/Qualifiers
misc_feature        1..15
                    note = Engineered antibody sequence
source              1..15
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 454
aactactaca tgcaa 15

SEQ ID NO: 455      moltype = DNA length = 42
FEATURE            Location/Qualifiers
misc_feature        1..42
                    note = Engineered antibody sequence
source              1..42
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 455
tgggtccgtc aggctccagg gaaggggctg gagtgggtcg ga 42

SEQ ID NO: 456      moltype = DNA length = 48
FEATURE            Location/Qualifiers
misc_feature        1..48
                    note = Engineered antibody sequence
source              1..48
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 456
gtcattggtg tgaatggtaa gagatactac gcgagctggg cgaaaggc 48

SEQ ID NO: 457      moltype = DNA length = 96
FEATURE            Location/Qualifiers
misc_feature        1..96
                    note = Engineered antibody sequence
source              1..96
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 457
cgattcacca tctccagaga caattccaag accacgggtg atcttcaaat gaacagcctg 60
agagctgagg aactgctgtg gtatttctgt gccaga 96

SEQ ID NO: 458      moltype = length =
SEQUENCE: 458
000

SEQ ID NO: 459      moltype = DNA length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = Engineered antibody sequence
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 459
tggggccaag ggaccctcgt caccgtctcg agc 33

SEQ ID NO: 460      moltype = DNA length = 993
FEATURE            Location/Qualifiers
misc_feature        1..993

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source          note = Engineered antibody sequence
                1..993
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 460
gctccacca agggcccatc ggtcttcccc ctggcaccct cctccaagag cacctctggg 60
ggcacagcgg ccttgggctg cctgggtcaag gactacttcc ccgaaccggt gacggtgtcg 120
tggaactcag gcgccttgac cagcggcggtg cacaccttcc cggtgtctct acagtctca 180
ggactctact ccctcagcag cgtgggtgacc gtgccctcca gcagcttggg caccagacc 240
tacatctgca acgtgaatca caagccagc aacaccaagg tggacaagag agttgagccc 300
aaatcttgtg acaaaaactca cacatgccc cegtgccag cacctgaact cctgggggga 360
ccgtcagttc tcctcttccc cccaaaaccc aaggacaccc toatgatctc ccggaccct 420
gaggtcacat gcgtgggtgt ggacgtgagc cacgaagacc ctgaggtcaa gttcaactgg 480
tacgtggagc gcgtggaggt gcataatgcc aagacaaagc cgcgggagga gcagtacgcc 540
agcacgtacc gtgtgggtcag cgtcctcacc gtccctgcacc aggactggct gaatggcaag 600
gagtacaagt gcaaggtctc caacaaagcc ctcccagccc ccacgagaa aacctctcc 660
aaagccaaag ggcagccccc agaaccacag gtgtacaccc tgccccatc ccgggaggag 720
atgaccaaga accaggtcag cctgacctgc ctgggtcaaag gcttctatcc cagcgacatc 780
gccgtggagt gggagagcaa tgggcagccg gagaacaact acaagaccac gcctcccggt 840
ctggactcag acggctctct ctctctctac agcaagctca ccgtggacaa gagcagggtg 900
cagcagggga acgtctctct atgctccgtg atgcatgagg ctctgcacaa ccactacacg 960
cagaagagcc tctccctgtc tccgggtaaa tga 993

SEQ ID NO: 461      moltype = AA length = 219
FEATURE            Location/Qualifiers
REGION            1..219
                  note = Engineered antibody sequence
source            1..219
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 461
QVLTQSPSSL SASVGDRVTI NCRASQSVYY NNYLAWYQQK PGKVPKQLIY STSTLASGVP 60
SRFSGSGSGT DFTLTISLQ PEDVATYYCL GSYDCSNGDC FVFGGGTKVE IKRTVAAPSV 120
FIFFPSDEQL KSGTASVVCL LNNFYPREAK VQWKVDNALQ SGNSQESVTE QDSKDYSTYSL 180
SSTLTLSKAD YEKHKVYACE VTHQGLSSPV TKSPNRGEC 219

SEQ ID NO: 462      moltype = AA length = 113
FEATURE            Location/Qualifiers
REGION            1..113
                  note = Engineered antibody sequence
source            1..113
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 462
QVLTQSPSSL SASVGDRVTI NCRASQSVYY NNYLAWYQQK PGKVPKQLIY STSTLASGVP 60
SRFSGSGSGT DFTLTISLQ PEDVATYYCL GSYDCSNGDC FVFGGGTKVE IKR 113

SEQ ID NO: 463      moltype = AA length = 22
FEATURE            Location/Qualifiers
REGION            1..22
                  note = Engineered antibody sequence
source            1..22
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 463
QVLTQSPSSL SASVGDRVTI NC 22

SEQ ID NO: 464      moltype = AA length = 13
FEATURE            Location/Qualifiers
REGION            1..13
                  note = Engineered antibody sequence
source            1..13
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 464
RASQSVYYNN YLA 13

SEQ ID NO: 465      moltype = AA length = 15
FEATURE            Location/Qualifiers
REGION            1..15
                  note = Engineered antibody sequence
source            1..15
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 465
WYQKPGKVP KQLIY 15

SEQ ID NO: 466      moltype = AA length = 7
FEATURE            Location/Qualifiers

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REGION	1..7	
	note = Engineered antibody sequence	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 466		
STSTLAS		7
SEQ ID NO: 467	moltype = AA length = 32	
FEATURE	Location/Qualifiers	
REGION	1..32	
	note = Engineered antibody sequence	
source	1..32	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 467		
GVPSRFGSGG SGTDFTLTIS SLQPEDVATY YC		32
SEQ ID NO: 468	moltype = AA length = 13	
FEATURE	Location/Qualifiers	
REGION	1..13	
	note = Engineered antibody sequence	
source	1..13	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 468		
LGSYDCSNGD CFV		13
SEQ ID NO: 469	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Engineered antibody sequence	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 469		
FGGGTKVEIK R		11
SEQ ID NO: 470	moltype = AA length = 106	
FEATURE	Location/Qualifiers	
REGION	1..106	
	note = Engineered antibody sequence	
source	1..106	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 470		
TVAAPSVFIF PPSDEQLKSG TASVVCLLNN FYPREAKVQW KVDNALQSGN SQESVTEQDS		60
KDSTYLSST LTLSKADYEK HKVYACEVTH QGLSSPVTKS FNRGEC		106
SEQ ID NO: 471	moltype = DNA length = 660	
FEATURE	Location/Qualifiers	
misc_feature	1..660	
	note = Engineered antibody sequence	
source	1..660	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 471		
caagtgtctga cccagctctcc atcctccctg tctgcatctg taggagacag agtcaccatc		60
aattgccggg ccagtcagag tgtttactat aacaactacc tagcctggta tcagcagaaa		120
ccagggaag ttcctaagca actgatctat tctacatcca ctctggcatc tggggtocca		180
tctcggttca gtggcagtg atctgggaca gatttcactc tcaccatcag cagcctgcag		240
cctgaagatg ttgcaactta ttactgtctg ggcagttatg attgtagtaa tggtgattgt		300
tttgttttcg gcggaggaac caaggtggaa atcaaacgta cggtggctgc accatctgtc		360
ttcatcttcc cgccatctga tgagcagttg aaatctggaa ctgcctctgt tgtgtgctg		420
ctgaataact tctatcccag agaggccaaa gtacagtggg aggtggataa cgccctccaa		480
tcgggtaact cccaggagag tgtcacagag caggacagca aggacagcac ctacagcctc		540
agcagcaccg tgacgctgag caaagcagac tacgagaaac acaaagtcta cgctgcgaa		600
gtcaccatc aggcctgag ctgcgccgtc acaaagagct tcaacagggg agagtgttag		660
SEQ ID NO: 472	moltype = DNA length = 339	
FEATURE	Location/Qualifiers	
misc_feature	1..339	
	note = Engineered antibody sequence	
source	1..339	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 472		
caagtgtctga cccagctctcc atcctccctg tctgcatctg taggagacag agtcaccatc		60
aattgccggg ccagtcagag tgtttactat aacaactacc tagcctggta tcagcagaaa		120

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ccagggaag ttcctaagca actgatctat tctacatcca ctctggcatc tggggtccca 180
tctcgtttca gtggcagtg atctgggaca gatttcaactc tcaccatcag cagcctgcag 240
cctgaagatg ttgcaactta ttactgtctg ggcagttatg attgtagtaa tgggtgattgt 300
ttgttttctg gcgagggaac caaggtggaa atcaaacgt 339

```

```

SEQ ID NO: 473      moltype = DNA length = 66
FEATURE            Location/Qualifiers
misc_feature        1..66
                    note = Engineered antibody sequence
source              1..66
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 473
caagtgtga cccagttctc atcctccttg tctgcatctg taggagacag agtcaccatc 60
aattgc 66

```

```

SEQ ID NO: 474      moltype = DNA length = 39
FEATURE            Location/Qualifiers
misc_feature        1..39
                    note = Engineered antibody sequence
source              1..39
                    mol_type = other DNA
                    organism = synthetic construct

```

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SEQUENCE: 474
cgggccagtc agagtgttta ctataacaac tacctagcc 39

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```

SEQ ID NO: 475      moltype = DNA length = 45
FEATURE            Location/Qualifiers
misc_feature        1..45
                    note = Engineered antibody sequence
source              1..45
                    mol_type = other DNA
                    organism = synthetic construct

```

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SEQUENCE: 475
tggatcagc agaaaccagg gaaagttcct aagcaactga tctat 45

```

```

SEQ ID NO: 476      moltype = DNA length = 21
FEATURE            Location/Qualifiers
misc_feature        1..21
                    note = Engineered antibody sequence
source              1..21
                    mol_type = other DNA
                    organism = synthetic construct

```

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SEQUENCE: 476
tctacatcca ctctggcatc t 21

```

```

SEQ ID NO: 477      moltype = DNA length = 96
FEATURE            Location/Qualifiers
misc_feature        1..96
                    note = Engineered antibody sequence
source              1..96
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 477
ggggtcccat ctcgtttcag tggcagtgga tctgggacag atttcaactct caccatcagc 60
agcctgcagc ctgaagatgt tgaacttat tactgt 96

```

```

SEQ ID NO: 478      moltype = DNA length = 39
FEATURE            Location/Qualifiers
misc_feature        1..39
                    note = Engineered antibody sequence
source              1..39
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 478
ctgggcagtt atgattgtag taatgggtgat tgttttgtt 39

```

```

SEQ ID NO: 479      moltype = DNA length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = Engineered antibody sequence
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 479
ttcggcggag gaaccaaggt ggaatcaaaa cgt 33

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```

SEQ ID NO: 480      moltype = DNA length = 321
FEATURE            Location/Qualifiers

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misc_feature      1..321
                  note = Engineered antibody sequence
source            1..321
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 480
acgggtggctg caccatctgt cttcatcttc cggccatctg atgagcagtt gaaatctgga 60
actgcctctg ttgtgtgect gctgaataac ttctatccca gagaggccaa agtacagtgg 120
aagggtgata acgcctctca atcgggtaac tcccaggaga gtgtcacaga gcaggacagc 180
aaggacagca cctacagcct cagcagcacc ctgacgctga gcaaagcaga ctacgagaaa 240
cacaaagtct acgcctgcga agtcacccat cagggcctga gctcgcccg cacaagagagc 300
ttcaacaggg gagagtgtta g                                     321

SEQ ID NO: 481      moltype = AA length = 441
FEATURE            Location/Qualifiers
REGION            1..441
                  note = Engineered antibody sequence
source            1..441
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 481
QSVEESGGGL VQPEGSLTLT CTASGDFDSS NAMWWVRQAP GKGLEWIGCI YNGDGSTYYA 60
SWVNGRFSIS KTSSTTVTLQ LNSLTVADTA TYCARDLDL WPGGTLVTVS SASTKGPSVF 120
PLAPSSKSTS GGTAALGCLV KDYPPEPVTV SWNSGALTSG VHTFPAYLQS SGLYSLSSV 180
TVPSSSLGTQ TYICNVNHKP SNTKVDKRVE PKCDKHTC PPCPAPELLG GPSVFLFPPK 240
PKDTLMISRT PEVTCVVVDV SHEDPEVKFN WYVDGVEVHN AKTKPREEQY ASTYRVVSVL 300
TVLHQDWLNG KEYKCKVSNK ALPAPIEKTI SKAKGQPREP QVYTLPPSRE EMTKNQVSLT 360
CLVKGFPYPSD IAVEWESNGQ PENNYKTTTP VLDSDGSFPL YSKLTVDKSR WQQGNVFS 420
VMHEALHNHY TQKSLSLSPG K                                     441

SEQ ID NO: 482      moltype = AA length = 111
FEATURE            Location/Qualifiers
REGION            1..111
                  note = Engineered antibody sequence
source            1..111
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 482
QSVEESGGGL VQPEGSLTLT CTASGDFDSS NAMWWVRQAP GKGLEWIGCI YNGDGSTYYA 60
SWVNGRFSIS KTSSTTVTLQ LNSLTVADTA TYCARDLDL WPGGTLVTVS S          111

SEQ ID NO: 483      moltype = AA length = 29
FEATURE            Location/Qualifiers
REGION            1..29
                  note = Engineered antibody sequence
source            1..29
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 483
QSVEESGGGL VQPEGSLTLT CTASGDFDSS                                     29

SEQ ID NO: 484      moltype = AA length = 5
FEATURE            Location/Qualifiers
REGION            1..5
                  note = Engineered antibody sequence
source            1..5
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 484
SNAMW                                                     5

SEQ ID NO: 485      moltype = AA length = 14
FEATURE            Location/Qualifiers
REGION            1..14
                  note = Engineered antibody sequence
source            1..14
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 485
WVRQAPGKGL EWIG                                         14

SEQ ID NO: 486      moltype = AA length = 17
FEATURE            Location/Qualifiers
REGION            1..17
                  note = Engineered antibody sequence
source            1..17
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 486

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CIYNGDGSTY YASWVNG 17

SEQ ID NO: 487 moltype = AA length = 31
 FEATURE Location/Qualifiers
 REGION 1..31
 note = Engineered antibody sequence
 source 1..31
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 487
 RFSISKTSST TVTLQLNSLT VADTATYYCA R 31

SEQ ID NO: 488 moltype = AA length = 4
 FEATURE Location/Qualifiers
 REGION 1..4
 note = Engineered antibody sequence
 source 1..4
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 488
 DLDL 4

SEQ ID NO: 489 moltype = AA length = 11
 FEATURE Location/Qualifiers
 REGION 1..11
 note = Engineered antibody sequence
 source 1..11
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 489
 WGPGLVTVS S 11

SEQ ID NO: 490 moltype = AA length = 330
 FEATURE Location/Qualifiers
 REGION 1..330
 note = Engineered antibody sequence
 source 1..330
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 490
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTPPAVLQSS 60
 GLYSLSSVVT VPSSSLGTQT YICNVNHNKPS NTKVDKRVEP KSCDKTHTCP PCPAPELLGG 120
 PSVFLFPPKP KDTLMISRTPEVTCVVVDVS HEDPEVKFNW YVDGVEVHNA KTKPREEQYA 180
 STYRVVSVLT VLHQDWLNGK EYKCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSREE 240
 MTKNQVSLTLC LVKGFYPSDI AVEWESNGQP ENNYKTTTPV LDSDGSFFLY SKLTVDKSRW 300
 QQGNVFSCSV MHEALHNHYT QKSLSLSPGK 330

SEQ ID NO: 491 moltype = DNA length = 1326
 FEATURE Location/Qualifiers
 misc_feature 1..1326
 note = Engineered antibody sequence
 source 1..1326
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 491
 cagtcggtgg aggagtcggtg gggaggcctg gtccagcctg agggatccct gacactcacc 60
 tgcacagcct ctggattcga cttcagtagc aatgcaatgt ggtgggtccg ccaggctcca 120
 gggaaggggc tggagtggtg cggatgcatt tacaatggtg atggcagcac atactacgcg 180
 agctgggtga atggccgatt ctccatctcc aaaacctcgt cgaccacggt gactctgcaa 240
 ctgaatagtc tgacagtcgc ggacacggcc acgtattatt gtgcgagaga tcttgacttg 300
 tggggcccggt gcaccctcgt caccgtctcg agcgccctcca ccaaggggccc atcggtcttc 360
 cccctggcac cctcctccaa gagcacctct gggggcacag cggccctggg ctgcctggtc 420
 aaggactact tccccgaacc ggtgacggtg tctgtggaact caggcgccct gaccagcggc 480
 gtgcacacct tcccggctgt cctacagtc tccagactct actccctcag cagcgtggtg 540
 accgtgcctt ccagcagctt gggcaccacg acctacatct gcaacgtgaa tcacaagccc 600
 agcaacacca aggtggacaa gagagttgag cccaaatctt gtgacaaaac tcacacatgc 660
 ccaccgtgcc cagcacctga actcctgggg ggaccgtcag tcttctctct ccccccaaaa 720
 cccaaggaca cctccatgat ctcccggacc cctgaggtca catgcgtggt ggtggacgtg 780
 agccaagcga accctgaggt caagttcaac tggtagctgg acggcgtgga ggtgcataat 840
 gccaaagcaa agccgcggga ggagcagtag gccagcagct accgtgtggt cagcgtcctc 900
 accgtcctgc accaggactg gctgaatggc aaggagtaga agtgcaaggt ctccaacaaa 960
 gccctccag ccccatcga gaaaaccatc tccaaagcca aagggcagcc ccgagaacca 1020
 caggtgtaca ccttgccccc atcccgagg gagatgacca agaaccaggt cagcctgacc 1080
 tgctgtgta aaggtctcta tcccagcgac atcgccgtgg agtgggagag caatggggcag 1140
 cgggagaaca actacaagac cagcctccc gtgctggact ccgacggctc cttcttctc 1200
 tacagcaagc tcaccgtgga caagagcagg tggcagcagg ggaacgtctt ctcatgctcc 1260
 gtgatgcatt aggtctctga caaccactac acgcagaaga gcctctcctt gtctccgggt 1320
 aatatga 1326

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SEQ ID NO: 492      moltype = DNA length = 333
FEATURE            Location/Qualifiers
misc_feature        1..333
                    note = Engineered antibody sequence
source              1..333
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 492
cagtcggtgg aggagtcgaggaggcctg gtccagcctg agggatccct gacactcacc 60
tgcacagcct ctggattcga cttcagtagc aatgcaatgt ggtgggtccg ccaggctcca 120
gggaaggggc tggagtggat cggatgcatt tacaatgggt atggcagcac atactacgcg 180
agctgggtga atggccgatt ctccatctcc aaacactcgt cgaccacggt gactctgcaa 240
ctgaatagtc tgacagtcgc ggacacggcc acgtattatt gtgcgagaga tcttgacttg 300
tggggcccg gcacccctcgt caccgtctcg agc 333

SEQ ID NO: 493      moltype = DNA length = 87
FEATURE            Location/Qualifiers
misc_feature        1..87
                    note = Engineered antibody sequence
source              1..87
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 493
cagtcggtgg aggagtcgaggaggcctg gtccagcctg agggatccct gacactcacc 60
tgcacagcct ctggattcga cttcagtc 87

SEQ ID NO: 494      moltype = DNA length = 15
FEATURE            Location/Qualifiers
misc_feature        1..15
                    note = Engineered antibody sequence
source              1..15
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 494
agcaatgcaa tgtgg 15

SEQ ID NO: 495      moltype = DNA length = 42
FEATURE            Location/Qualifiers
misc_feature        1..42
                    note = Engineered antibody sequence
source              1..42
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 495
tgggtccgcc aggetccagg gaaggggctg gagtggatcg ga 42

SEQ ID NO: 496      moltype = DNA length = 51
FEATURE            Location/Qualifiers
misc_feature        1..51
                    note = Engineered antibody sequence
source              1..51
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 496
tgcatttaca atggtgatgg cagcacatac tacgcgagct gggatgaatgg c 51

SEQ ID NO: 497      moltype = DNA length = 93
FEATURE            Location/Qualifiers
misc_feature        1..93
                    note = Engineered antibody sequence
source              1..93
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 497
cgattctcca tctccaaaac ctgcgtcgacc acggtgactc tgcaactgaa tagtctgaca 60
gtcgcgga cggccacgta ttattgtgcg aga 93

SEQ ID NO: 498      moltype = DNA length = 12
FEATURE            Location/Qualifiers
misc_feature        1..12
                    note = Engineered antibody sequence
source              1..12
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 498
gatcttgact tg 12

SEQ ID NO: 499      moltype = DNA length = 33
FEATURE            Location/Qualifiers

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misc_feature      1..33
                  note = Engineered antibody sequence
source            1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 499
tggggcccgg gcaccctcgt caccgtctcg agc                               33

SEQ ID NO: 500      moltype = DNA length = 993
FEATURE            Location/Qualifiers
misc_feature        1..993
                  note = Engineered antibody sequence
source              1..993
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 500
gcctccacca agggcccac cgtcttcccc ctggcaccct cctccaagag cacctctggg  60
ggcacagcgg ccctgggctg cctgggtcaag gactacttcc ccgaaccggt gacgggtgctg 120
tggaactcag gcgccttgac cagcggcgctg cacaccttcc cggctgtcct acagtccctca 180
ggactctact ccctcagcag cgtgggtgacc gtgccctcca gcagcttggg caccagacc 240
tacatctgca acgtgaatca caagcccagc aacaccaagg tggacaagag agttgagccc 300
aaatcttggt acaaaactca cacatgcccc ccgtgcccag cacctgaact cctgggggga 360
ccgtcagtc tctcttcccc cccaaaaccc aaggacaccc tcatgatctc ccggaccct 420
gaggtcacat gcgtgggtgt ggacgtgagc cacgaagacc ctgaggtcaa gttcaactgg 480
taactgtgac gcgtgggtgt gcataatgcc aagacaaagc cgcgggagga gcagtacgcc 540
agcacgtacc gtgtgggtcag cgtctcacc gtccctgacc aggactgggt gaatggcaag 600
gagtacaagt gcaaggtctc caacaaagcc ctcccagccc ccacgagaa aaccatctcc 660
aaagccaaag ggcagccccc agaaccacag gtgtacaccc tgcccccac cgggaggag 720
atgaccaaga accaggtcag cctgacctgc ctgggtcaag gcttctatcc cagcgacatc 780
gccgtggagt gggagagcaa tgggcagccg gagaacaact acaagaccac gcctcccgtg 840
ctggactcgg acgggtcctt ctctctctac agcaagctca ccgtggacaa gagcaggtgg 900
cagcagggga acgtctcttc atgctccgtg atgcatgagg ctctgcacaa ccactacacg 960
cagaagagcc tctccctgtc tccgggtaaa tga                               993

SEQ ID NO: 501      moltype = AA length = 219
FEATURE            Location/Qualifiers
REGION              1..219
                  note = Engineered antibody sequence
source              1..219
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 501
AIVMTQTPSS KSVPVGDTVT INCQASESLY NNNALAWFQQ KPGQPPKRLI YDASKLASGV  60
PSRFSGGGSG TQFTLTISGV QCDDAATYYC GGYRSDSDVG VAFAGGTEVV VKRTVAAPSV 120
FIFPPSDEQL KSGTASVVCL LNNFYFPREK VQWKVDNALQ SGNSQESVTE QDSKDYSTYL 180
SSTLTLSKAD YEKHKVYACE VTHQLSSPV TKSFNRGEC                               219

SEQ ID NO: 502      moltype = AA length = 113
FEATURE            Location/Qualifiers
REGION              1..113
                  note = Engineered antibody sequence
source              1..113
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 502
AIVMTQTPSS KSVPVGDTVT INCQASESLY NNNALAWFQQ KPGQPPKRLI YDASKLASGV  60
PSRFSGGGSG TQFTLTISGV QCDDAATYYC GGYRSDSDVG VAFAGGTEVV VKR                               113

SEQ ID NO: 503      moltype = AA length = 23
FEATURE            Location/Qualifiers
REGION              1..23
                  note = Engineered antibody sequence
source              1..23
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 503
AIVMTQTPSS KSVPVGDTVT INC                               23

SEQ ID NO: 504      moltype = AA length = 13
FEATURE            Location/Qualifiers
REGION              1..13
                  note = Engineered antibody sequence
source              1..13
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 504
QASESLYNNN ALA                               13

SEQ ID NO: 505      moltype = AA length = 15

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FEATURE	Location/Qualifiers	
REGION	1..15	
source	note = Engineered antibody sequence 1..15 mol_type = protein organism = synthetic construct	
SEQUENCE: 505		
WFQQKPGQPP KRLIY		15
SEQ ID NO: 506	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
source	note = Engineered antibody sequence 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 506		
DASKLAS		7
SEQ ID NO: 507	moltype = AA length = 32	
FEATURE	Location/Qualifiers	
REGION	1..32	
source	note = Engineered antibody sequence 1..32 mol_type = protein organism = synthetic construct	
SEQUENCE: 507		
GVPSRFSGGG SGTQFTLTIS GVQCDDAATY YC		32
SEQ ID NO: 508	moltype = AA length = 12	
FEATURE	Location/Qualifiers	
REGION	1..12	
source	note = Engineered antibody sequence 1..12 mol_type = protein organism = synthetic construct	
SEQUENCE: 508		
GGYRSDSVDG VA		12
SEQ ID NO: 509	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
source	note = Engineered antibody sequence 1..11 mol_type = protein organism = synthetic construct	
SEQUENCE: 509		
FAGGTEVVVK R		11
SEQ ID NO: 510	moltype = AA length = 106	
FEATURE	Location/Qualifiers	
REGION	1..106	
source	note = Engineered antibody sequence 1..106 mol_type = protein organism = synthetic construct	
SEQUENCE: 510		
TVAAPSVFIF PPSDEQLKSG TASVVCLLNN FYPREAKVQW KVDNALQSGN SQESVTEQDS		60
KDSTYLSST LTLKADYEK HKVYACEVTH QGLSSPVTKS FNRGEC		106
SEQ ID NO: 511	moltype = DNA length = 660	
FEATURE	Location/Qualifiers	
misc_feature	1..660	
source	note = Engineered antibody sequence 1..660 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 511		
gccatcgtga tgacccagac tccatcttcc aagtctgtcc ctgtgggaga cacagtcacc		60
atcaattgcc aggccagtga gagtctttat aataacaacg ccttggcctg gtttcagcag		120
aaaccagggc agcctcccaa gcgcctgcat tatgatgcat ccaaactggc atctggggtc		180
ccatcgcggt tcagtggcgg tgggtctggg acacagttca ctctcaccat cagtggcgtg		240
cagtgtgacg atgctgccac ttactactgt ggaggctaca gaagtgatag tgttgatggt		300
gttgctttcg ccggaggggac cgaggtggtg gtcaaacgta cggtggtgc accatctgtc		360
ttcatcttcc cgccatctga tgagcagttg aaatctggaa ctgcctctgt tgttgacctg		420
ctgaataact tctatcccaa agaggccaaa gtacagtggg aggtggataa cgccctccaa		480
tcgggtaact cccaggagag tgtcacagag caggacagca aggacagcac ctacagcctc		540
agcagcacc tgacgctgag caaagcagac tacgagaaac acaaagtcta cgccctgcga		600
gtcaccatc agggcctgag ctgcgccgtc acaaagagct tcaacagggg agagtgttag		660

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SEQ ID NO: 512 moltype = DNA length = 339
 FEATURE Location/Qualifiers
 misc_feature 1..339
 note = Engineered antibody sequence
 source 1..339
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 512
 gccatcgtga tgacccagac tccatcttcc aagtctgtcc ctgtgggaga cacagtcacc 60
 atcaattgcc aggcocagtga gagtctttat aataacaacg ccttggcctg gtttcagcag 120
 aaaccagggc agcctcccaa gcgcctgatc tatgatgcat ccaaactggc atctggggtc 180
 ccacgcgggt tcagtggcgg tgggtctggg acacagtcca ctctcaccat cagtggcgtg 240
 cagtgtgacg atgctgccac ttactactgt ggaggctaca gaagtgatag tgttgatggt 300
 gttgctttcg ccgaggggac cgaggtggtg gtcaaacgt 339

SEQ ID NO: 513 moltype = DNA length = 69
 FEATURE Location/Qualifiers
 misc_feature 1..69
 note = Engineered antibody sequence
 source 1..69
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 513
 gccatcgtga tgacccagac tccatcttcc aagtctgtcc ctgtgggaga cacagtcacc 60
 atcaattgc 69

SEQ ID NO: 514 moltype = DNA length = 39
 FEATURE Location/Qualifiers
 misc_feature 1..39
 note = Engineered antibody sequence
 source 1..39
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 514
 caggccagtg agagtcttta taataacaac gccttggcc 39

SEQ ID NO: 515 moltype = DNA length = 45
 FEATURE Location/Qualifiers
 misc_feature 1..45
 note = Engineered antibody sequence
 source 1..45
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 515
 tggtttcagc agaaaccagg gcagcctccc aagcgctga tctat 45

SEQ ID NO: 516 moltype = DNA length = 21
 FEATURE Location/Qualifiers
 misc_feature 1..21
 note = Engineered antibody sequence
 source 1..21
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 516
 gatgcatcca aactggcatc t 21

SEQ ID NO: 517 moltype = DNA length = 96
 FEATURE Location/Qualifiers
 misc_feature 1..96
 note = Engineered antibody sequence
 source 1..96
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 517
 ggggtcccat cgcggttcag tggcggtggg tctgggacac agttcaactct caccatcagt 60
 ggcgtgcagt gtgacgatgc tgccacttac tactgt 96

SEQ ID NO: 518 moltype = DNA length = 36
 FEATURE Location/Qualifiers
 misc_feature 1..36
 note = Engineered antibody sequence
 source 1..36
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 518
 ggaggctaca gaagtgatag tgttgatggt gttgct 36

SEQ ID NO: 519 moltype = DNA length = 33

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FEATURE Location/Qualifiers
 misc_feature 1..33
 note = Engineered antibody sequence
 source 1..33
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 519
 ttcgccggag ggaccgaggt ggtgggtcaaa cgt 33

SEQ ID NO: 520 moltype = DNA length = 321
 FEATURE Location/Qualifiers
 misc_feature 1..321
 note = Engineered antibody sequence
 source 1..321
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 520
 acgggtggctg caccatctgt cttcatcttc ccgccatctg atgagcagtt gaaatctgga 60
 actgcctctg ttgtgtgctt gctgaataac ttctatccca gagaggccaa agtacagtgg 120
 aagggtgata acgcctctca atcgggtaac tcccaggaga gtgtcacaga gcaggacagc 180
 aaggacagca cctacagcct cagcagcacc ctgacgctga gcaaagcaga ctacagagaa 240
 cacaaagtct acgcctgcga agtcacccat cagggcctga gctcgcccg cacaagagc 300
 ttcaacaggg gagagtgtta g 321

SEQ ID NO: 521 moltype = AA length = 441
 FEATURE Location/Qualifiers
 REGION 1..441
 note = Engineered antibody sequence
 source 1..441
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 521
 EVQLVESGGG LVQPGGSLRL SCAVSGIGLS SYMQWVRQA PGKGLEWVG IGS DGKTYIA 60
 TWAKGRFTIS RDNSKTTVYL QMNSLRAEDT AVYFCTRGDI WQGT LVTVS SASTKGPSVF 120
 PLAPSSKSTS GGTAAAGCLV KDFPEPPTV SWNSGALTSG VHTFPAVLQS SGLYSLSSV 180
 TVPSSSLGTQ TYICNVNHPK SNTKVDARVE PKSCDKTHTC PPCPAPELLG GPSVFLFPPK 240
 PKDTLMISRT PEVTCVVVDV SHEDPEVKFN WYVDGVEVHN AKTKPREEQY ASTYRVVSVL 300
 TVLHQDWLNG KEYKCKVSNK ALPAPIEKTI SKAKGQPREP QVYTLPPSRE EMTKNQVSLT 360
 CLVKGFPYPSD IAVEWESNGQ PENNYKTTTP VLDS DGSFFL YSKLTVDKSR WQQGNV FSCS 420
 VMHEALHNHY TQKSLSLSPG K 441

SEQ ID NO: 522 moltype = AA length = 111
 FEATURE Location/Qualifiers
 REGION 1..111
 note = Engineered antibody sequence
 source 1..111
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 522
 EVQLVESGGG LVQPGGSLRL SCAVSGIGLS SYMQWVRQA PGKGLEWVG IGS DGKTYIA 60
 TWAKGRFTIS RDNSKTTVYL QMNSLRAEDT AVYFCTRGDI WQGT LVTVS S 111

SEQ ID NO: 523 moltype = AA length = 30
 FEATURE Location/Qualifiers
 REGION 1..30
 note = Engineered antibody sequence
 source 1..30
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 523
 EVQLVESGGG LVQPGGSLRL SCAVSGIGLS 30

SEQ ID NO: 524 moltype = AA length = 5
 FEATURE Location/Qualifiers
 REGION 1..5
 note = Engineered antibody sequence
 source 1..5
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 524
 SYMQ 5

SEQ ID NO: 525 moltype = AA length = 14
 FEATURE Location/Qualifiers
 REGION 1..14
 note = Engineered antibody sequence
 source 1..14
 mol_type = protein
 organism = synthetic construct

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SEQUENCE: 525
WVRQAPGKGL EWVG 14

SEQ ID NO: 526 moltype = AA length = 16
FEATURE Location/Qualifiers
REGION 1..16
 note = Engineered antibody sequence
source 1..16
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 526
VIGSDGKTTY ATWAKG 16

SEQ ID NO: 527 moltype = AA length = 32
FEATURE Location/Qualifiers
REGION 1..32
 note = Engineered antibody sequence
source 1..32
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 527
RPTISRDN SK TTVYLQMN SL RAEDTAVYFC TR 32

SEQ ID NO: 528 moltype = length =
SEQUENCE: 528
000

SEQ ID NO: 529 moltype = AA length = 11
FEATURE Location/Qualifiers
REGION 1..11
 note = Engineered antibody sequence
source 1..11
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 529
WGQGTLLVTVS S 11

SEQ ID NO: 530 moltype = AA length = 330
FEATURE Location/Qualifiers
REGION 1..330
 note = Engineered antibody sequence
source 1..330
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 530
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 60
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDARVEP KSCDKTHTCP PCPAPELLGG 120
PSVFLFPPPKP KDTLMISRTP EVTCVVVDVS HEDPEVKFNW YVDGVEVHNA KTKPREEQYA 180
STYRVVSVLT VLHQDWLNGK EYKCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSREE 240
MTKNQVSLTC LVKGFYPSDI AVEWESNGQP ENNYKTTTPV LDSDGSFFLY SKLTVDKSRW 300
QQGNVFSCSV MHEALHNHYT QKSLSLSPGK 330

SEQ ID NO: 531 moltype = DNA length = 1326
FEATURE Location/Qualifiers
misc_feature 1..1326
 note = Engineered antibody sequence
source 1..1326
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 531
gagggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctgggggggc cctgagactc 60
tctctgtcag tctctggaat cggcctcagt agctactaca tgcaatgggt ccgtcaggct 120
ccagggaagg ggctggagtg ggtcggagtc attggtagtg atggtaaagc atactacgcg 180
acctgggcga aaggccgatt caccatctcc agagacaatt ccaagaccac ggtgtatctt 240
caaatgaaca gcctgagagc ttaggacact gctgtgtatt tctgtaccag aggggacatc 300
tggggccaag ggaccctcgt caccgtctcg agcgcctcca ccaagggccc atcgggtctt 360
ccccctggcac cctcctccaa gaggcacctct gggggcacag cggccctggg ctgcctggtc 420
aaggactact tcccgaacc ggtgacggtg tctggaact caggcgccct gaccagcggc 480
gtgcacacct tcccggtgt cctacagtcc tcaggactct actccctcag cagcgtggtg 540
accgtgcctt ccagcagctt gggcacccag acctacatct gcaacgtgaa tcacaagccc 600
agcaacacca aggtggacgc gagagttgag cccaatctt gtgacaaaac tcacacatgc 660
ccaccgtgcc cagcacctga actcctgggg ggaccgtcag tcttctctct ccccccaaaa 720
cccaaggaca cctcatgat ctcccgagc cctgaggtca catgcgtggt ggtggacgtg 780
agccacgaag accctgaggt caagttcaac tggtagctgg acggcgctgga ggtgcataat 840
gccaaagaaa agccgcggga ggagcagtag gccagcacgt accgtgtggt cagcgtcttc 900
accgtcctgc accaggactg gctgaatgac aaggagtaca agtgcaaggt ctccaacaaa 960
gccctccag ccccatcga gaaaaccatc tccaaagcca aagggcagcc ccgagaacca 1020
cagggtgata cctgcccccc atcccgggag gagatgacca agaaccaggt cagcctgacc 1080
tgctgtgtca aaggttcta tcccagcgac atcgccgtgg agtgggagag caatgggcag 1140

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ccggagaaca actacaagac cacgcctccc gtgctggact cgcacgggtc cttcttctc 1200
tacagcaagc tcacogtggg caagagcagg tggcagcagg ggaacgtctt ctcatgctcc 1260
gtgatgcatg aggctctgca caaccactac acgcagaaga gcctctccct gtctccgggt 1320
aatga 1326

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```

SEQ ID NO: 532      moltype = DNA length = 333
FEATURE            Location/Qualifiers
misc_feature        1..333
                    note = Engineered antibody sequence
source              1..333
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 532
gaggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctgggggggtc cctgagactc 60
tcctgtgcag tctctggaat cggcctcagt agctactaca tgcaatgggt ccgtcaggct 120
ccagggaagg ggctggagtg ggtcggagtc attggtagtg atggtgaagc atactacgcg 180
acctgggcga aaggccgatt caccatctcc agagacaatt ccaagaccac ggtgtatctt 240
caaatgaaca gcctgagagc tgaggacact gctgtgtatt tctgtaccag aggggacatc 300
tggggccaag ggaccctcgt caccgtctcg agc 333

```

```

SEQ ID NO: 533      moltype = DNA length = 90
FEATURE            Location/Qualifiers
misc_feature        1..90
                    note = Engineered antibody sequence
source              1..90
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 533
gaggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctgggggggtc cctgagactc 60
tcctgtgcag tctctggaat cggcctcagt 90

```

```

SEQ ID NO: 534      moltype = DNA length = 15
FEATURE            Location/Qualifiers
misc_feature        1..15
                    note = Engineered antibody sequence
source              1..15
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 534
agctactaca tgcaa 15

```

```

SEQ ID NO: 535      moltype = DNA length = 42
FEATURE            Location/Qualifiers
misc_feature        1..42
                    note = Engineered antibody sequence
source              1..42
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 535
tgggtccgtc aggtccagg gaaggggctg gagtgggtcg ga 42

```

```

SEQ ID NO: 536      moltype = DNA length = 48
FEATURE            Location/Qualifiers
misc_feature        1..48
                    note = Engineered antibody sequence
source              1..48
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 536
gtcattggta gtgatggtaa gacatactac gcgacctggg cgaaaggc 48

```

```

SEQ ID NO: 537      moltype = DNA length = 96
FEATURE            Location/Qualifiers
misc_feature        1..96
                    note = Engineered antibody sequence
source              1..96
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 537
cgattcacca tctccagaga caattccaag accacgggtg atcttcaa at gaacagcctg 60
agagctgagg aactgctgtg gtatttctgt accaga 96

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SEQ ID NO: 538      moltype = length =
SEQUENCE: 538
000

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SEQ ID NO: 539      moltype = DNA length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33

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source          note = Engineered antibody sequence
                1..33
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 539
tggggcccaag ggaccctcgt caccgtctcg agc                               33

SEQ ID NO: 540      moltype = DNA length = 993
FEATURE            Location/Qualifiers
misc_feature        1..993
                    note = Engineered antibody sequence
source              1..993
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 540
gcctccacca agggcccacg ggtcttcccc ctggcaccct cctccaagag caccctctggg 60
ggcacagcgg ccttggggtg cctgggtcaag gactacttcc ccgaaccggg gacgggtgtcg 120
tggaactcag gcgccttgac cagcggcgctg cacaccttcc cggctgtcct acagtctcctca 180
ggactctact ccttcagcag cgtgggtgacc gtgccctcca gcagcttggg caccagagacc 240
tacatctgca acgtgaatca caagcccagc aacaccaaagc tggacgcgag agttgagccc 300
aaatcttgtg acaaaactca cacatgccca ccgtgcccag cactgaact cctgggggga 360
ccgtcagtc tctctctccc cccaaaaccc aaggacaccc tcatgatctc ccggaccocct 420
gaggtcacat gcgtgggtgt ggacgtgagc cacgaagacc ctgaggtcaa gttcaactgg 480
tacgtggacg gcgtggaggt gcataatgcc aagacaaagc cgcgggagga gcagtacgcc 540
agcacgtacc gtgtgggtcag cgtcctcacc gtctgcacc aggactggct gaatggcaag 600
gagtagaagt gcaagggtct caacaaagcc ctcccagccc ccacgcagaa aaccatctcc 660
aaagccaaag ggcagccccc agaaccacag gtgtacaccc tgcccccatc ccgggaggag 720
atgaccaaga accagggtcag cctgacctgc ctgggtcaaag gcttctatcc cagcgacatc 780
gccgtggagt gggagagcaa tgggcagccg gagaacaact acaagaccac gcctcccgtg 840
ctggactccg acggctcctt ctctctctac agcaagctca ccgtggacaa gagcagggtg 900
cagcagggga acgtctcttc atgctccgtg atgcatgagg ctctgcacaa ccactacacg 960
cagaagagcc tctccctgtc tccgggtaaa tga                               993

SEQ ID NO: 541      moltype = AA length = 219
FEATURE            Location/Qualifiers
REGION              1..219
                    note = Engineered antibody sequence
source              1..219
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 541
QVLTQSPSSL SASVGDRVTI NCQASQNVYN NNYLAWYQQK PGKVPKQLIY STSTLASGVP 60
SRFSGSGSGT DFTLTISLQ PEDVATYYCL GSYDCSRGDC FVFGGGTKVE IKRTVAAPSV 120
FIFPPSDEQL KSGTASVVCL LNNFYPREAK VQWKVDNALQ SGNSQESVTE QDSKSTYSL 180
SSTLTLSKAD YEKHKVYACE VTHQGLSSPV TKSFNRGEC                               219

SEQ ID NO: 542      moltype = AA length = 113
FEATURE            Location/Qualifiers
REGION              1..113
                    note = Engineered antibody sequence
source              1..113
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 542
QVLTQSPSSL SASVGDRVTI NCQASQNVYN NNYLAWYQQK PGKVPKQLIY STSTLASGVP 60
SRFSGSGSGT DFTLTISLQ PEDVATYYCL GSYDCSRGDC FVFGGGTKVE IKR          113

SEQ ID NO: 543      moltype = AA length = 22
FEATURE            Location/Qualifiers
REGION              1..22
                    note = Engineered antibody sequence
source              1..22
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 543
QVLTQSPSSL SASVGDRVTI NC                               22

SEQ ID NO: 544      moltype = AA length = 13
FEATURE            Location/Qualifiers
REGION              1..13
                    note = Engineered antibody sequence
source              1..13
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 544
QASQNVYNNN YLA                               13

SEQ ID NO: 545      moltype = AA length = 15
FEATURE            Location/Qualifiers

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REGION	1..15	
	note = Engineered antibody sequence	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 545		
WYQKPQKVP KQLIY		15
SEQ ID NO: 546	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = Engineered antibody sequence	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 546		
STSTLAS		7
SEQ ID NO: 547	moltype = AA length = 32	
FEATURE	Location/Qualifiers	
REGION	1..32	
	note = Engineered antibody sequence	
source	1..32	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 547		
GVPSRFSGSG SGTDFTLTIS SLQPEDVATY YC		32
SEQ ID NO: 548	moltype = AA length = 13	
FEATURE	Location/Qualifiers	
REGION	1..13	
	note = Engineered antibody sequence	
source	1..13	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 548		
LGSYDCSRGD CFV		13
SEQ ID NO: 549	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Engineered antibody sequence	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 549		
FGGGTKVEIK R		11
SEQ ID NO: 550	moltype = AA length = 106	
FEATURE	Location/Qualifiers	
REGION	1..106	
	note = Engineered antibody sequence	
source	1..106	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 550		
TVAAPSVFIF PPSDEQLKSG TASVVCLLNN FYPREAKVQW KVDNALQSGN SQESVTEQDS		60
KDSTYLSLST LTLKADYK HKVYACEVTH QGLSPVTKS FNRGEC		106
SEQ ID NO: 551	moltype = DNA length = 660	
FEATURE	Location/Qualifiers	
misc_feature	1..660	
	note = Engineered antibody sequence	
source	1..660	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 551		
caagtgtctga cccagtctcc atcctccctg tctgcatctg taggagacag agtcaccatc		60
aattgccagg ccagtcagaa tgtttacaat aacaactacc tagcctggta tcagcagaaa		120
ccagggaag ttcttaagca actgatctat tctacatcca ctctggcctc tgggggtccca		180
tctcgtttca gtggcagtg atctgggaca gatttcactc tcaccatcag cagcctgcag		240
cctgaagatg ttgcaactta ttactgtctg ggcagttatg attgtagtcg tgggtgattgt		300
tttgttttcg gcggaggaac caaggtggaa atcaaacgta cggtggctgc accatctgtc		360
ttcatcttcc cgccatctga tgagcagttg aaatctggaa ctgcctctgt tgtgtgacctg		420
ctgaataact tctatccag agaggccaaa gtacagtggg aggtggataa cgccctccaa		480
tcgggtaact ccaggagag tgtcacagag caggacagca aggacagcac ctacagcctc		540
agcagcaccg tgacgctgag caaagcagac tacgagaaac acaaagtcta cgccctgcgaa		600
gtcaccatc agggcctgag ctgcgccgtc acaaagagct tcaacagggg agagtgttag		660

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SEQ ID NO: 552      moltype = DNA length = 339
FEATURE            Location/Qualifiers
misc_feature        1..339
                    note = Engineered antibody sequence
source              1..339
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 552
caagtgctga cccagtctcc atcctccctg tctgcatctg taggagacag agtcaccatc 60
aattgccagg ccagtcagaa tgtttacaat aacaactacc tagcctggta tcagcagaaa 120
ccagggaagag ttcctaagca actgatctat tctacatcca ctctggcatc tgggggtcca 180
tctcgtttca gtggcagtg atctgggaca gatttcactc tcaccatcag cagcctgcag 240
cctgaagatg ttgcaactta ttactgtctg ggcagttatg attgtagtcg tggtgattgt 300
ttgttttcg gcggaggaac caaggtggaa atcaaactg 339

SEQ ID NO: 553      moltype = DNA length = 66
FEATURE            Location/Qualifiers
misc_feature        1..66
                    note = Engineered antibody sequence
source              1..66
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 553
caagtgctga cccagtctcc atcctccctg tctgcatctg taggagacag agtcaccatc 60
aattgc 66

SEQ ID NO: 554      moltype = DNA length = 39
FEATURE            Location/Qualifiers
misc_feature        1..39
                    note = Engineered antibody sequence
source              1..39
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 554
caggccagtc agaattgtta caataacaac tacctagcc 39

SEQ ID NO: 555      moltype = DNA length = 45
FEATURE            Location/Qualifiers
misc_feature        1..45
                    note = Engineered antibody sequence
source              1..45
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 555
tggatcagc agaaaccagg gaaagttcct aagcaactga tctat 45

SEQ ID NO: 556      moltype = DNA length = 21
FEATURE            Location/Qualifiers
misc_feature        1..21
                    note = Engineered antibody sequence
source              1..21
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 556
tctacatcca ctctggcatc t 21

SEQ ID NO: 557      moltype = DNA length = 96
FEATURE            Location/Qualifiers
misc_feature        1..96
                    note = Engineered antibody sequence
source              1..96
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 557
gggggtccat ctcgtttcag tggcagtgga tctgggacag atttcaactc caccatcagc 60
agcctgcagc ctgaagatgt tgcaacttat tactgt 96

SEQ ID NO: 558      moltype = DNA length = 39
FEATURE            Location/Qualifiers
misc_feature        1..39
                    note = Engineered antibody sequence
source              1..39
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 558
ctgggcagtt atgattgtag tctgtgtgat tgttttgtt 39

SEQ ID NO: 559      moltype = DNA length = 33
FEATURE            Location/Qualifiers

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misc_feature      1..33
                  note = Engineered antibody sequence
source            1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 559
ttcggcggag gaaccaaggt ggaaatcaaa cgt                               33

SEQ ID NO: 560      moltype = DNA length = 321
FEATURE            Location/Qualifiers
misc_feature        1..321
                  note = Engineered antibody sequence
source              1..321
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 560
acggtggtctg caccatctgt cttcatcttc cgcctatctg atgagcagtt gaaatctgga 60
actgcctctg  ttgtgtgctt gctgaataac ttctatccca gagaggccaa agtacagtgg 120
aaggtggata  acgcctctcca atcgggtaac tcccaggaga gtgtcacaga gcaggacagc 180
aaggacagca  cctacagcct  cagcagcacc ctgacgctga gcaaagcaga ctacgagaaa 240
cacaaagtct  acgcctgcga  agtcacccat cagggcctga gctgcgccgt cacaaagagc 300
ttcaacaggg  gagagtgtta  g                               321

SEQ ID NO: 561      moltype = AA length = 37
FEATURE            Location/Qualifiers
VARIANT            37
source              1..37
                  mol_type = protein
                  organism = Homo sapiens

SEQUENCE: 561
ACDTATCVTH RLAGLLSRSG GVVKNFVPT NVGSKAF                               37

SEQ ID NO: 562      moltype = AA length = 37
FEATURE            Location/Qualifiers
VARIANT            37
source              1..37
                  mol_type = protein
                  organism = Homo sapiens

SEQUENCE: 562
ACNTATCVTH RLAGLLSRSG GMVKS NFVPT NVGSKAF                               37

SEQ ID NO: 563      moltype = AA length = 106
FEATURE            Location/Qualifiers
source              1..106
                  mol_type = protein
                  organism = Homo sapiens

SEQUENCE: 563
TVAAPSVFIF PPSDEQLKSG TASVVCLLNN FYPREAKVQW KVDNALQSGN SQESVTEQDS 60
KDYSLSSST LTLKADYK HKVYACEVTH QGLSSPVTKS FNRGEC                     106

SEQ ID NO: 564      moltype = AA length = 330
FEATURE            Location/Qualifiers
REGION            1..330
                  note = Engineered antibody sequence
source              1..330
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 564
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 60
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKRVPE KCDKTHTCP PCPAPELLGG 120
PSVFLFPPKP KDTLMISRTPEVTCVVVDVS HEDPEVKFNW YVDGVEVHNA KTKPREEQYA 180
STYRVVSVLT VHLQDWLNGK EYCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSREE 240
MTKNQVSLTC LVKGFYPSDI AVEWESNGQP ENNYKTTTPV LDSGGSFFLY SKLTVDKSRW 300
QQGNVFSCSV MHEALHNHYT QKSLSLSPGK                                     330

SEQ ID NO: 565      moltype = AA length = 329
FEATURE            Location/Qualifiers
REGION            1..329
                  note = Engineered antibody sequence
source              1..329
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 565
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 60
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKRVPE KCDKTHTCP PCPAPELLGG 120
PSVFLFPPKP KDTLMISRTPEVTCVVVDVS HEDPEVKFNW YVDGVEVHNA KTKPREEQYA 180
STYRVVSVLT VHLQDWLNGK EYCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSREE 240
MTKNQVSLTC LVKGFYPSDI AVEWESNGQP ENNYKTTTPV LDSGGSFFLY SKLTVDKSRW 300
QQGNVFSCSV MHEALHNHYT QKSLSLSPG                                     329

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SEQ ID NO: 566      moltype = AA  length = 440
FEATURE            Location/Qualifiers
REGION             1..440
                   note = Engineered antibody sequence
source             1..440
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 566
EVQLVESGGG LVQPGGSLRL SCAVSGIDLS GYYMNWVRQA PGKGLEWVG V IGINGATYYA 60
SWAKGRFTIS RDNSKTTVYL QMNSLRAEDT AVYFCARGDI WQQGLTVTVS SASTKGPSVF 120
PLAPSSKSTS GGTAAALGCLV KDYPPEPVTV SWNSGALTSG VHTFPVLQ S SGLYSLSSV 180
TVPSSSLGTQ TYICNVNHPK SNTKVDARVE PKSCDKHTC PPCPAPELLG GPSVFLFPPK 240
PKDTLMISRT PEVTCVVVDV SHEDPEVKFN WYVDGVEVHN AKTKPREEQY ASTYRVVSVL 300
TVLHQDWLNG KEYCKVSNK ALPAPIEKTI SKAKGQPREP QVYTLPPSRE EMTKNQVSLT 360
CLVKGFPYPSD IAVFWEESNGQ PENNYKTPPP VLDSDGSFPL YSKLTVDKSR WQGNVVFSCS 420
VMHEALHNHY TQKSLSLSPG                                     440

SEQ ID NO: 567      moltype = DNA  length = 1323
FEATURE            Location/Qualifiers
misc_feature        1..1323
                   note = Engineered antibody sequence
source              1..1323
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 567
gaggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctgggggggc cctgagactc 60
tctgtgcagc tctctggaat cgacctcagt ggctactaca tgaactgggt ccgtcaggct 120
ccagggaagg ggctggagtg ggtcggagtc attggtatta atggtgccac atactacgcg 180
agctgggcga aaggccgatt caccatctcc agagacaatt ccaagaccac ggtgatctt 240
caaatgaaca gcctgagagc tgaggacact gctgtgtatt tctgtgctag aggggacatc 300
tggggccaa gggacctcgt caccgtctcg agcgctcca ccaagggccc atcggtcttc 360
cccctggcac cctcctccaa gagcacctct gggggcacag cggccctggg ctgcctggtc 420
aaggactact tcccgaaccc ggtgacggtg tctgtggaact caggcgccct gaccagcggc 480
gtgcacacct tcccggctgt cctacagtcc tcaggactct actccctcag cagcggtggtg 540
accgtgccct ccagcagctt gggcacccag acctacatct gcaacgtgaa tcacaagccc 600
agcaaacacca aggtggagcg gagagttgag cccaaatctt gtgacaaaac tcacacatgc 660
ccaccgtgcc cagcacctga actcctgggg ggaccgtcag tcttctctct ccccccaaaa 720
cccaaggaca cctcatgat ctcccgagac cctgaggtca catgcgtggt ggtggacgtg 780
agccacgaag accctgaggt caagttcaac tggtagctgg acggcgtgga ggtgcataat 840
gccaaagcaa agccgcggga ggagcagtag gccagcaggt accgtgtggt cagcgctctc 900
accgtcctgc accaggactg gctgaatggc aaggagtaca agtgcaaggt ctccaacaaa 960
gccctccag cccccaatcg gaaaaccatc tccaaagcca aagggcagcc cggagaacca 1020
caggtgtaca cctgcctccc atccccggag gagatgacca agaaccaggt cagcctgacc 1080
tgccctggtc aaggtctcta tcccagcgac atcgccgtgg agtgggagag caatggggag 1140
ccggagaaca actacaagac caccgctccc gtgctggact ccgacggctc cttctctctc 1200
tacagcaagc tcaccgtgga caagagcagg tggcagcagg ggaacgtctt ctcatgctcc 1260
gtgatgcatg aggtcttgcg caaccactac acgcagaaga gcctctccct gtctccgggt 1320
tga                                     1323

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What is claimed is:

1. A method of individual therapy in a patient suffering from migraine, comprising:

- (a) identifying a most bothersome symptom (MBS) of the patient, wherein the MBS is a symptom which is associated with the migraine of the patient and is most bothersome to the patient;
- (b) intravenously administering to the patient an effective amount of an anti-calcitonin gene related peptide (CGRP) antibody;
- (c) assessing changes in the MBS after the administration of (a); and
- (d) if the MBS is improved in the assessment of (c), further intravenously administering to the patient an effective amount of the anti-CGRP antibody,

wherein the anti-CGRP antibody comprises:

- (A) a heavy chain variable domain (VH) comprising heavy chain complementarity-determining region (CDR) 1, 2, and 3 polypeptide sequences of SEQ ID NOS: 204, 206, and 208, respectively; and
- (B) a light chain variable domain (VL) comprising light chain CDR 1, 2, and 3 polypeptide sequences of SEQ ID NO: 224, 226, and 228, respectively.

2. The method of claim 1, wherein the assessing in (c) is performed at 1-12 hours, within 1 month, within 3 months, or within 6 months from the administering in (a).

3. The method of claim 1, wherein:

- (A) the amino acid sequence of the VH comprises SEQ ID NO: 202; and/or
- (B) the amino acid sequence of the VL comprises SEQ ID NO: 222.

4. The method of claim 1, wherein the anti-CGRP comprises:

- (A) a heavy chain polypeptide comprising SEQ ID NO: 201 or SEQ ID NO: 566; and/or
- (B) a light chain polypeptide comprising SEQ ID NO: 221.

5. The method of claim 1, wherein the MBS identified in (a) is not headache and not pain.

6. The method of claim 1, wherein the MBS identified in (a) is selected from the group consisting of: sensitivity to light (photophobia); nausea and/or vomiting; sensitivity to sound (phonophobia); aura; throbbing and/or pulsation; cognitive disruption; fatigue; mood changes; sensitivity to smell (osmophobia or olfactophobia); visual impact; pressure and/or tightness; dizziness; inactivity; sensory disturbance; sleep disturbance; and speech difficulty.

7. The method of claim 1, wherein the MBS identified in (a) is selected from the group consisting of: vomiting; throbbing and/or pulsation; cognitive disruption; fatigue; mood changes; sensitivity to smell (osmophobia or olfactophobia); visual impact; pressure and/or tightness; dizziness; inactivity; sensory disturbance; sleep disturbance; and speech difficulty. 5

8. The method of claim 1, wherein the effective amount in (b) and/or (d) is between about 100 mg and about 300 mg, optionally about 100 mg, about 125 mg, about 150 mg, about 175 mg, about 200 mg, about 225 mg, about 250 mg, about 275 mg, or about 300 mg. 10

9. The method of claim 1, wherein the administering in (d) is performed about 10-14 weeks, optionally 11-13 weeks, further optionally about 12 weeks or about 3 months, after the administering in (b). 15

10. The method of claim 1, wherein the administering in (d) is performed every 10-14 weeks, optionally every 11-13 weeks, further optionally every 12 weeks.

11. The method of claim 1, wherein the patient has chronic migraine when the administering in (b) is performed. 20

12. The method of claim 1, wherein the patient has episodic migraine when the administering in (b) is performed. 25

* * * * *